

Generative AI for Software Development





Facebook interface for the page **somkiat.cc**. The top navigation bar includes the Facebook logo, the page name **somkiat.cc**, a search icon, and user profile icons for **Somkiat** and **Home**. The main navigation bar features **Page**, **Messages**, **Notifications** (with a red badge showing 3), **Insights**, **Publishing Tools**, **Settings**, and **Help**.

The left sidebar shows the page name **somkiat.cc** and handle **@somkiat.cc**. Below this is a menu with **Home**, **Posts**, **Videos**, and **Photos**.

The main content area displays a large video of a man in a white Superman t-shirt with "SOMKIAT.CC" printed on it, posing against a white wall. A blue call-to-action button is overlaid on the video, reading "Help people take action on this Page." with a close icon. Below the video, there are buttons for **Liked**, **Following**, **Share**, and a three-dot menu. A blue button labeled **+ Add a Button** is also visible.



**[https://github.com/up1/
workshop-ai-with-technical-team](https://github.com/up1/workshop-ai-with-technical-team)**



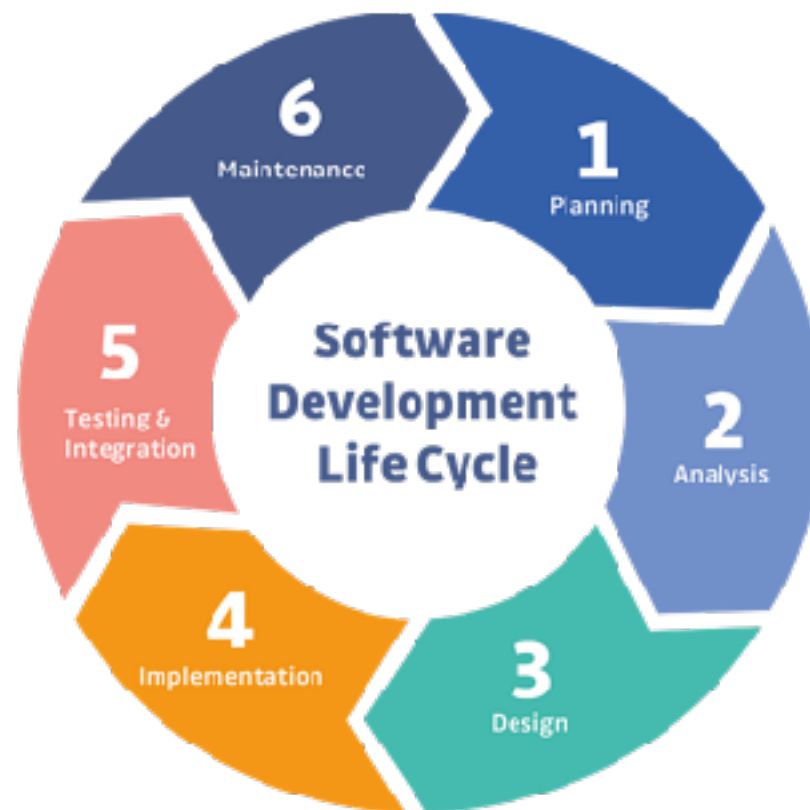
Goals

Integrate Generative AI in Software Development

Optimize code quality

Team up with AI on coding tasks

Develop innovative solutions



Learning Path

AI/ML

AI Model

Prompt Engineer

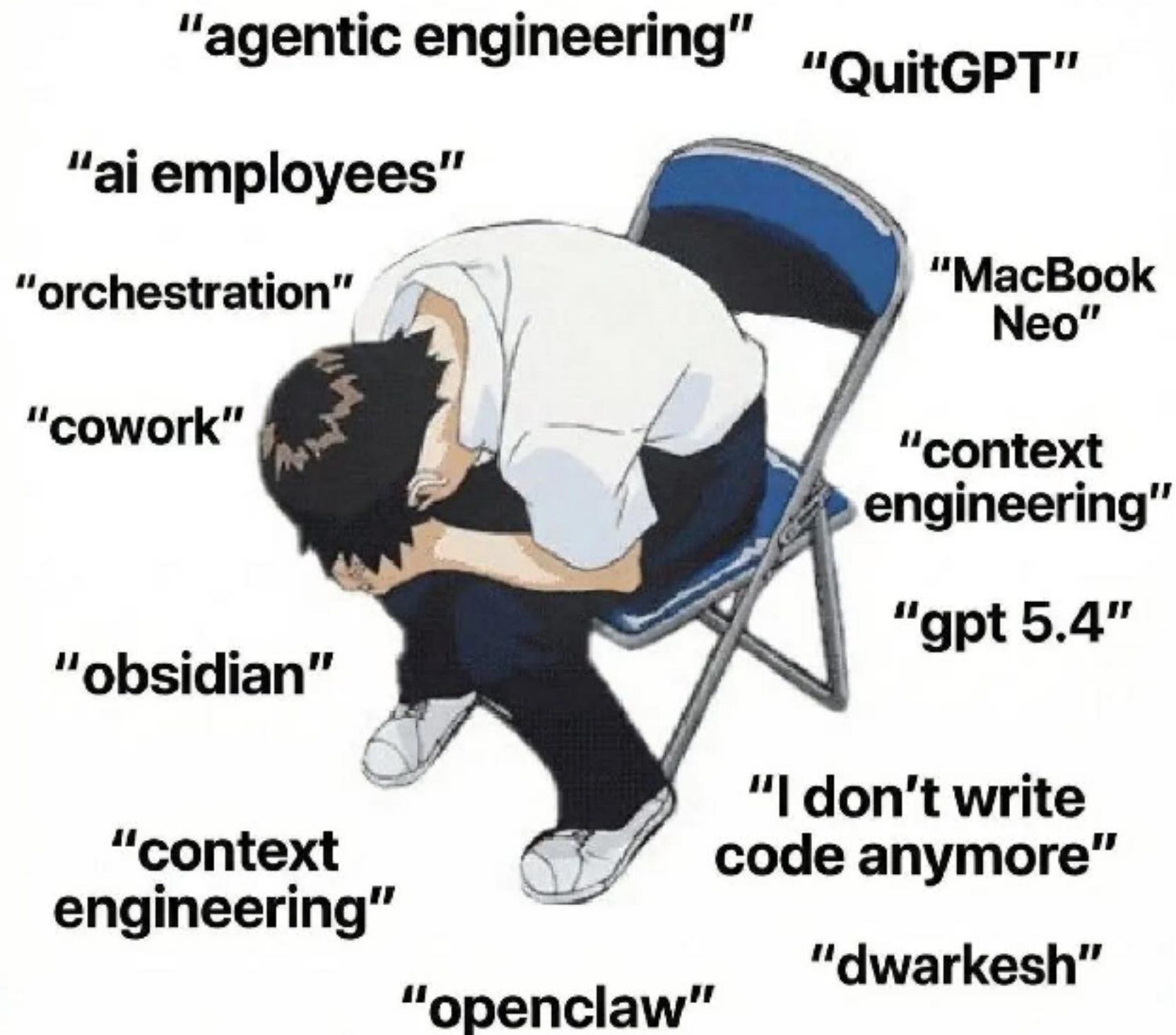
AI in Software
development

Develop AI/LLM
app

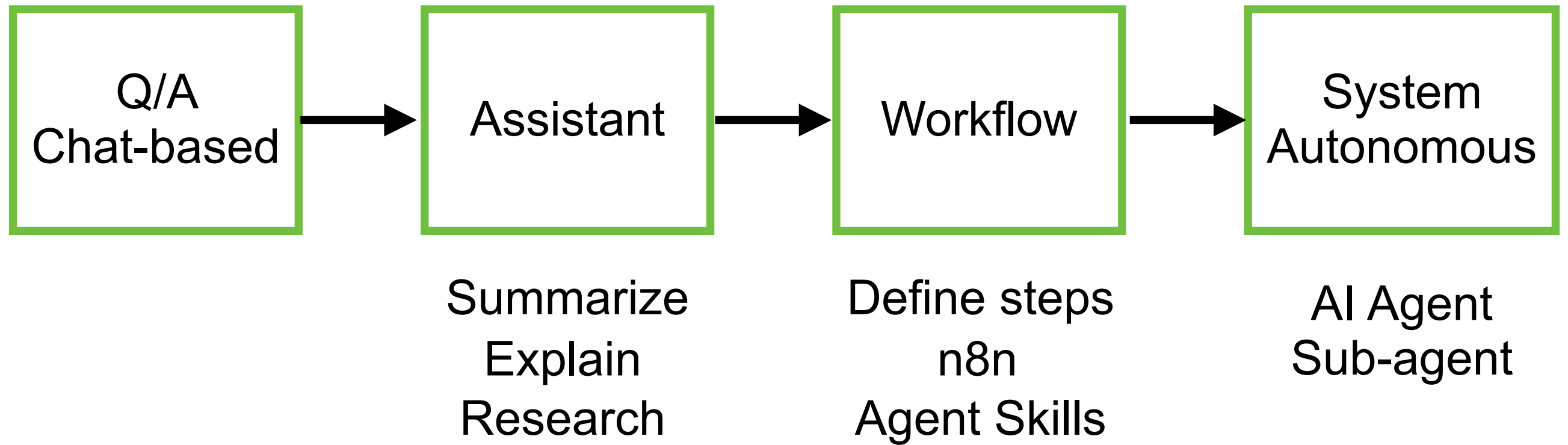
RAG

Use cases and workshops

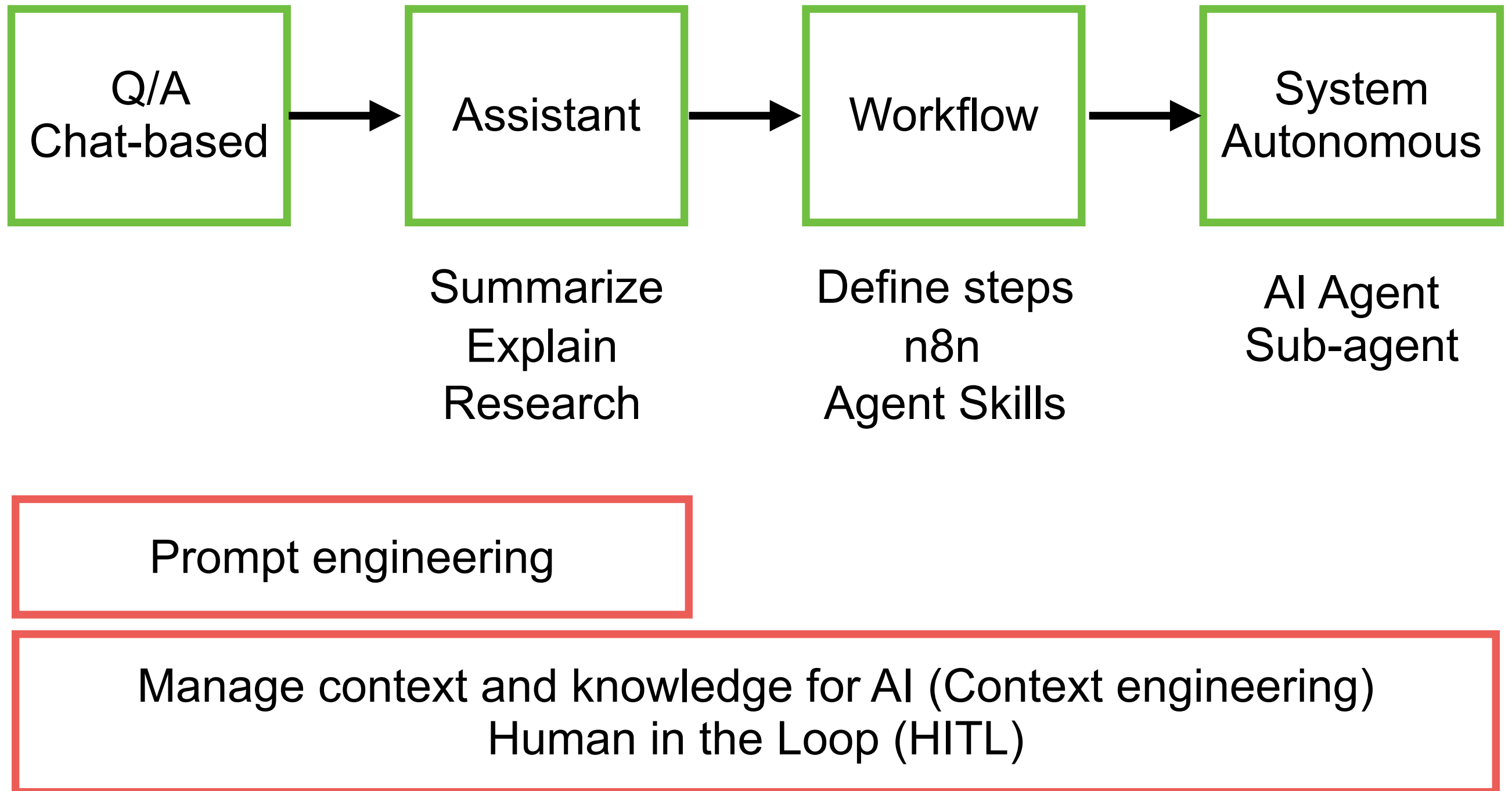




Steps to use AI ?



Steps to use AI ?



Software Development

Requirement

Design

Develop

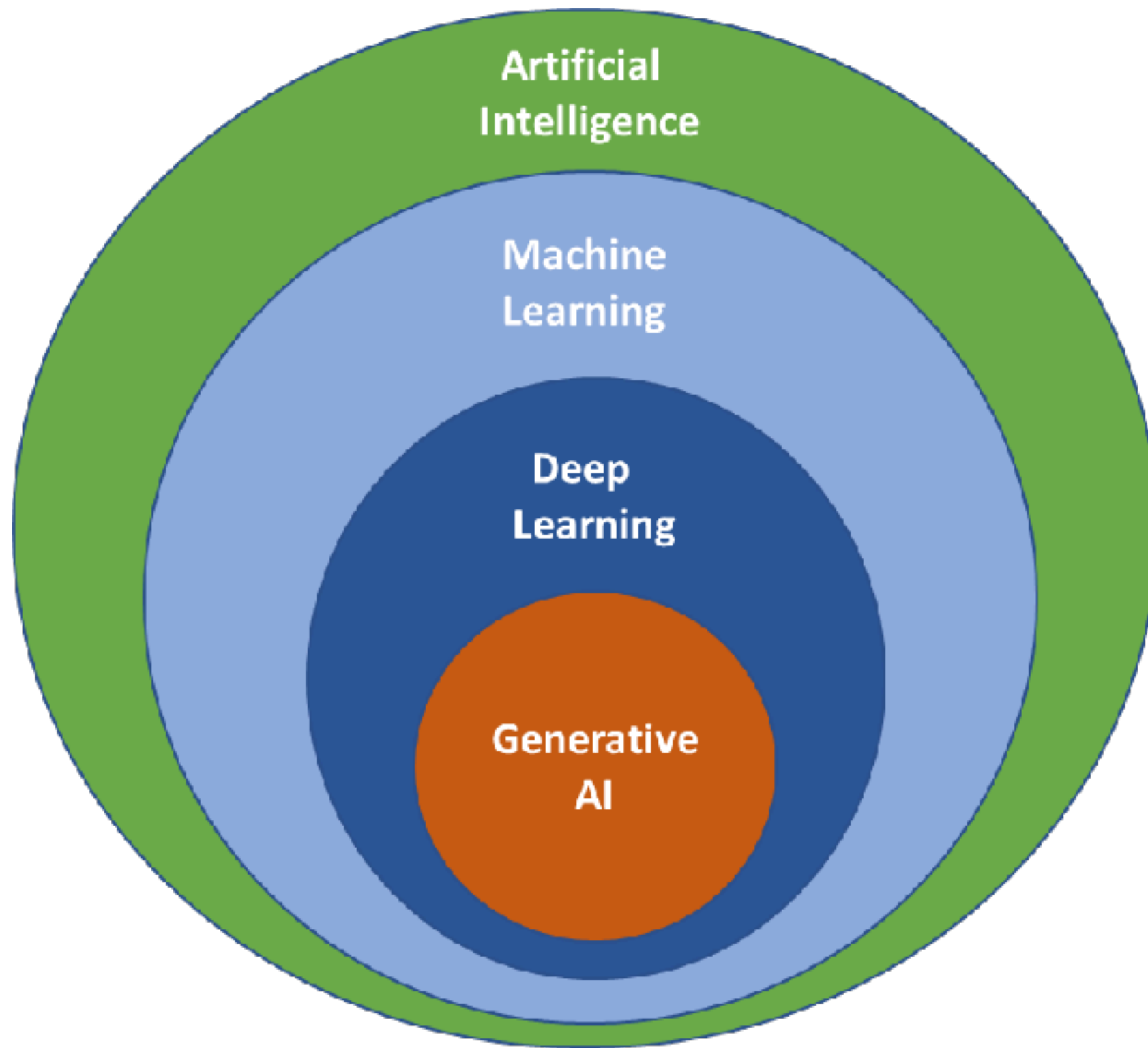
Testing

Deploy

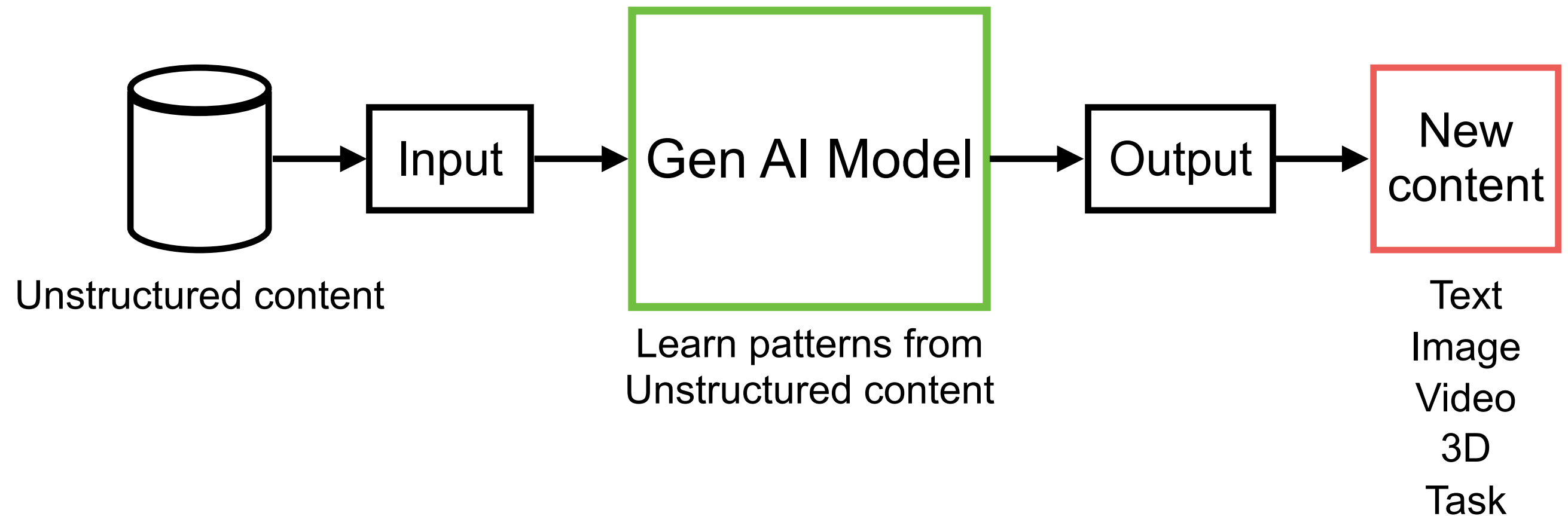
Generative AI

Improve Productivity ... (Replace human !!)





Generative AI



<https://grow.google/ai-essentials/>



Generative AI

LLMs

Large Language Models

Text generation
Code generation
Chatbot
Conversation AI

GANs

Generative Adversarial Network

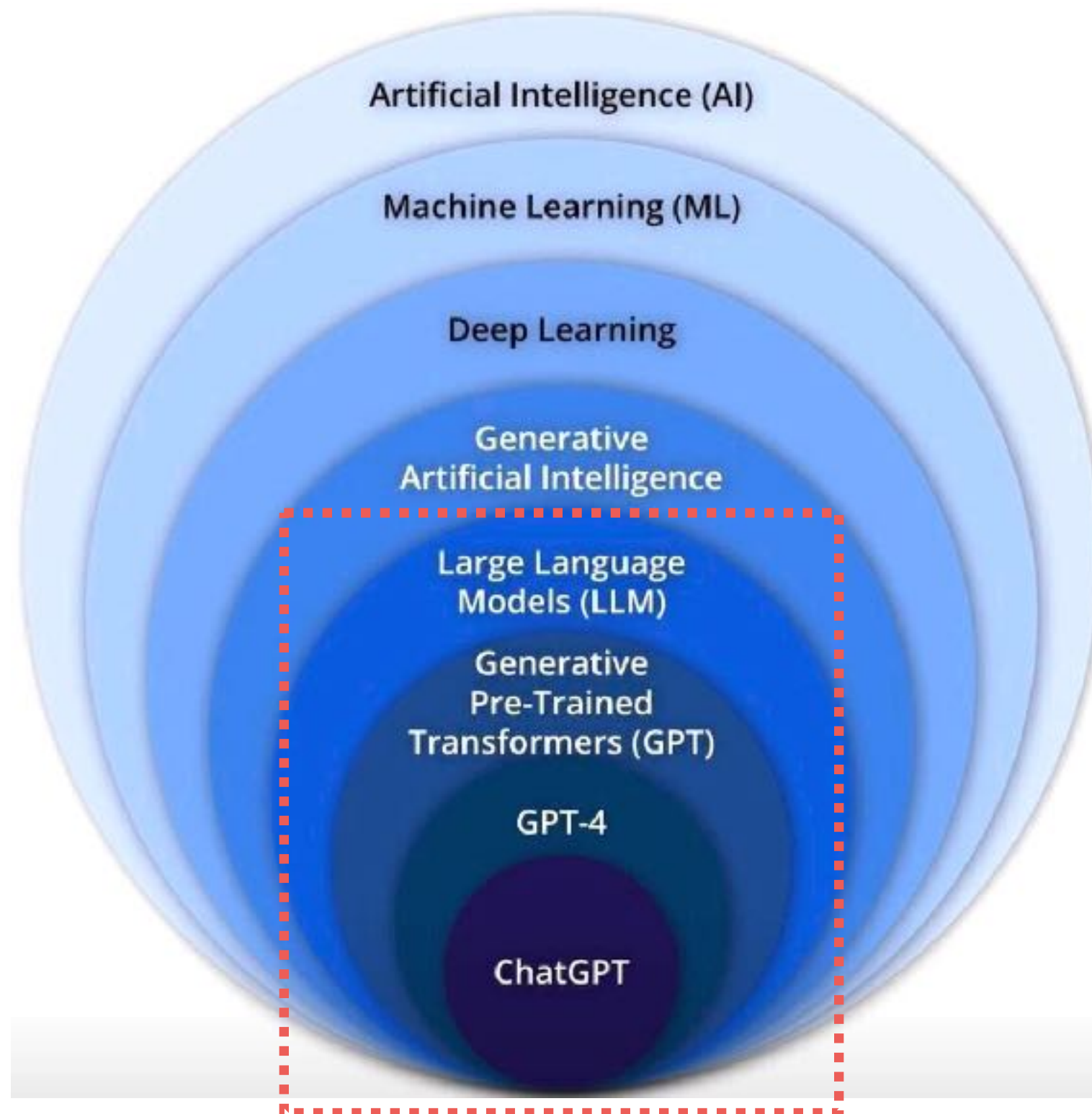
Image generation
Deep fake
Art creation
Simulate financial market

VAEs

Variational Autoencoders

Data compression
Synthetic data generation
Image reconstruction





Large Language Model (LLM)

Type of AI model
Process, understand
Generate human readable data

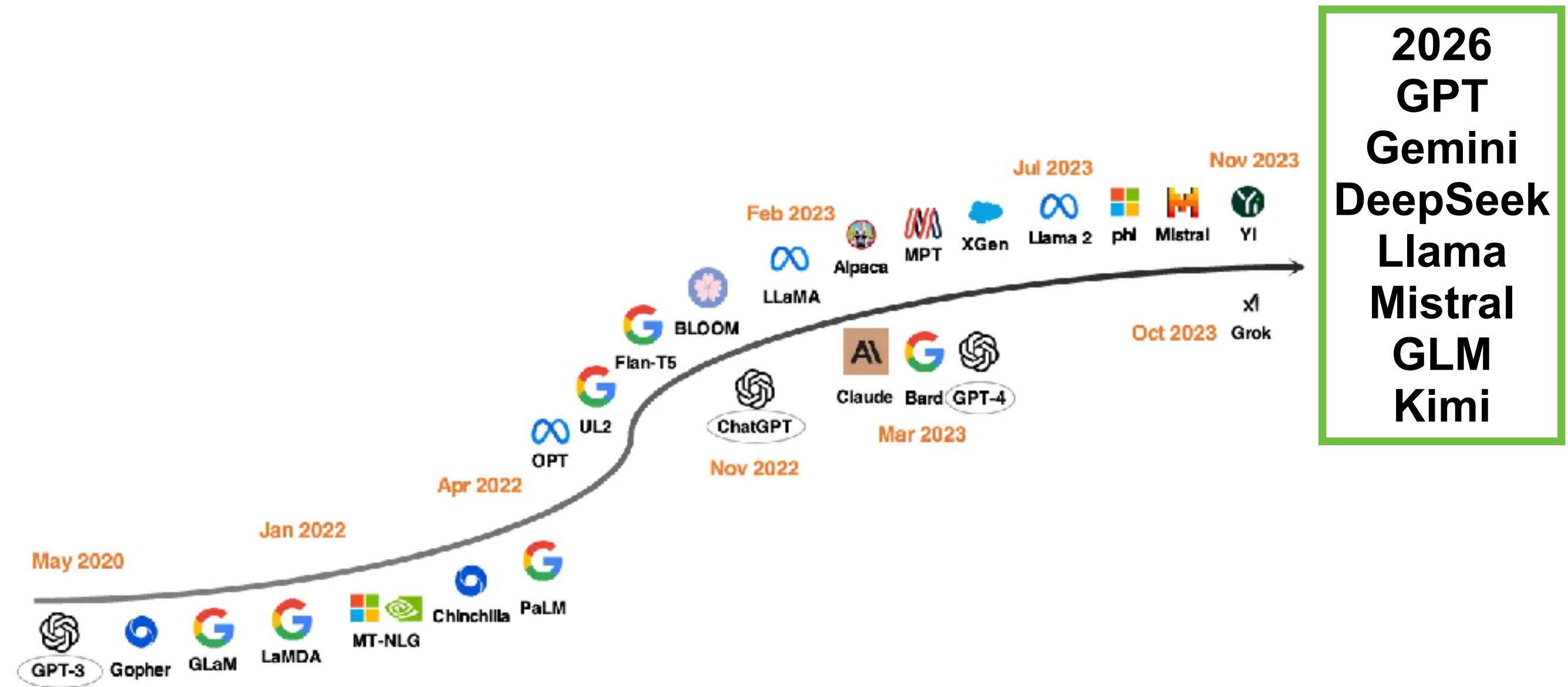


LLMs in industry

Model name	Company
Bidirectional Encoder Representation from Transformers (BERT)	Google AI
Generative Pre-trained transformer-5 (GPT-5)	OpenAI
Pathways Language Model-E (PaLM-E)	Google AI
BLOOM	NVIDIA AI
Llama 4	Facebook
Claude 4.5 Sonnet/Opus	Anthropic



LLM Development timeline



2026
GPT
Gemini
DeepSeek
Llama
Mistral
GLM
Kimi

Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

<https://arxiv.org/abs/2311.16989>



Let's go !!



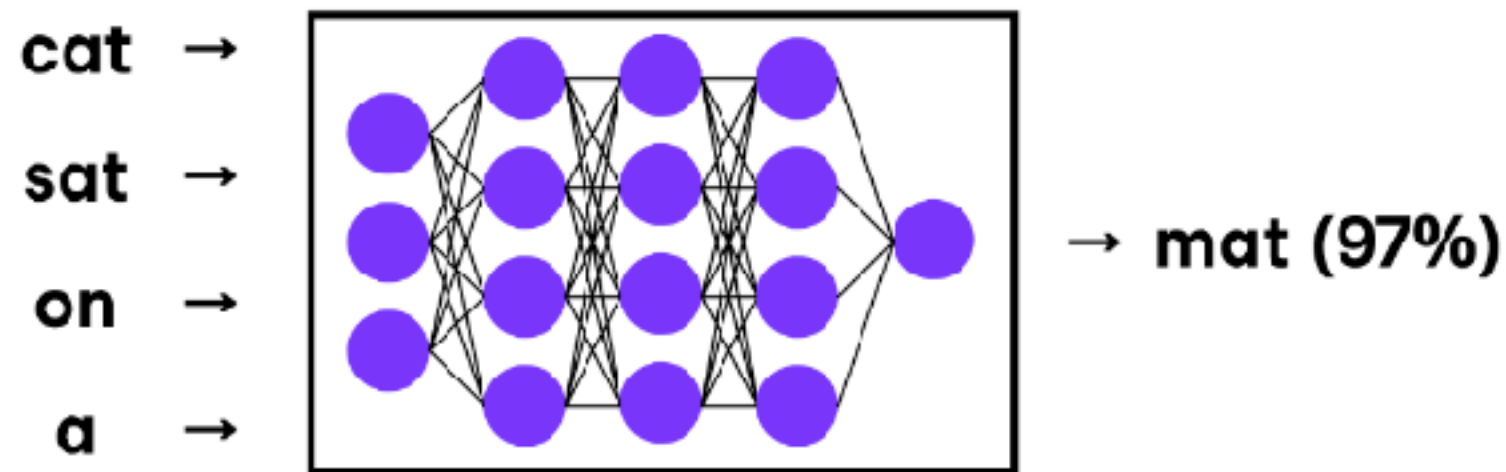
สี่เหลืง



Large Language Model (LLM)

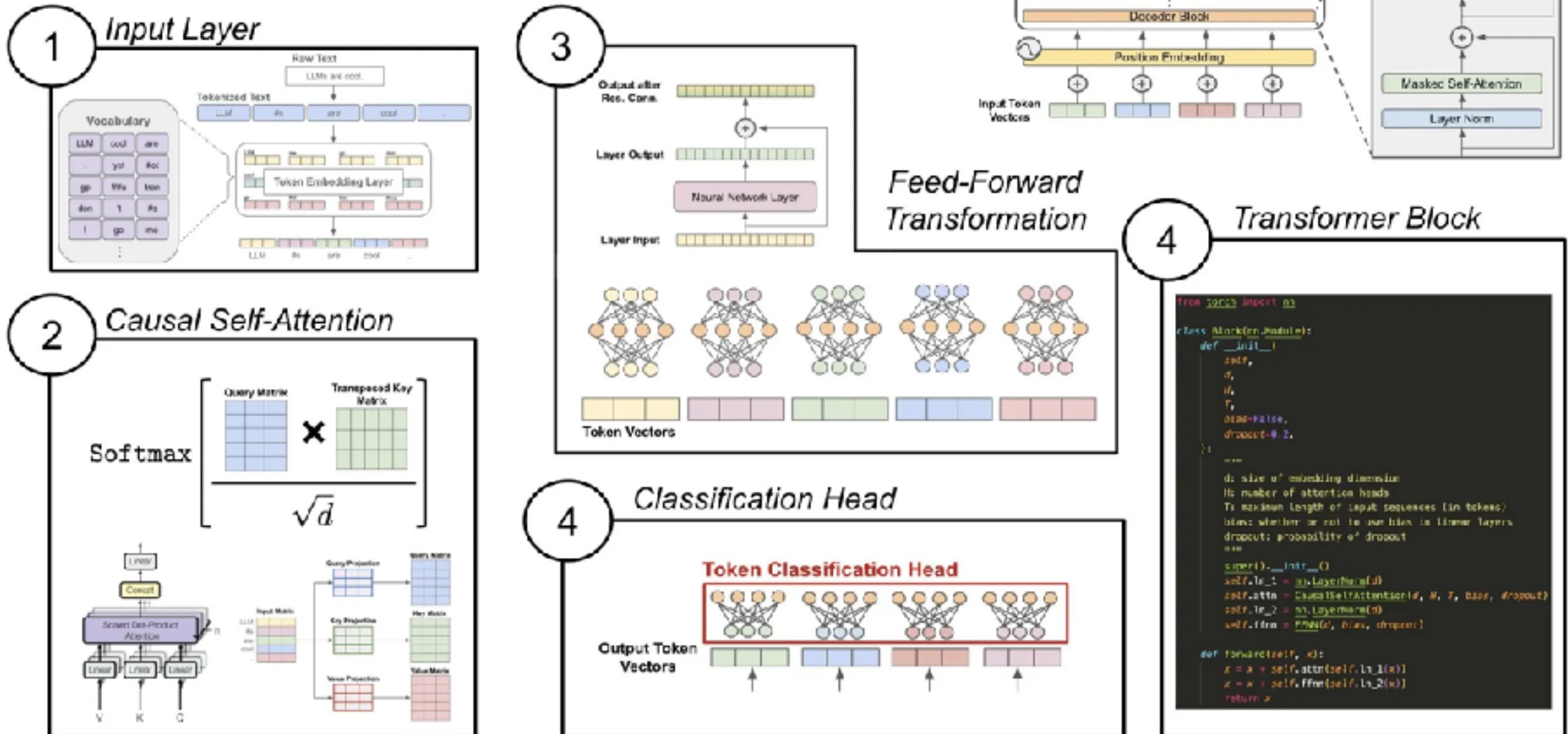
Neural network

Predicts the next word in a sequence



LLM components !!

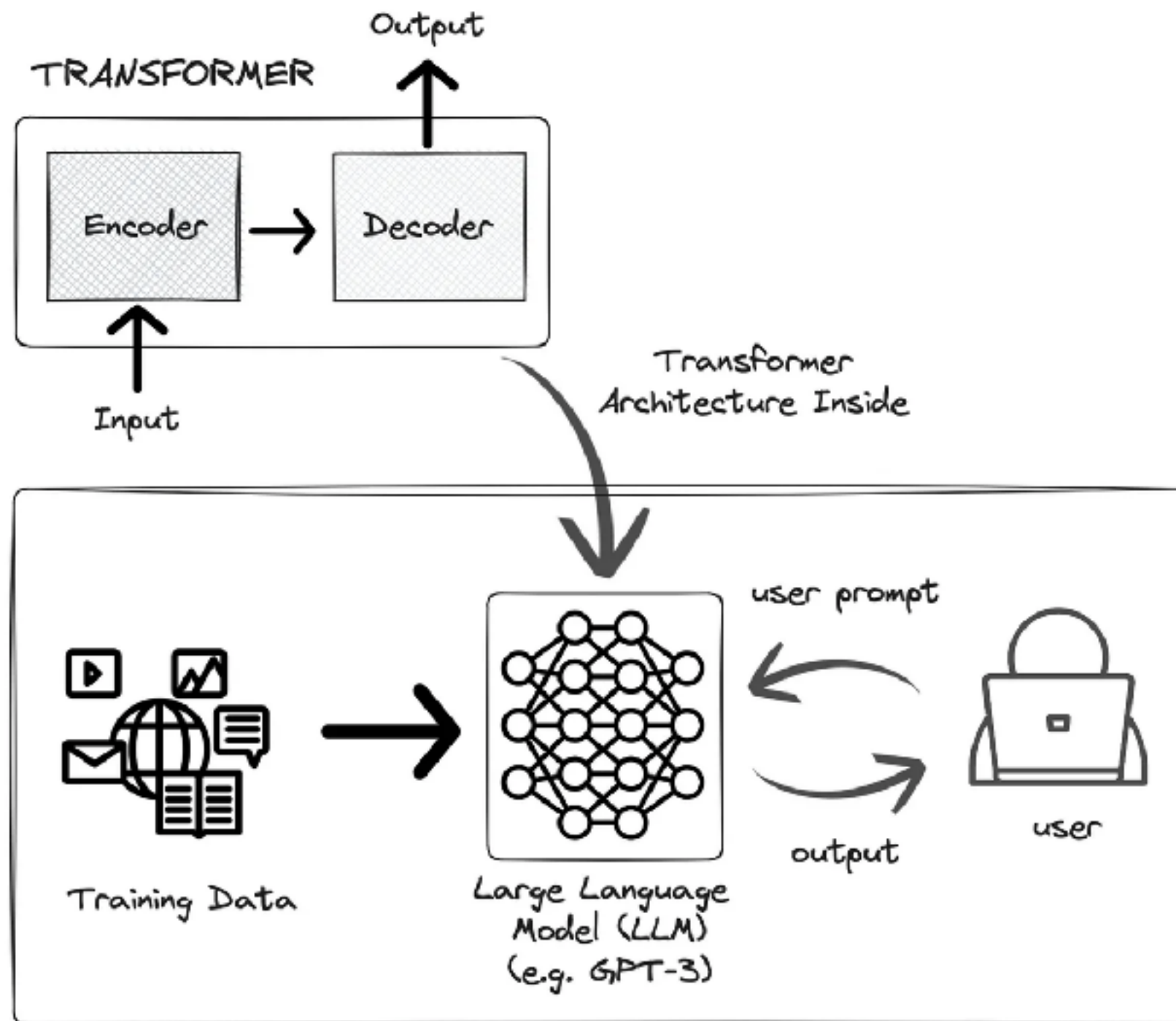
Components of the Decoder-only Transformer



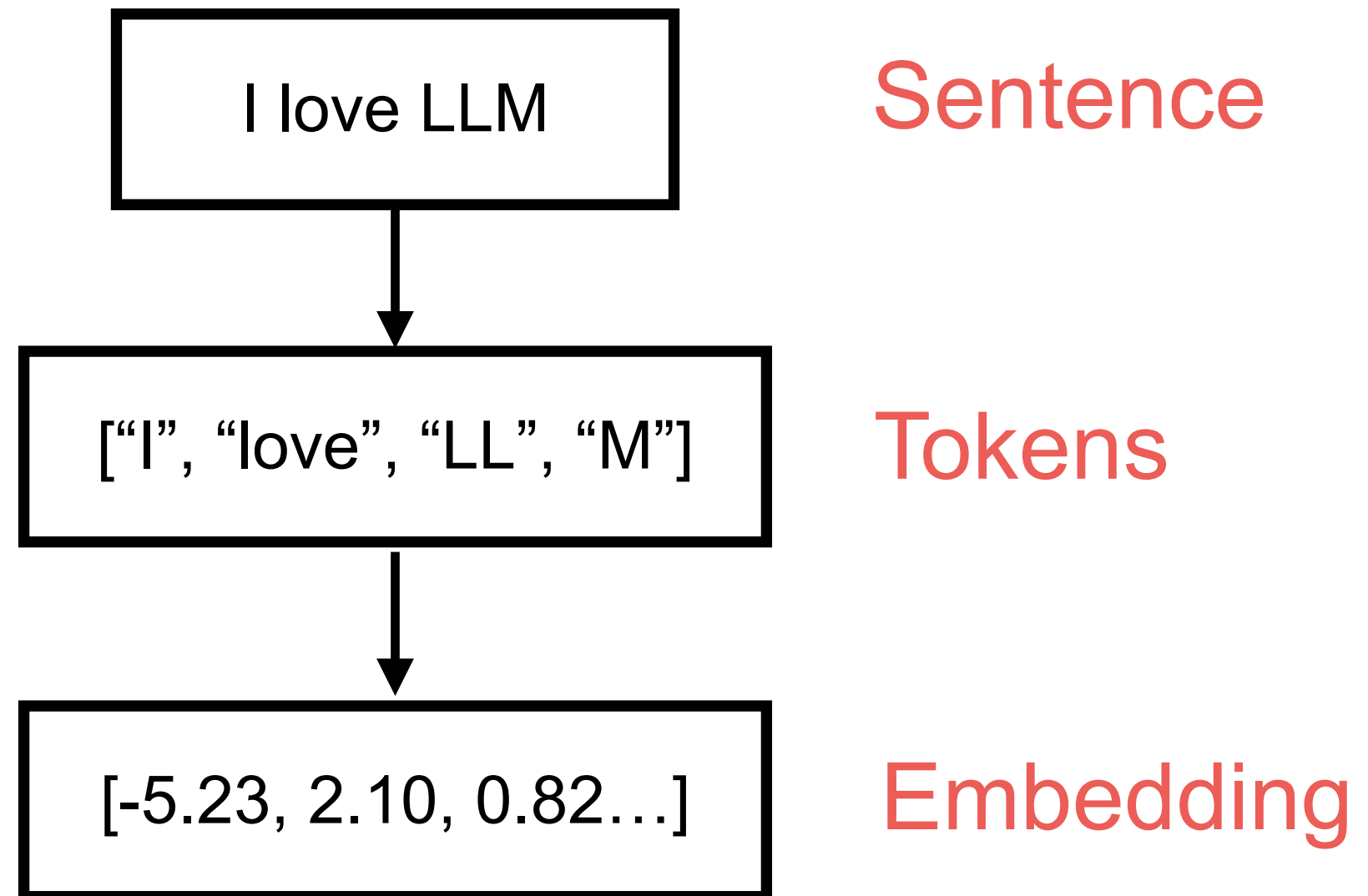
<https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much>



Transformer inside



Transformer process



OpenAI Tokenizer

GPT-4o & GPT-4o mini (coming soon) **GPT-3.5 & GPT-4** GPT-3 (Legacy)

ประเทศไทย

Clear Show example

Tokens Characters
10 9

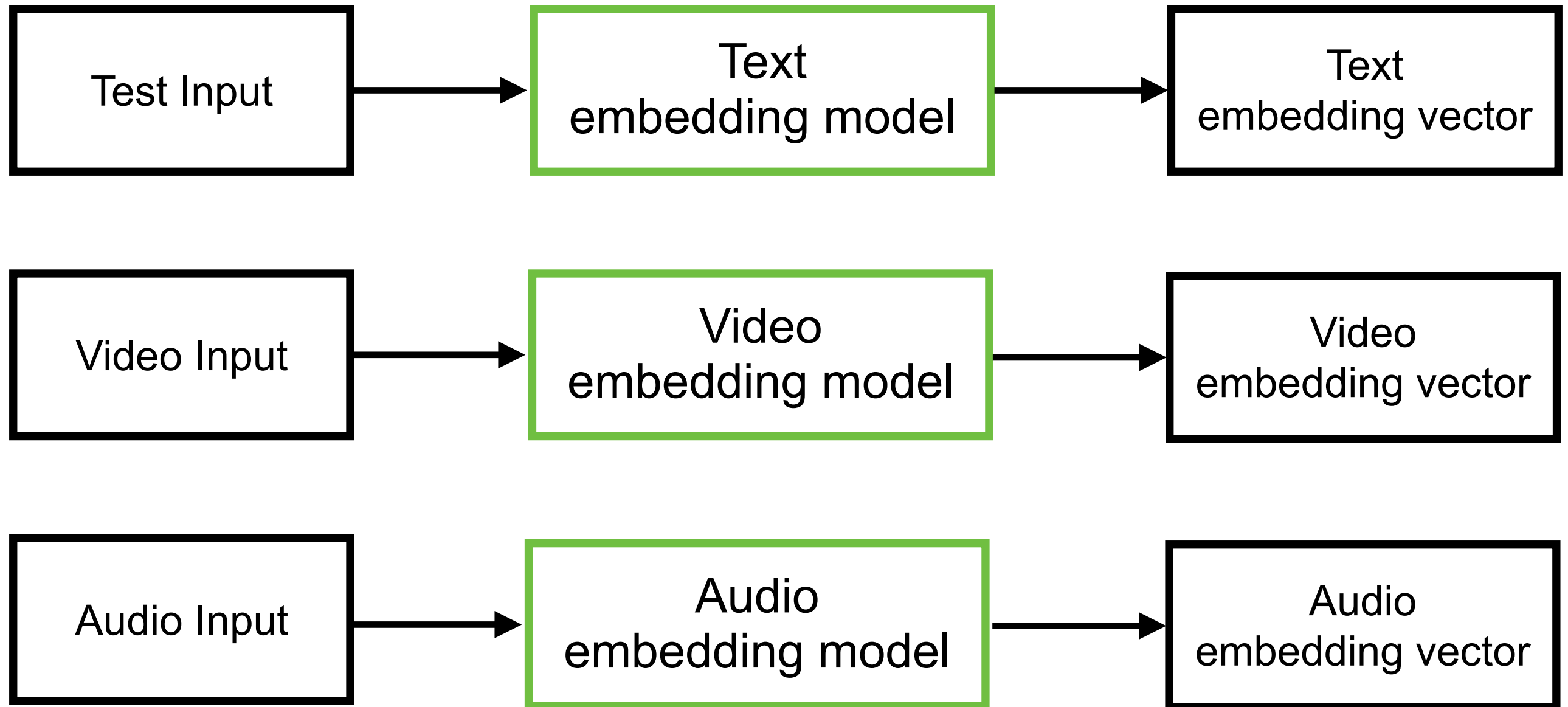
ประเทศไทย

Text Token IDs

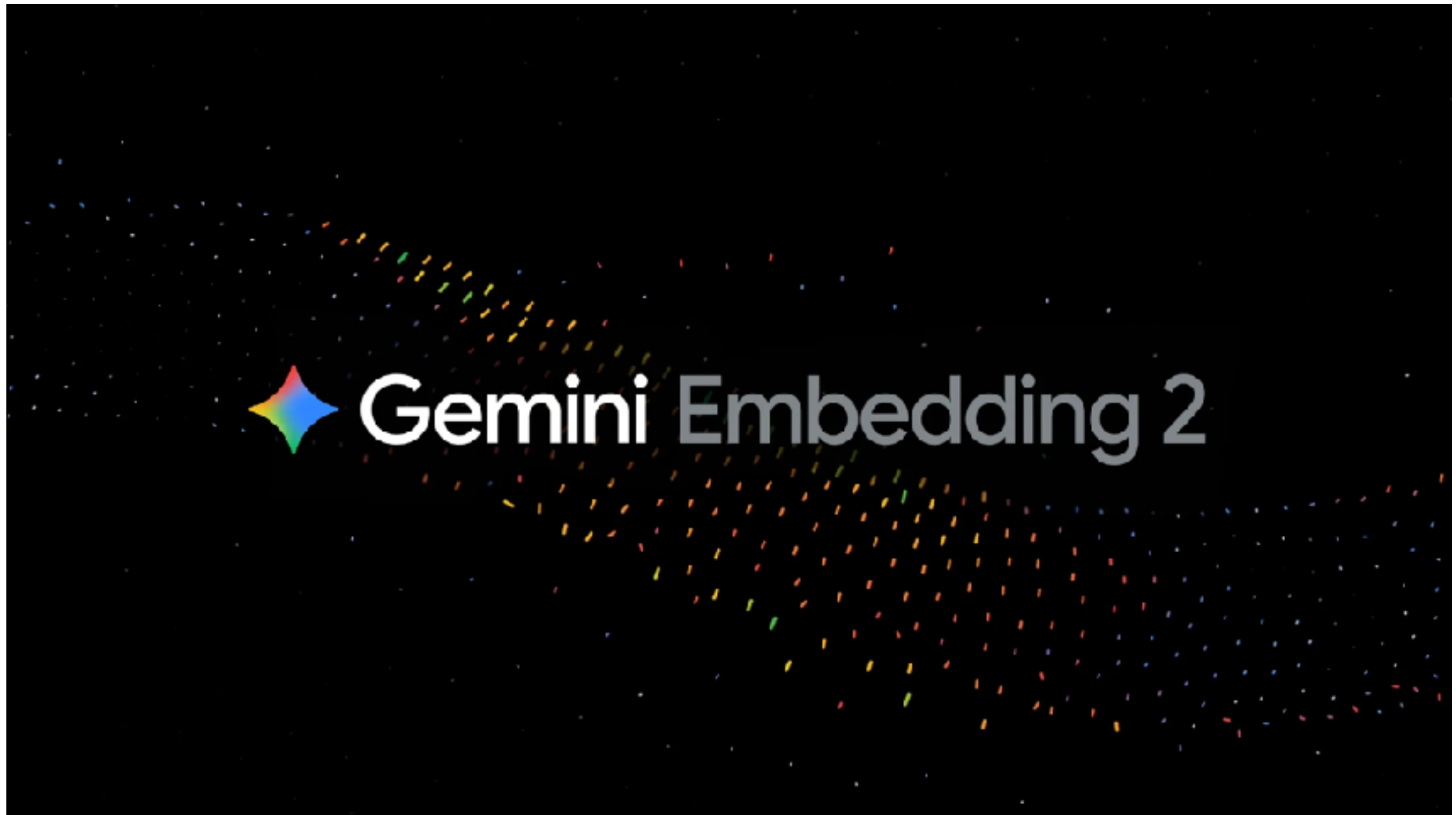
<https://platform.openai.com/tokenizer>



Embedding



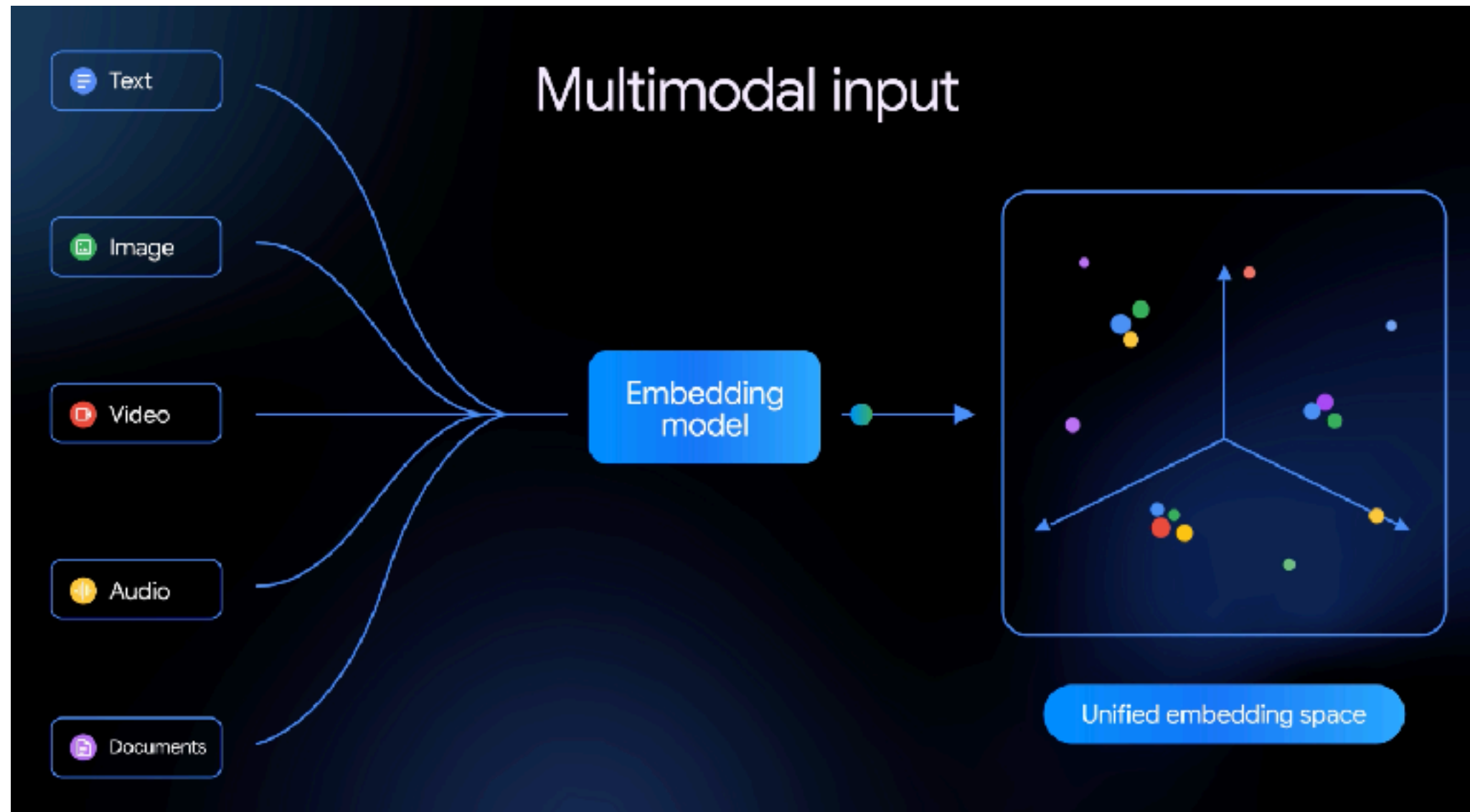
Multimodal Embedding Model



<https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/>



Multimodal Embedding Model



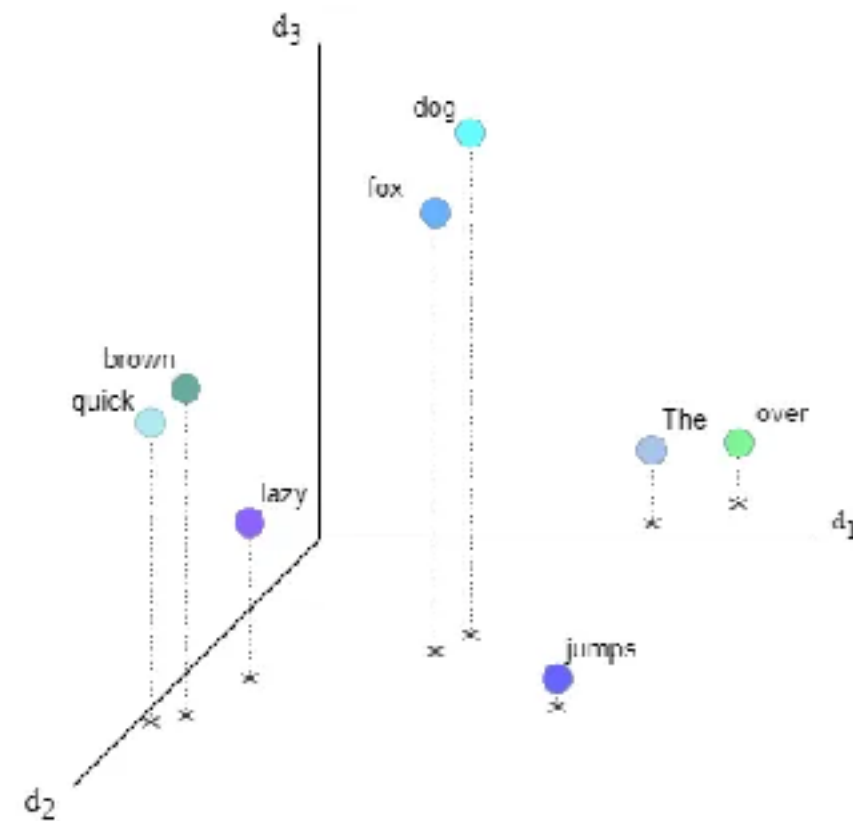
<https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-embedding-2/>



Embedding ?

Map items of **unstructured data** to high-dimensional real vectors

		0	1	2		d_{model}
The	→	0.64	-0.09	0.23		0.005
quick	→	0.05	0.79	0.47		-0.54
brown	→	0.12	0.74	0.51		-0.3
fox	→	0.42	0.52	0.83		0.01
jumps	→	0.88	0.69	0.02		0.27
over	→	0.84	0.15	0.13		0.05
the	→	0.64	0.09	0.23		0.005
lazy	→	0.1	0.65	0.28		-0.19
dog	→	0.54	0.49	0.90		0.000



<https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca>



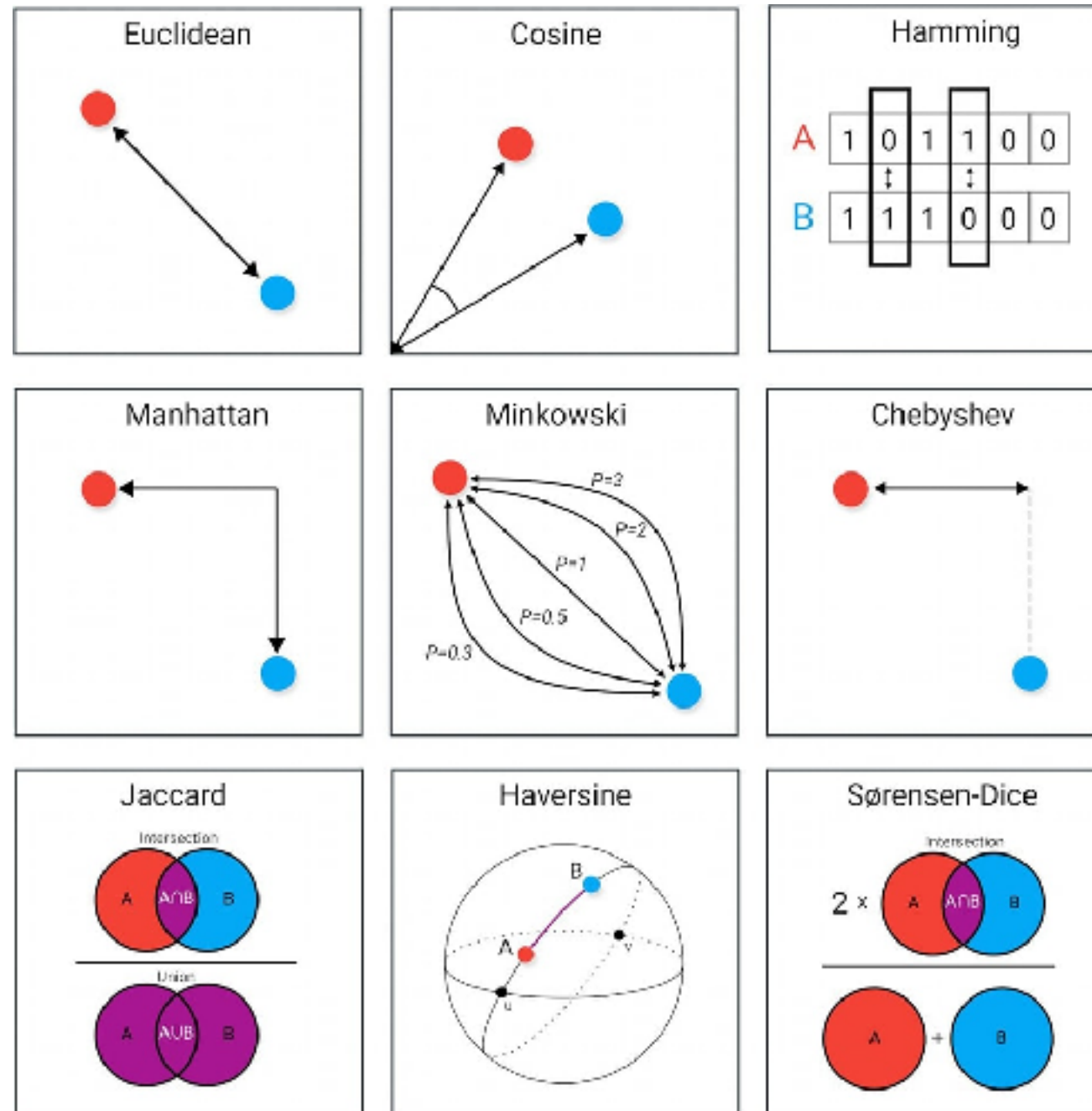
Embedding Leaderboard

Rank (Boxed)	Model	Embedding Dim	Max Tokens	Mean (T...	Mean (TaskT...	Bitext ...	Classification	Clustering	Instructi
1	Kalm-Embedding-Gemma3-12B-2511	3840	32768	72.32	62.51	83.76	77.88	55.77	5.49
2	llama-embed-nemotron-8b	4096	32768	69.46	61.69	81.72	73.21	54.35	10.82
3	Qwen3-Embedding-8B	4096	32768	70.53	61.69	80.89	74.60	57.65	10.86
4	gemini-embedding-001	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18
5	Qwen3-Embedding-4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56
6	Octen-Embedding-8B	4096	32768	67.84	60.28	80.35	66.68	55.68	8.98
7	F2LLM-v2-14B	5120	40960	68.74	59.45	77.15	73.60	60.91	0.62
8	F2LLM-v2-8B	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85
9	Seed1.6-embedding-1215	2048	32768	70.26	61.34	78.68	76.75	56.78	-0.82
10	F2LLM-v2-4B	2560	40960	67.06	58.25	74.49	70.73	59.53	2.25
11	jina-embeddings-v5-text-small	1024	32768	67.00	58.90	69.71	71.32	53.41	1.35

<https://huggingface.co/spaces/mteb/leaderboard>



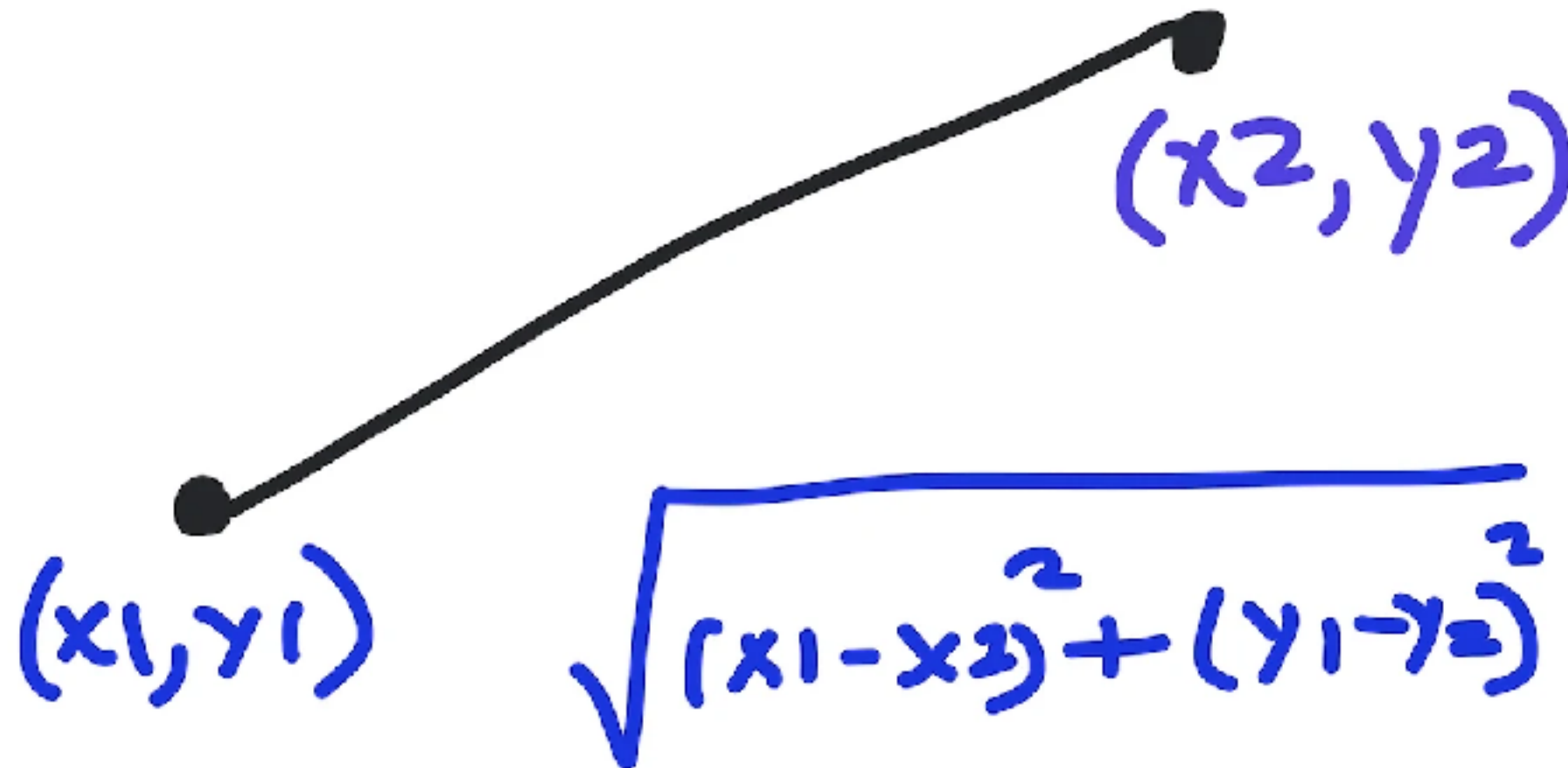
Distance measure in Data Science



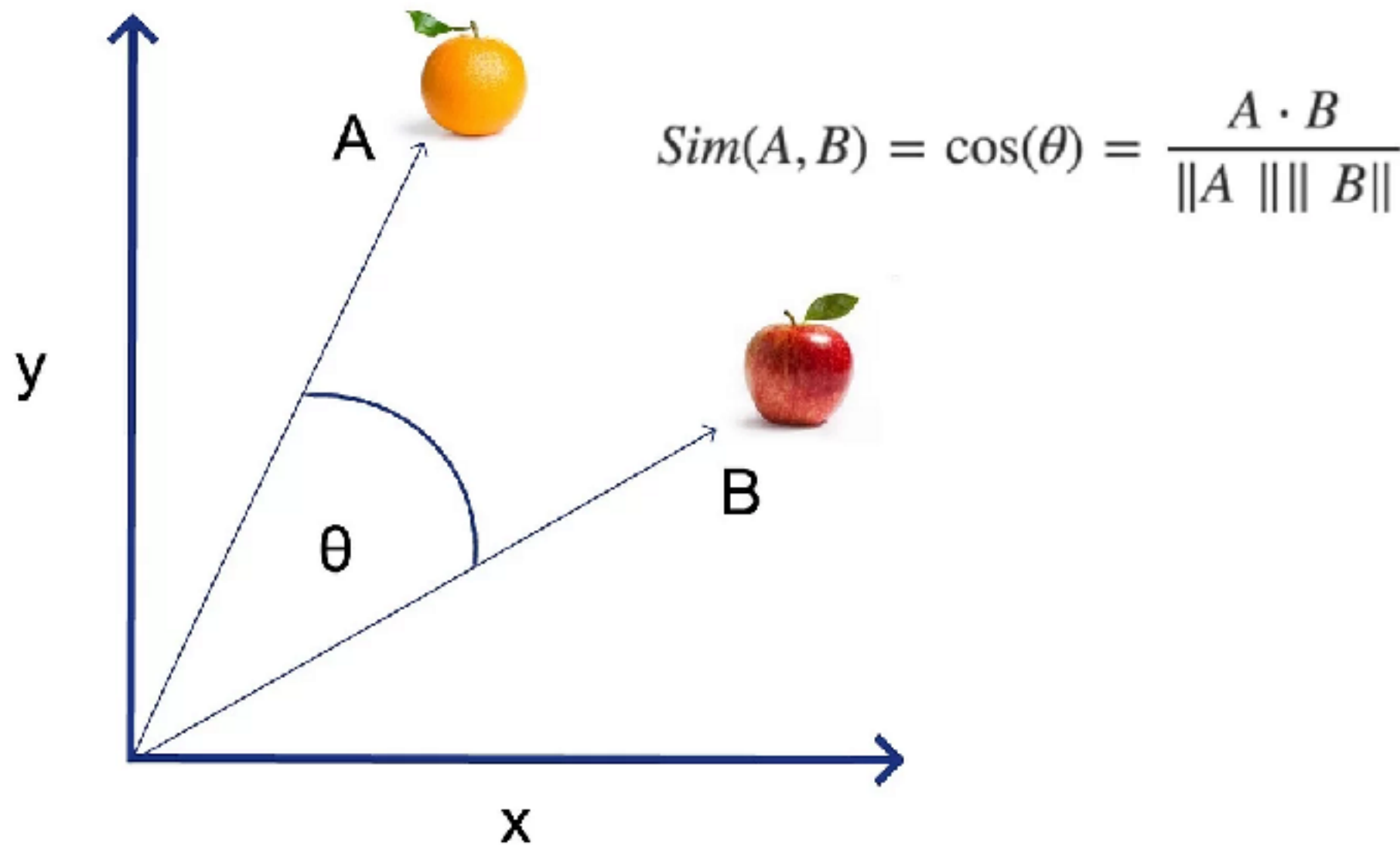
<https://www.maartengrootendorst.com/blog/distances/>



Euclidean Distance



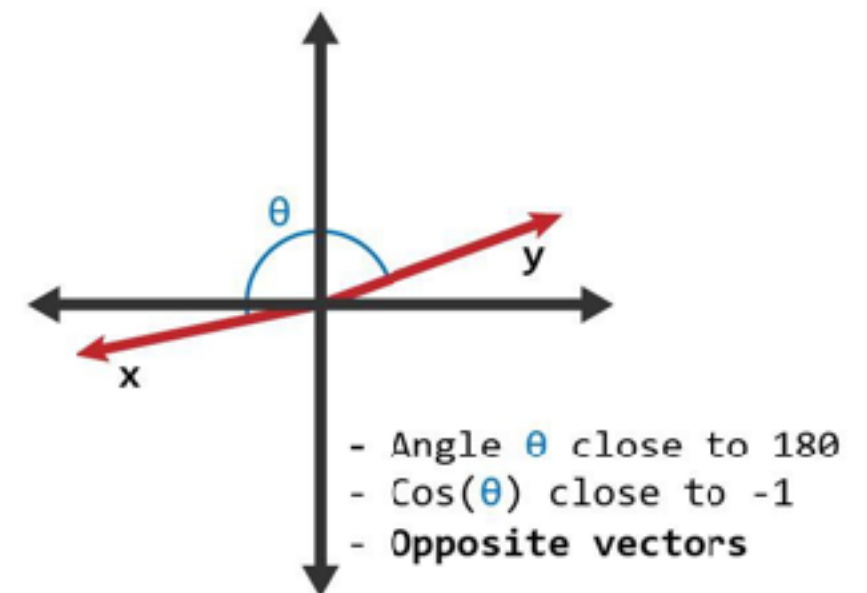
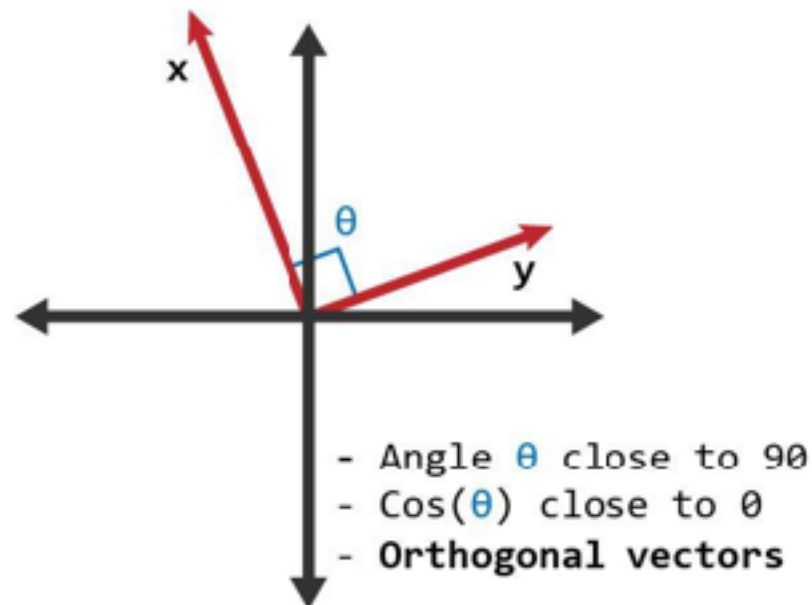
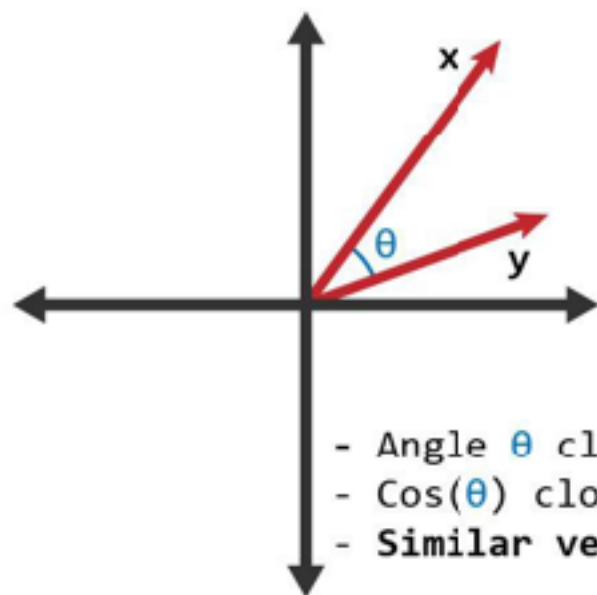
Cosine Similarity



<https://python.plainenglish.io/mastering-cosine-metrics-a-data-scientists-essential-toolkit-3ce95eda91f9>



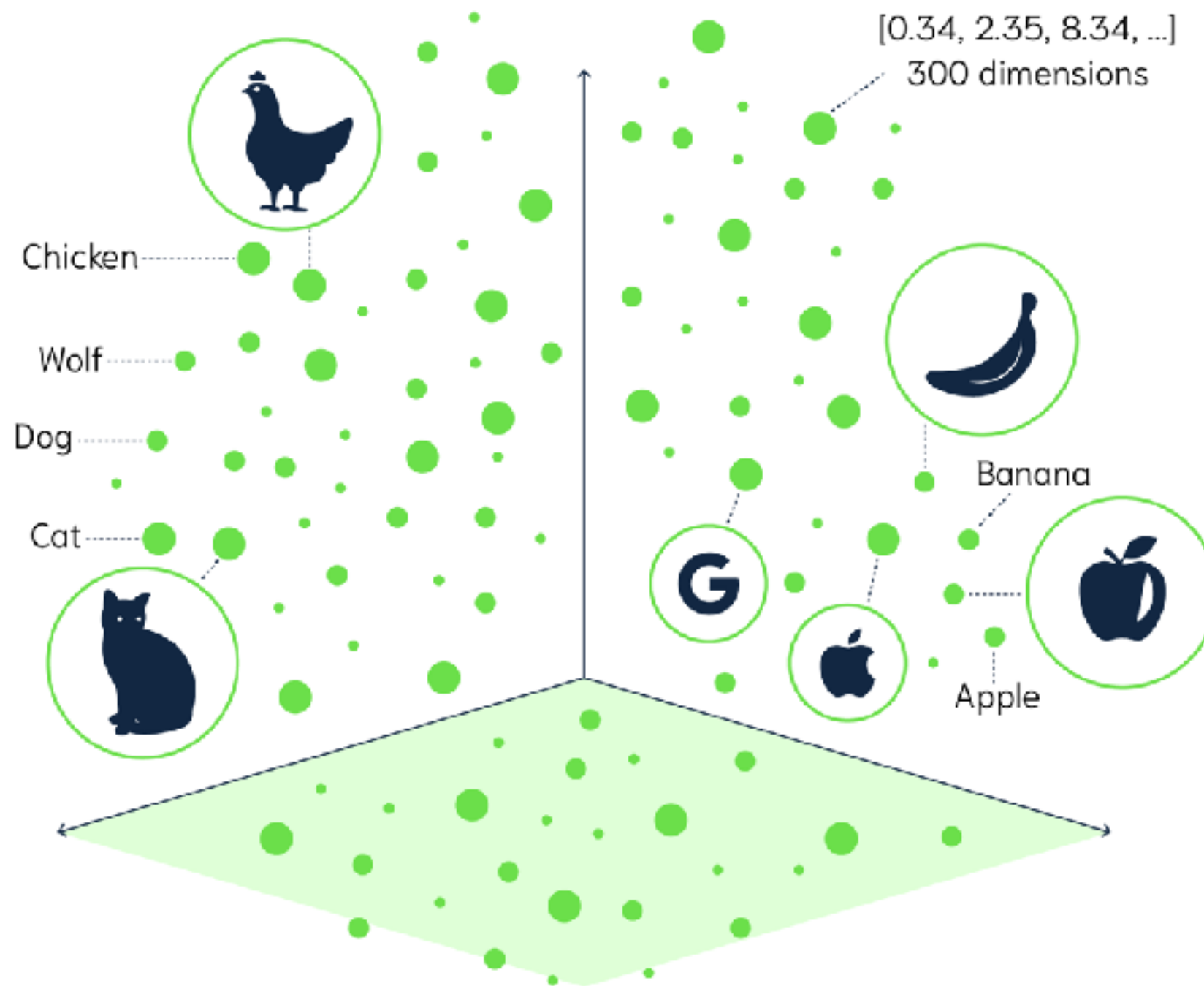
Cosign Similarity



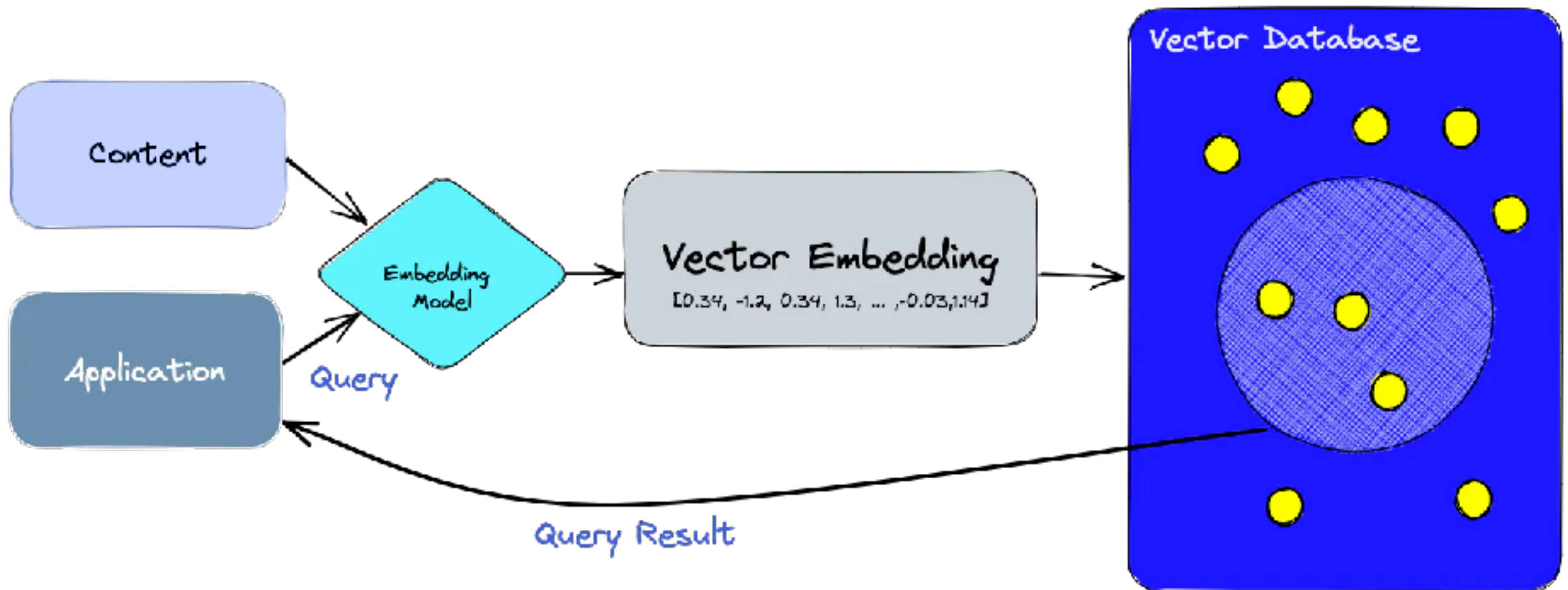
<https://www.learndatasci.com/glossary/cosine-similarity/>



Visual of Vector space

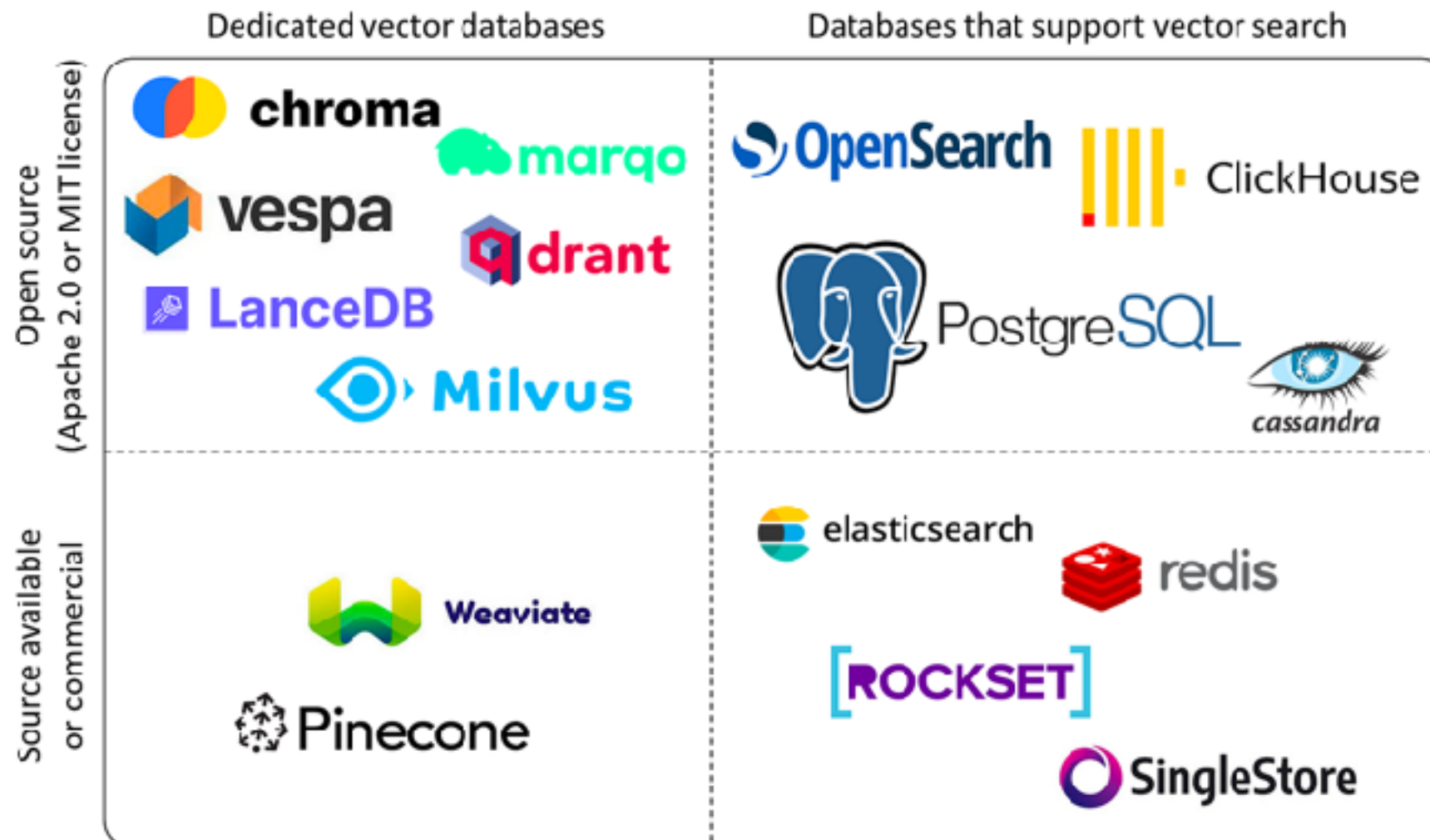


Store data in Vector Database



Vector Database ?

Map items of unstructured data to high-dimensional
real **vectors**



<https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca>



Application

Infrastructure

Model



Application

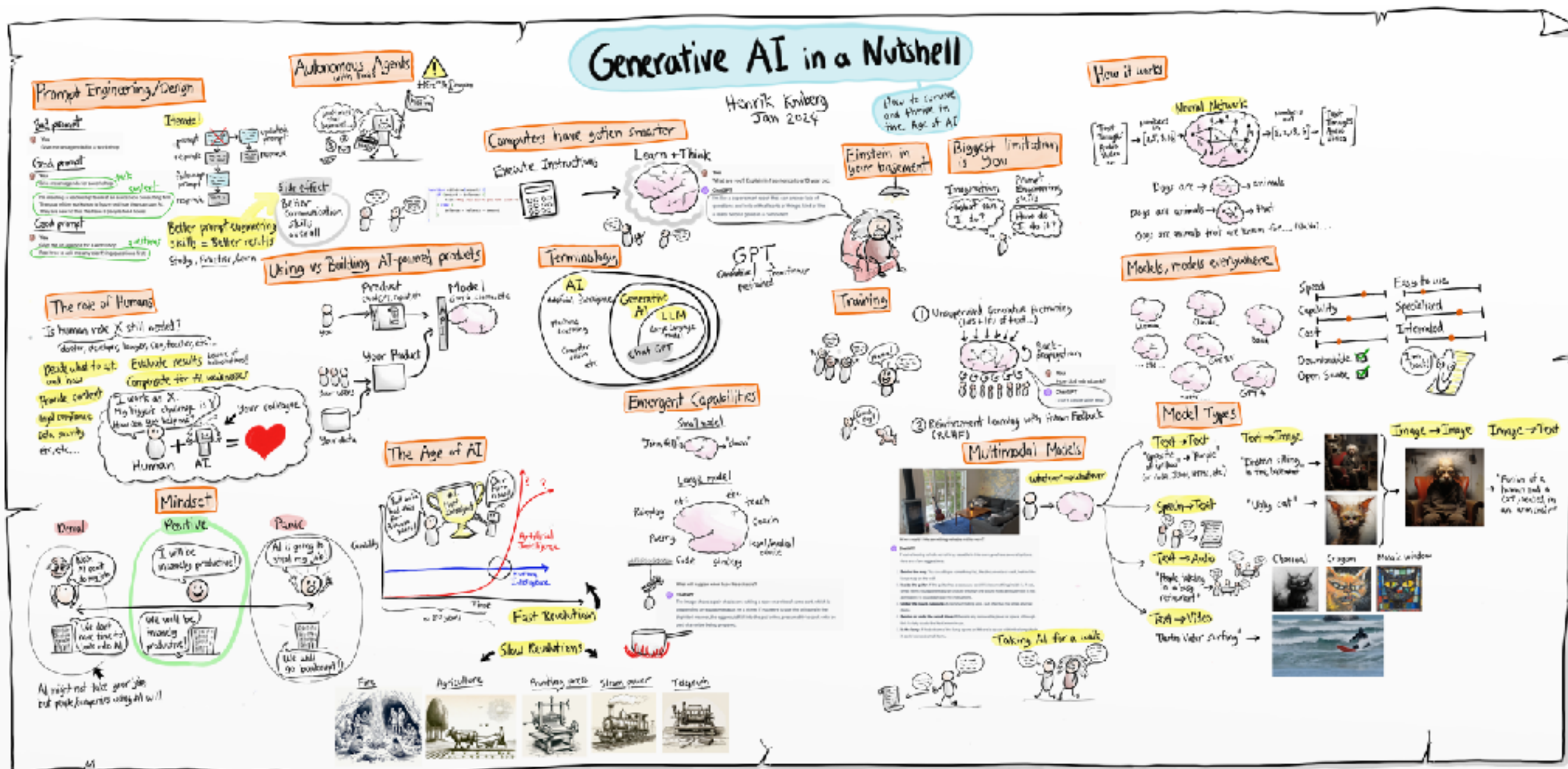
REST API

Infrastructure

Model



Generative AI in Nutshell

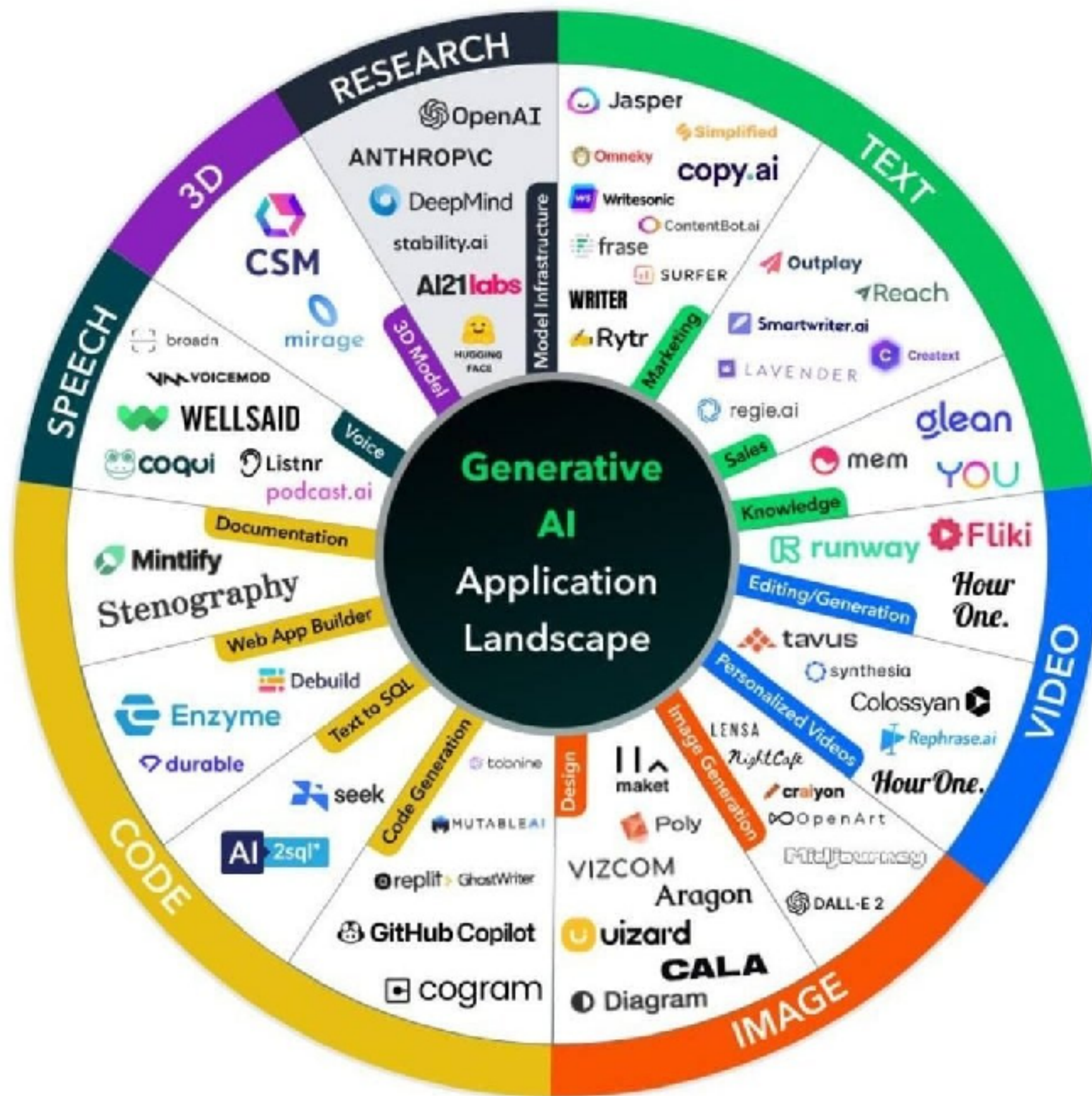


<https://www.youtube.com/watch?v=2IK3DFHRFfw>



Application Tool chains





Tool chains category

Assist
tasks

Interaction
modes

Prompt
composition

Properties of
model

LLM models



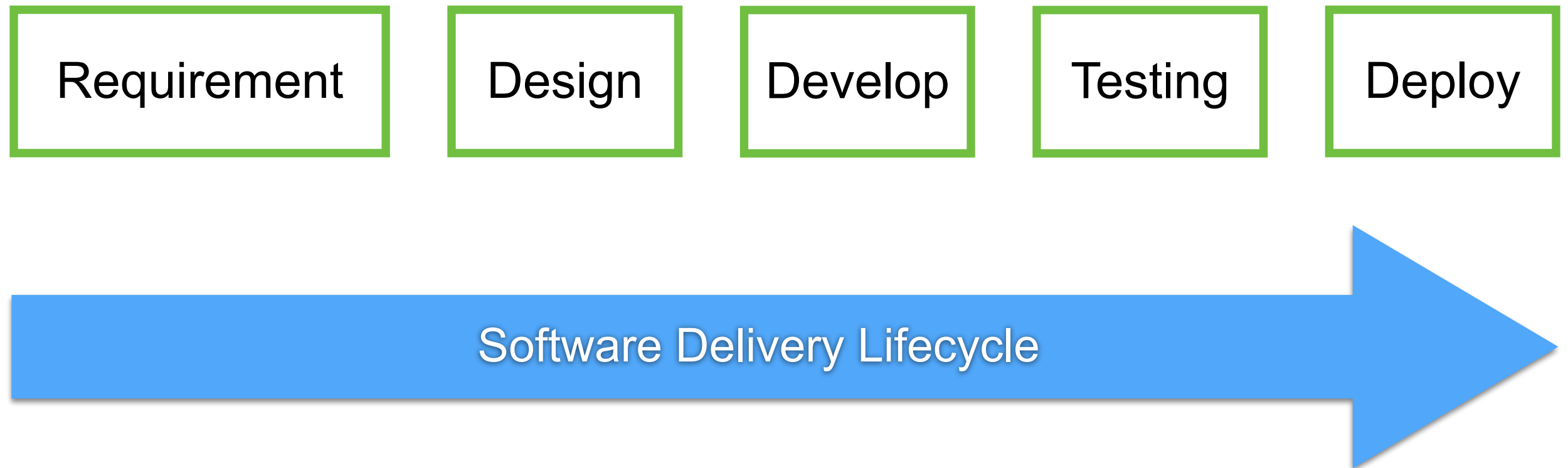
Assist tasks

Finding information faster in context

Generating code

Reasoning about code

Transforming code into something ..



Interaction modes

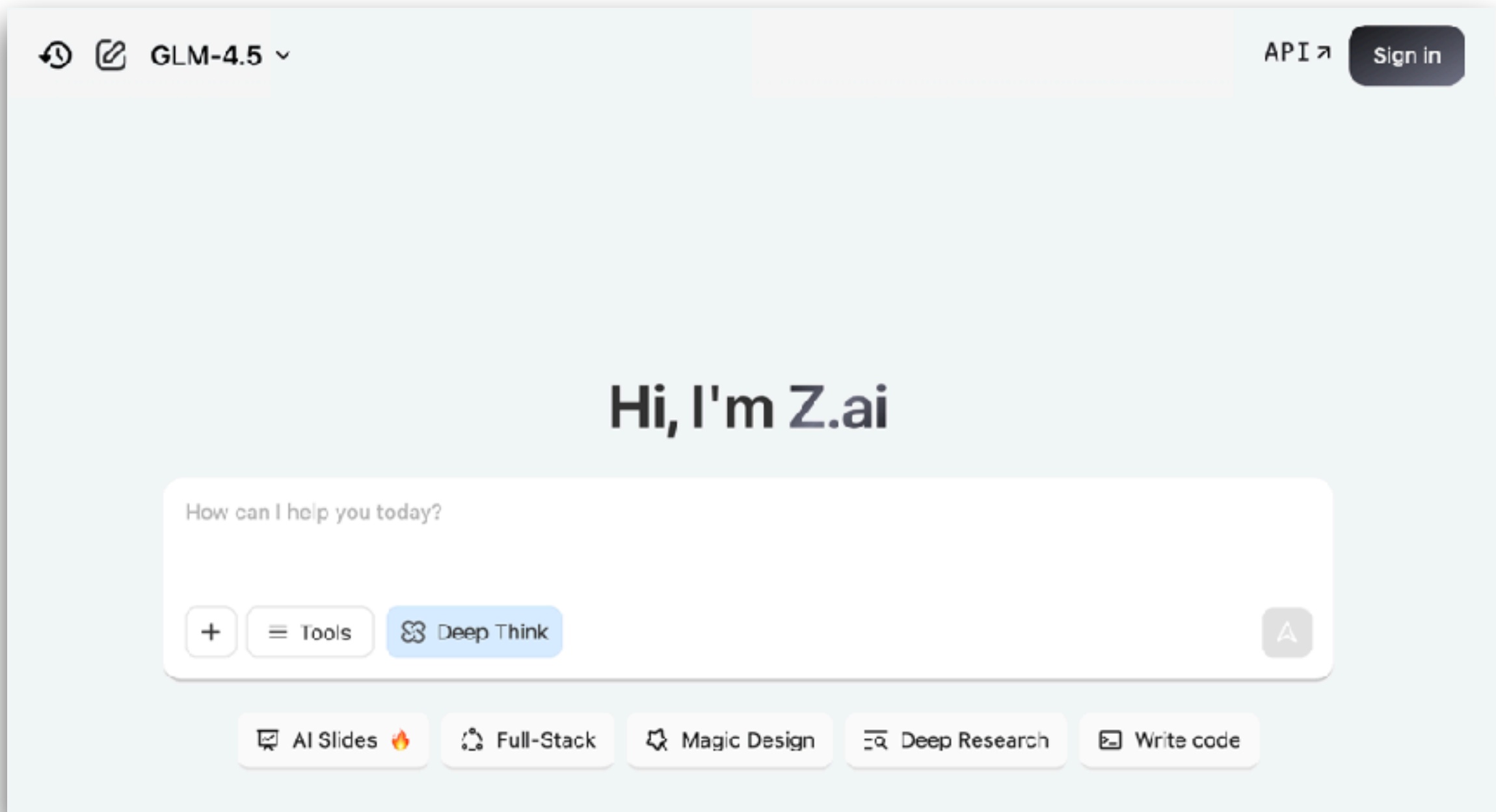
Chat interfaces

In-line assistance (typing in code editor)

CLI (command-line interface)



Z.ai



<https://z.ai/>



Models

The screenshot displays the 'Models' section of the Z.ai website. It features a dark theme with a header 'Models' and three filter tabs: 'Language Models', 'Image Models', and 'Video Generation Models'. Below the filters, there are five model cards, each marked with a 'New' badge in the top right corner. Each card includes a category tag (Reasoning Model, Visual Reasoning Model, or Language Model), the model name, a brief description, and a 'Learn More' link with a right-pointing arrow.

Model Name	Category	Description
GLM-4.6	Reasoning Model	Z.ai's latest model achieves SOTA among open-source models! Context window expanded to 200K. Brings you superior performance in real-world coding, reasoning, tool using and role-playing.
GLM-4.5-Air	Reasoning Model	Z.ai's new lightweight flagship model delivers SOTA performance with exceptional cost-effectiveness!
GLM-4.5-Flash	Reasoning Model	The free version of GLM-4.5 now stands as Z.ai's most powerful offering, delivering unparalleled performance at no cost.
GLM-4.5V	Visual Reasoning Model	Achieve the state-of-the-art (SOTA) performance among open-source VLMs of the same level in various benchmark tests.
GLM-4-32B-0414-128K	Language Model	A general-purpose, cost-efficient LLM for advanced Q&A, coding, search, and structured task automation across business and technical domains.

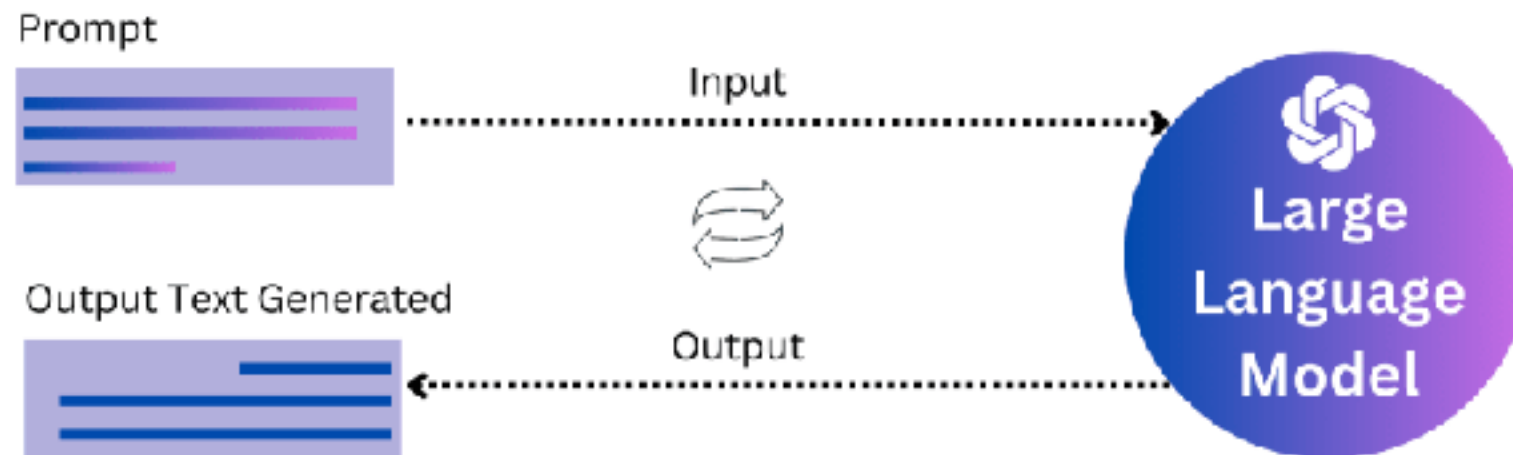
<https://z.ai/>



Prompt composition

Prompt engineering

Compose prompts from user inputs and context



<https://platform.openai.com/docs/guides/prompt-engineering>



Better Prompt

Write clear instructions

Provide reference text

Split complex tasks into simpler subtasks

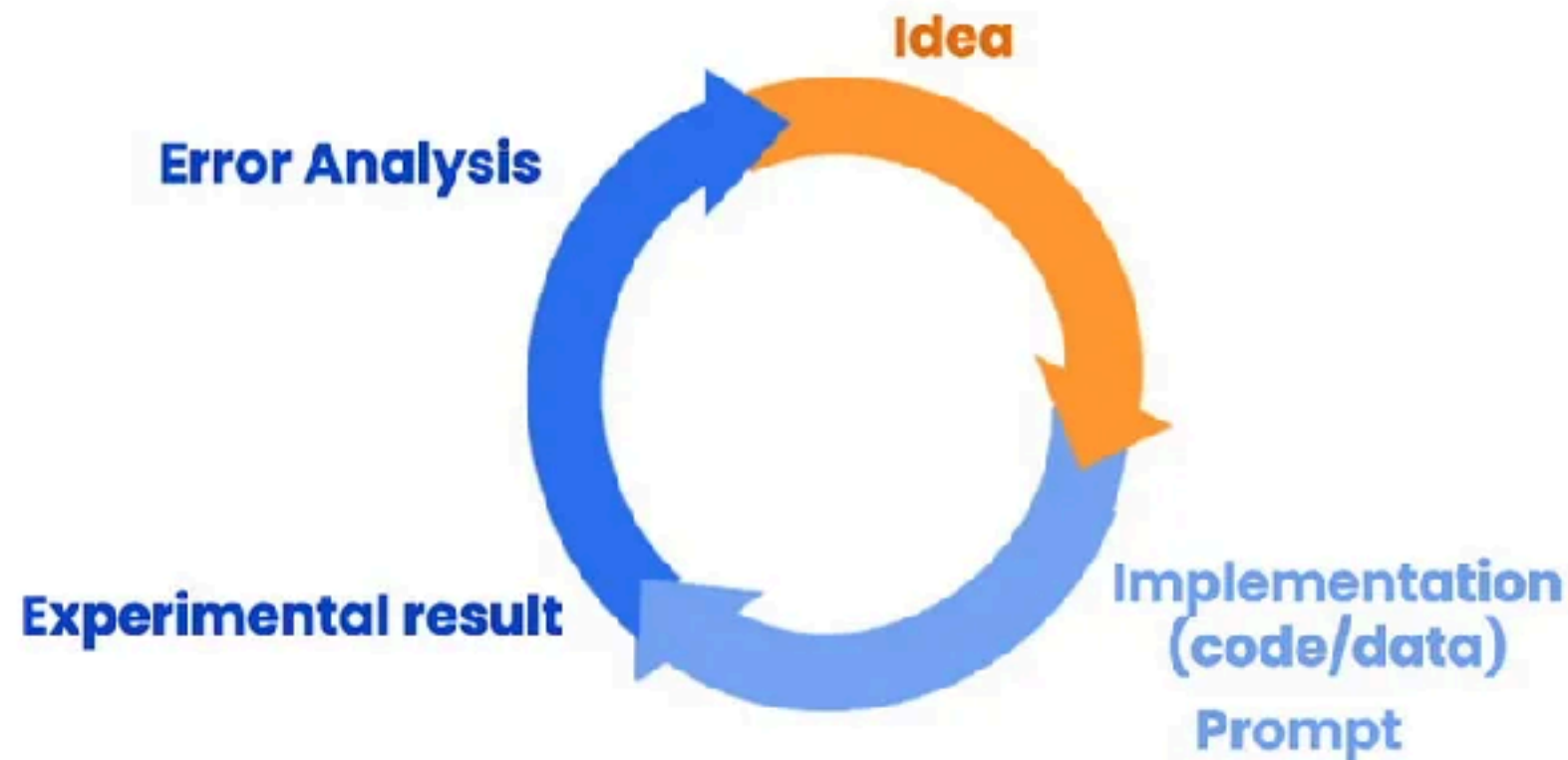
Give the model time to think

Testing and improve ...

<https://platform.openai.com/docs/guides/prompt-engineering>



Iterative Prompt Development



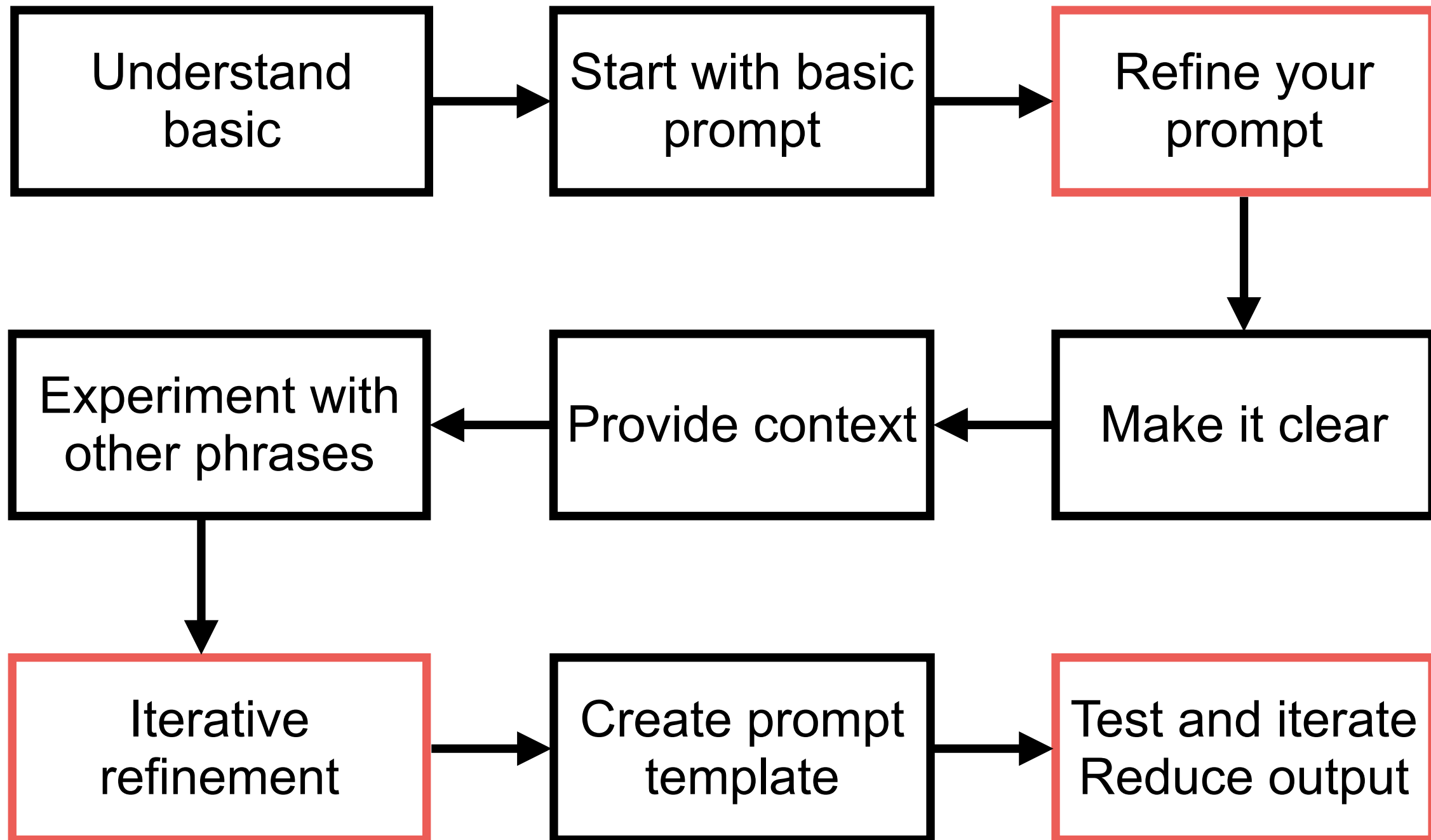
Iterative Process

- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

<https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>



Basic of Prompt Engineer



Better Prompt

Write clear instructions

Provide reference text

Split complex tasks into simpler subtasks

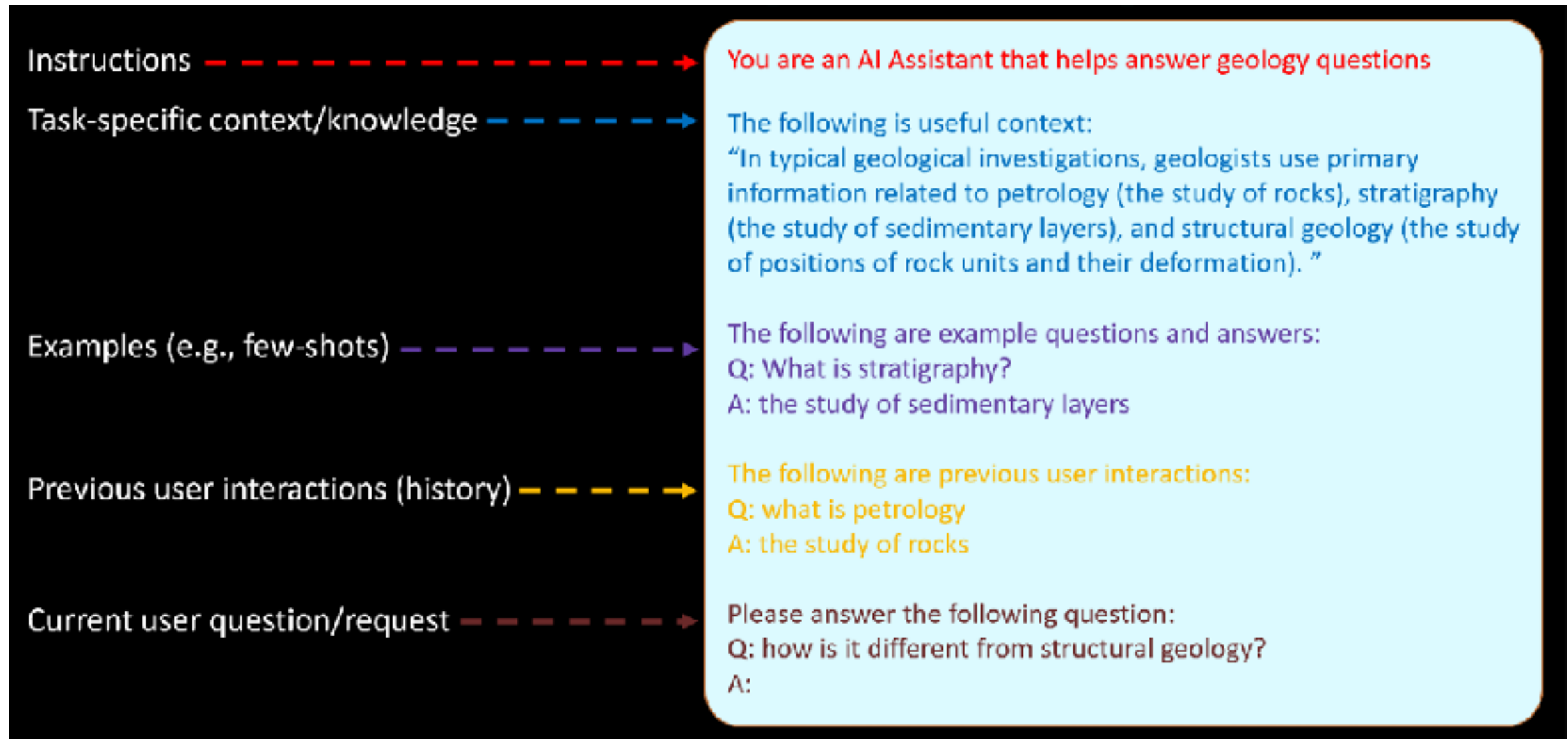
Give the model time to think

Testing and improve ...

<https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results>



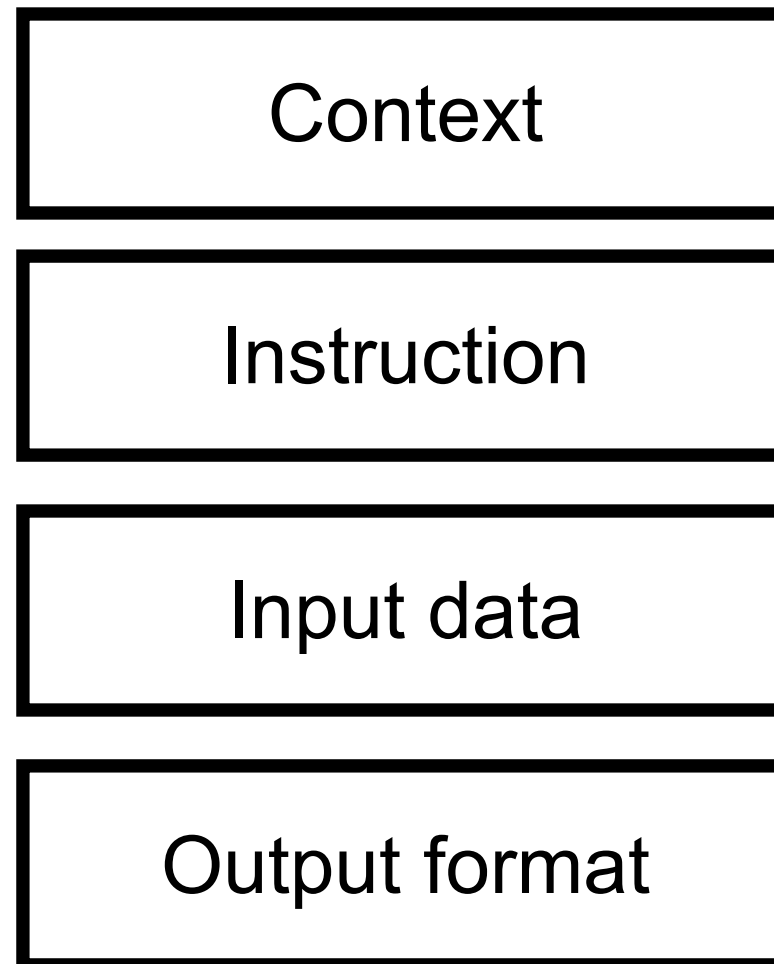
Prompt Structure



<https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering>



Structure of Prompt



<https://platform.openai.com/docs/guides/prompt-engineering>



Prompting Guide

Prompt Engineering

Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

<https://www.promptingguide.ai/>



Structure of Prompt !!

APE

Action, Purpose,
Expectation

RACE

Role, Action,
Context,
Expectation

TAG

Task, Action, Goal

COAST

Context, Objective,
Action, Scenario,
Task

RISE

Role, Input, Step,
Expectation

<https://twitter.com/pradeepeth/status/1673271866696544257>



Prompt Techniques

Zero-shot

Chain-of
Thought (CoT)

Few-shot

Meta or structure

<https://www.promptingguide.ai/techniques>



Chain of Thought Prompting (CoT)

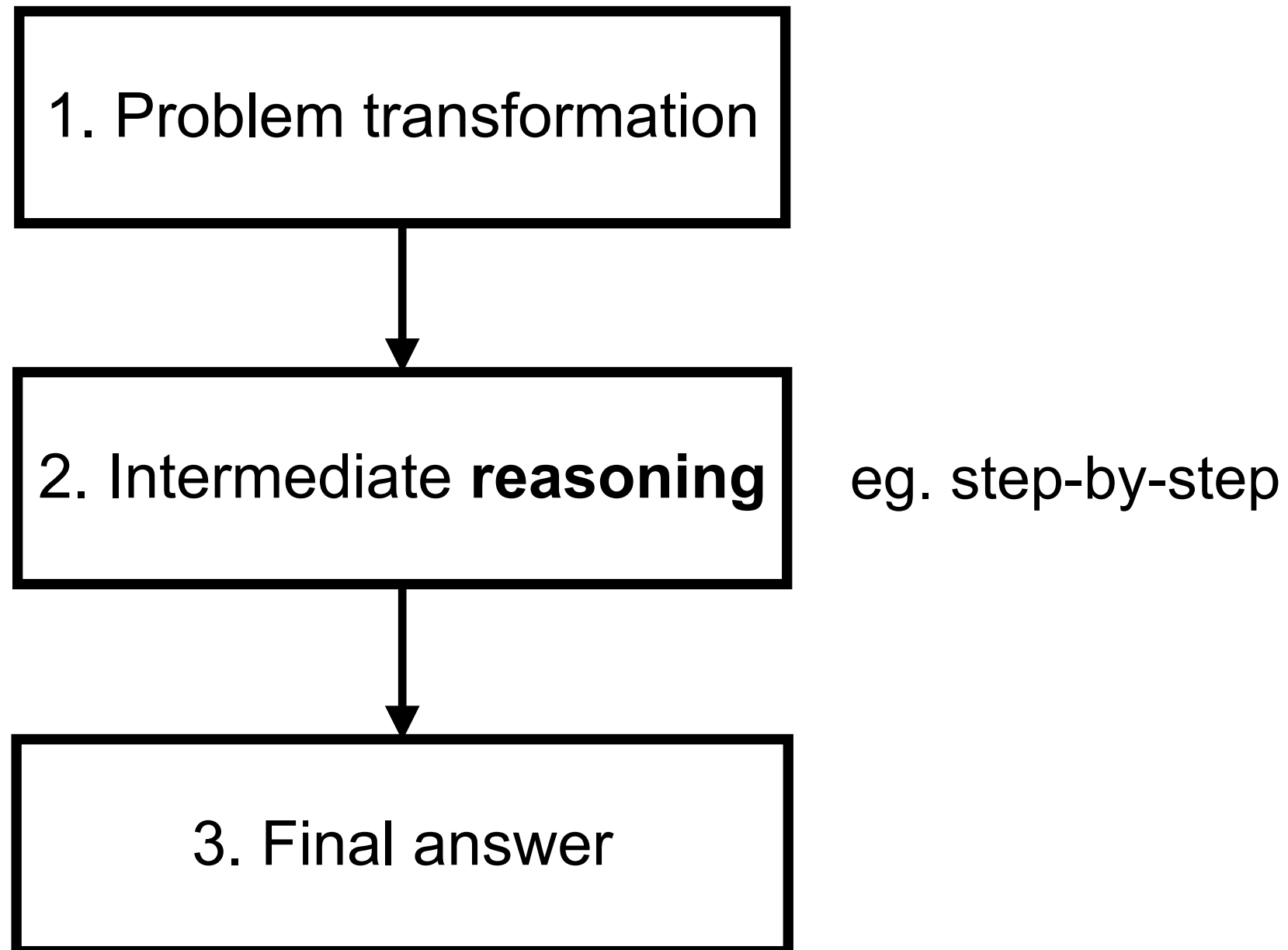
Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!



Chain of Thought Prompting (CoT)



Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

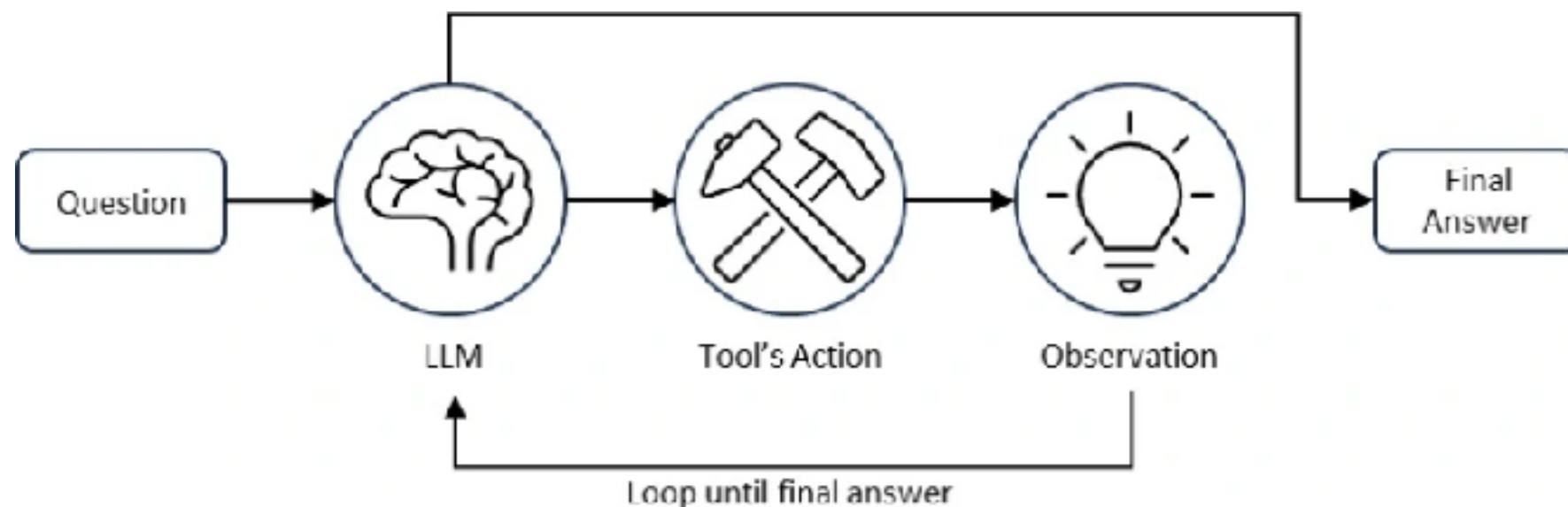
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

<https://www.promptingguide.ai/techniques/cot>



ReAct Prompt

LLM reasoning and additional tools (expert)
Improve better answer



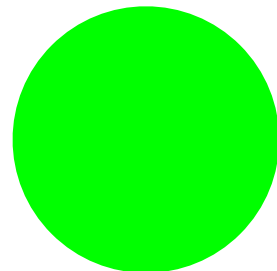
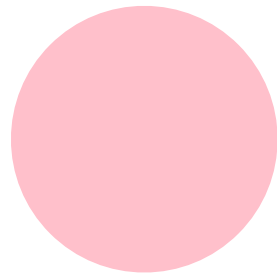
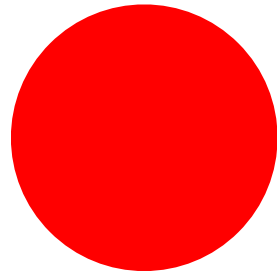
<https://www.promptingguide.ai/techniques/react>



Workshop



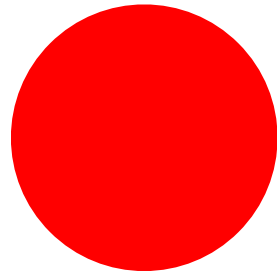
RGB ?



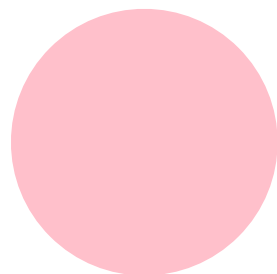
<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb>



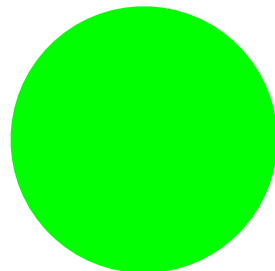
RGB ?



[255, 0, 0]



[255, 192, 203]

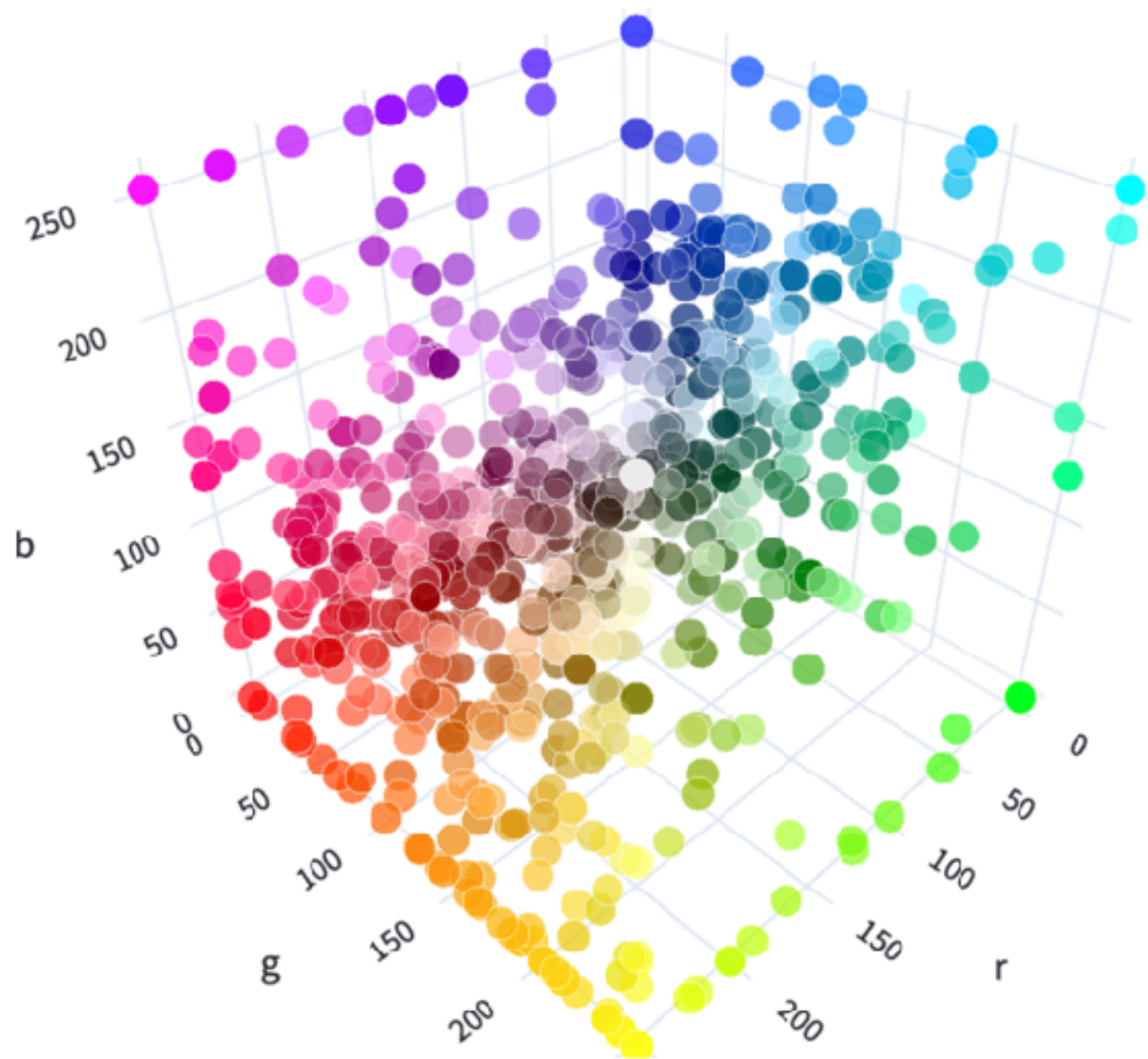


[0, 255, 0]

<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb>



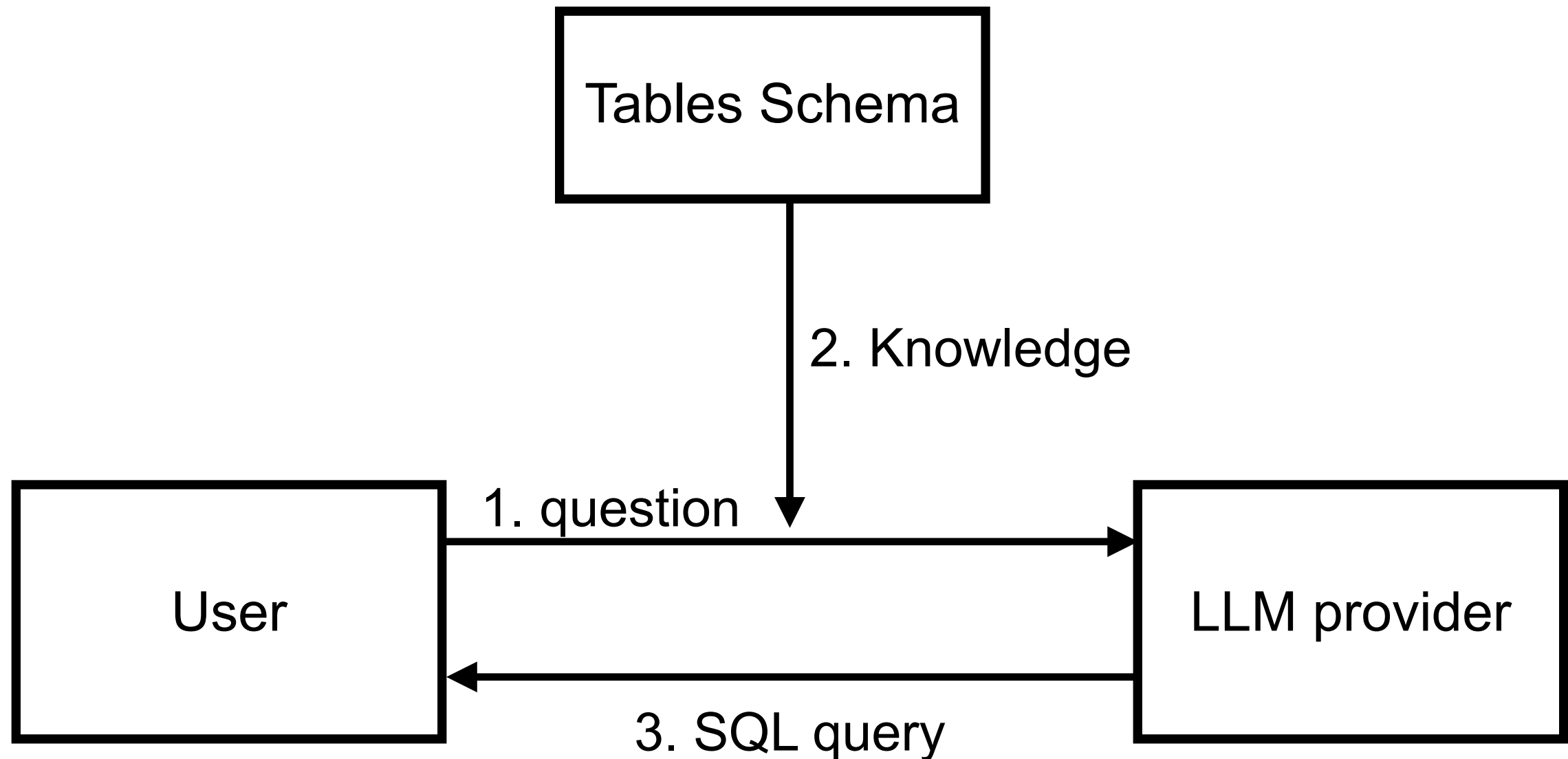
Visualize RGB



https://huggingface.co/spaces/jphwang/colorful_vectors



Text-to-SQL



<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql>



Software Development Life Cycle



SDLC

Requirement

Design

Develop

Testing

Deploy



Planning

Written planning process

Arch diagrams

Testing

Automated testing

TDD

Testing environments

Testing in prod

Performance testing

Load testing

Generate test data

Shipping

FF & experimentation

Logging

Monitoring & alerting

Staged rollouts

Development

Automated dev env

CI/CD

Prototyping

Code review

Code generation

Templates

Cross-platform dev

Preview env

Post-commit code review

Linting

Static code analysis

Project mgmt

Maintenance

Debug production

Documentation

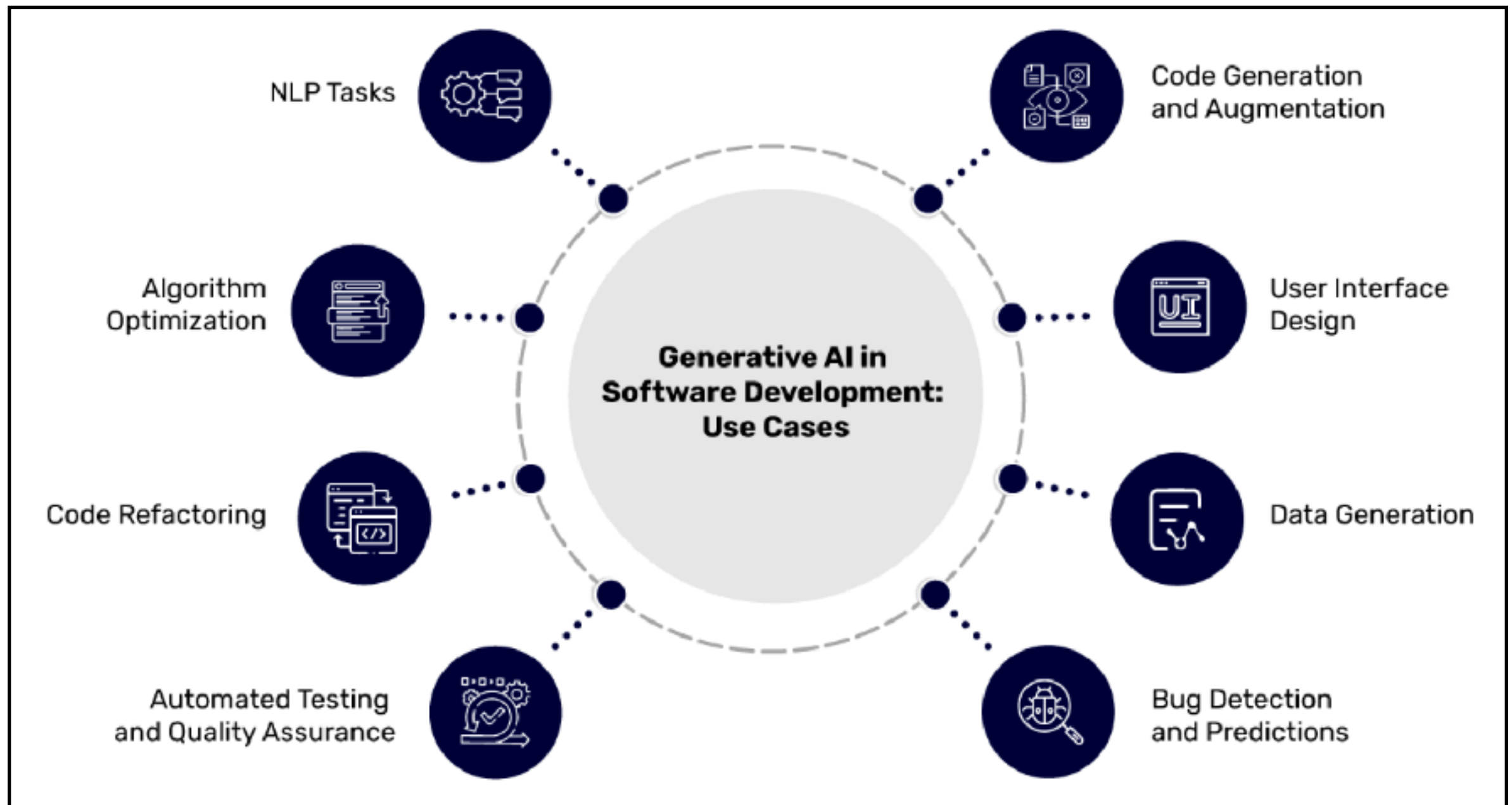
Runbook examples

Mitigation runbook

pragmaticengineer.com



SDLC



Impacts with productivity ?

Automated
simple tasks

Improve quality
and reliability

Improve
communication

Faster
prototype



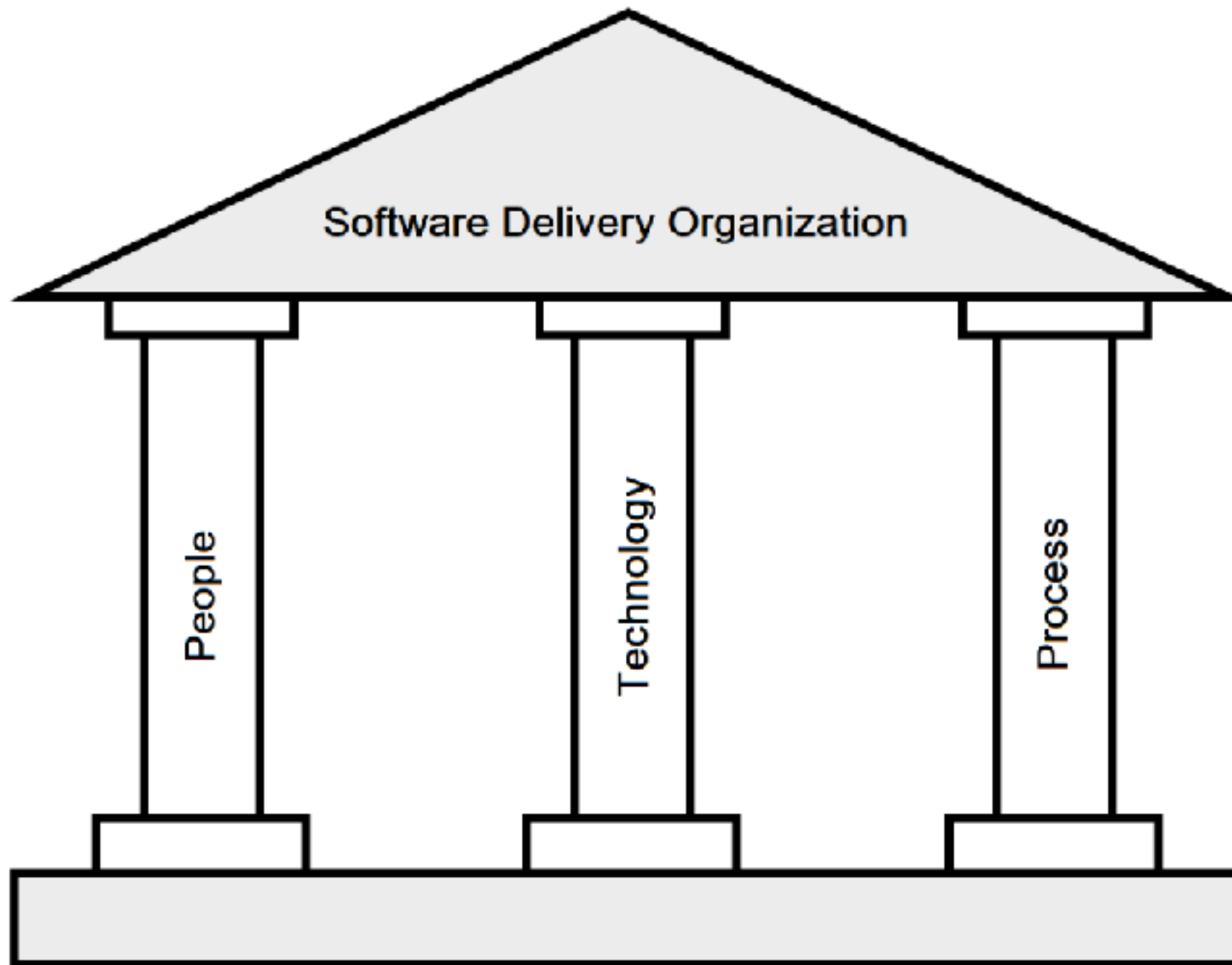
Generative AI isn't just a tool
it's your team member



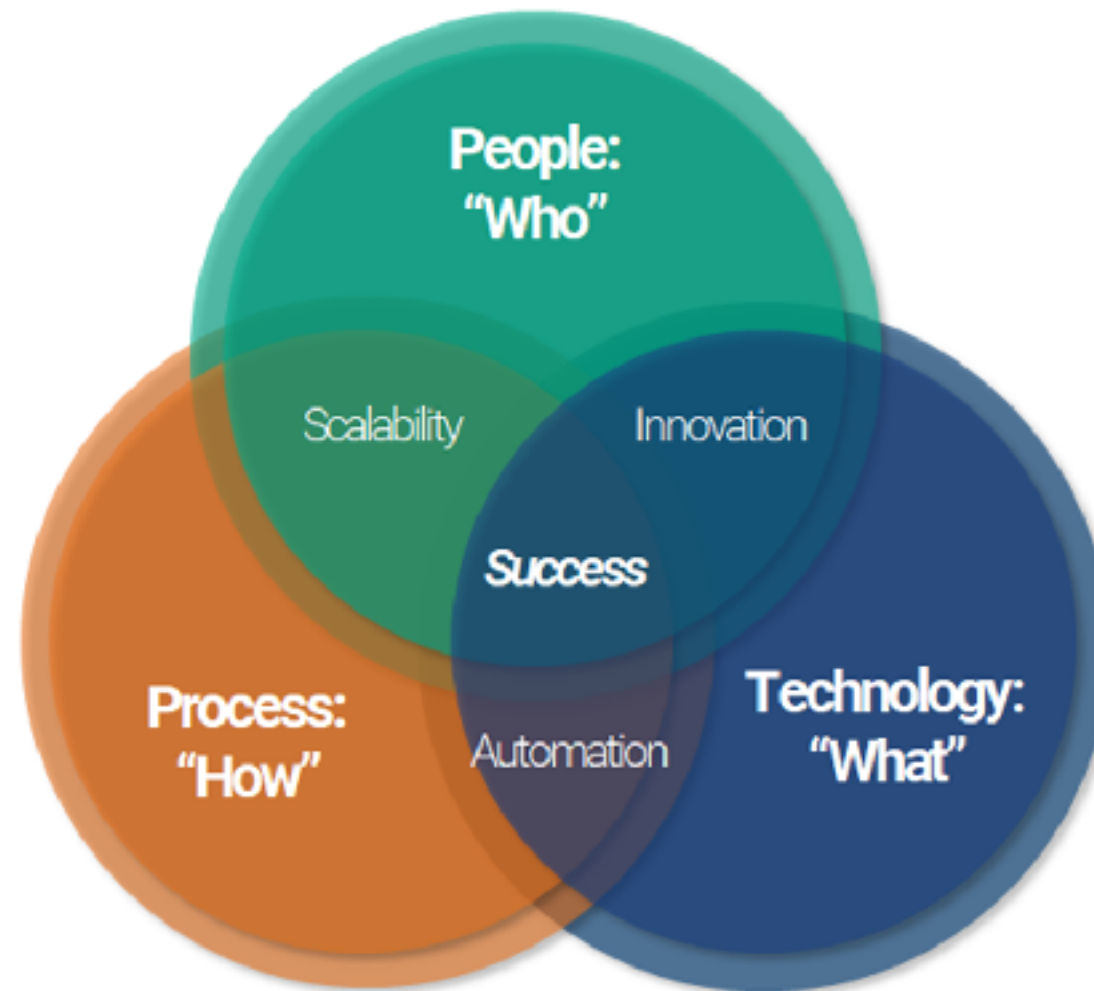
Trust, but *verify* output !!



3 Pillar of Software Development



3 Pillar of Software Development



Requirement and Analysis



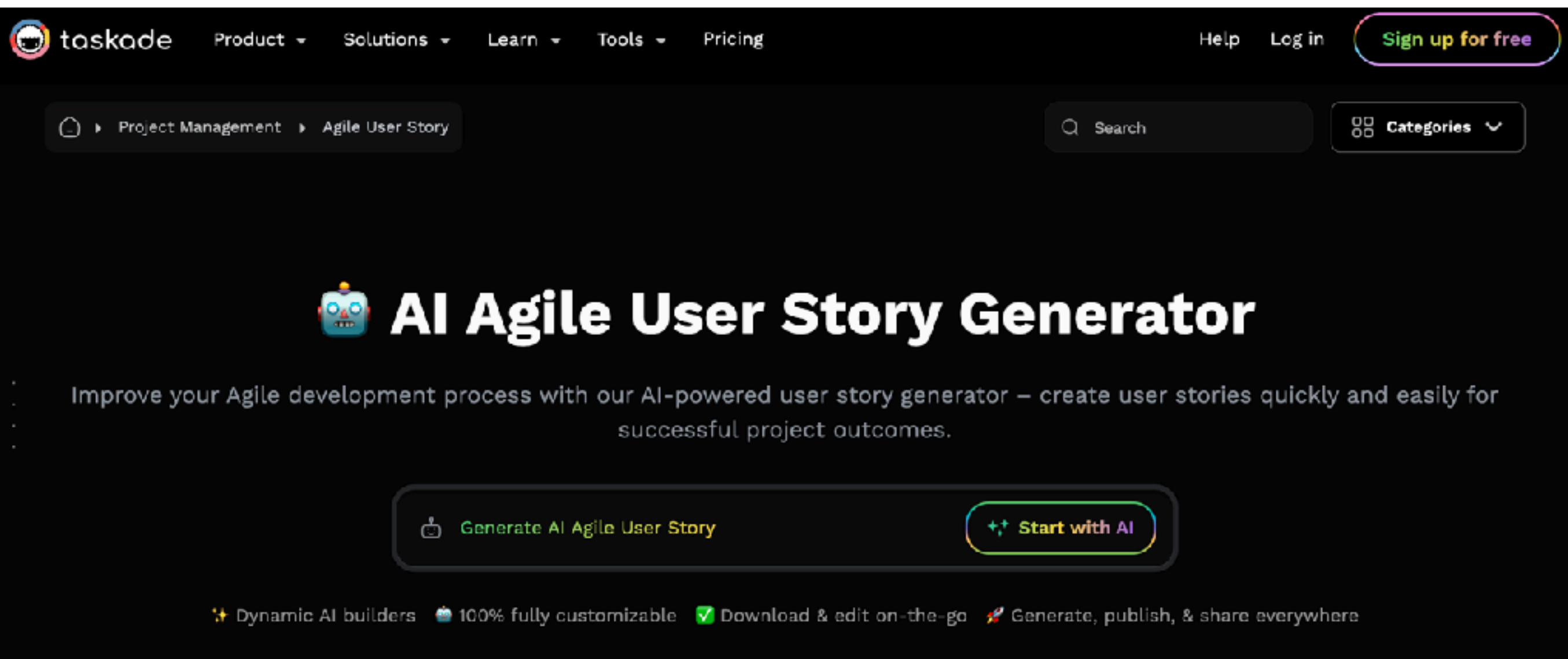
Requirement and Analysis



Requirements writing and analysis
User story generation



Agents AI for Automated tasks



The screenshot shows the Taskade website's navigation bar with links for Product, Solutions, Learn, Tools, and Pricing. A 'Sign up for free' button is in the top right. Below the navigation bar, a breadcrumb trail reads 'Project Management > Agile User Story'. A search bar and a 'Categories' dropdown are also visible. The main heading is 'AI Agile User Story Generator', accompanied by a robot icon. Below this, a descriptive text states: 'Improve your Agile development process with our AI-powered user story generator – create user stories quickly and easily for successful project outcomes.' Two primary buttons are present: 'Generate AI Agile User Story' and 'Start with AI'. At the bottom, a row of features is listed: 'Dynamic AI builders', '100% fully customizable', 'Download & edit on-the-go', and 'Generate, publish, & share everywhere'.

taskade Product Solutions Learn Tools Pricing Help Log in Sign up for free

Project Management > Agile User Story Search Categories

AI Agile User Story Generator

Improve your Agile development process with our AI-powered user story generator – create user stories quickly and easily for successful project outcomes.

 Generate AI Agile User Story  Start with AI

✦ Dynamic AI builders 🤖 100% fully customizable ✅ Download & edit on-the-go 🚀 Generate, publish, & share everywhere

<https://www.taskade.com/generate/project-management/agile-user-story>



Example with food delivery

Food Delivery Workflow Template

 **Order Processing #order**

- ☐ Check for new orders
 - ☐ Verify customer details
 - ☐ Confirm payment status
- ☐ Prepare order items
 - ☐ Gather ingredients
 - ☐ Cook or prepare food
 - ☐ Package items securely

 **Delivery Management #delivery**

- ☐ Assign delivery driver
- ☐ Plan delivery route
 - ☐ Prioritize multiple deliveries
 - ☐ Use GPS for directions
- ☐ Confirm delivery with customer
 - ☐ Send delivery notification
 - ☐ Obtain customer signature

 **Post-Delivery Tasks #postdelivery**

 What would you like to do next?

 Create project

 Continue writing

 Make longer



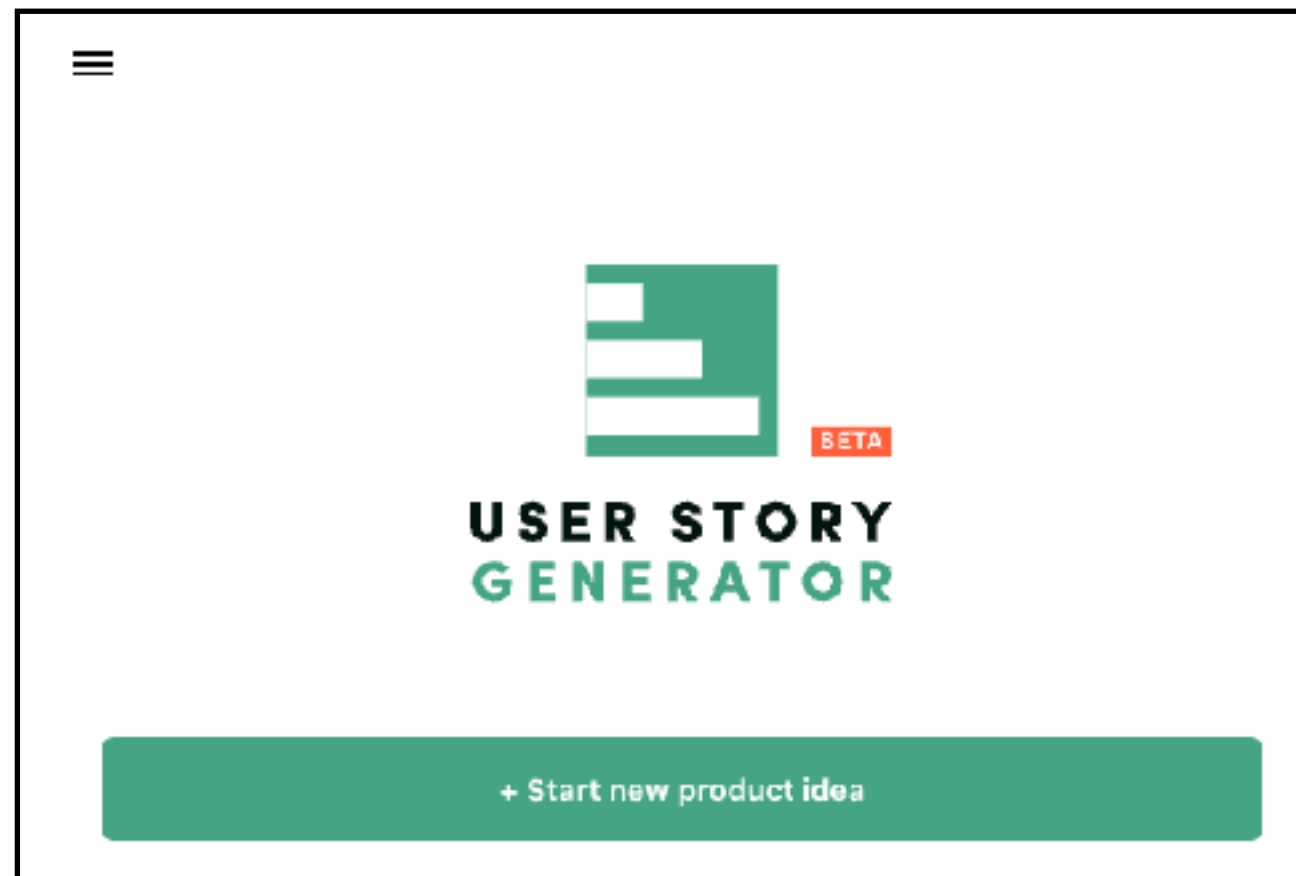
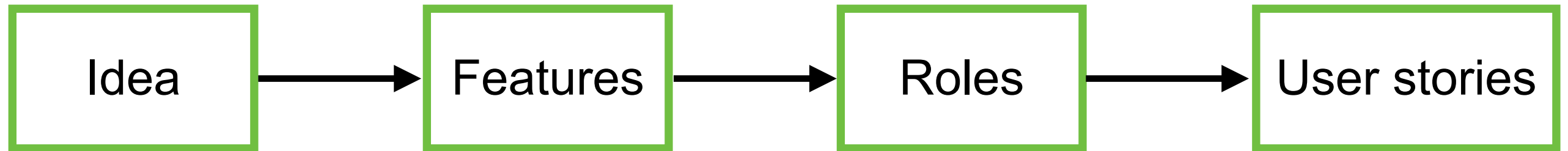
<https://www.taskade.com/generate/project-management/agile-user-story>

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AI for Software Development

78

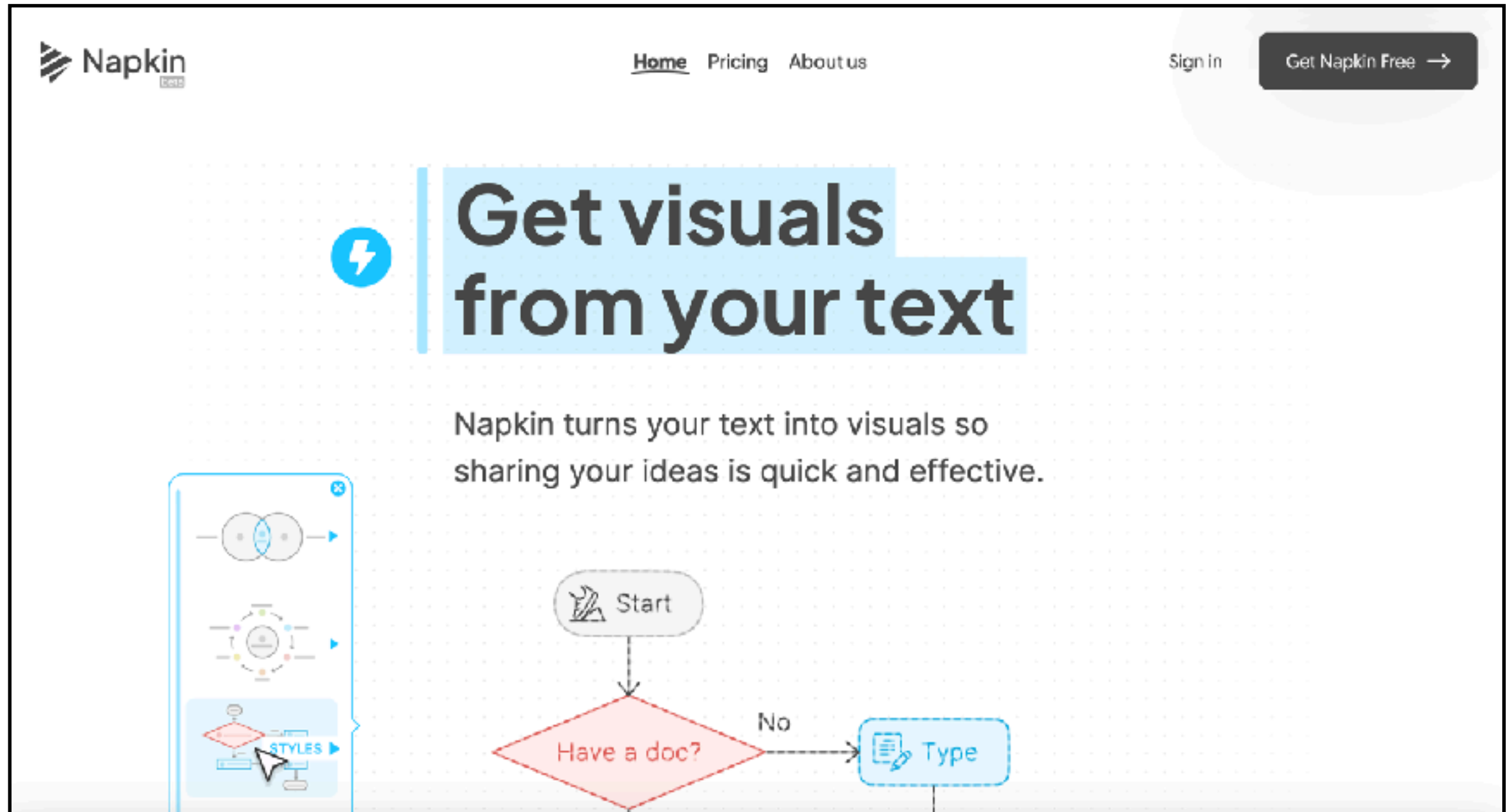
User Story Generator



<https://userstorygenerator.ai/>



Napkin



The screenshot shows the Napkin AI website. At the top left is the Napkin logo. The navigation bar includes links for Home, Pricing, and About us. On the right, there are links for Sign in and a button labeled 'Get Napkin Free' with a right arrow. The main heading, 'Get visuals from your text', is displayed in large, bold letters, with the word 'visuals' highlighted in a light blue box. Below the heading, a subtext states: 'Napkin turns your text into visuals so sharing your ideas is quick and effective.' To the left of this text is a vertical panel containing three visual examples: a Venn diagram, a circular flow diagram, and a flowchart. The flowchart is highlighted with a blue border and a 'STYLES' button. Below the panel, a sample flowchart is shown on a grid background. It begins with an oval 'Start' node, followed by a red diamond decision node labeled 'Have a doc?'. A 'No' path leads to a blue rounded rectangle 'Type' node.

Get visuals from your text

Napkin turns your text into visuals so sharing your ideas is quick and effective.

Start

Have a doc?

No

Type

<https://www.napkin.ai/>



Requirement analysis

Clarify of User requirement ?

<https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis>



Data Analysis

The screenshot shows the Julius AI website. At the top is a navigation bar with links for Product, Use Cases, Resources, Security, Connectors, Community, and Pricing. On the right are 'Log In' and 'Sign Up' buttons. The main heading is 'Add data, ask questions', followed by the tagline 'Ask for what you want and Julius analyzes the data for you'. Below this is a three-step process:

- 1 Connect all your data sources**
Connect with data sources like databases, spreadsheets, and more. The visual shows icons for Google Drive, XLSX, PDF, and a database.
- 2 Ask for analysis**
You provide the questions, Julius handles the analysis. The visual shows a text box with the example question: 'Predict customer churn from purchase history' and an upward arrow icon.
- 3 Get results, instantly**
Choose from charts, tables or full reports tailored to your data. The visual shows various data visualization charts like bar charts and a heatmap.

At the bottom center is a blue button that says 'Get started for free >'.

<https://julius.ai/>



Design Process



Design



Architecture writing assistance

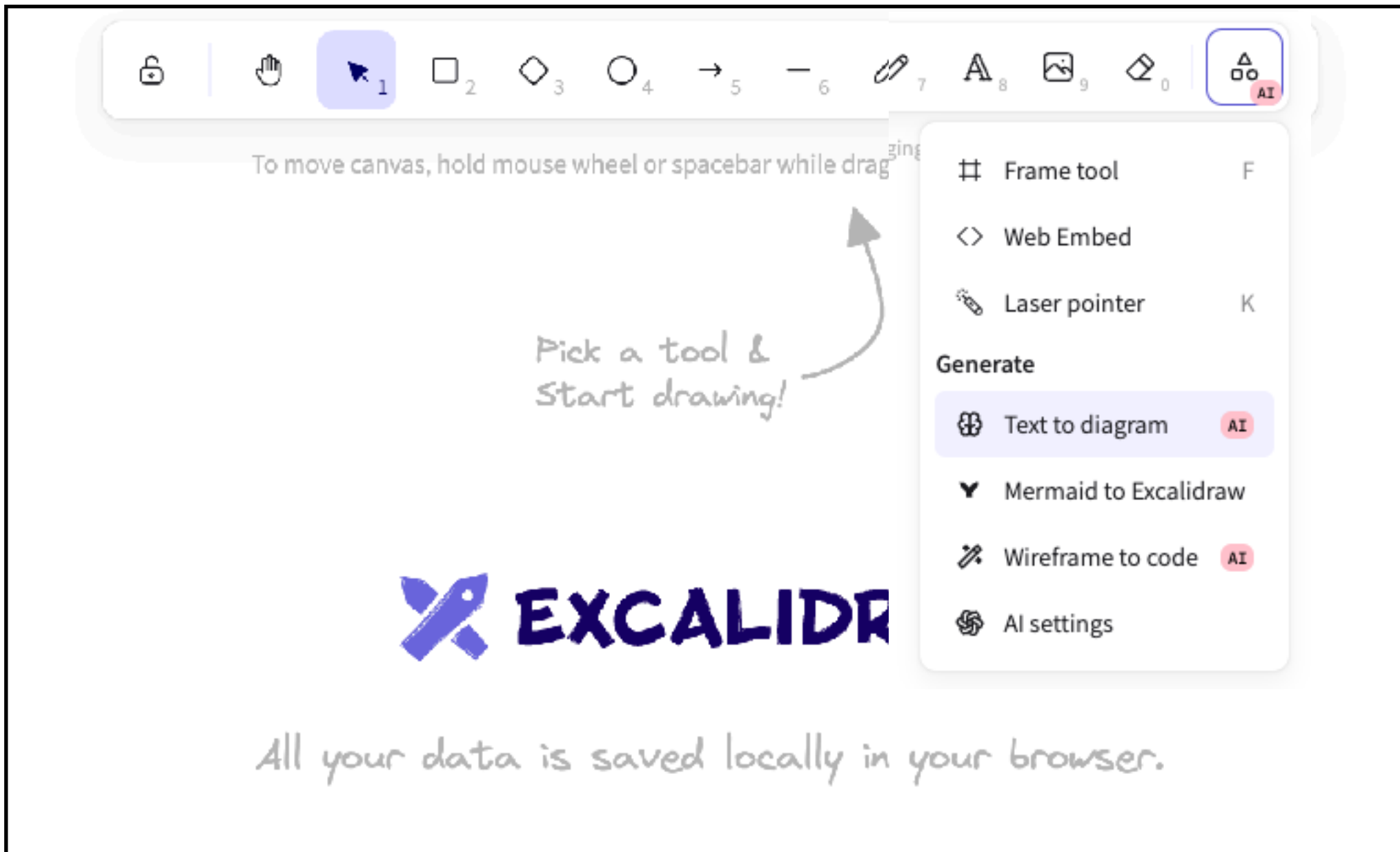
Sequence flow diagram generation

Data modeling

UX/UI design assistance



Excalidraw with AI



<https://excalidraw.com/>



Demo

Text to diagram **AI Builder** Mermaid

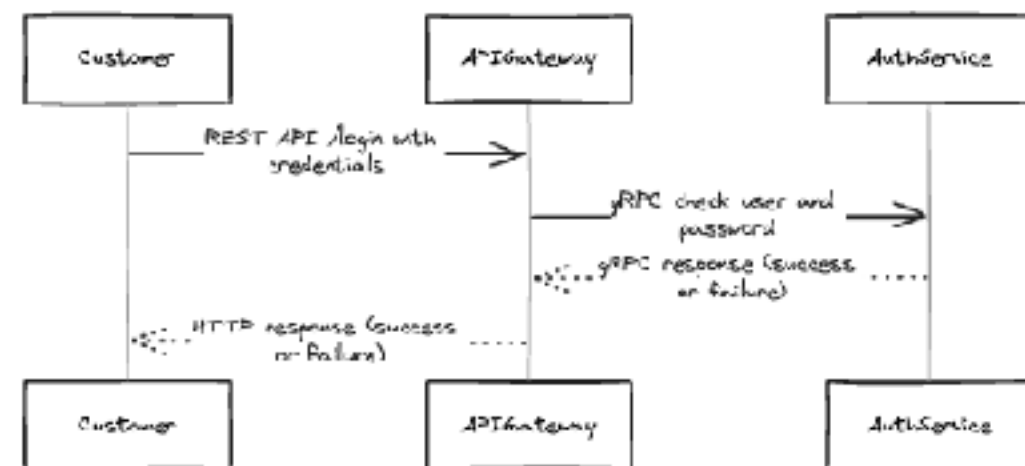
Currently we use Mermaid as a middle step, so you'll get best results if you describe a diagram, workflow, flow chart, and similar.

Prompt

9 requests left today

Try to generate authentication service from below
1. Customer call api gateway with REST API /login
2. Api gateway check user and password from auth service via gRPC
3. API gateway send response to client

Preview



Generate →

Cmd Enter

View as Mermaid →

Insert →



Mermaid Diagram

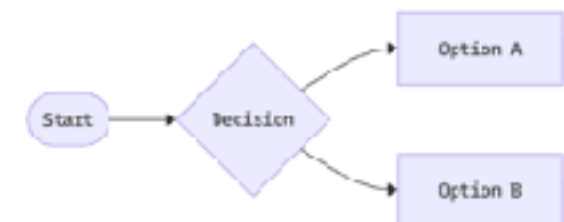
Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

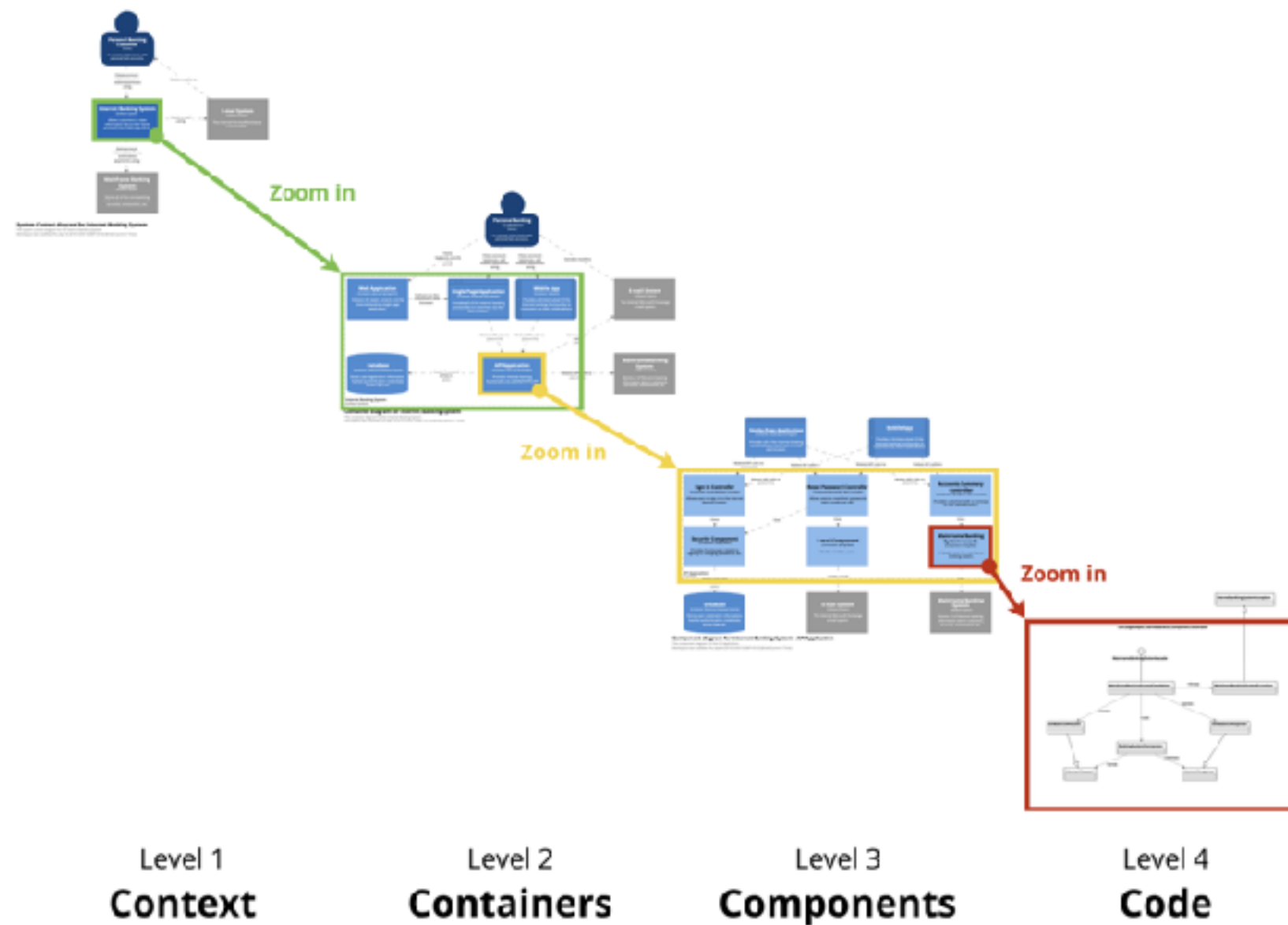
```
1 flowchart LR
2   A(["Start"])
3   A --> B{"Decision"}
4   B --> C["Option A"]
5   B --> D["Option B"]
```



<https://mermaid.js.org/>



Software Architecture



<https://c4model.com/>



Database

The screenshot shows the Database Build interface. On the left, there's a sidebar with a 'New database' button and a list of databases: 'User Management Database ...' and 'User and Shipping Address ...'. The main area is titled 'Diagram' and shows a database diagram with two tables: 'shipping_addresses' and 'users'. The 'shipping_addresses' table has columns: 'id', 'user_id', 'address_line1', 'address_line2', 'city', 'state', 'postal_code', and 'country'. The 'users' table has columns: 'id', 'name', and 'email'. A message on the right says: 'Create table of user that have many shipping address'. Below this, there's a section titled 'Executed SQL' with a dropdown arrow. The text below says: 'The tables `users` and `shipping\_addresses` have been created successfully. Here's a brief overview:'. Below this, there's a list of tables: 'Users Table: Contains user information with columns for `id`, `name`, and `email`.' and 'Shipping Addresses Table: Stores multiple addresses for each user, with columns for `id`, `user\_id` (foreign key referencing `users`), `address\_line1`, `address\_line2`, `city`, `state`, `postal\_code`, and `country`.' At the bottom right, there's a message: 'Conversation renamed to User and Shipping Address Management change'. At the bottom left, there's a 'Toggle theme' button. At the bottom center, there's a 'PG 18 | Local-only database' label and a row of icons: 'Primary key', 'Identity', 'Unique', and 'Nullable'. At the bottom right, there's a 'Message AI or write SQL' button with an upward arrow.

New database

User Management Database ...

User and Shipping Address ...

Diagram

Migrations

shipping_addresses

Column	Type
id	int8
user_id	int8
address_line1	text
address_line2	text
city	text
state	text
postal_code	text
country	text

users

Column	Type
id	int8
name	text
email	text

Create table of user that have many shipping address

Executed SQL

The tables `users` and `shipping\_addresses` have been created successfully. Here's a brief overview:

- Users Table:** Contains user information with columns for `id`, `name`, and `email`.
- Shipping Addresses Table:** Stores multiple addresses for each user, with columns for `id`, `user\_id` (foreign key referencing `users`), `address\_line1`, `address\_line2`, `city`, `state`, `postal\_code`, and `country`.

Conversation renamed to User and Shipping Address Management change

Toggle theme

PG 18 | Local-only database

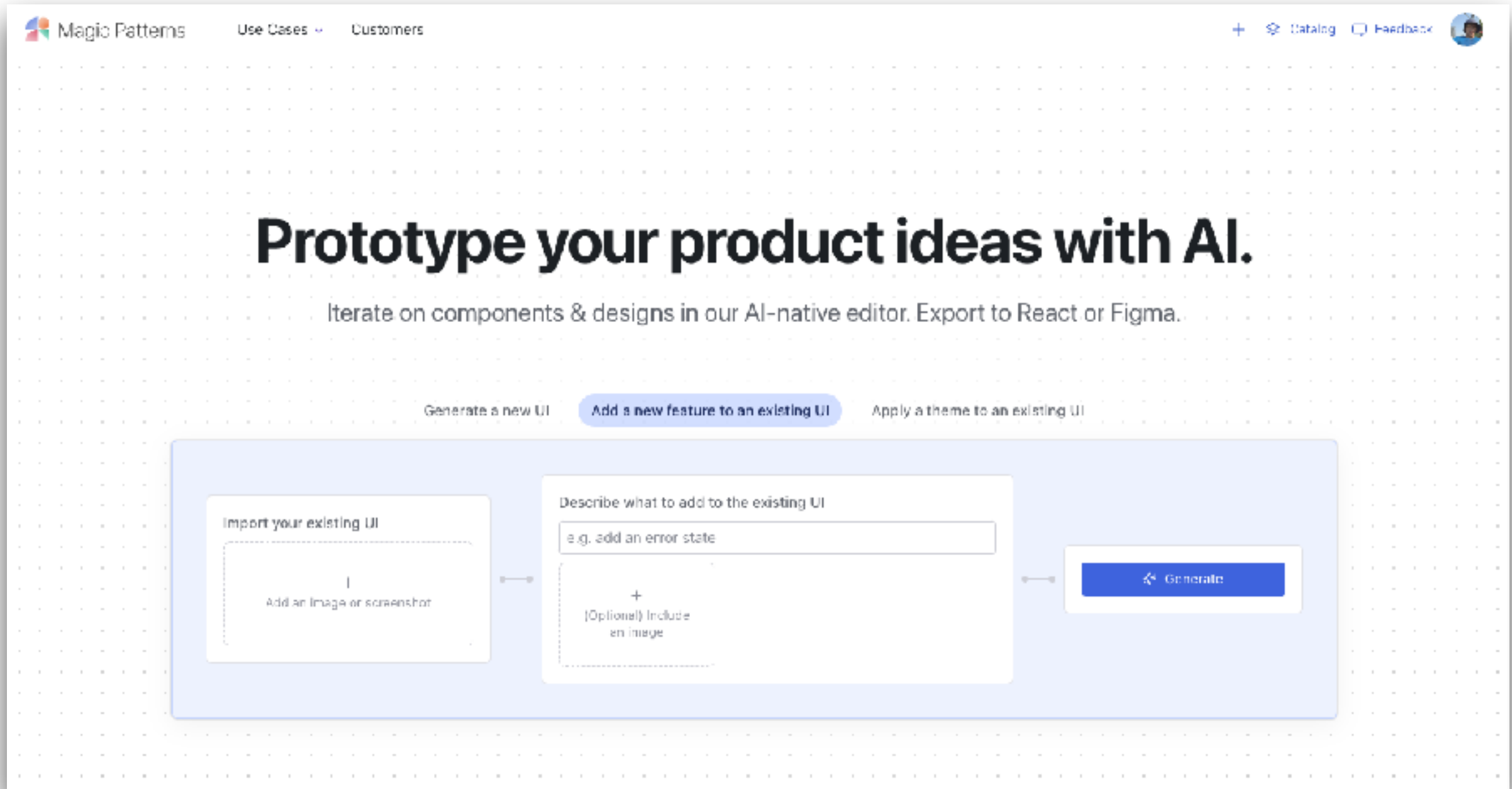
Primary key Identity Unique Nullable

Message AI or write SQL

<https://database.build/>



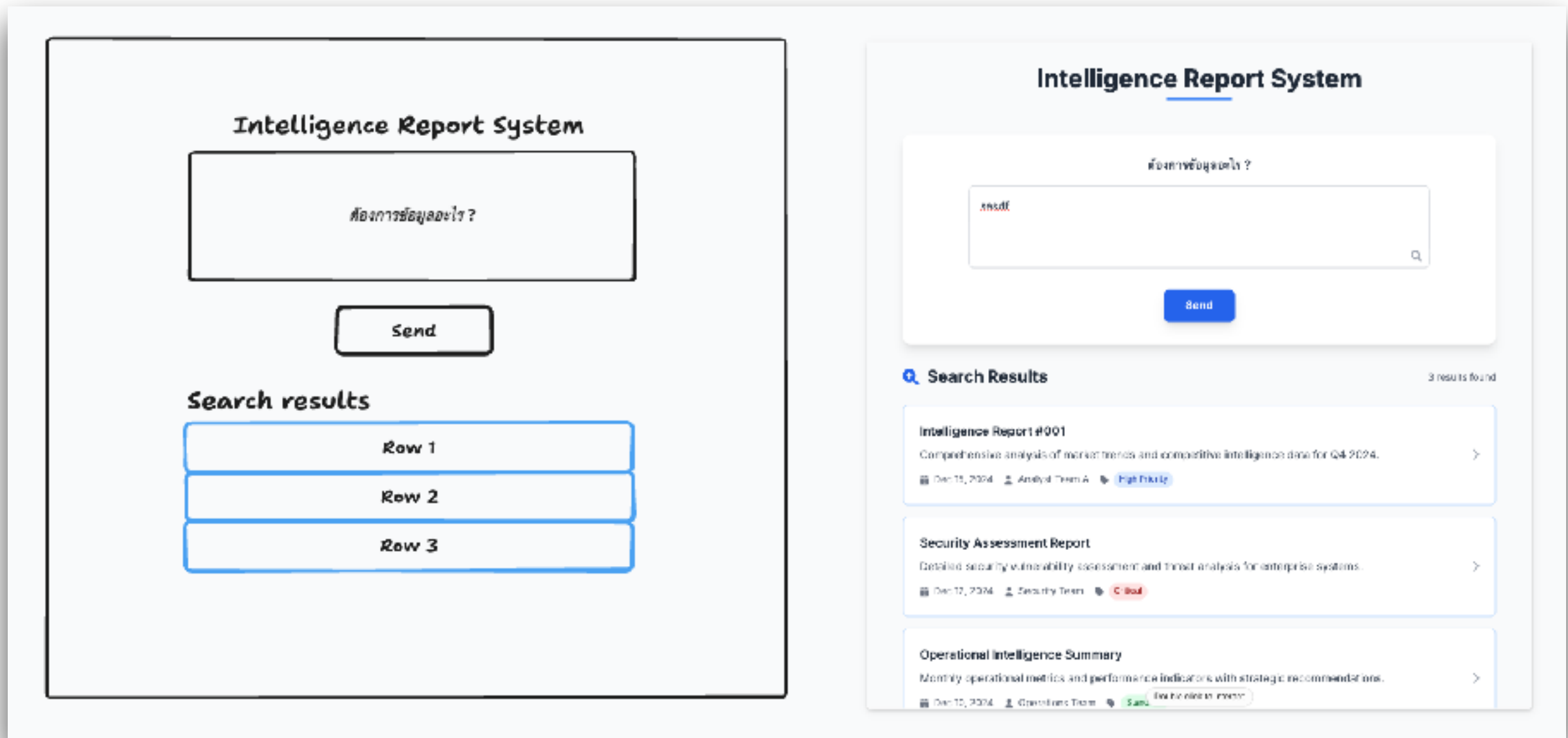
Magic Pattern



<https://www.magicpatterns.com/>



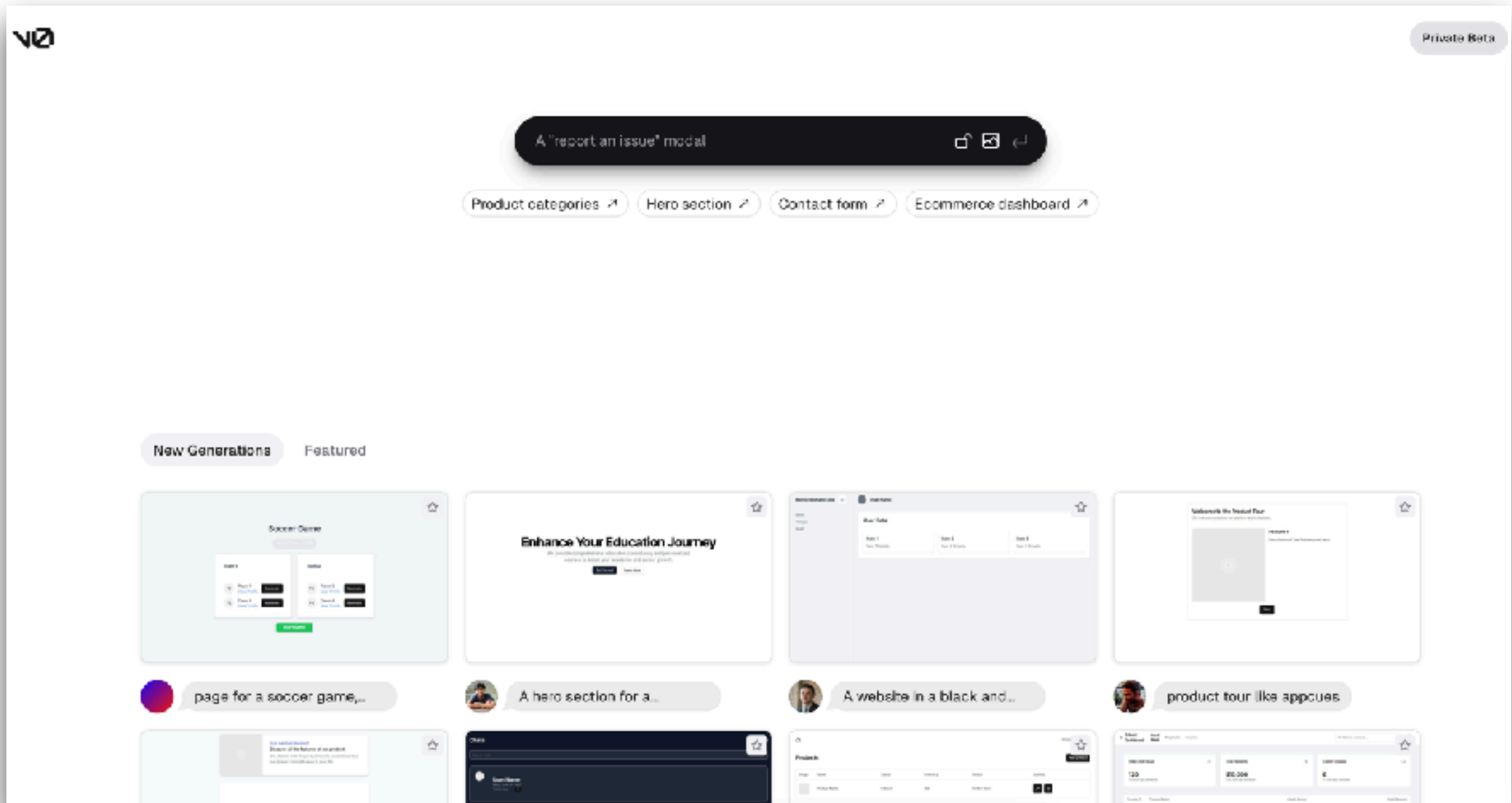
Make Real



<https://github.com/tldraw/make-real>



V0.dev



<https://v0.dev/>



Bolt.new

What do you want to build?

Prompt, run, edit, and deploy full-stack web apps.

How can Bolt help you today?



Start a blog with Astro

Build a mobile app with NativeScript

Create a docs site with Vitepress

Scaffold UI with shadcn

Draft a presentation with Slidify

Code a video with Remotion

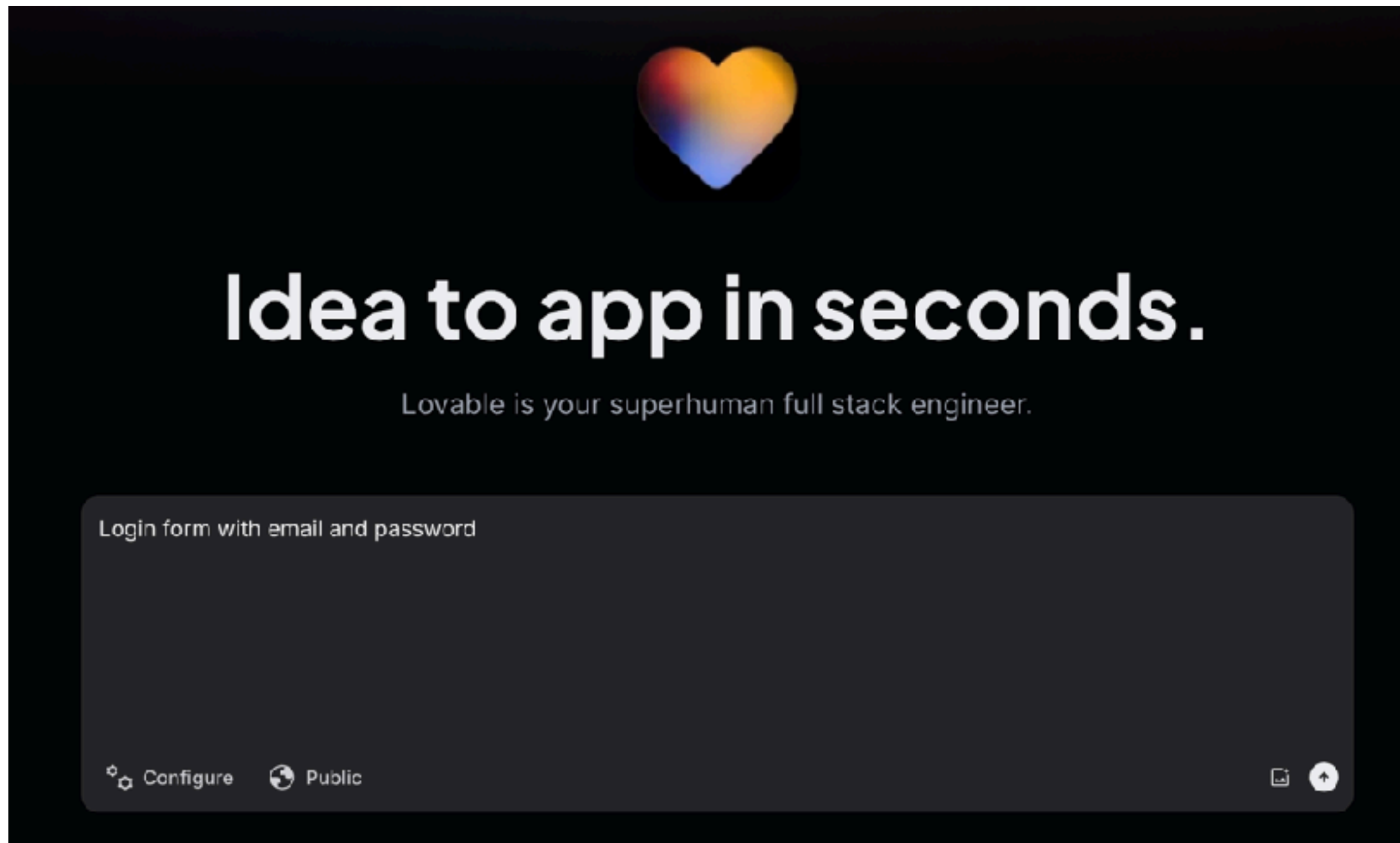
or start a blank app with your favorite stack



<https://bolt.new/>



Lovable



<https://lovable.dev/>



SuperDesign

Vibe it. Design it.

Explore freely, iterate fast. Your design, AI-powered.

Design on web

Integrate with   

Describe what you want to create...

  /   Gemini 3 Flash 

 Use AI Designer  

 Recreate Screenshot

 Import from Site

 Explore Effects

<https://app.superdesign.dev/>

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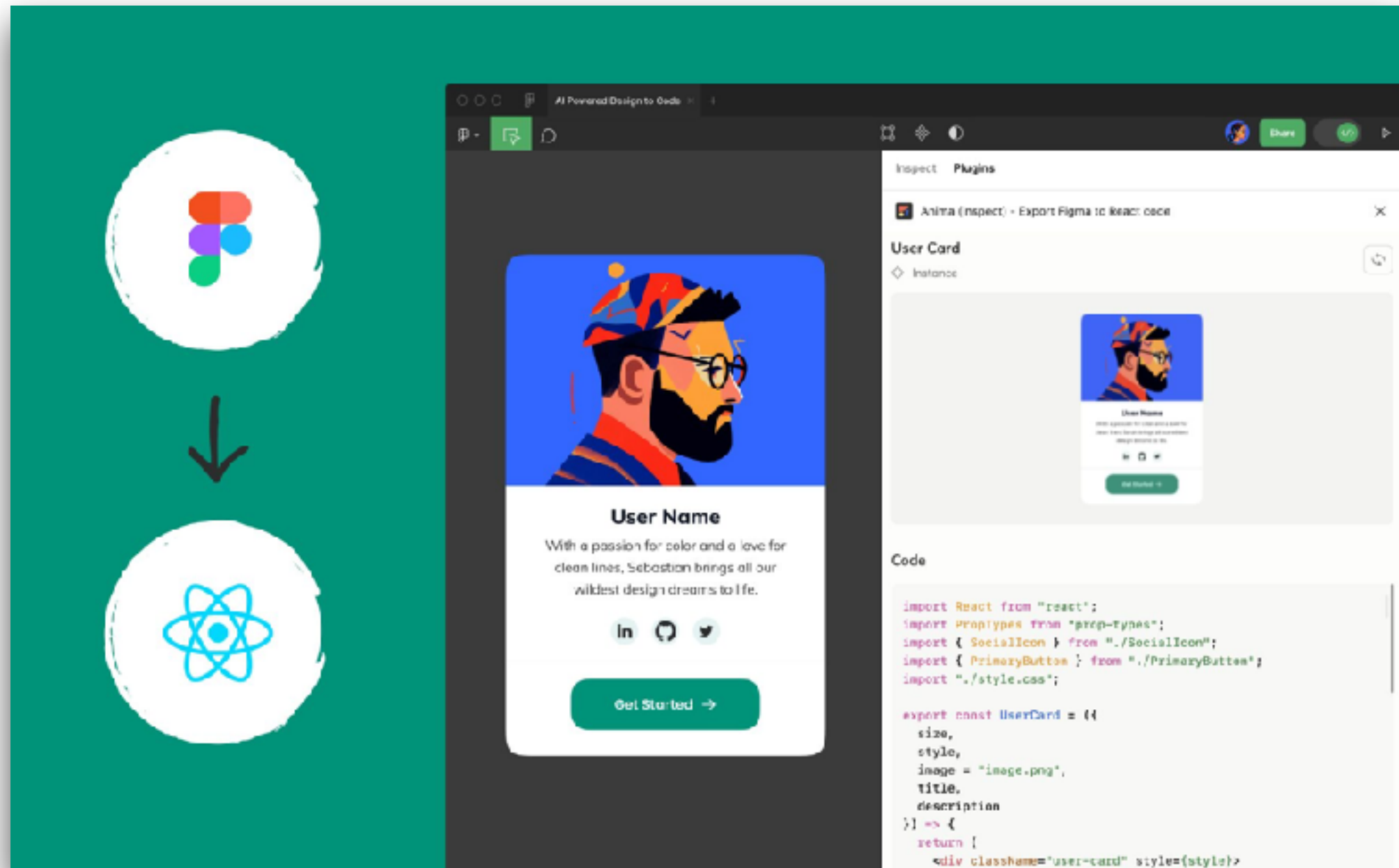
AI for Software Development

95

Design to Code



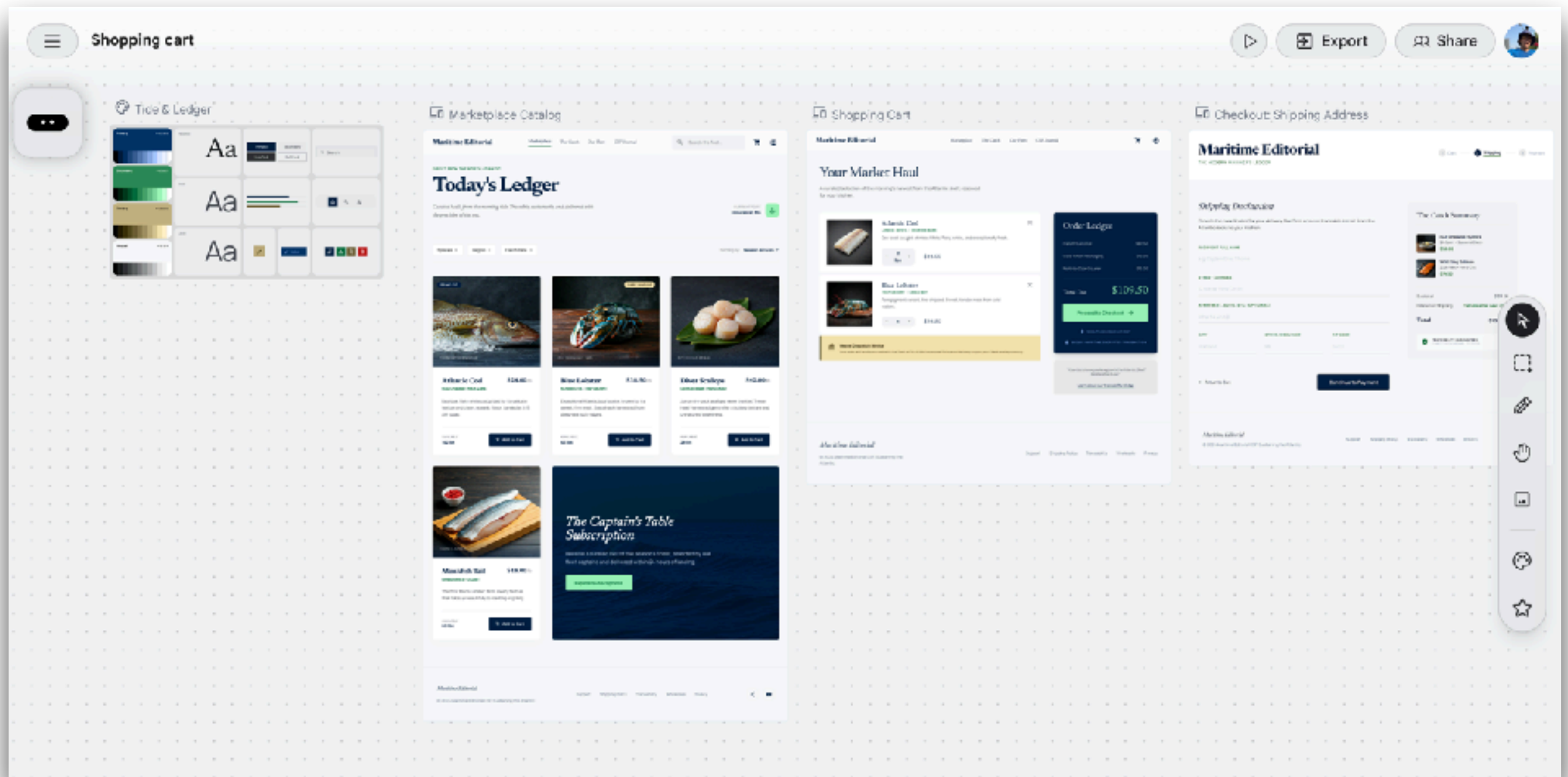
Figma



<https://www.figma.com/mcp-catalog/>



Google Stitch

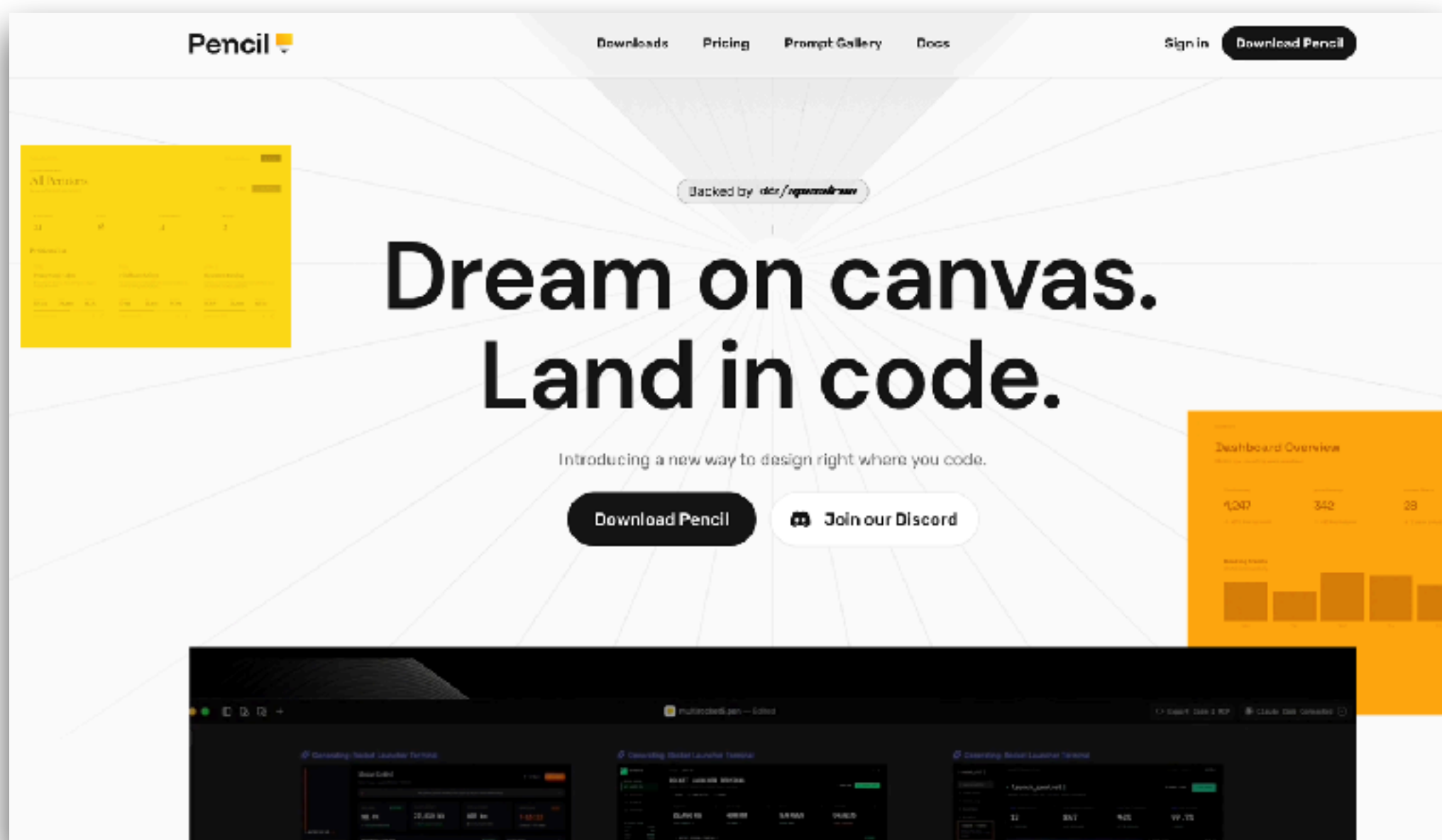


<https://stitch.withgoogle.com/>



Pencil

Bridge the gap between design and development



<https://docs.pencil.dev/getting-started/ai-integration>



Development Process



Develop



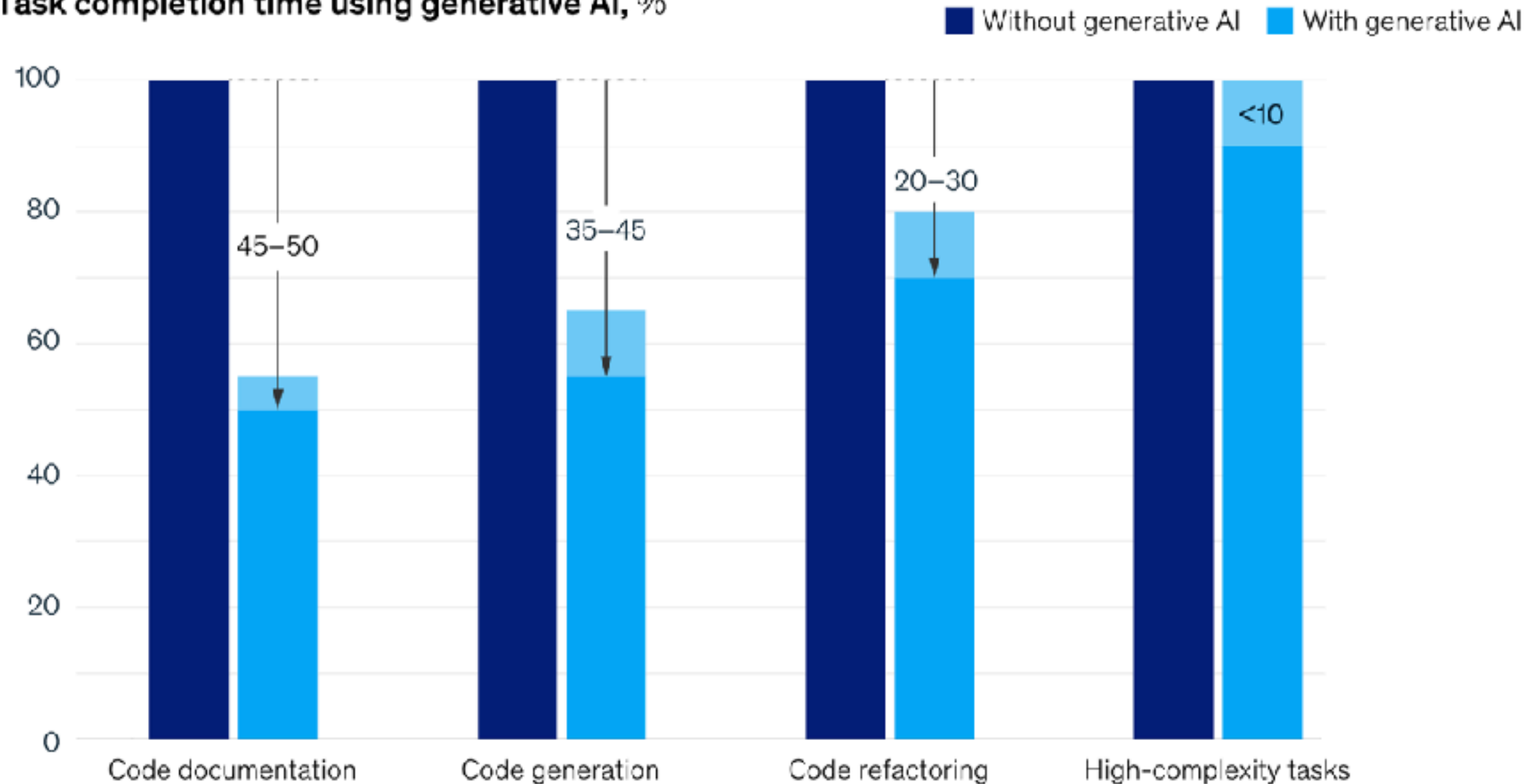
Code generation
Review and explain code
Debugging code
Improve consistency
Code translation



Development

Generative AI can increase developer speed, but less so for complex tasks.

Task completion time using generative AI, %



<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai>



Category of Tools

Chat AI

ChatGPT
Gemini
Claude.ai
DeepSeek

Code AI

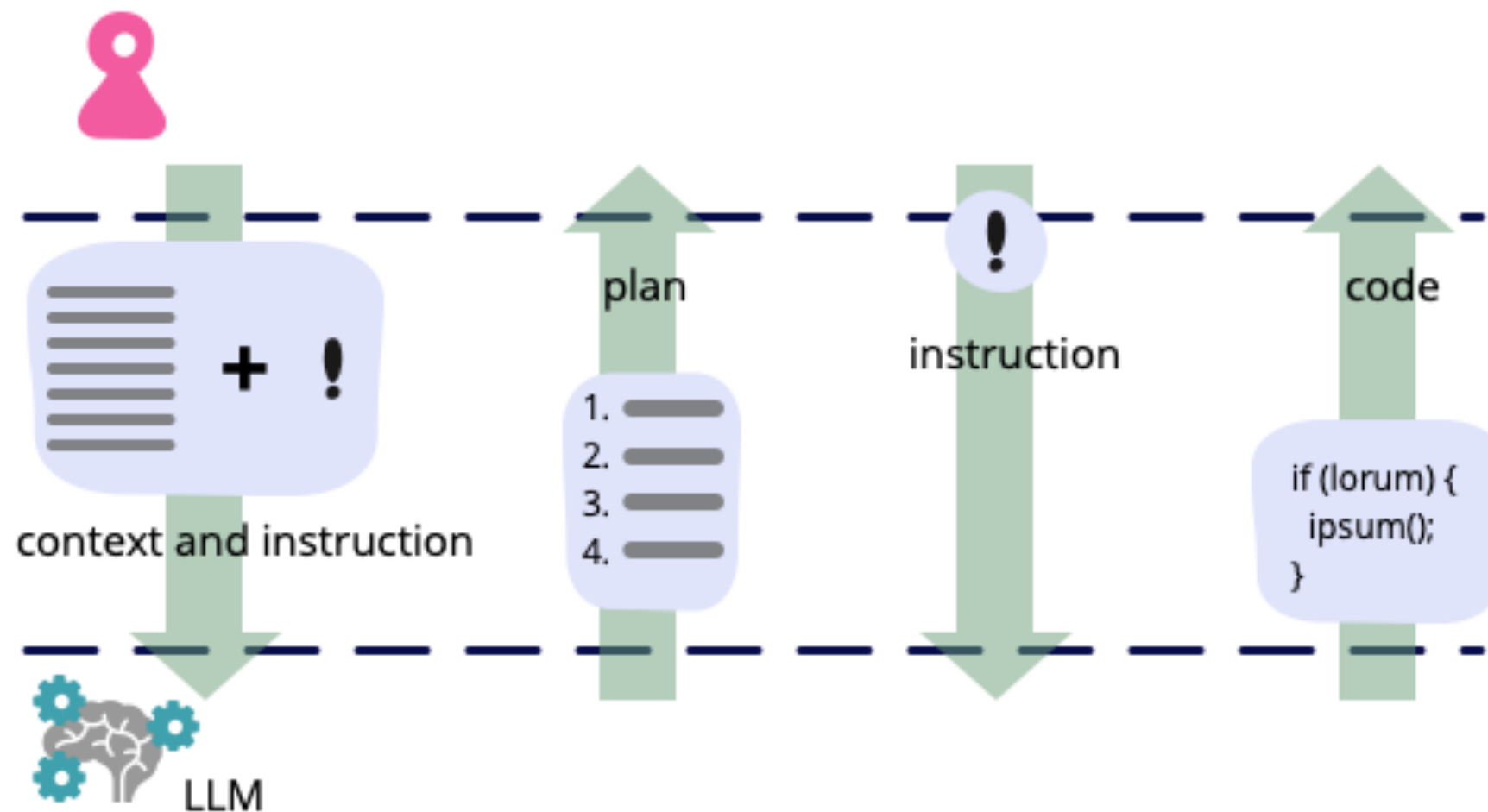
Copilot in VSCode
Cursor IDE
Windsurf
Zed

AI Agent Public and Local

Aider
Claude code
Gemini CLI
Codex CLI
Qwen3-coder



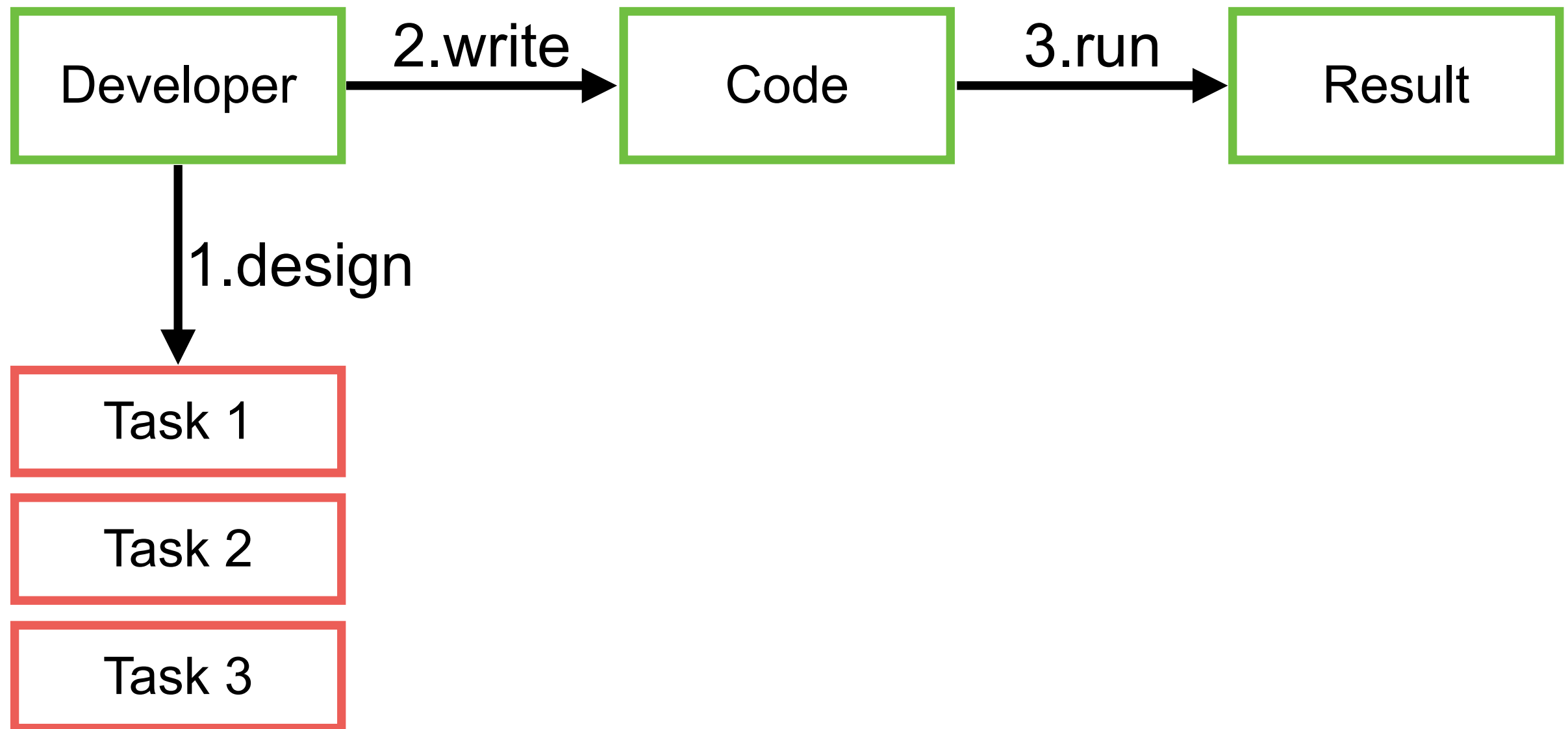
Development



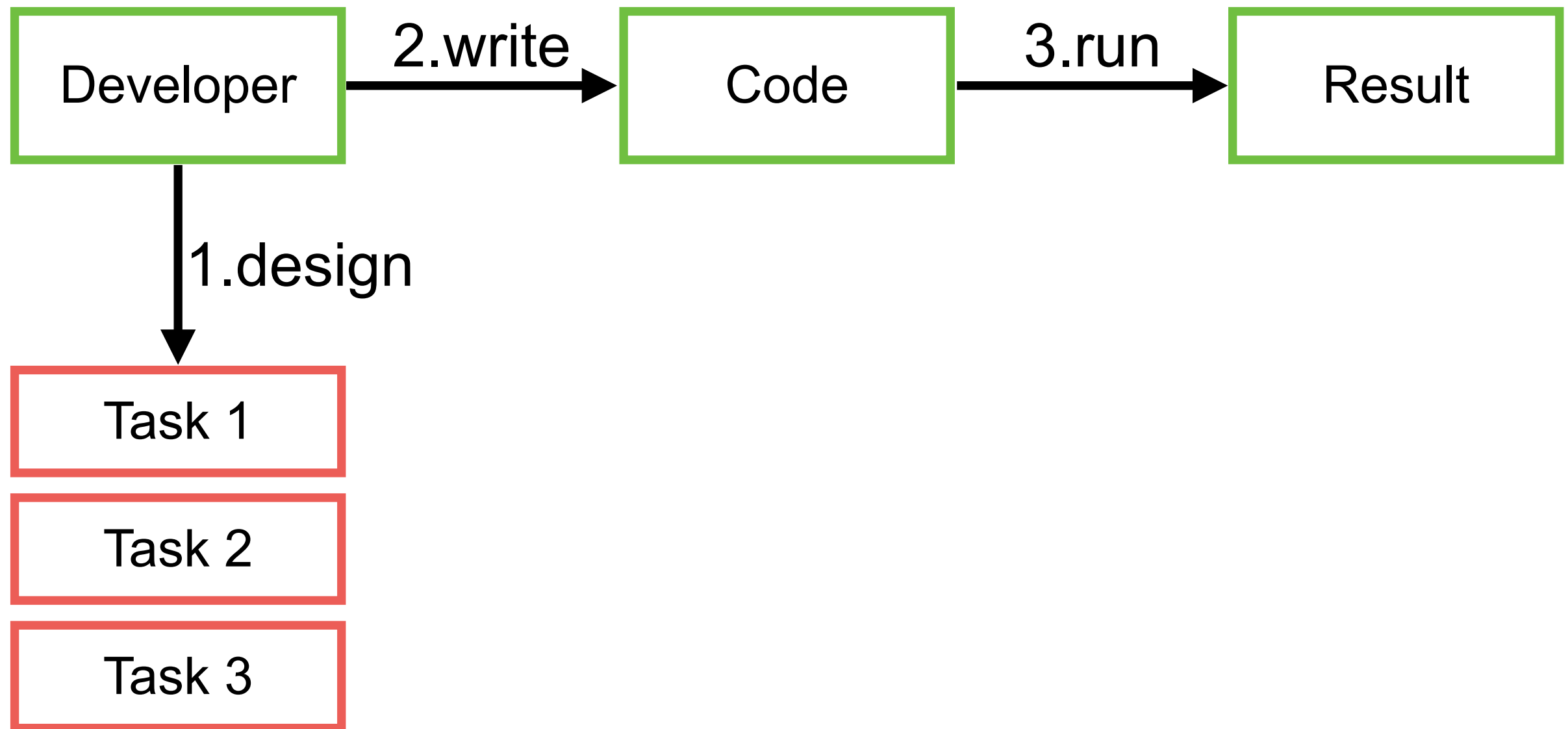
<https://martinfowler.com/articles/2023-chatgpt-xu-hao.html>



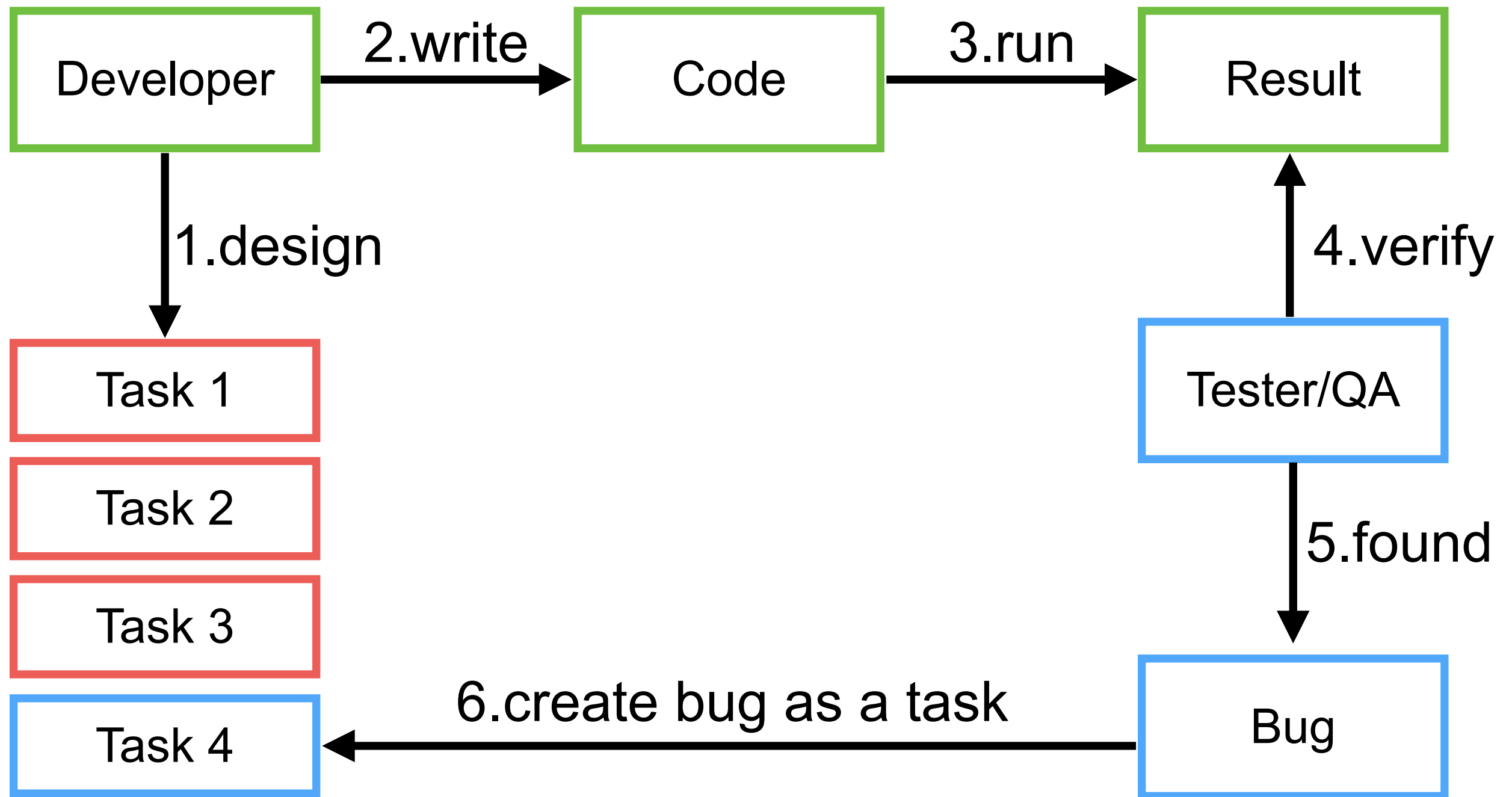
Development



Development



Development + Testing



Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code



Pair programming with AI



AI Pair Programming

Aider is AI pair programming in your terminal

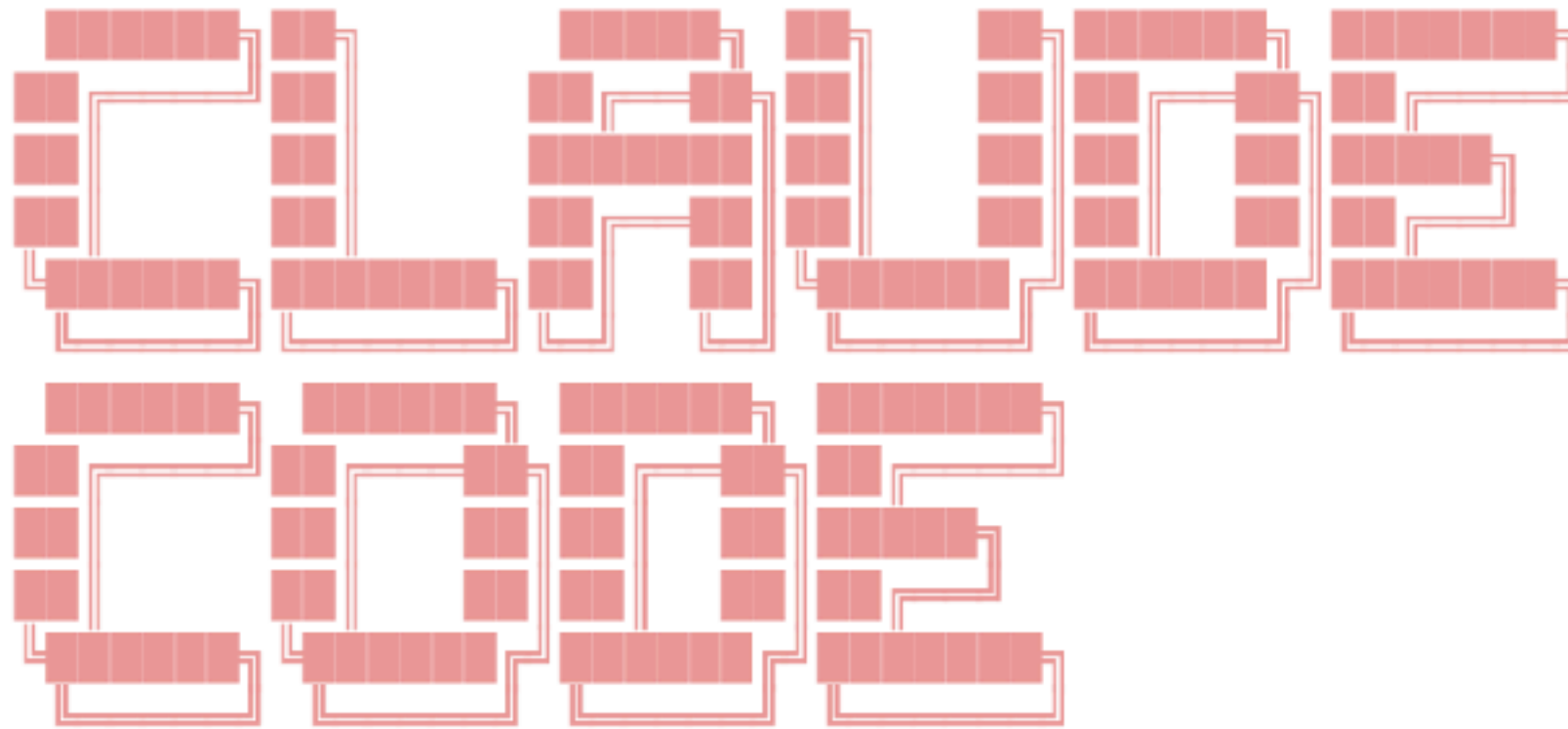
Aider lets you pair program with LLMs, to edit code in your local git repository. Start a new project or work with an existing git repo. Aider works best with GPT-4o & Claude 3.5 Sonnet and can [connect to almost any LLM](#).



<https://github.com/paul-gauthier/aider>



* Welcome to **Claude Code** research preview!



Claude Code is billed based on API usage through your Anthropic Console account.

Pricing may evolve as we move towards general availability.

Press **Enter** to login to your Anthropic Console account...

2025/02

<https://www.anthropic.com/news/claude-3-7-sonnet>





2025/06

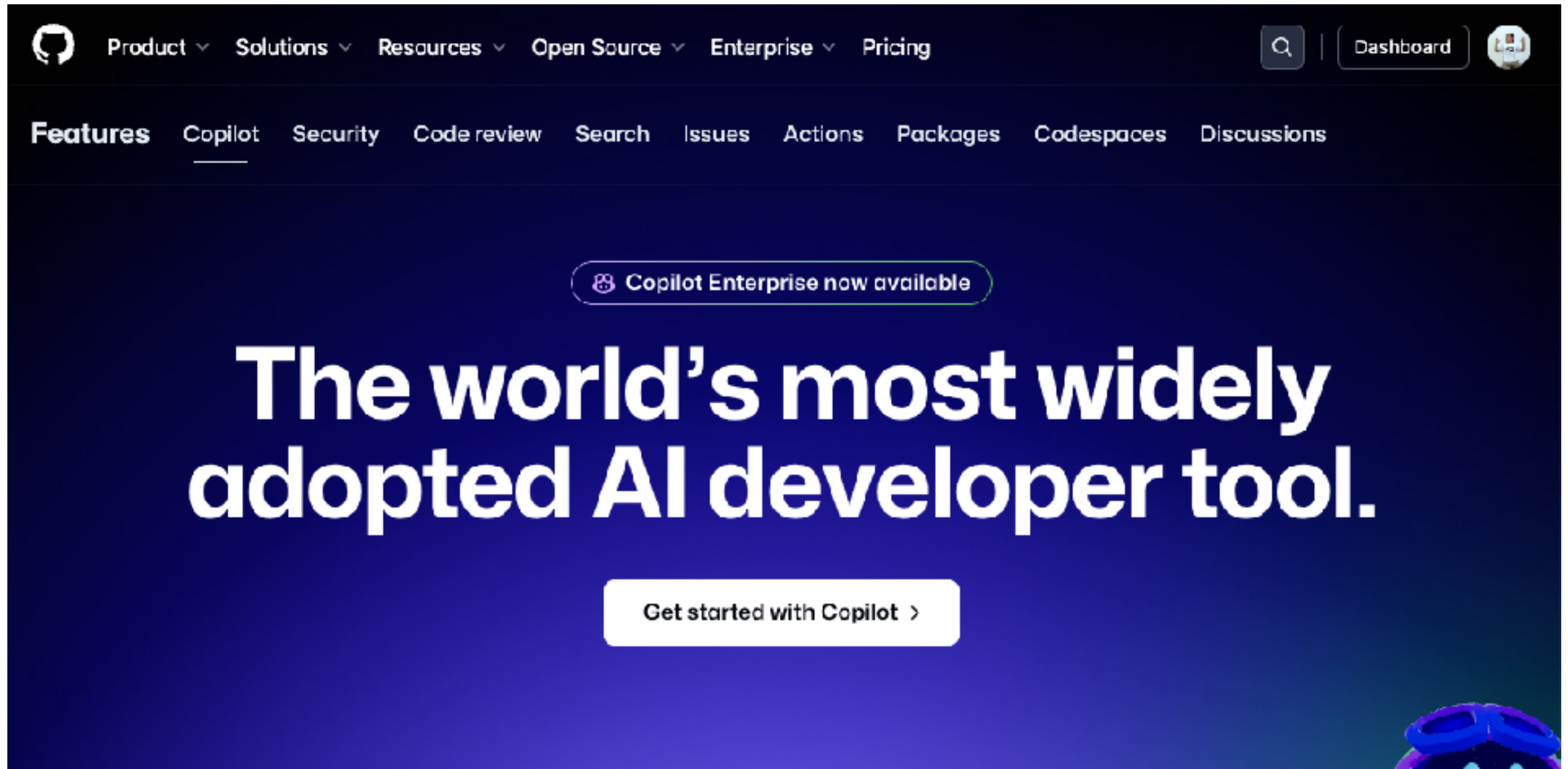
<https://github.com/google-gemini/gemini-cli>



AI in IDE



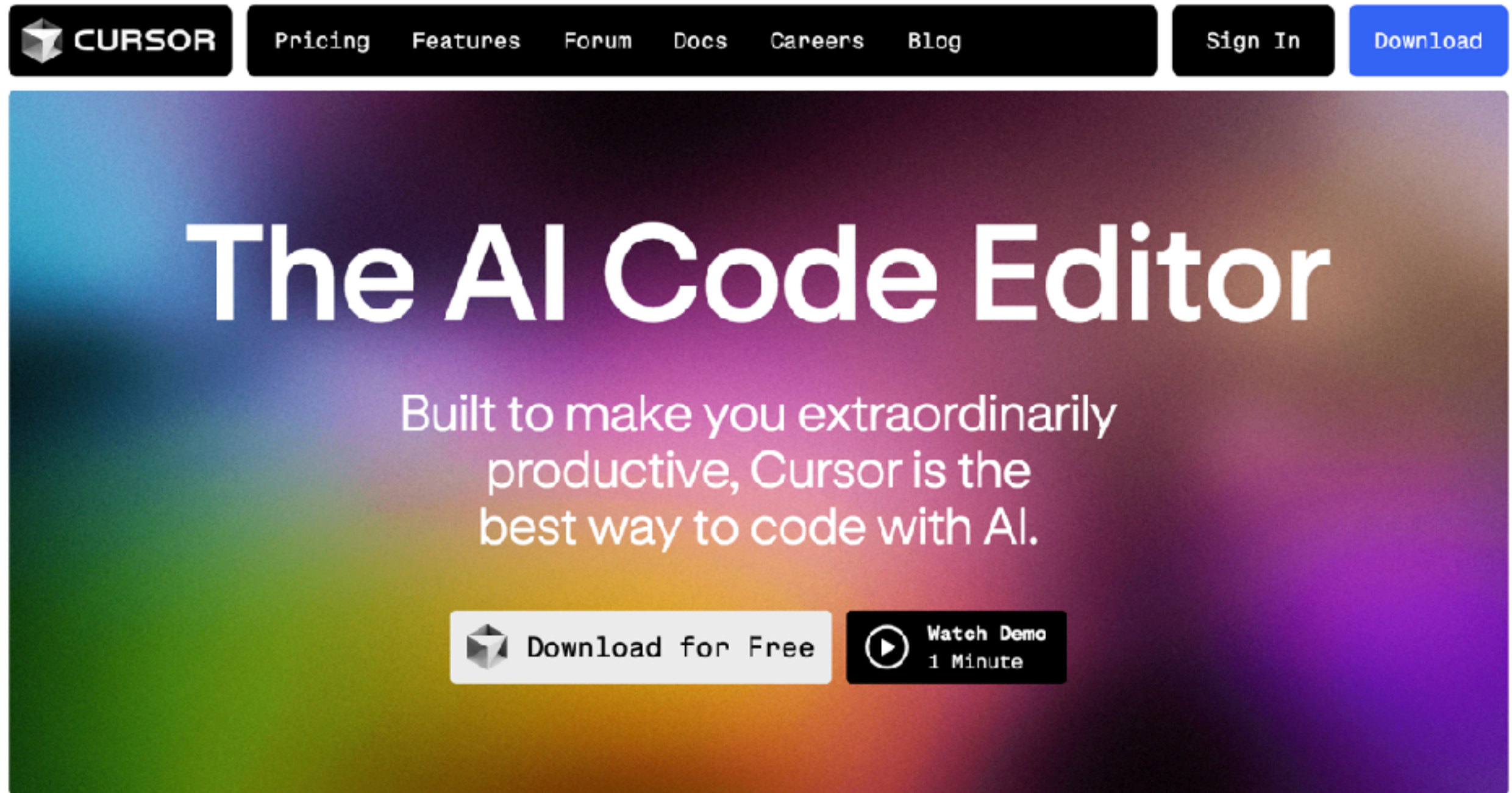
GitHub Copilot



<https://github.com/features/copilot>



Cursor.sh



The image shows the top portion of the Cursor.sh website. At the top is a dark navigation bar with the Cursor logo on the left and links for Pricing, Features, Forum, Docs, Careers, and Blog in the center. On the right of the navigation bar are 'Sign In' and 'Download' buttons. Below the navigation bar is a large hero section with a colorful gradient background. The main heading 'The AI Code Editor' is prominently displayed in white. Below it, a sub-headline states: 'Built to make you extraordinarily productive, Cursor is the best way to code with AI.' At the bottom of the hero section are two buttons: 'Download for Free' with the Cursor logo and 'Watch Demo 1 Minute' with a play button icon.

CURSOR Pricing Features Forum Docs Careers Blog Sign In Download

The AI Code Editor

Built to make you extraordinarily productive, Cursor is the best way to code with AI.

 Download for Free  Watch Demo 1 Minute

<https://www.cursor.com/>



Cursor Directory

The screenshot shows the Cursor Directory website. At the top, there's a navigation bar with 'cursor.directory', 'Get latest updates', 'Subscribe', 'Live', 'Learn', and 'About'. Below this is a search bar and a filter dropdown set to 'All'. A sidebar on the left lists various technologies with their respective counts: TypeScript (15), Python (9), Next.js (9), React (9), PHP (5), C# (4), Expo (4), React Native (4), Tailwind (4), Supabase (4), Web Development (3), Game Development (3), JavaScript (3), and Laravel (3). The main content area is titled 'TypeScript' and displays a grid of developer profiles. Each profile includes a bio, a list of skills, and a 'Submit +' button. The developers shown are Krish Kalaria, Mohammadali Karimi, and Nathan Brachette. The bio for Krish Kalaria reads: 'You are an expert in TypeScript, React Native, Expo, and Mobile UI development. Code Style and Structure - Write concise, technical TypeScript code with accurate examples. - Use functional and declarative programming patterns; avoid classes. - Prefer iteration and modularization over code duplication. - Use descriptive variable names with auxiliary verbs (e.g., isLoading, hasError). - Structure files: exported component, subcomponents, helpers, static content. - Follow Expo's official documentation for...'. The bio for Mohammadali Karimi reads: 'You are a Senior Front-End Developer and an Expert in ReactJS, NextJS, JavaScript, TypeScript, HTML, CSS and modern UI/UX frameworks (e.g., TailwindCSS, Shadcn, Radix). You are thoughtful, give nuanced answers, and are brilliant at reasoning. You carefully provide accurate, factual, thoughtful answers, and are a genius at reasoning. - Follow the user's requirements carefully & to the letter. - First think step-by-step - describe your plan for what to build in pseudocode, written out in great...'. The bio for Nathan Brachette reads: 'You are an expert in TypeScript, Gatsby, React and Tailwind. Code Style and Structure - Write concise, technical TypeScript code. - Use functional and declarative programming patterns; avoid classes. - Prefer iteration and modularization over code duplication. - Use descriptive variable names with auxiliary verbs (e.g., isLoading, hasError). - Structure files: exported...'. Each profile also has a 'Submit +' button and a small icon representing the developer's profile picture.

<https://cursor.directory/>



Windsurf by Codeium



Built to keep you in *flow state*

The first agentic IDE, and then some. The Windsurf Editor is where the work of developers and AI truly flow together, allowing for a coding experience that feels like literal magic.

 Download the Windsurf Editor

[See all download options](#)

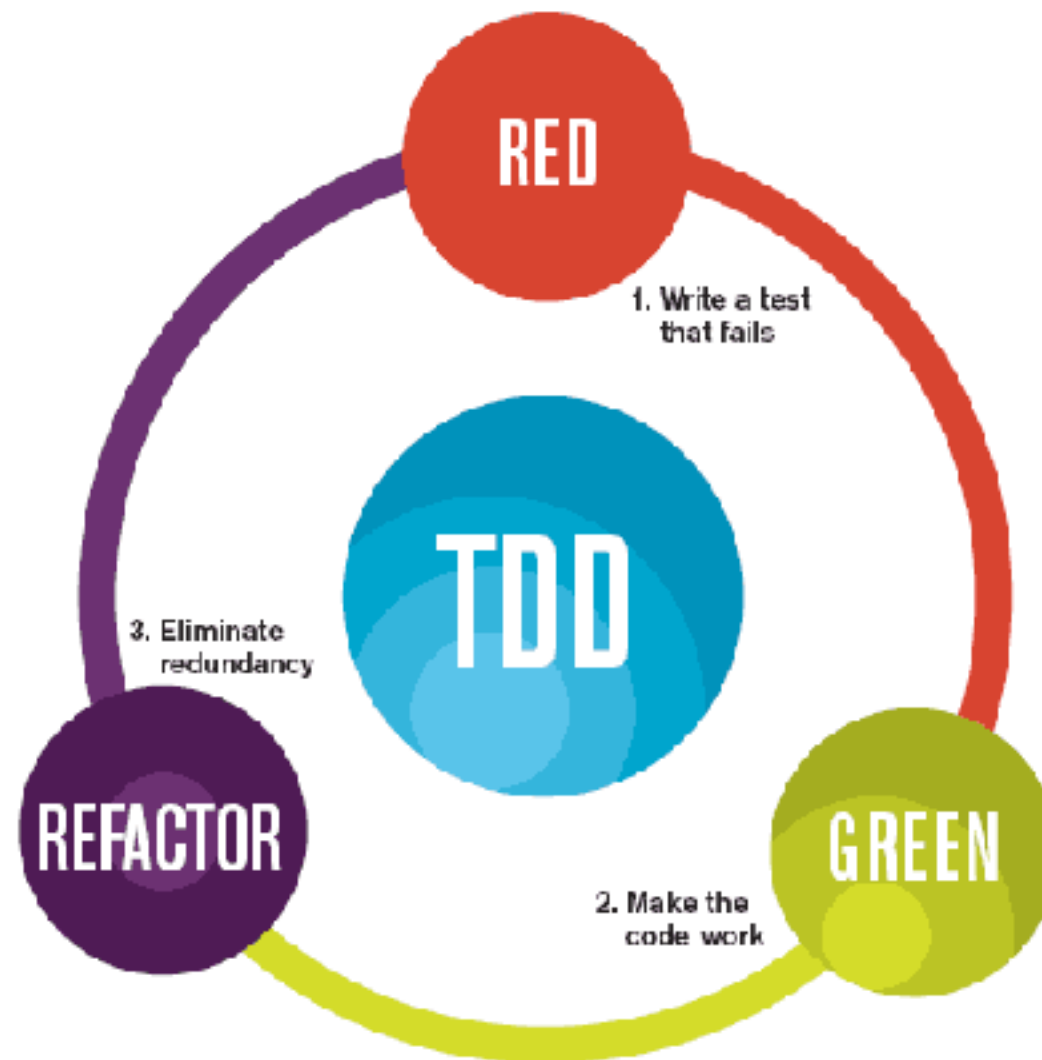
<https://codeium.com/windsurf>



Test-Driven Development



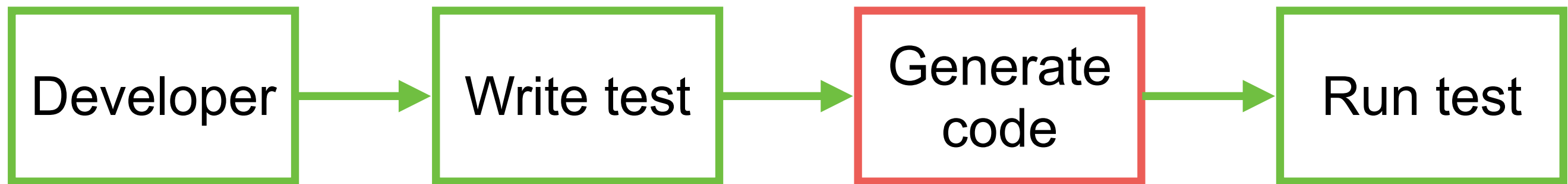
Test-Driven-Development



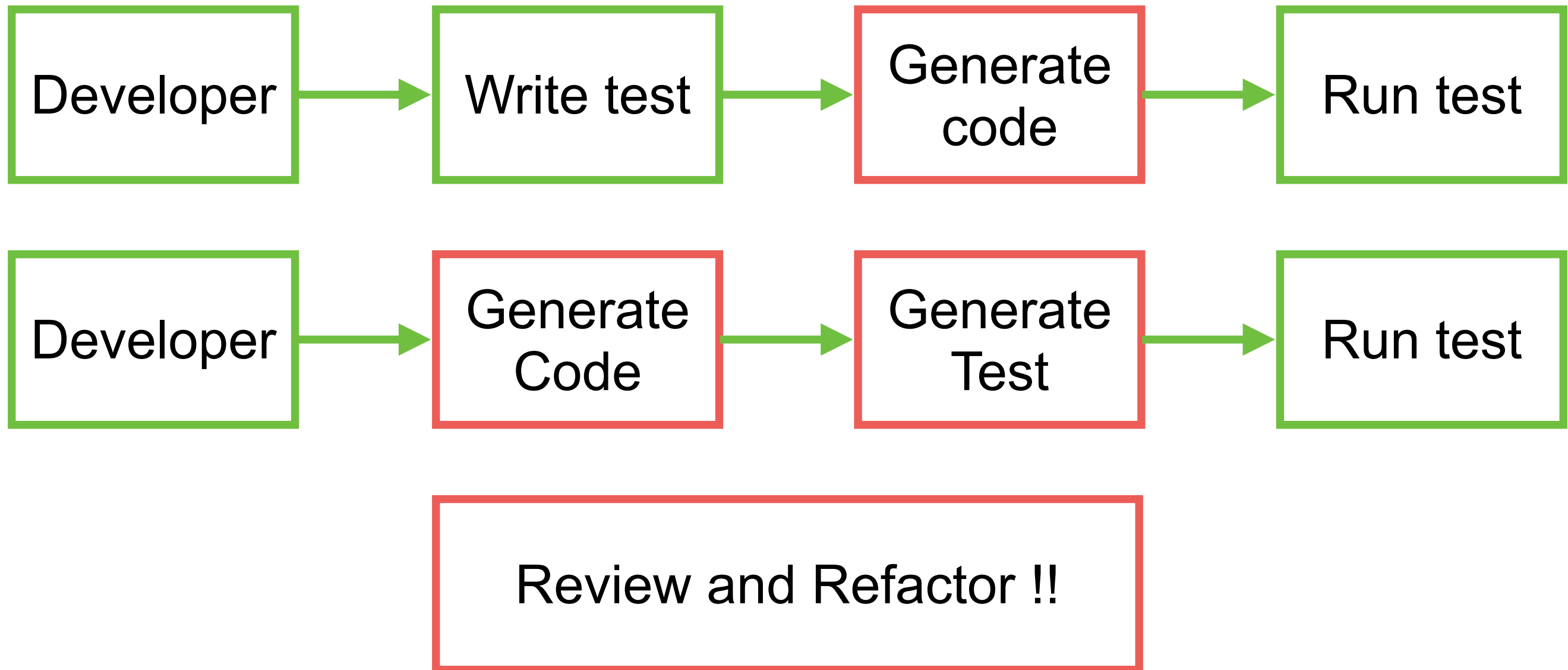
The mantra of Test-Driven Development (TDD) is "red, green, refactor."



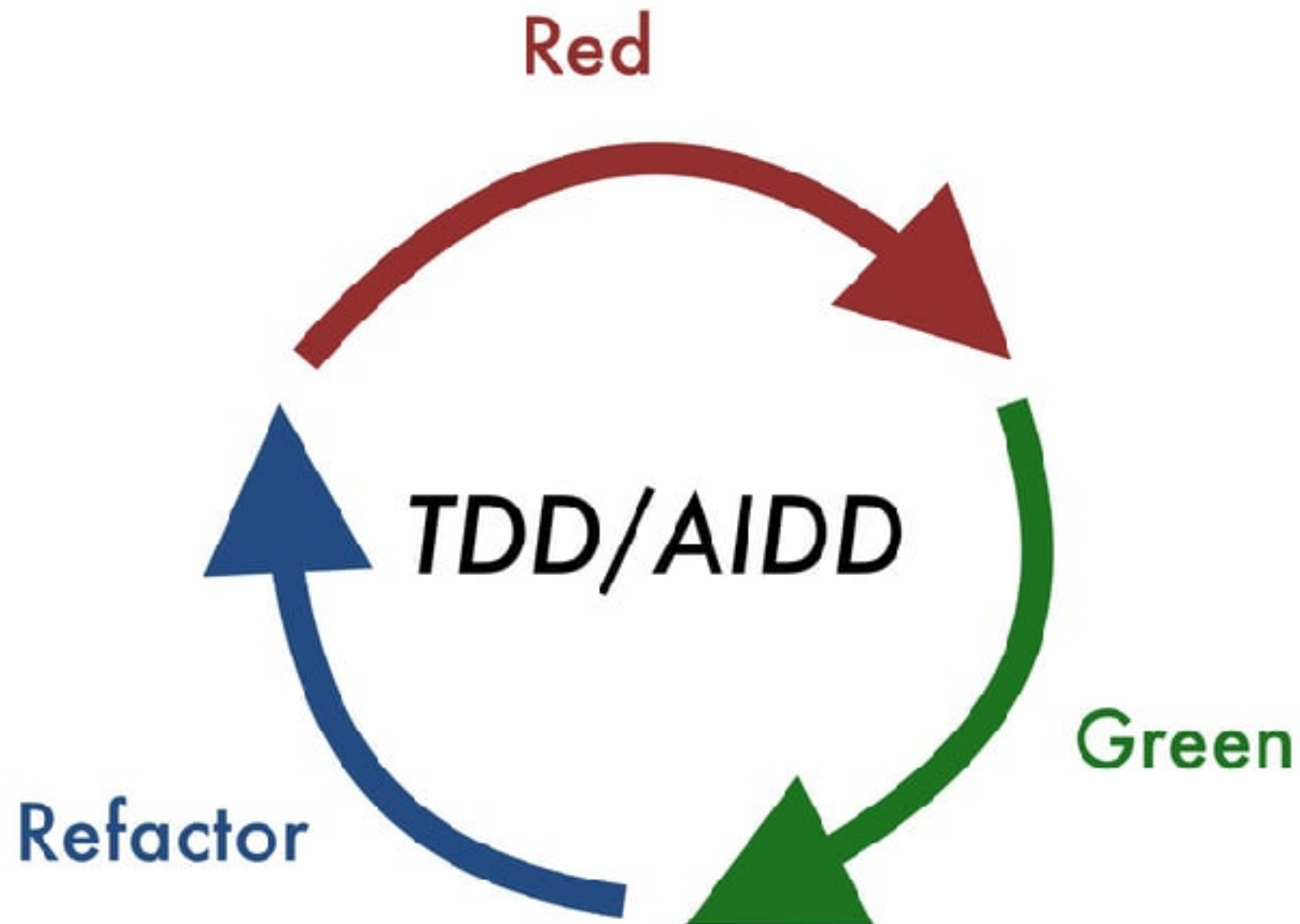
Test-Driven-Development



Test-Driven-Development



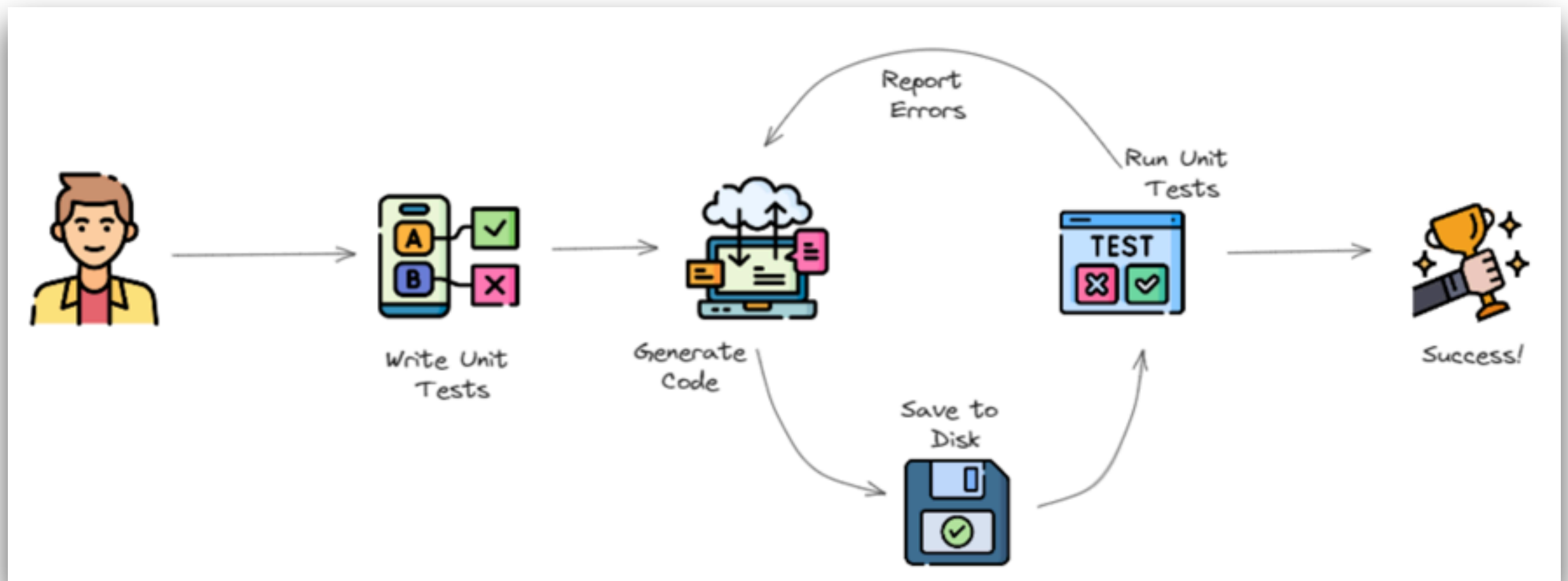
AI-DD



<https://dev.to/dawiddahl/ai-is-changing-the-way-we-code-ai-driven-development-aid-2ngo>



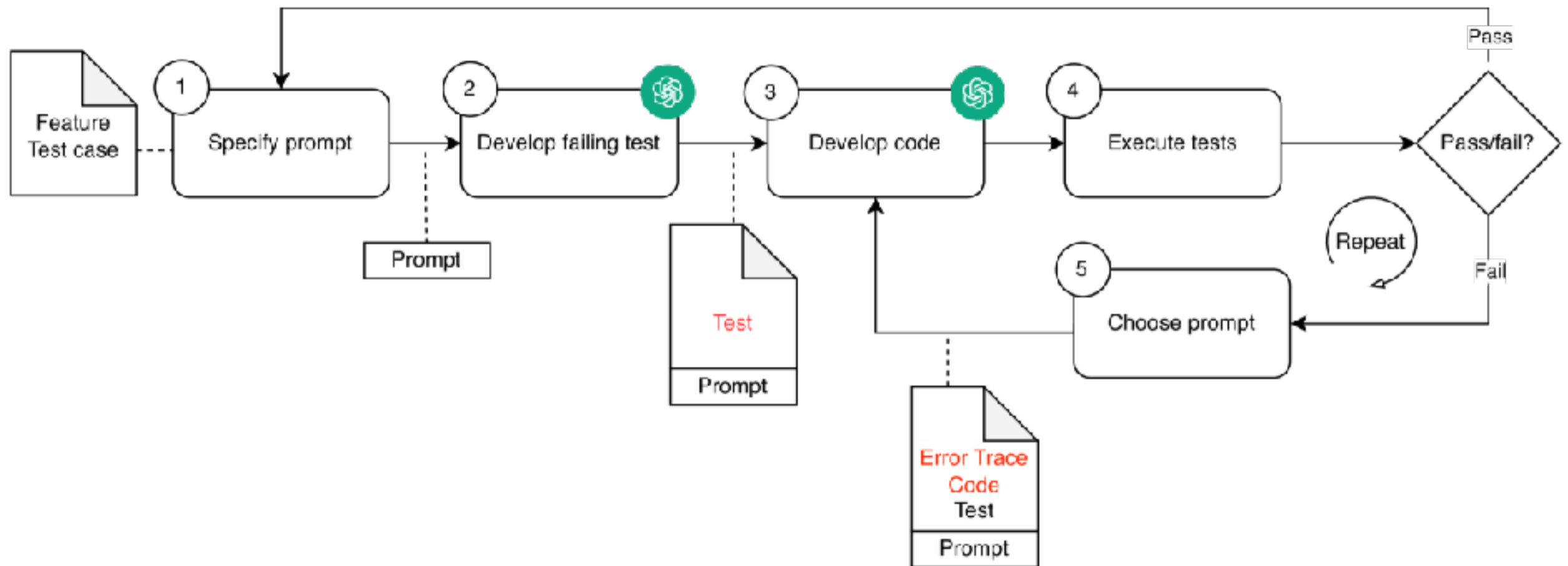
TDD with AI



<https://github.com/allenheltondev/tdd-ai>



TDD with AI



<https://arxiv.org/html/2405.10849v1>



Test-Driven-Generation (TDG)



Test-Driven-Generation (TDG)

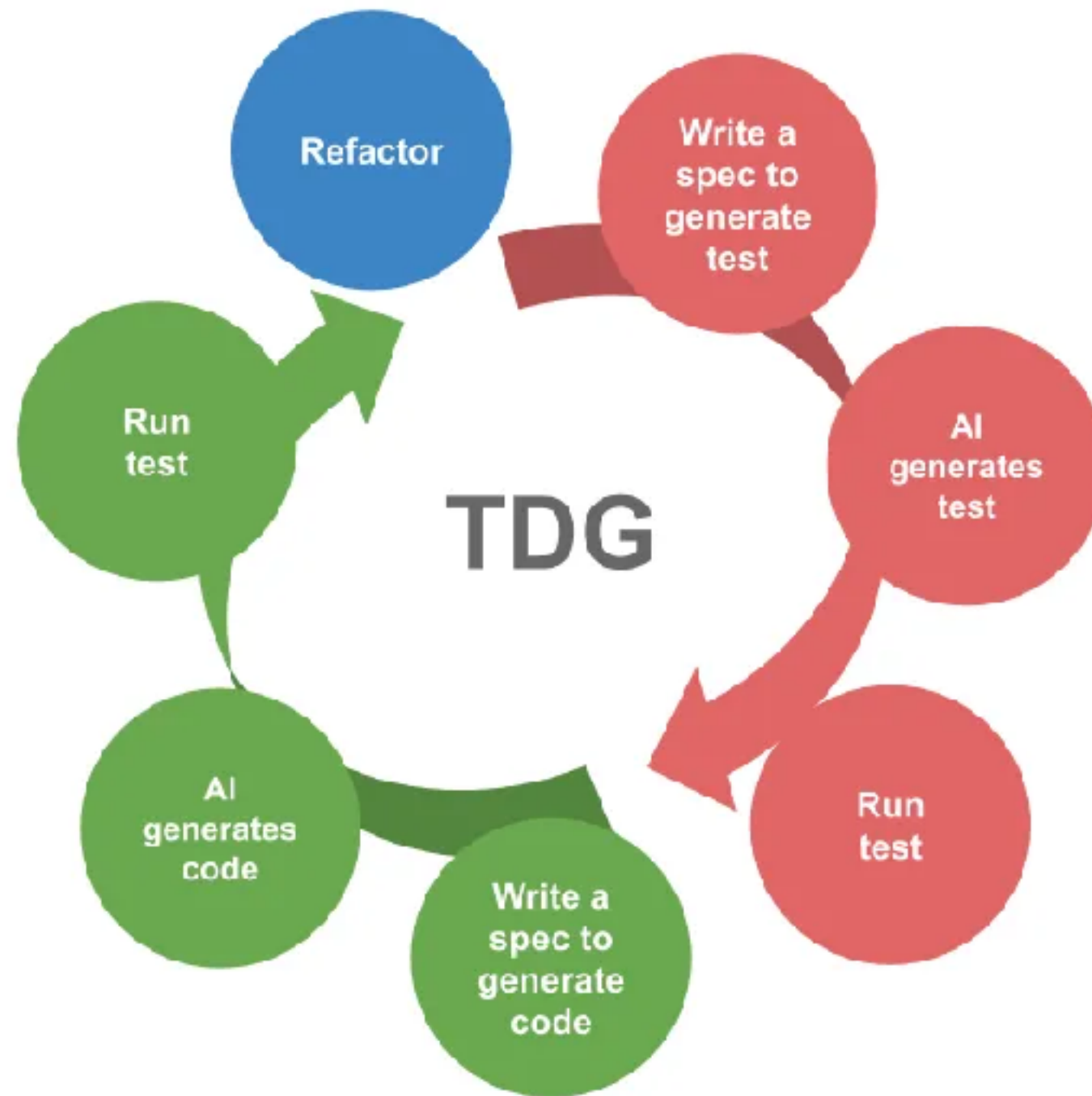
Development practice that integrate Generative AI into the development life-cycle



<https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8>



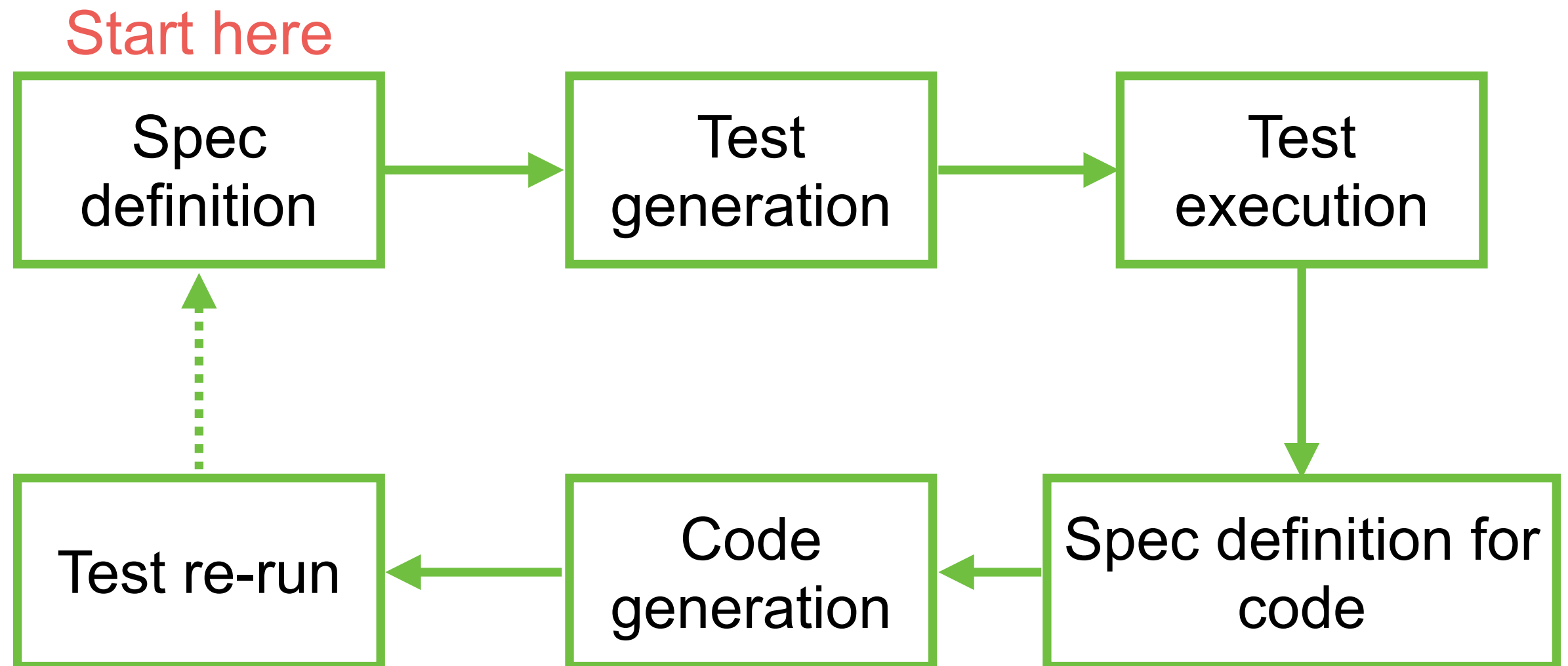
Test-Driven-Generation (TDG)



<https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8>



Test-Driven-Generation (TDG)



Tips and Techniques with AI

Scope of prompt

Error in result

Setup and config
project

Latest information

Use technical
keywords



Specification-Driven Development (SDD)

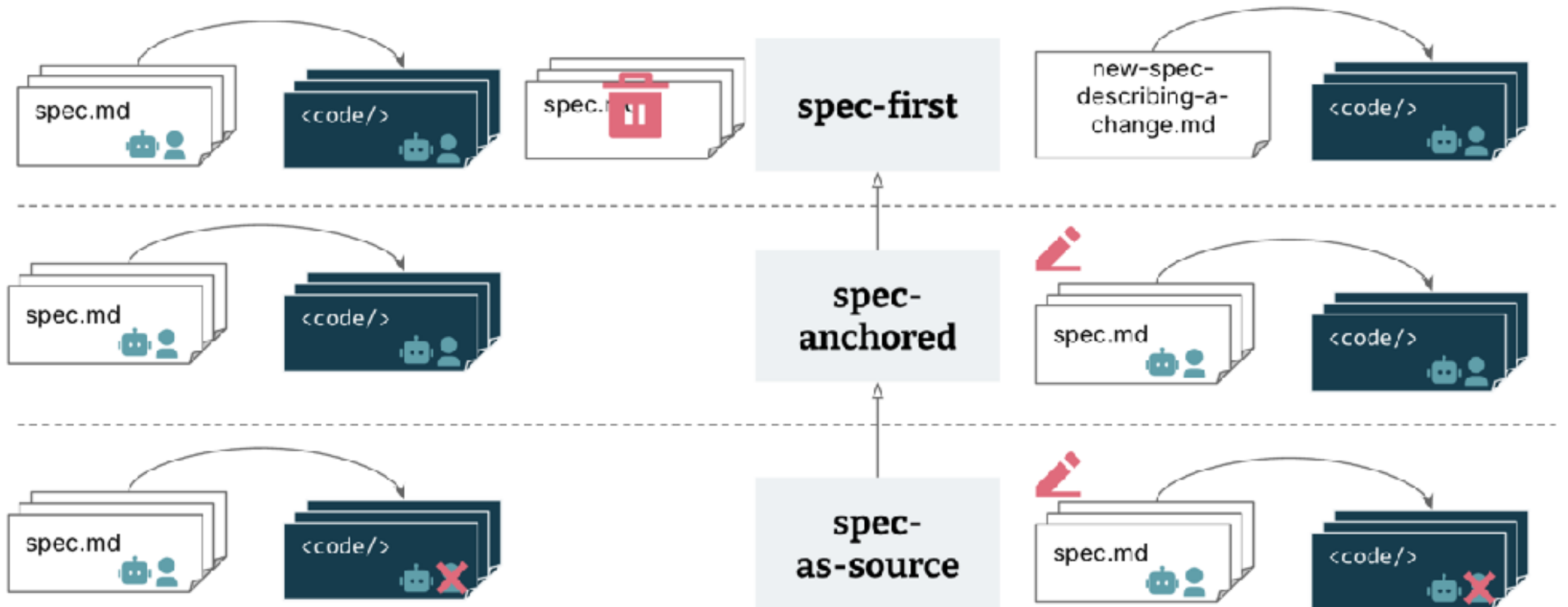


Workflow SDD

Levels of “spec-driven”

Creation of feature

Evolution and maintenance of feature

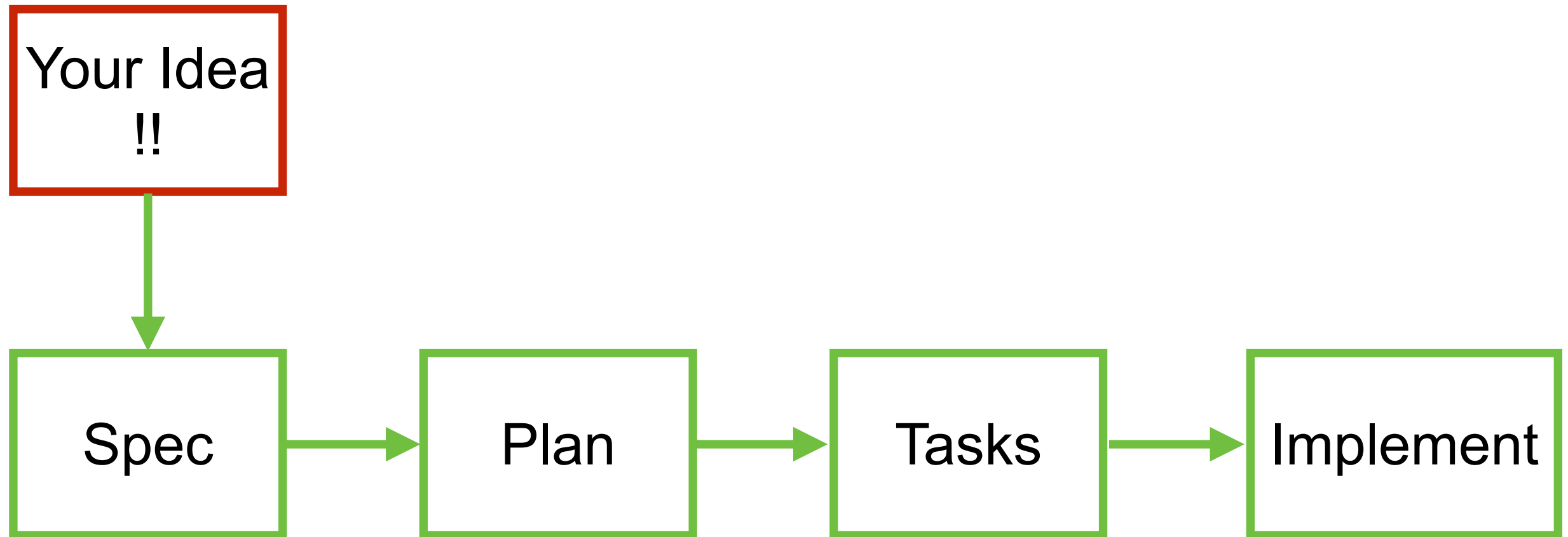


<https://martinfowler.com/articles/exploring-gen-ai.html>

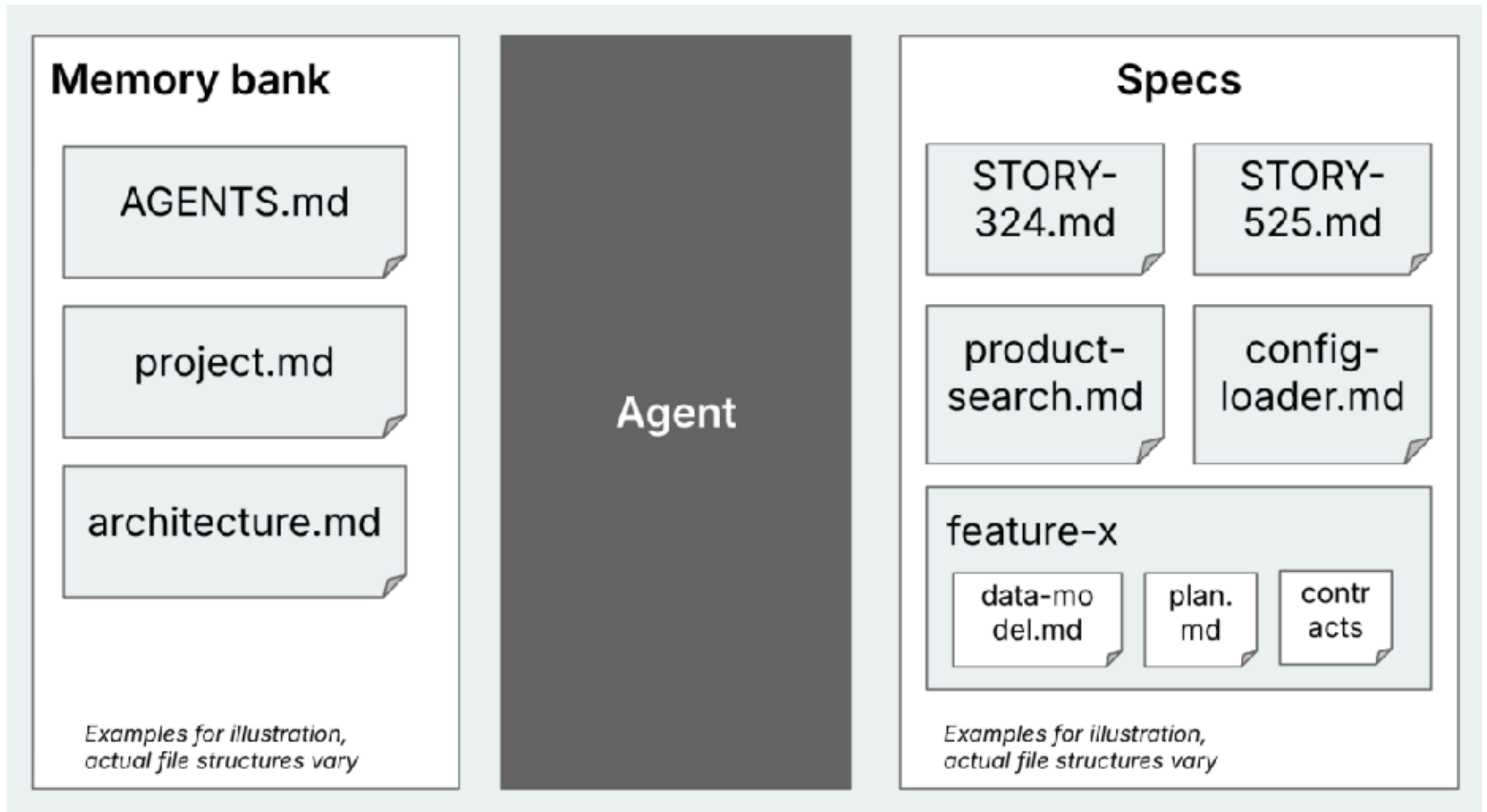
<https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html>



Workflow SDD



Memory Bank



Workflow Tools

AWS Kiro

Requirement
Design
Tasks

Spec-kit

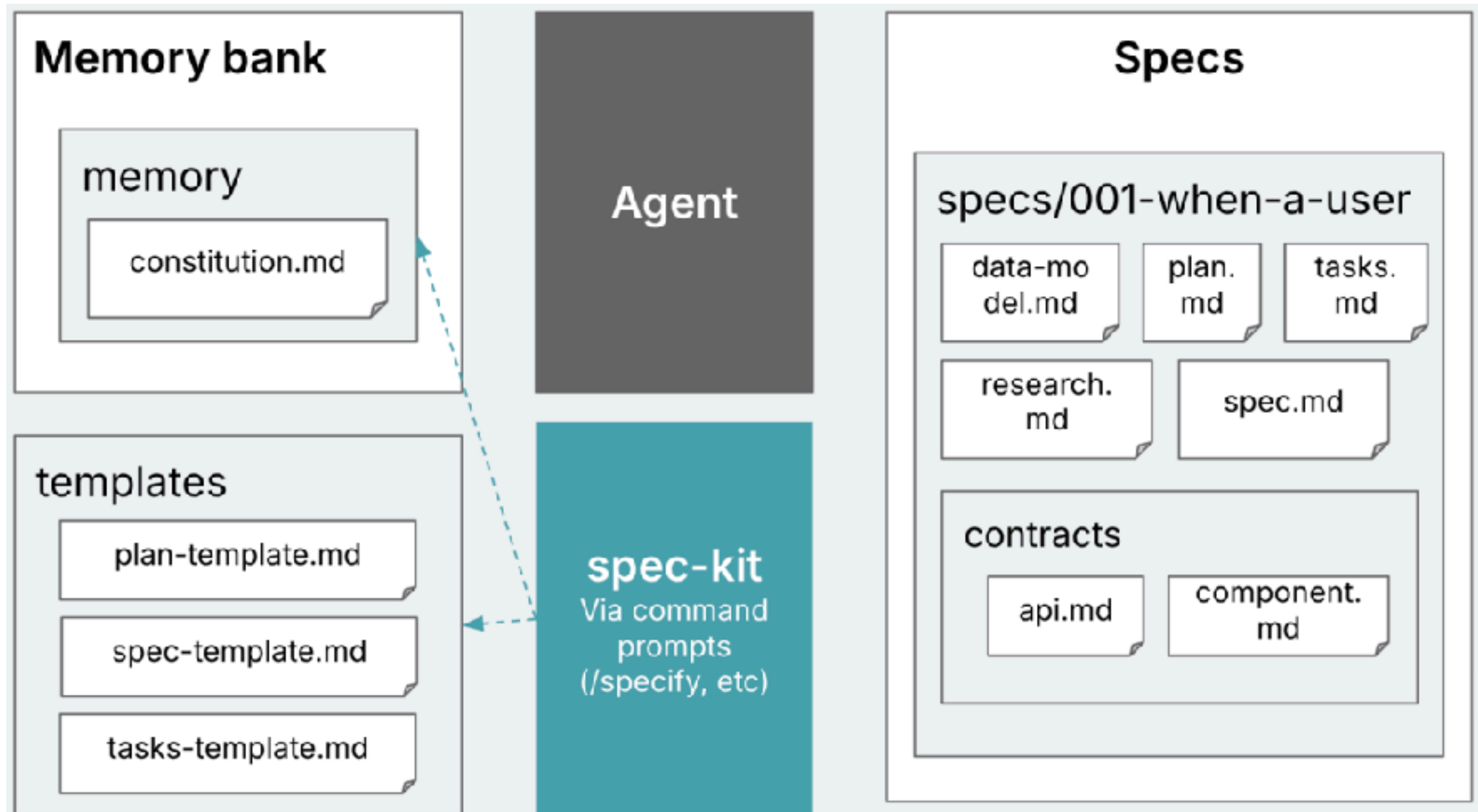
Specify
Plan
Tasks

Custom by IDE

VSCode rules
Cursor rules
Github rules



Spec-Kit



Workshop with Coding

Chat

Text Editor with AI

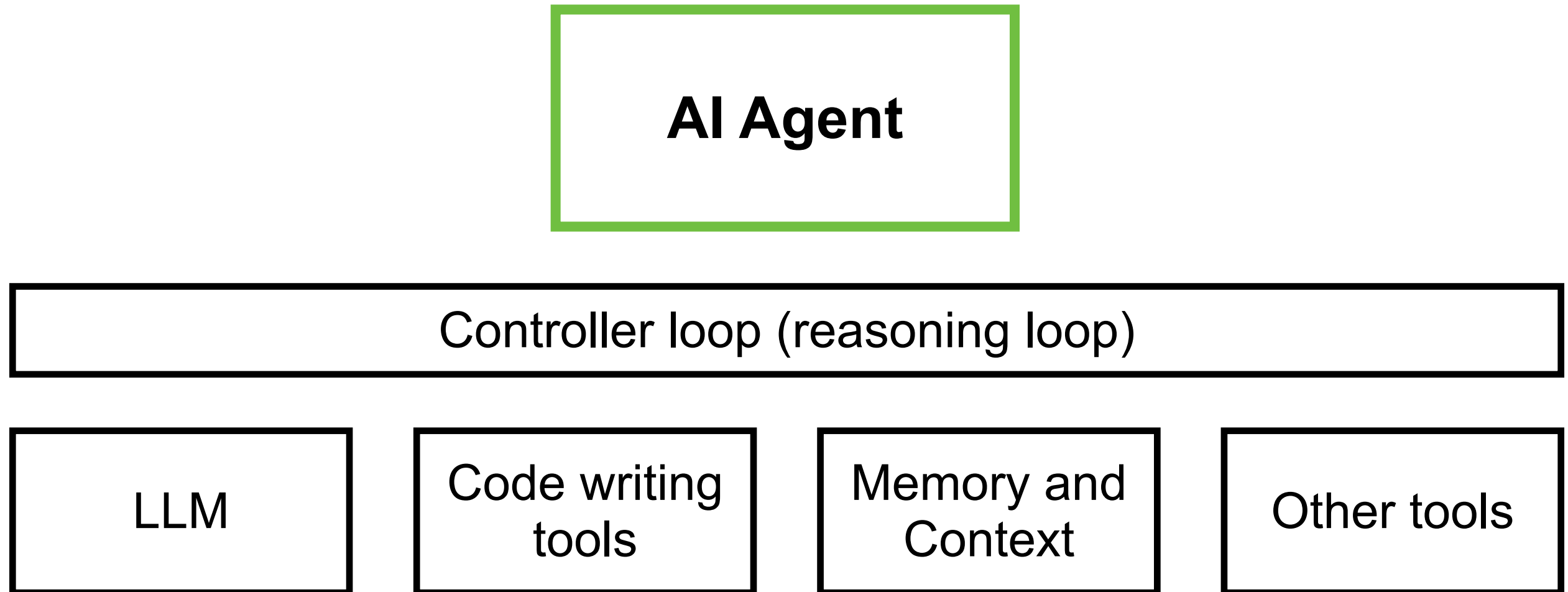
Pair programming
with AI

SDD and Rules

BMAD-METHOD



AI Agent for Coding



AI Agent for Coding

Claude Code
Github Copilot Chat
Cline
Qwen code
Gemini CLI
Pi coding agent



Testing Process with High quality process

Functional

Non-Functional



Testing



Test cases writing
Test code generation
Bug detection
Test planning
Data test generation



6 Keys Software Quality

Defect density

Code duplication

Hardcode token/key

Security
vulnerabilities

Outdated package

Non-permissive
opensource libraries

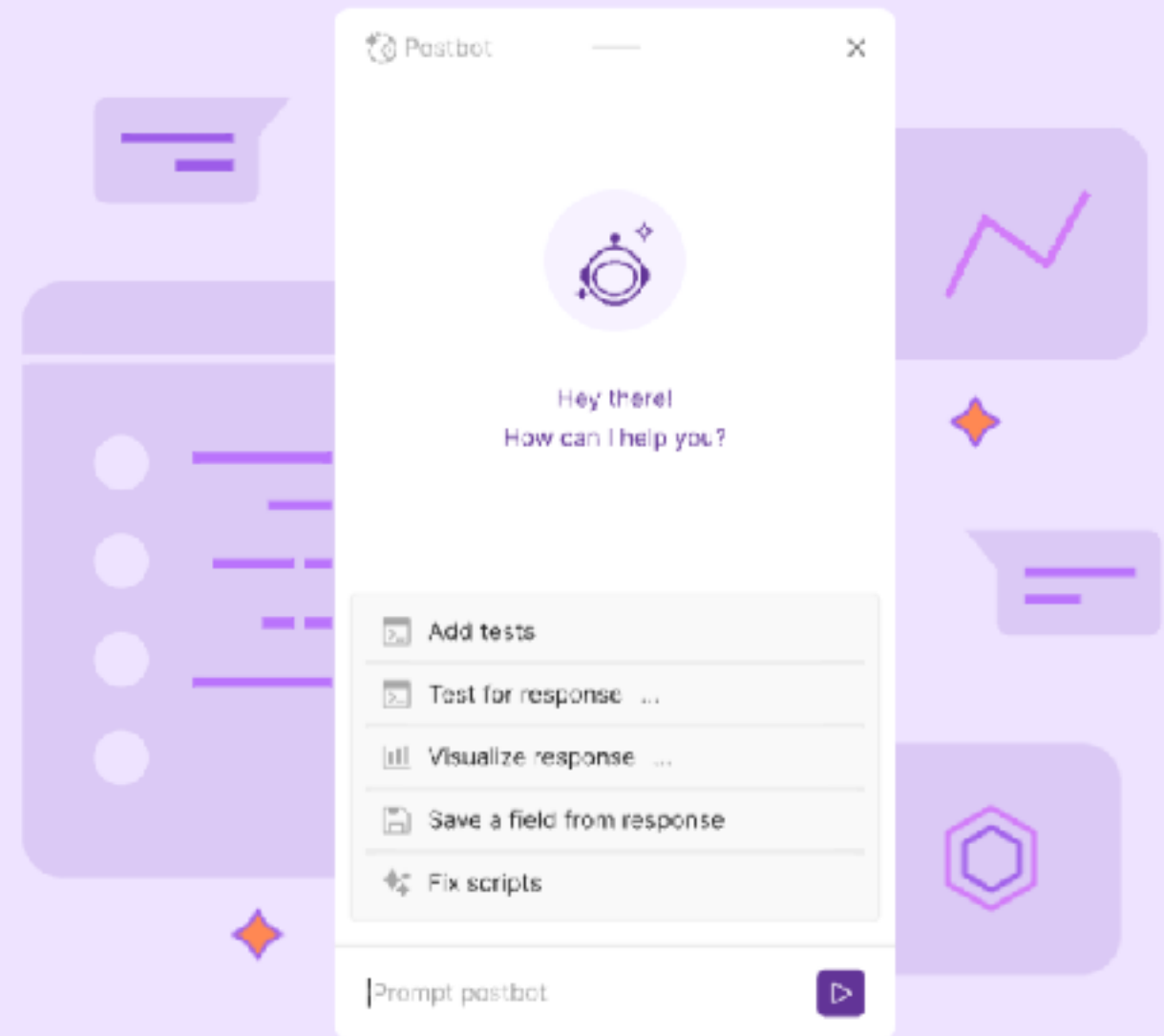


Testing with Postman

Postbot, our AI-powered assistant, will supercharge your API development.

Speed up your most common API development workflows with natural-language input, conversational interactions, and contextual suggestions.

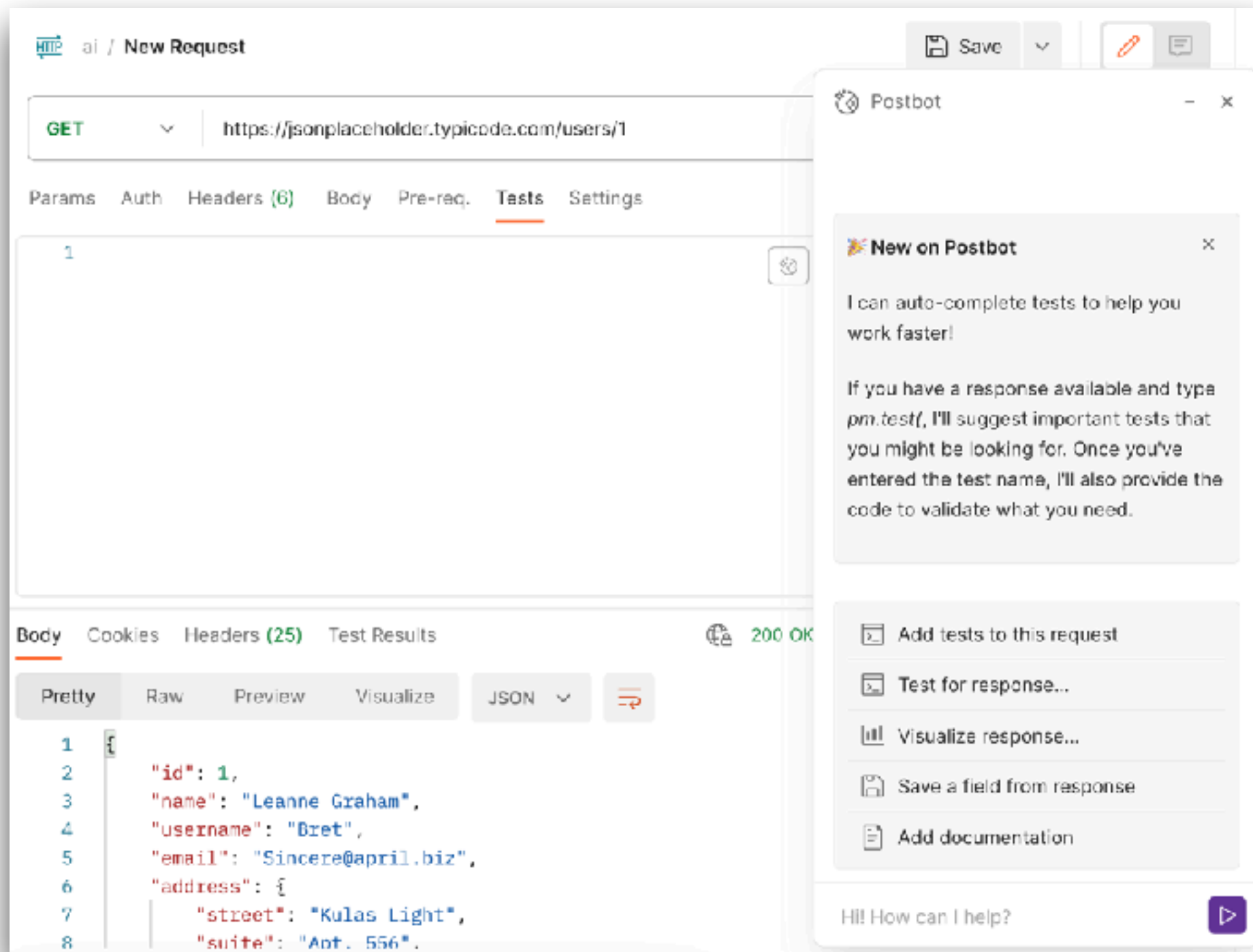
Get Started



<https://www.postman.com/product/postbot/>



PostBot



<https://www.postman.com/product/postbot/>



Browser Use

[FEATURES](#)[PRICING](#)[BLOG](#)[DOCUMENTATION](#) 72,343 26.3K 23.5K[CLOUD](#)

[BETA] THE MOST STEALTH BROWSER INFRASTRUCTURE

The AI browser agent

Repetitive work is dead. Browser Use empowers anyone to automate repetitive online tasks, no code required. No barriers. Simply tell it what you want done.

<https://browser-use.com/>



Playwright Test Agent



<https://playwright.dev/docs/test-agents>



AI Testing Tools

The screenshot displays the homepage of the AI Testing Tools website. At the top, the title "AI Testing Tools" is prominently displayed in purple. Below it, a subtitle reads: "Your one-stop destination for AI-powered tools for software testing, test automation, test management, and more." A navigation bar contains four buttons: "All" (highlighted in purple), "Test Automation", "Test Management", and "MCP Servers".

Below the navigation bar, there is a search bar with the placeholder text "Search among 75 items". To the right of the search bar are several filter dropdowns: "Category", "Pricing", "AI Capabilities", "Automation Capabilities", and a "Reset Filters" link.

The main content area features three featured tool cards:

- Test Collab**: Labeled "Featured" with a checkmark icon. The card shows a screenshot of the Test Collab interface, which includes a table of test cases with columns for ID, Name, Status, and Priority. Below the screenshot, the text reads: "Test Collab is AI-powered test management software for planning, executing, and tracking".
- Lynqa**: Labeled "New" and "Featured" with a checkmark icon. The card has a dark blue background with the Lynqa logo (a stylized blue fox head) and the text: "Execute your Xray tests, as-is, with AI". Below the screenshot, the text reads: "Lynqa is an AI Agent for Manual Test Execution."
- Test Management By Testsigma**: Labeled "Featured" with a checkmark icon. The card shows a screenshot of the Testsigma interface, which includes a sidebar with navigation options and a main area with a list of test cases. Below the screenshot, the text reads: "Test Management by Testsigma is a fully".

<https://www.testingtools.ai/>



Deployment Process



Deploy and manage



CI/CD pipeline

Infrastructure as a code

Automated script

Performance and monitoring suggestion

Document generation

AI-assist support

ChatOps, AIOps



K8sGPT



CLOUD NATIVE
SANDBOX

K8sGPT joins the CNCF Sandbox

K8sGPT is a tool for scanning your kubernetes clusters, diagnosing and triaging issues in simple english. It has SRE experience codified into its analyzers and helps to pull out the most relevant information to enrich it with AI.

Get it now!

<https://k8sgpt.ai/>



PromptOps



Solutions ▾

Resources ▾

Contact us

Log In

ChatGPT for your DevOps Teams

Turn DevOps tasks into automated workflows
with a single prompt straight from Slack

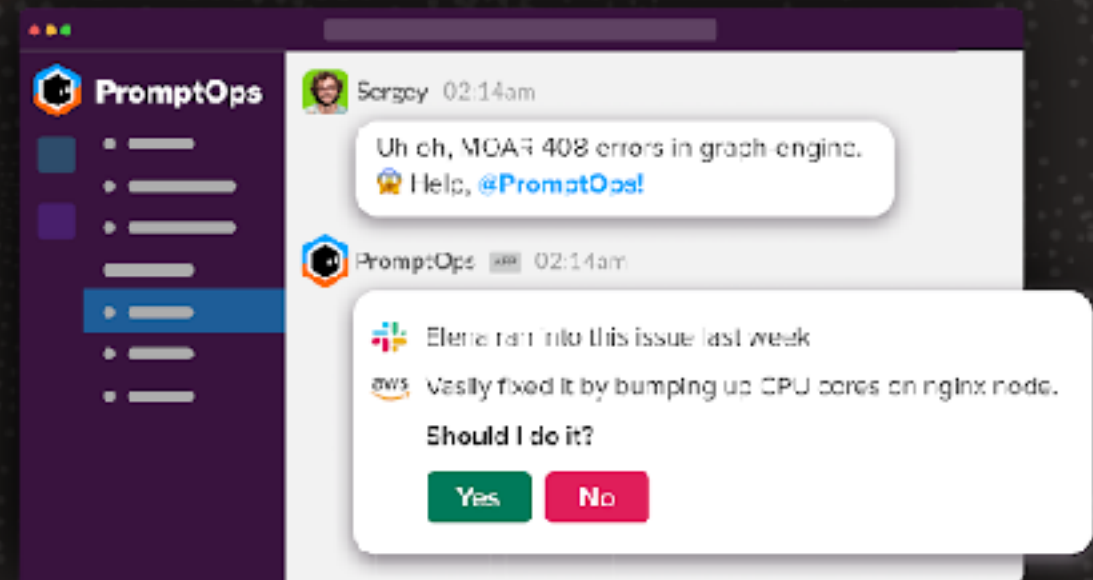
Get started

Learn More



PromptOps APP

Hello, I'm PromptOps, your DevOps virtual assistant for managing, troubleshooting, and running DevOps tasks directly from Slack!



Acknowledge

Resolve

Run a play ▾

Edit code

<https://www.promptops.com/devops/>



ChatOps for DevOps

[Product](#)[How It Works](#)[Learn](#)[Company](#)[uptime](#)[Sign In](#)[Book a demo](#)[Sign Up](#)

> ChatGPT for DevOps

Converse with your engineering platforms, powered by LLM.
A virtual teammate to handle DevOps requests so you can handle the rest.

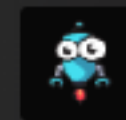
[Add Kubi to Slack](#)

Kubi (DevOps)

@Alerts I got an alert from Prometheus:

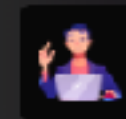


Deployment 'alert-manager' on namespace 'Openfaas' is experiencing high traffic



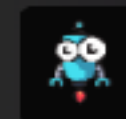
Kubi (DevOps)

@Alerts Should I increase the number of replicas on 'alert-manager'?



Jeff (R&D)

Yes



Kubi (DevOps)



The following deployment has been updated:

Deployment: alert-manager

Namespace: Openfaas

Replicas: 3

<https://www.kubiya.ai/>



Risks when using Generative AI



Risks

Quality of output generated

Explainability of decisions

Security policy !!

Sensitive data !!



Tips

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code



Local LLM



Local LLM

Run LLM on local machine/device
Try to customize with your requirement

Reduce cost

Data privacy

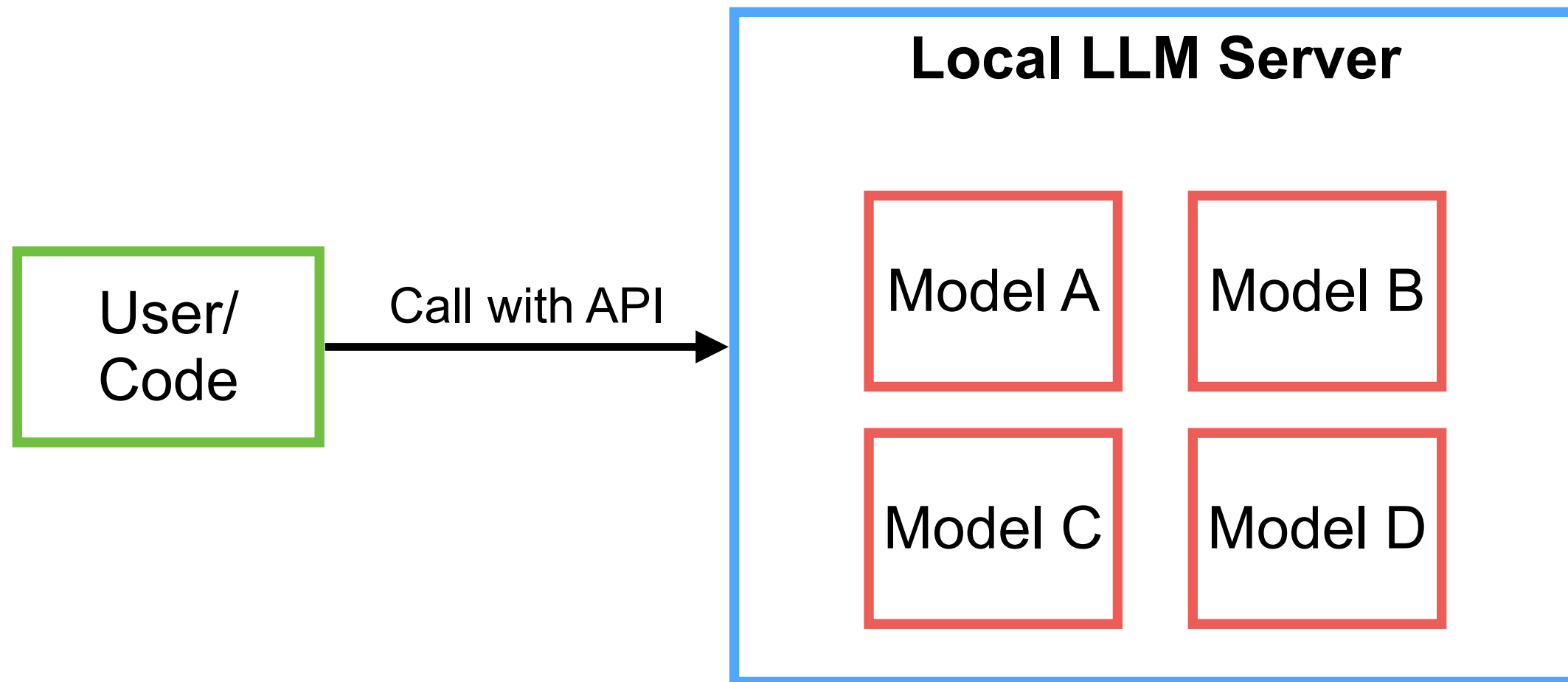
Responsive

Offline mode



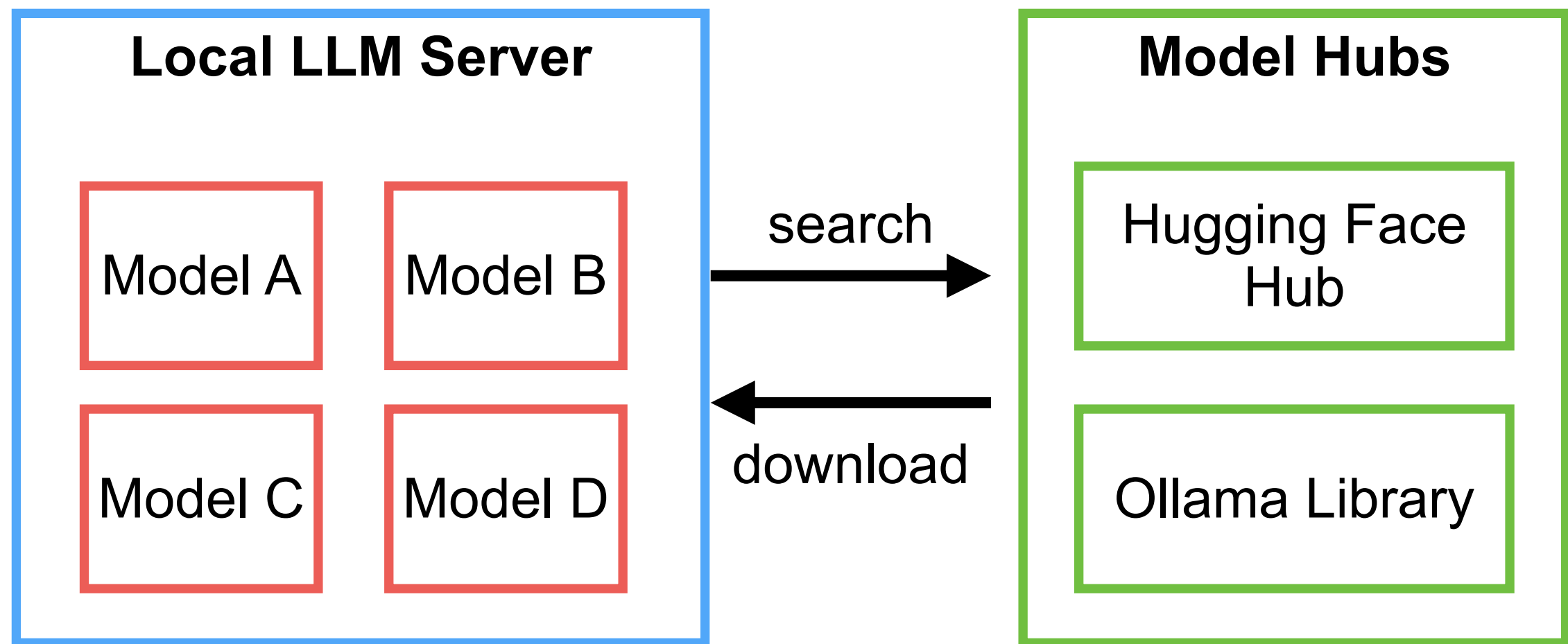
Local LLM

Improve your LLM models, more accurate answer



Models ?

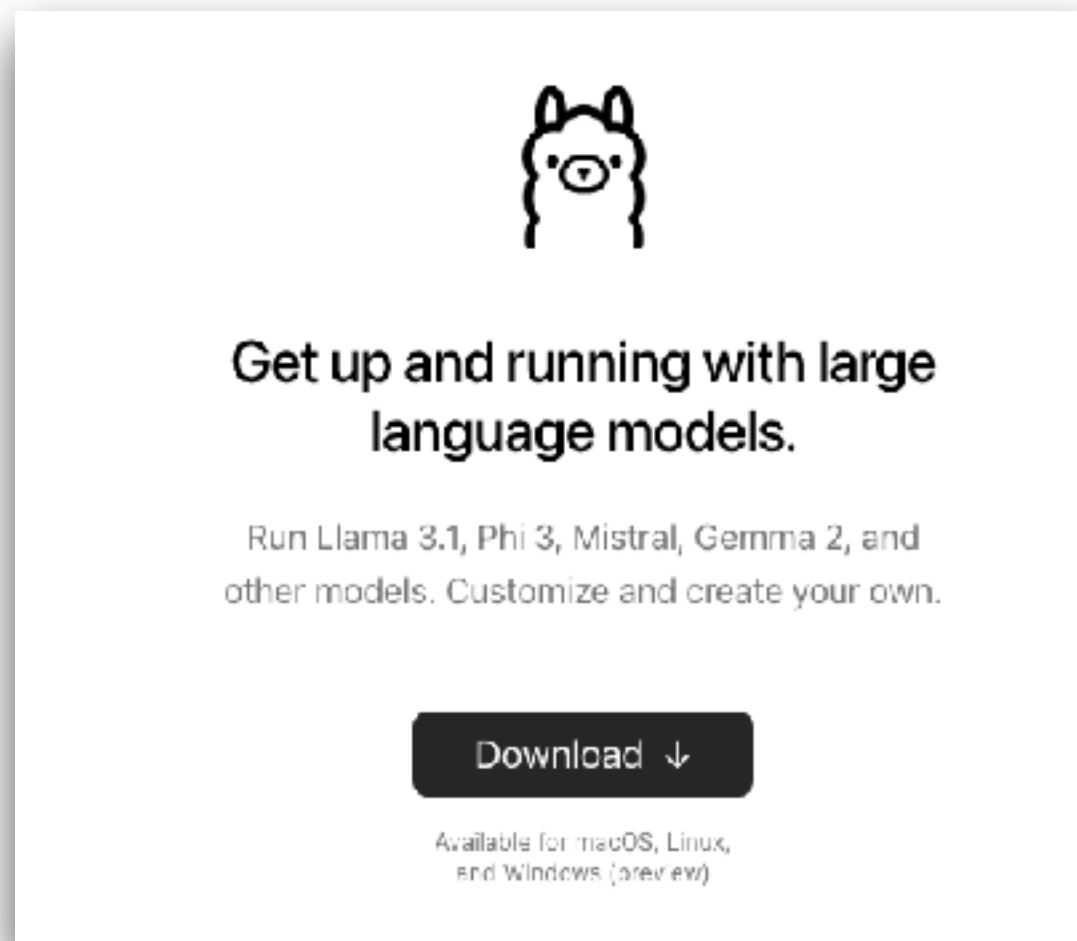
How to download models ?



Local LLM with Ollama

\$ollama run **llama3.2**

LLM model



<https://ollama.com/>



Local LLM with LM Studio

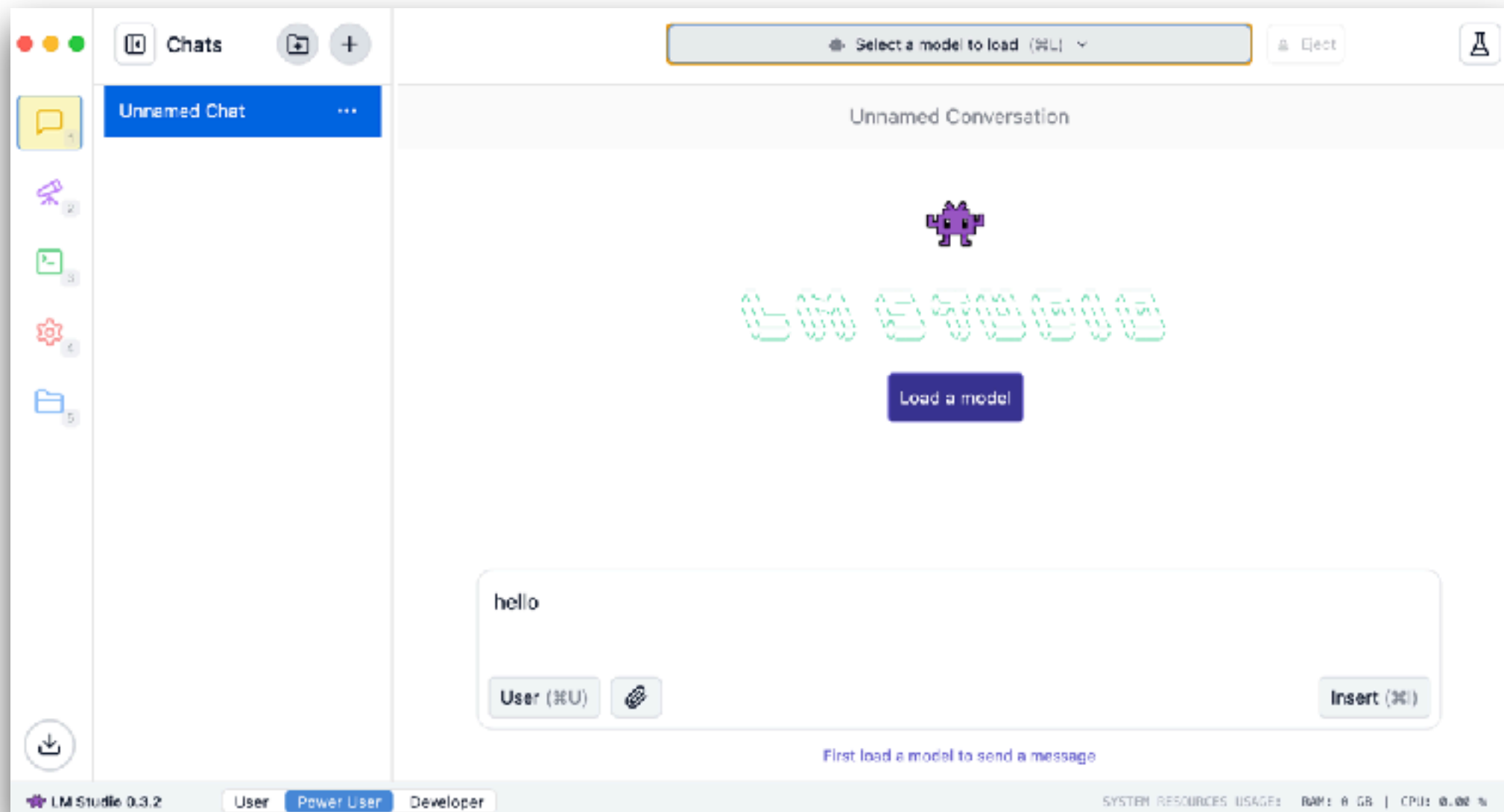
The image shows the LM Studio website and a preview of the application. The website header includes the LM Studio logo, navigation links for 'Docs', 'Blog', and a 'Download' button. The main heading is 'Discover, download, and run local LLMs'. Below this, a green box contains the text 'LM Studio v0.3.0 is finally here! 🥳🥳🥳 Read the announcement'. A row of buttons allows users to 'Run' different models: LLaMa, Phi, Gemma, DeepSeek, Qwen, and Mistral. A note states 'Built with open source projects like llama.cpp and lmstudio-is'. Three download buttons are provided for Mac (M1/M2/M3), Windows (x64), and Linux (x86), all version 0.3.2. At the bottom, it says 'LM Studio is provided under the terms of use'. The application preview on the right shows a code editor with a Lua script, a chat interface with a text input and 'Send' button, and an 'Advanced Configuration' sidebar on the right.

<https://lmstudio.ai/>



Local LLM with LM Studio

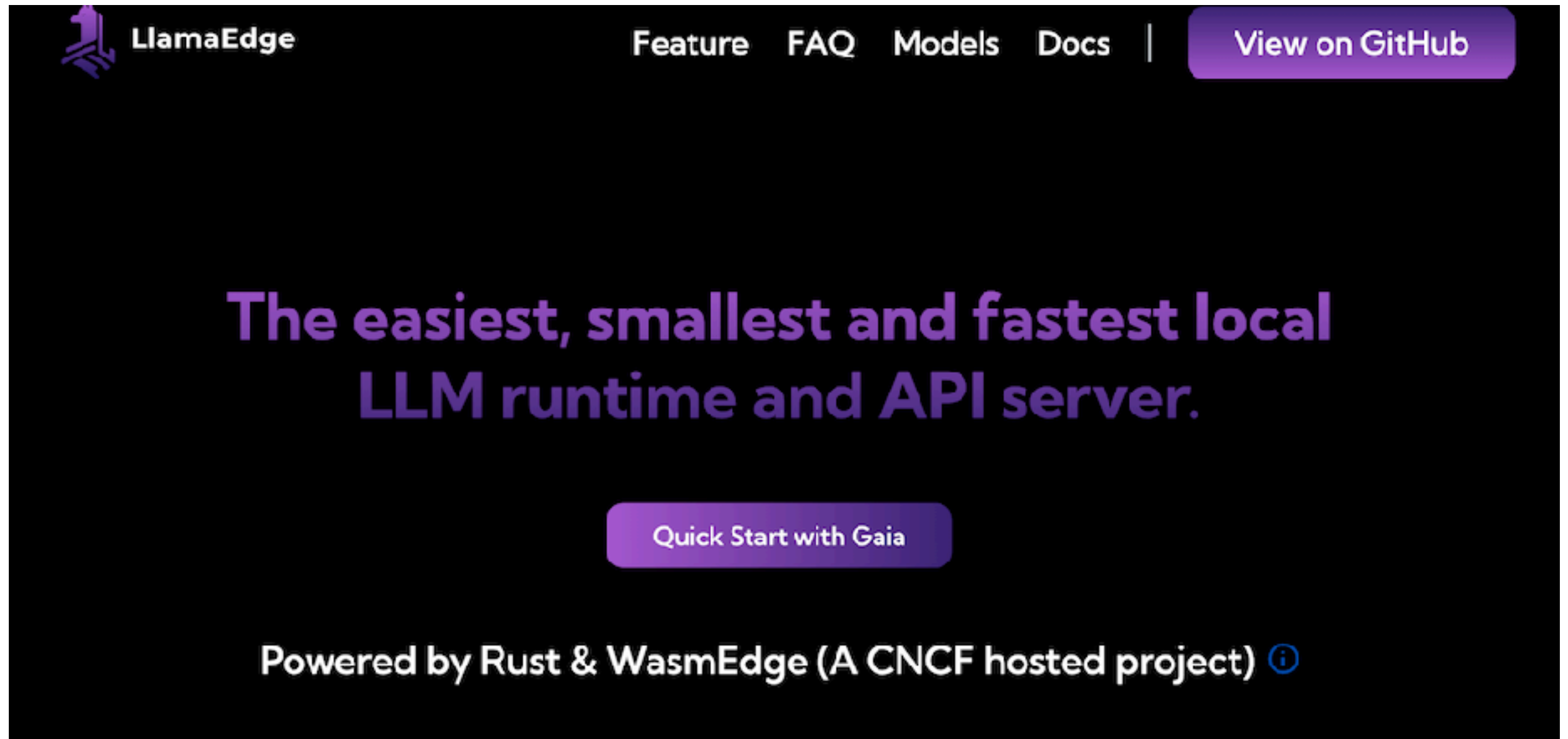
Load model from Hugging Face



<https://lmstudio.ai/>



Local LLM with LlamaEdge



<https://llamaedge.com/>



Local LLM with LocalAI



<https://localai.io/>



More

GPT4AI

LlamaFile

Jan.ai

NextChat

Anything LLM

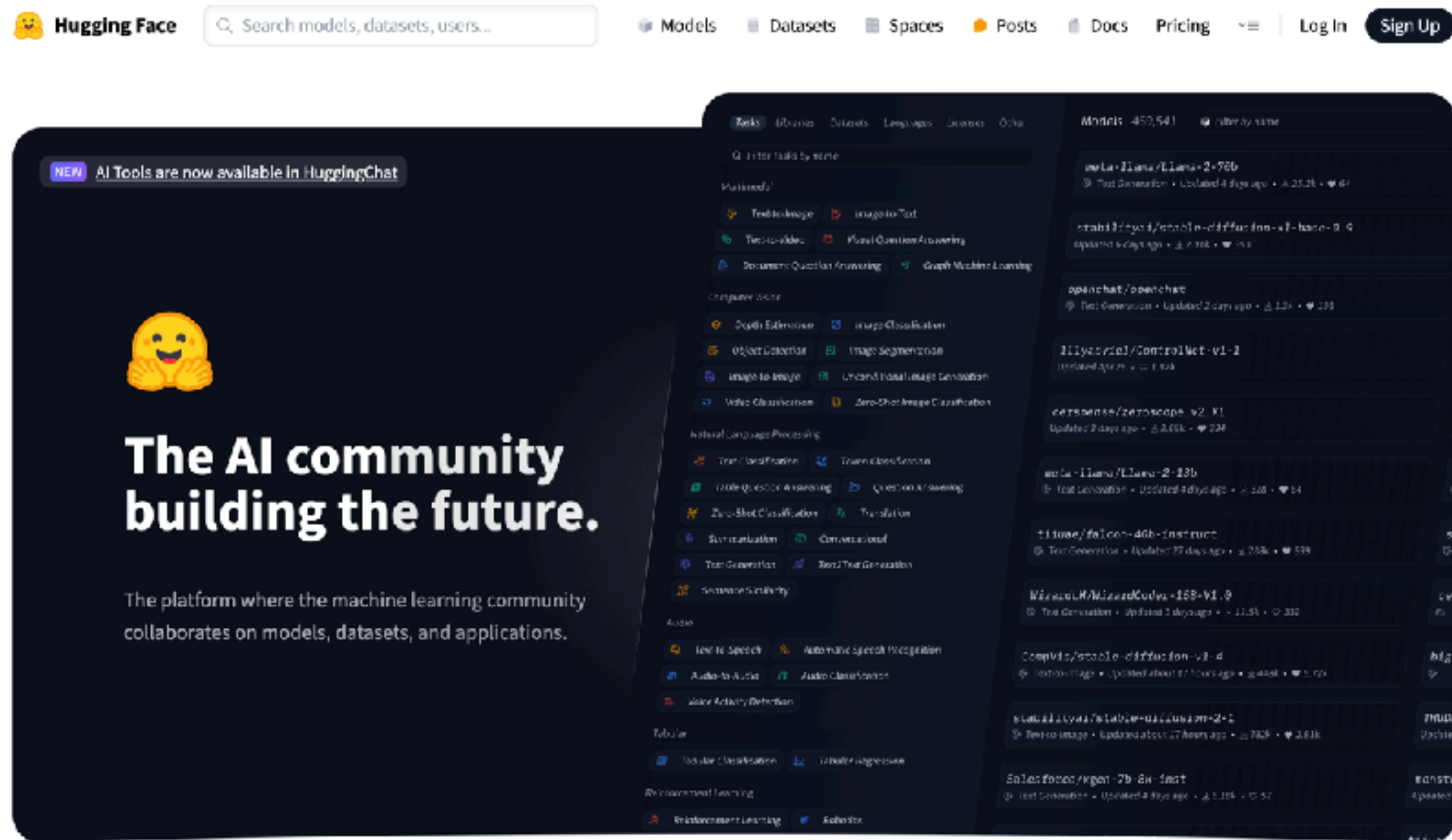
<https://github.com/Hannibal046/Awesome-LLM>



LLM Models



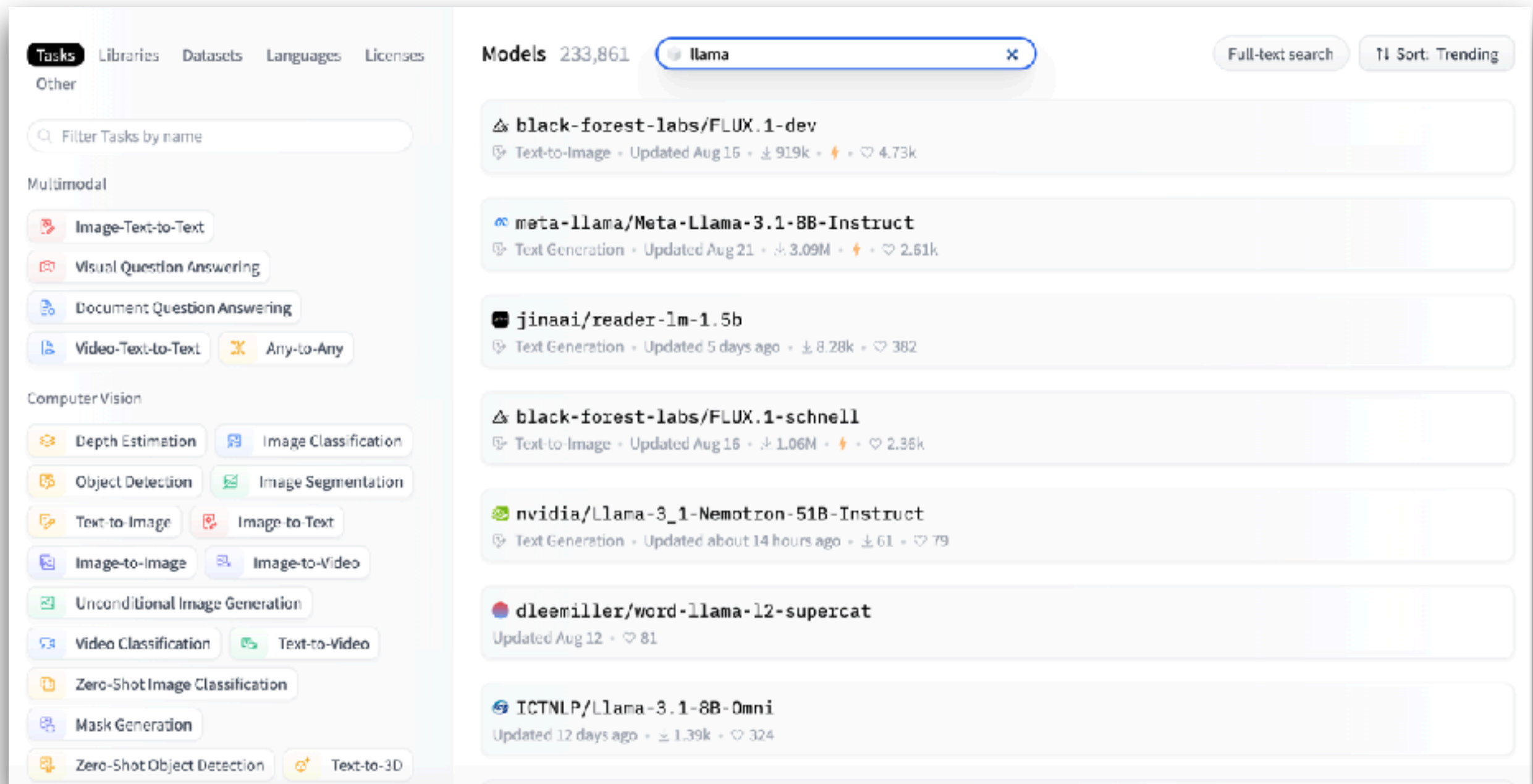
Hugging Face Model Hub



<https://huggingface.co/>



Hugging Face :: Model



The screenshot shows the Hugging Face Models interface. On the left, there's a sidebar with navigation tabs: Tasks, Libraries, Datasets, Languages, Licenses, and Other. Under 'Tasks', there are filters for Multimodal (Image-Text-to-Text, Visual Question Answering, Document Question Answering, Video-Text-to-Text, Any-to-Any) and Computer Vision (Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D). The main area displays a list of models filtered by 'llama'. The list includes:

- black-forest-labs/FLUX.1-dev**: Text-to-Image, Updated Aug 16, 919k downloads, 4.73k likes.
- meta-llama/Meta-Llama-3.1-8B-Instruct**: Text Generation, Updated Aug 21, 3.09M downloads, 2.61k likes.
- jinaai/reader-lm-1.5b**: Text Generation, Updated 5 days ago, 8.28k downloads, 382 likes.
- black-forest-labs/FLUX.1-schnell**: Text-to-Image, Updated Aug 16, 1.06M downloads, 2.35k likes.
- nvidia/Llama-3_1-Nemotron-51B-Instruct**: Text Generation, Updated about 14 hours ago, 61 downloads, 79 likes.
- dleemiller/word-llama-12-supercat**: Updated Aug 12, 81 likes.
- ICTNLP/Llama-3.1-8B-Omni**: Updated 12 days ago, 1.39k downloads, 324 likes.

<https://huggingface.co/>



Big Code model leaderboard

★ Big Code Models Leaderboard

Inspired from the 🤖 Open LLM Leaderboard and 🤖 Open LLM-Perf Leaderboard, we compare performance of base multilingual code generation models on HumanEval benchmark and MultiPL-E. We also measure throughput and provide information about the models. We only compare open pre-trained multilingual code models, that people can start from as base models for their trainings.

🔍 Evaluation table 📊 Performance Plot 📄 About 🚀 Submit results

🔍 See All Columns

🔍 Search for your model and press ENTER...

🏷️ Filter model types

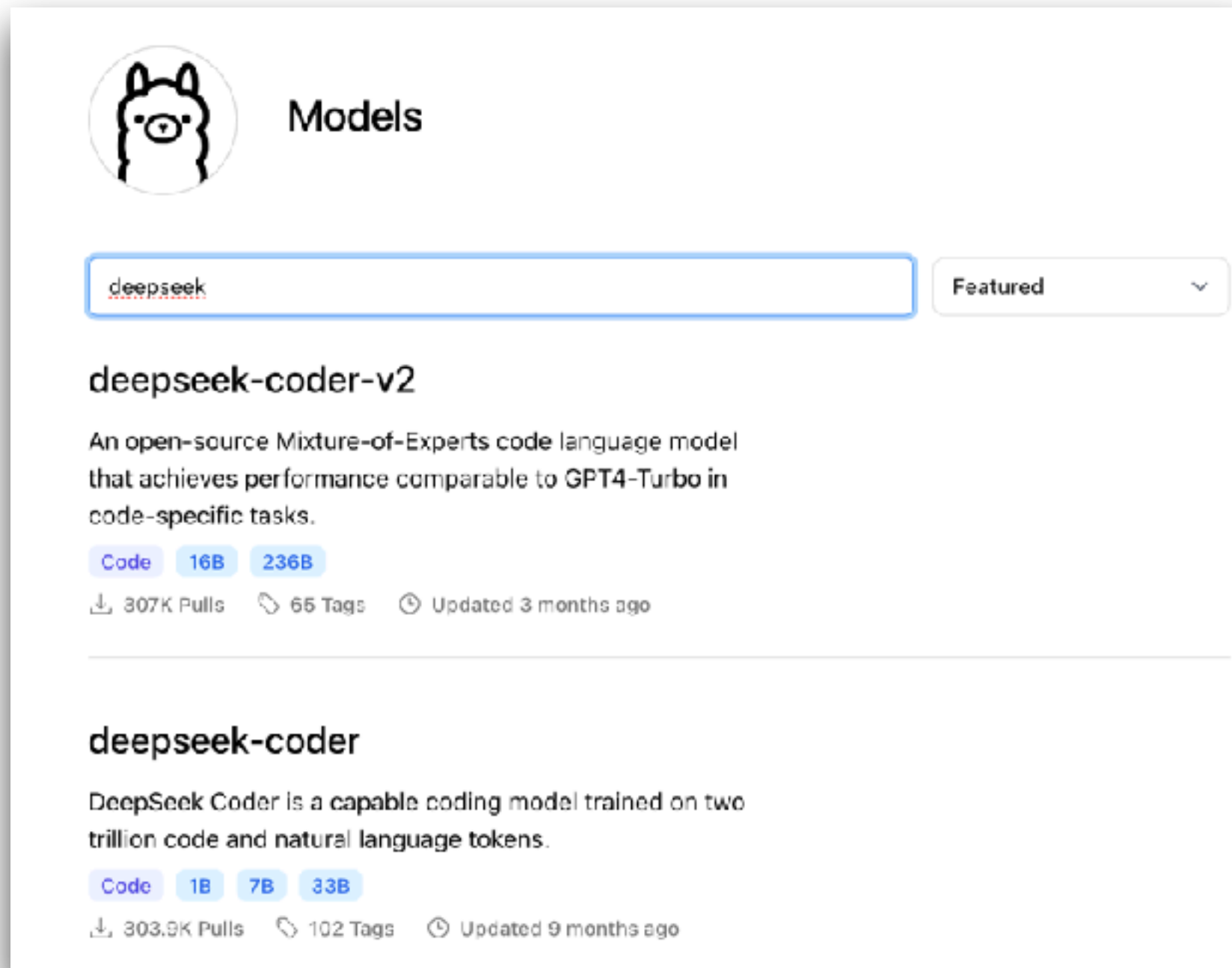
☒ all ☐ base ☐ instruction-tuned ☐ EXT external-evaluation

T	Model	Win Rate	humaneval-python	java	javascript	cpp
EXT	OpenCodeInterpreter-DS-33B	55.83	75.23	54.8	69.06	64.47
EXT	Nxcode-CO-7B-orpo	55.42	87.23	60.91	71.69	68.04
	CodeQwen1.5-7B-Chat	55.08	87.2	61.04	70.31	67.85
EXT	CodeFuse-DeepSeek-33b	54.33	76.83	60.76	66.46	65.22
EXT	DeepSeek-Coder-33b-instruct	52	80.02	52.03	65.13	62.36
EXT	Artigonz-Coder-DS-6.7B	51.5	70.89	56.84	66.16	59.75
EXT	DeepSeek-Coder-7b-instruct	50.33	80.22	53.34	65.8	59.66
EXT	OpenCodeInterpreter-DS-6.7B	49.67	73.2	51.41	63.85	60.81

<https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard>



Model in Ollama

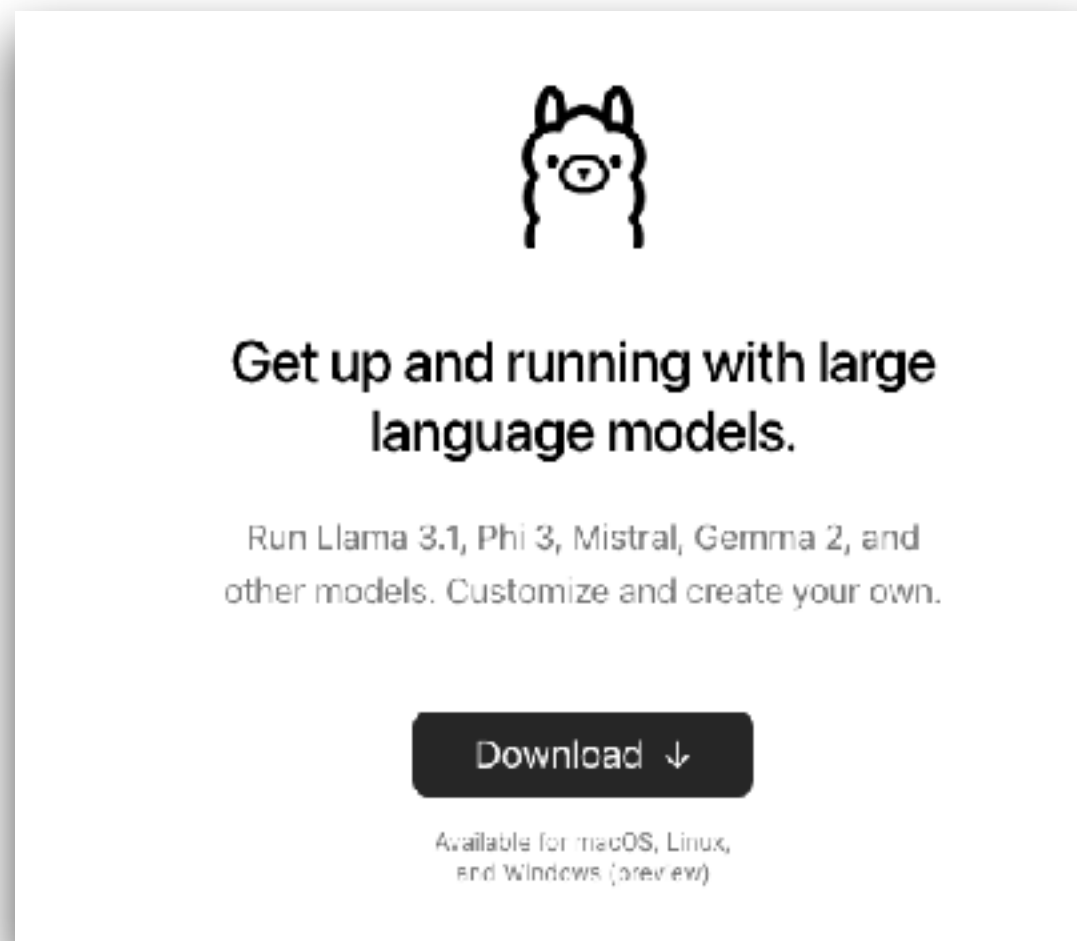


<https://ollama.com/library>



Workshop with Ollama

\$ollama run **llama3.1**



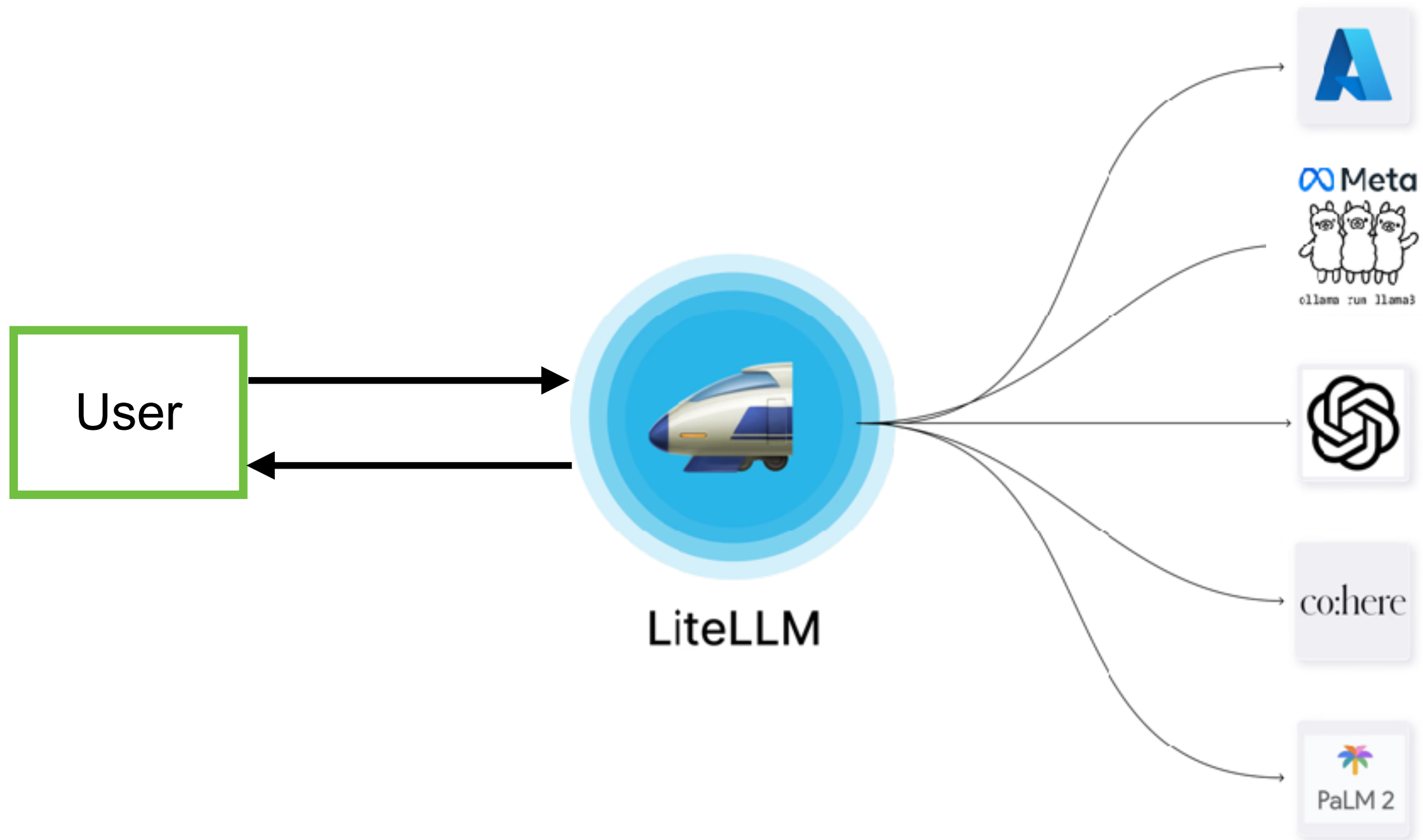
<https://github.com/up1/workshop-ai-with-technical-team/wiki/Local-LLM-with-Ollama>



LiteLLM as a Proxy



LiteLLM as a Proxy

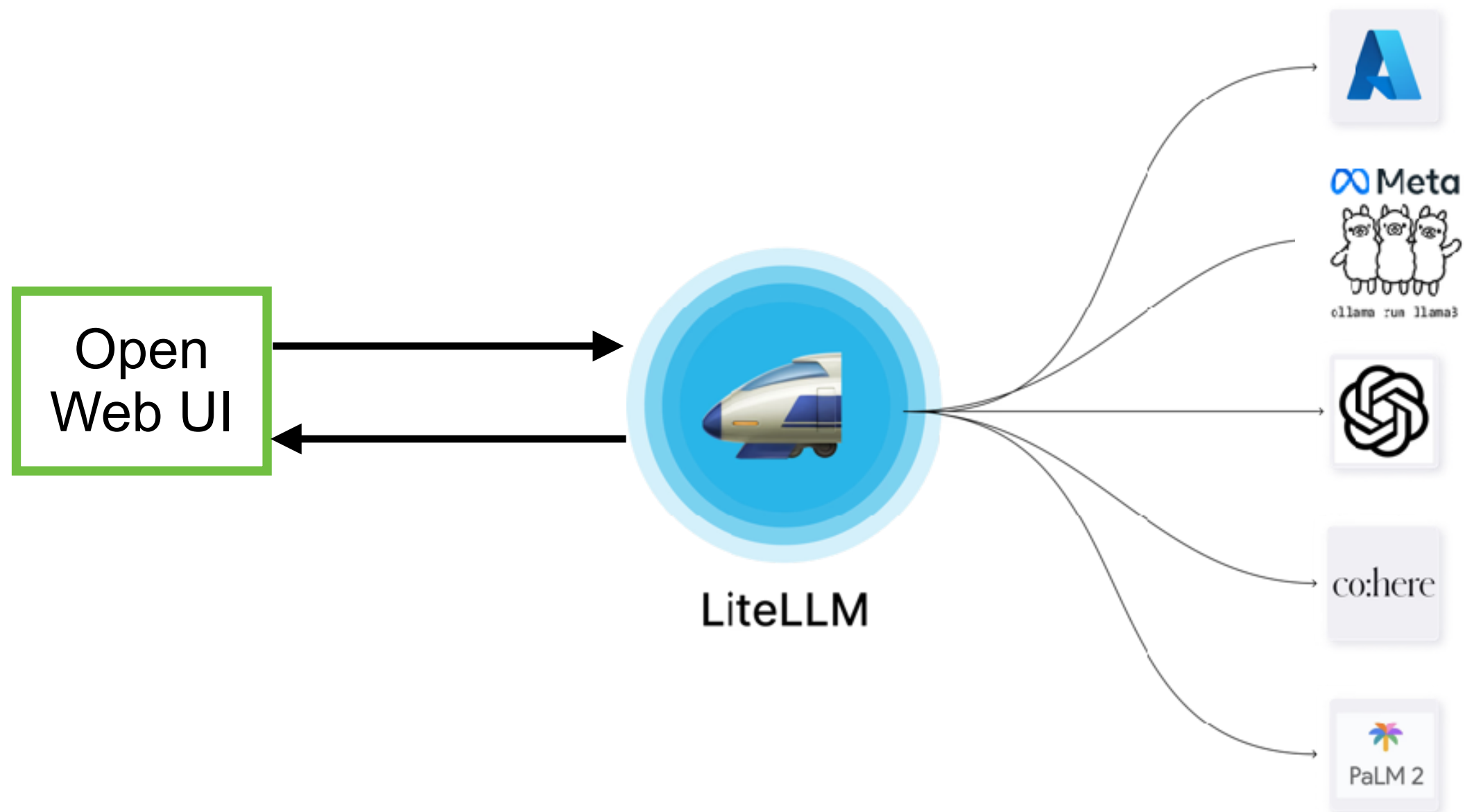


<https://www.litellm.ai/>



Workshop

Use docker compose to build and run



<https://github.com/up1/workshop-ai-with-technical-team/wiki/LiteLLM-and-WebUI>

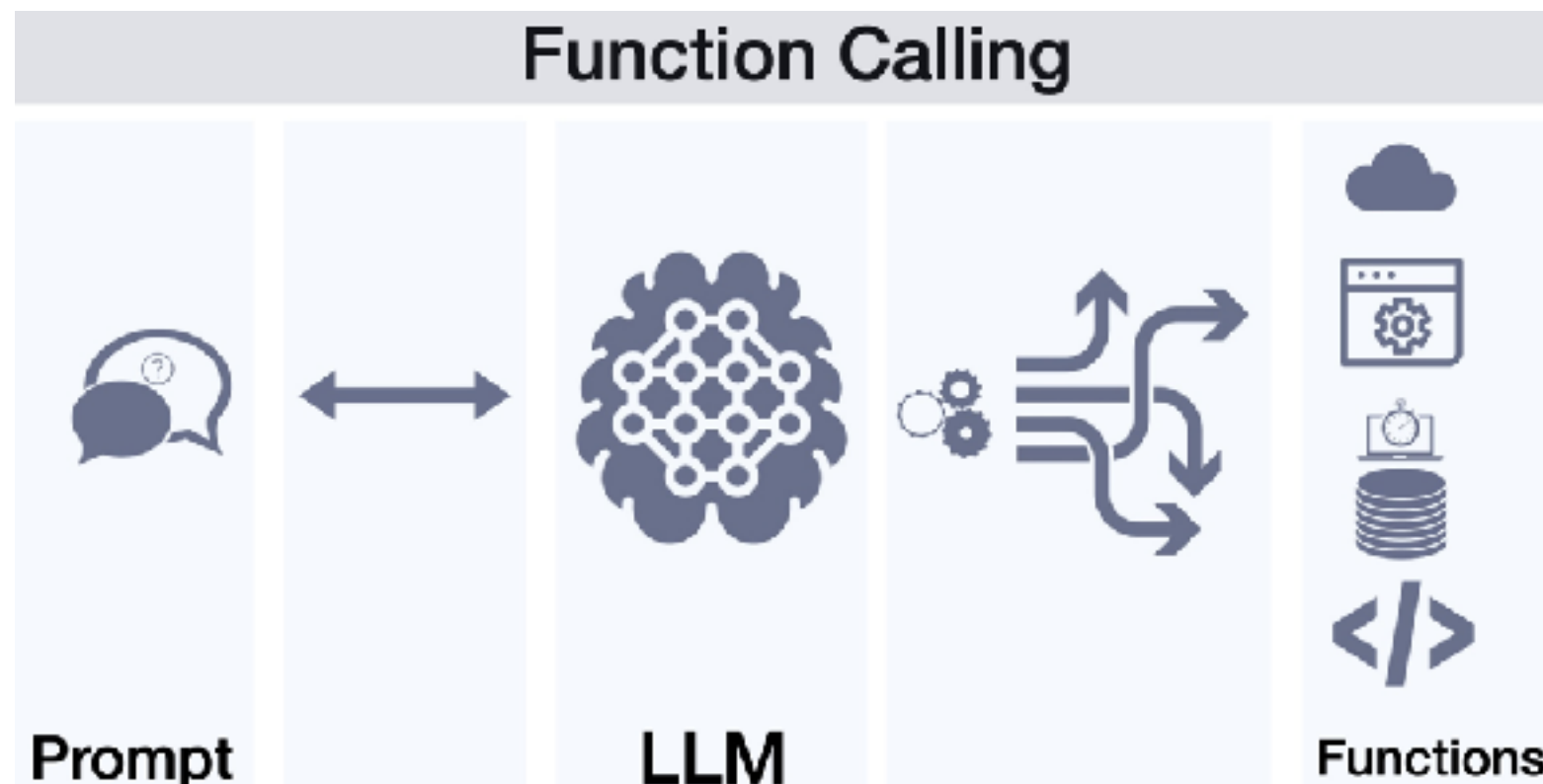


Function Calling ?

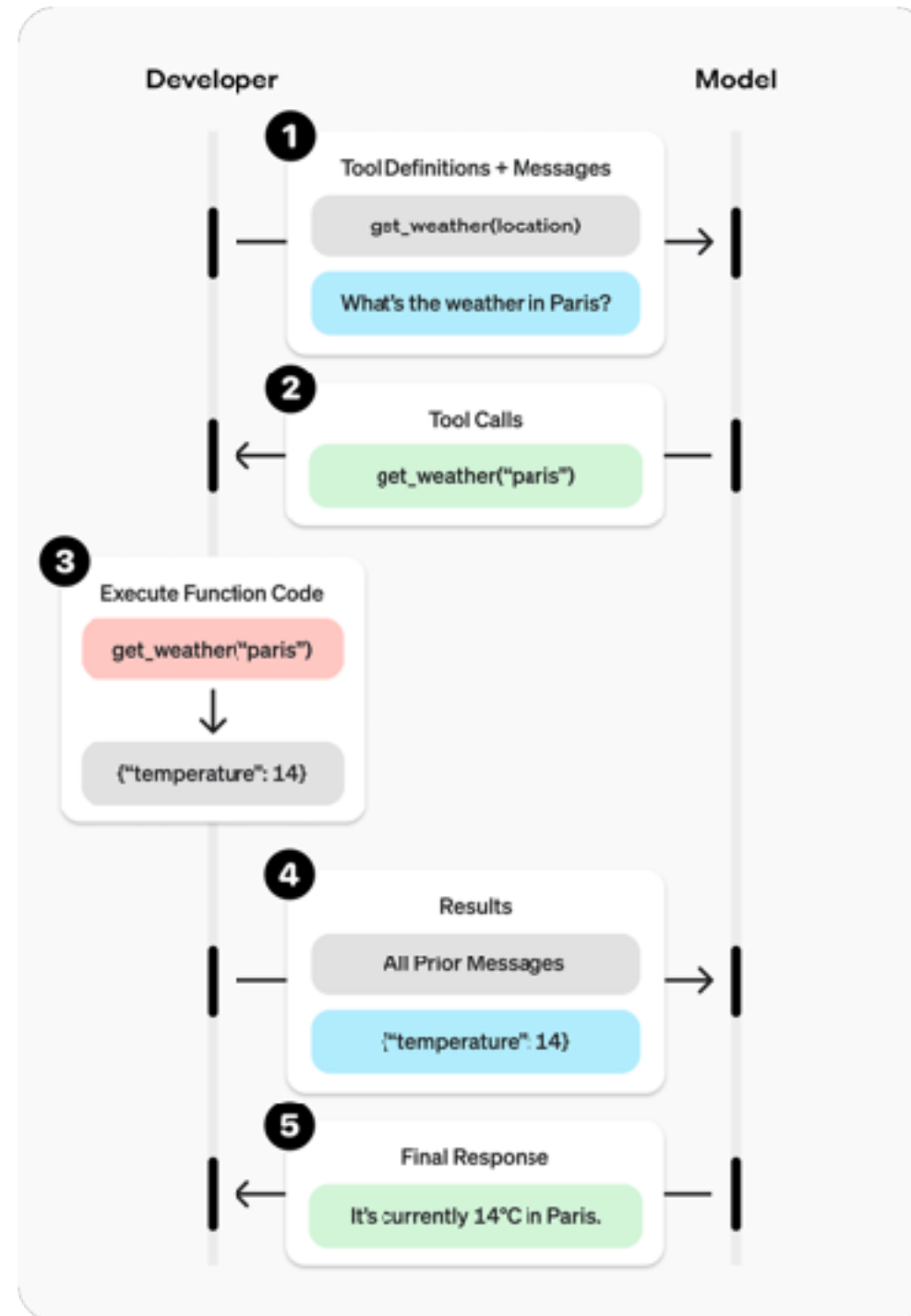


Function Calling ?

Allow LLM to recognize what tool it need based on user's input and when to invoke it.



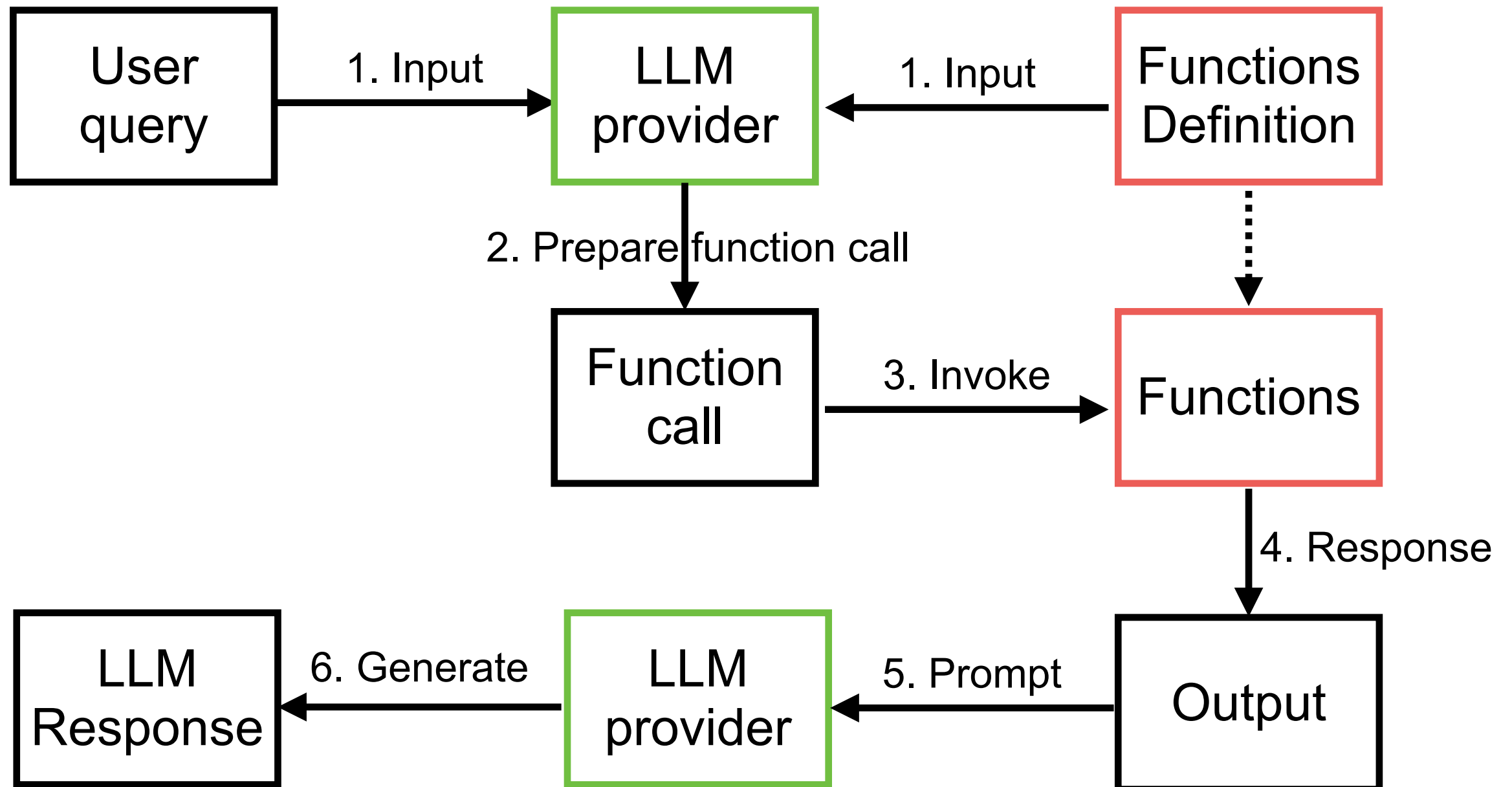
Function calling ?



<https://platform.openai.com/docs/guides/function-calling?api-mode=responses>



Function Calling

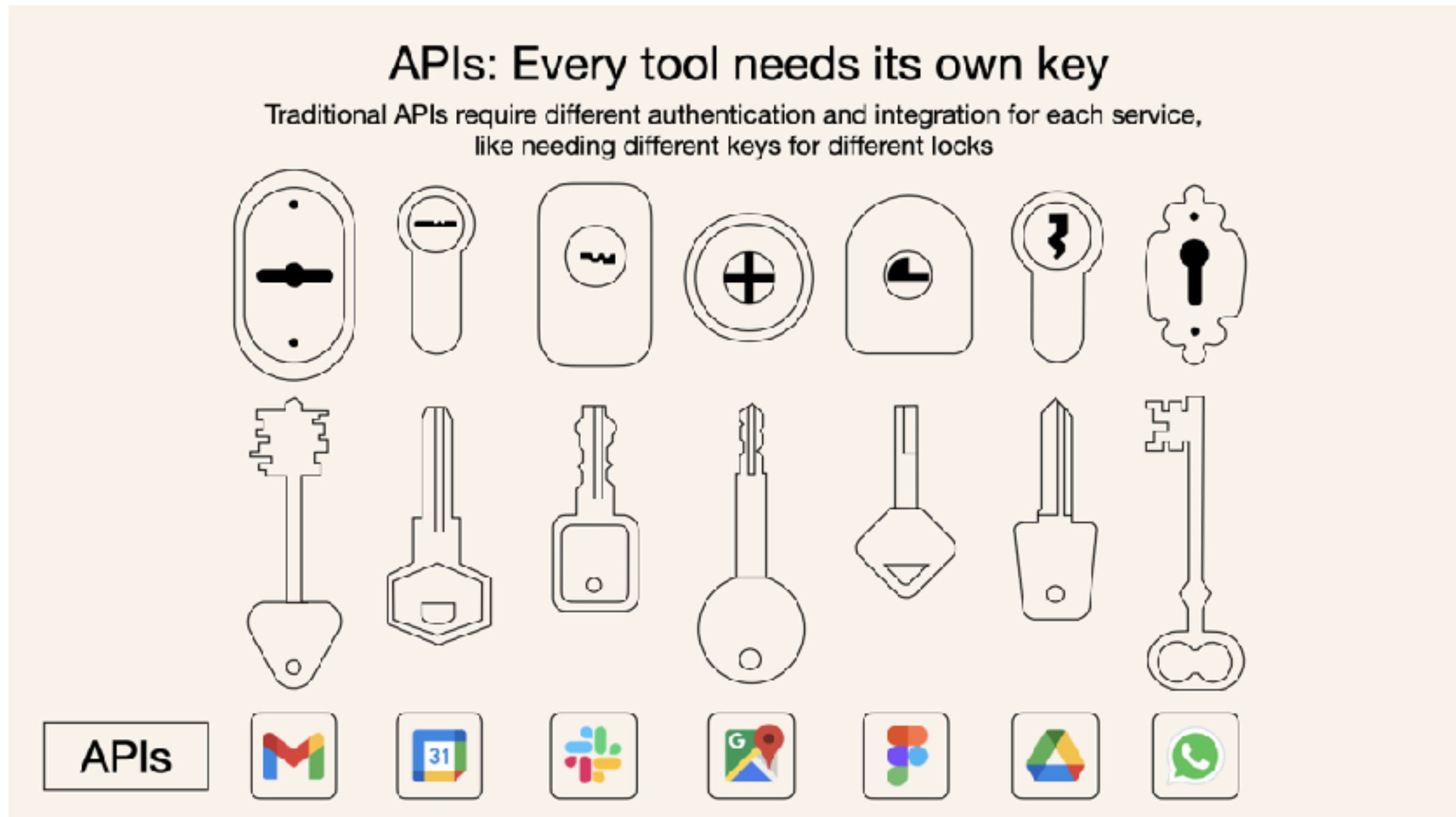


Support function calling

Provider name	Model name
OpenAI	GPT 4
Anthropic	Claude 3 (Sonnet, Haiku, Opus)
Google	Gemini



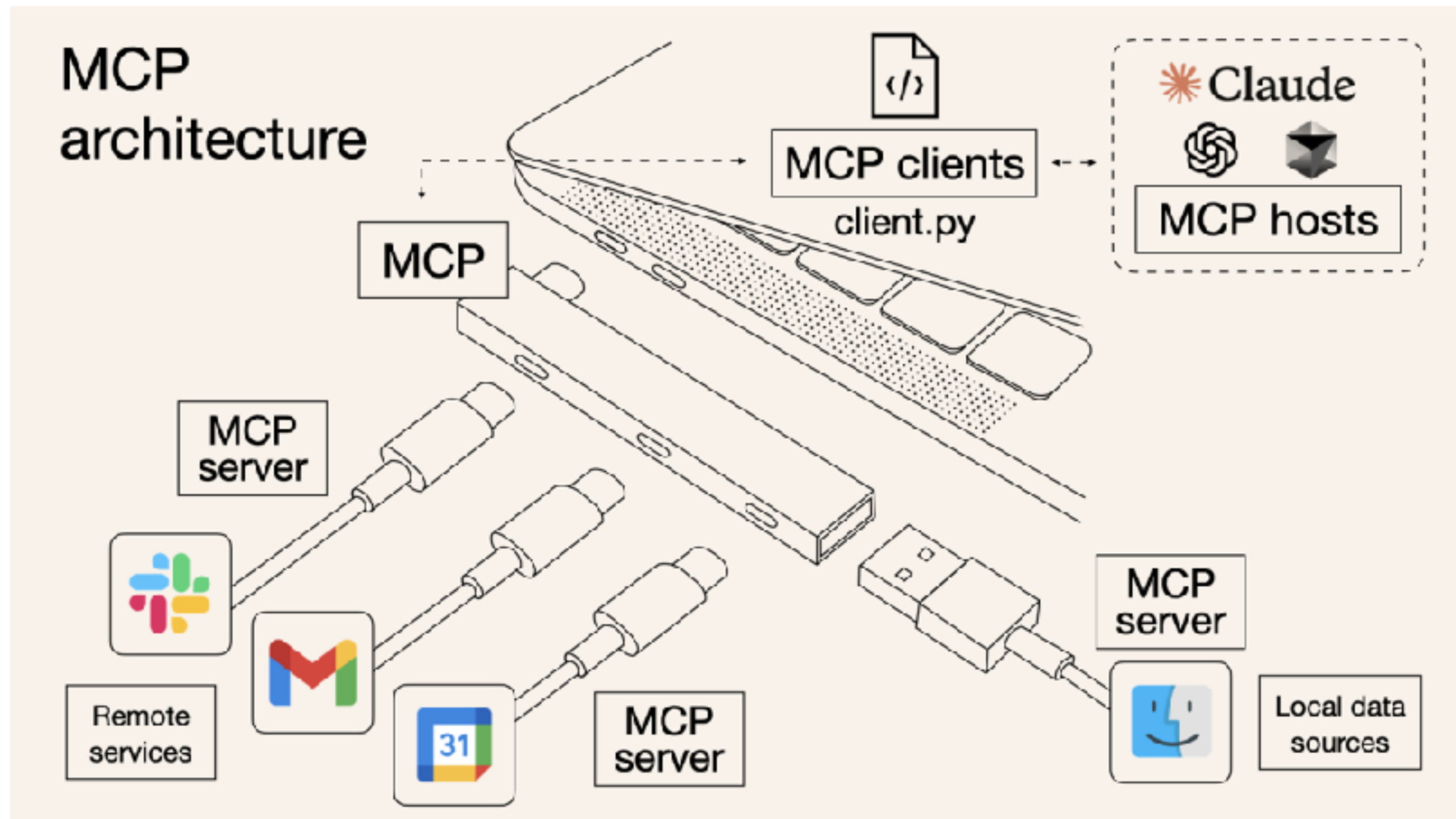
Function calling !!



<https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/>



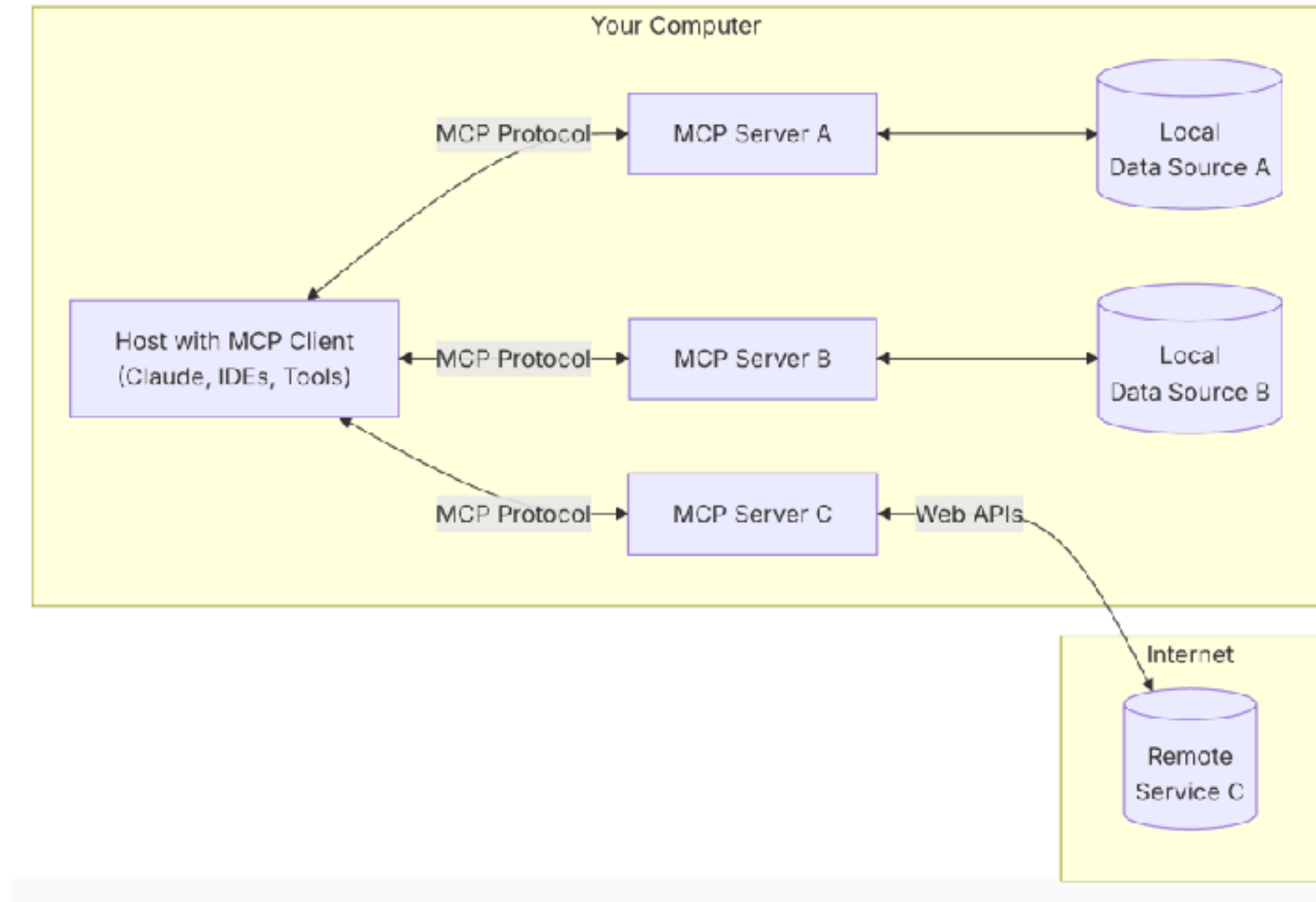
Standardized function calling !!



<https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/>



Model Context Protocol



<https://modelcontextprotocol.io/>



Awesome MCP

Find Awesome MCP Servers and Clients

MCP.so is a third-party MCP Marketplace with **17495** MCP Servers collected.

[ShipAny](#): NextJS boilerplate for building AI SaaS startups.

Today

Featured

Latest

Clients

Hosted

Official

Featured MCP Servers

View All →

**EdgeOne Pages MCP**

An MCP service designed for deploying HTML content to...

★

**AlphaVantage**

Bring enterprise-grade stock market data to agents and LLMs

★

Sponsor

**Time**

A Model Context Protocol server that provides time and timezone...

★

**Zhipu Web Search**

Zhipu Web Search MCP Server is a search engine specifically...

★

**MCP Advisor**

MCP Advisor & installation - Use the right MCP server for your...

★

**HowtoCook Mcp**

基于Anduin2017 / HowToCook (程序员在家做饭指南) 的mcp server, ...

★

**MiniMax MCP**

Official MiniMax Model Context Protocol (MCP) server that enabl...

★

**Serper MCP Server**

A Serper MCP Server

★

**Jina AI MCP Tools**

A Model Context Protocol (MCP) server that integrates with Jina AI...

★

**Amap Maps**

高德地图官方 MCP Server

★

**Playwright Mcp**

Playwright MCP server

★

**Baidu Map**

百度地图核心API现已全面兼容MCP协议，是国内首家兼容MCP协议的地图...

★

<https://mcp.so/>



Retrieval-Augmented Generation (RAG)



RAG

Enhances LLMs by retrieving external knowledge before generating response

Improve accuracy

Reduce
hallucinations

Real-time
knowledge updates



Core Components

Retriever

Fetch relevant documents from a knowledge base

Generator

Uses the retrieved information to generate a response

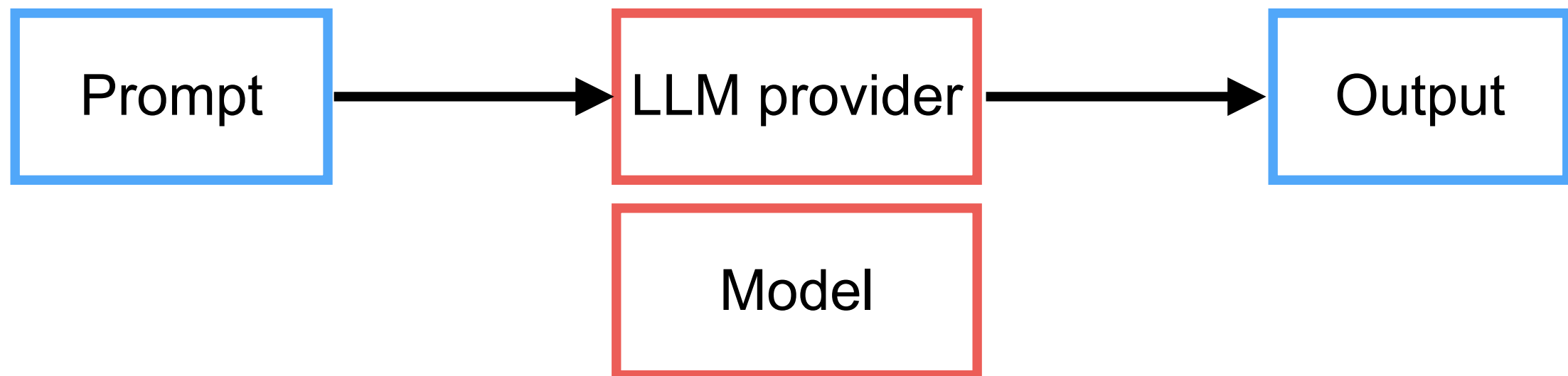
Enhancements

Different RAG architectures modify how retrieval and generation interact to improve performance



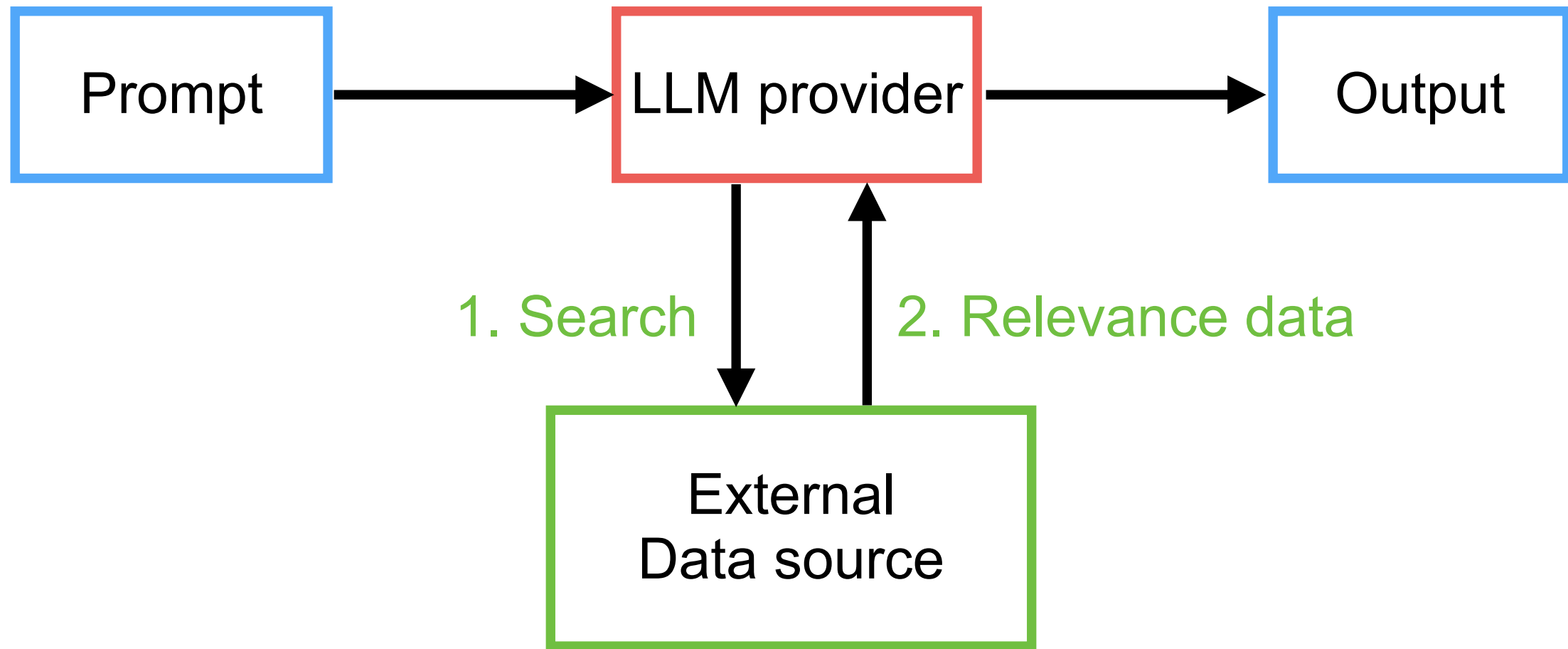
Limitation of LLM

Fixed-knowledge !!



RAG ?

Fixed-knowledge !!



How to search/retrieve data ?

Data source

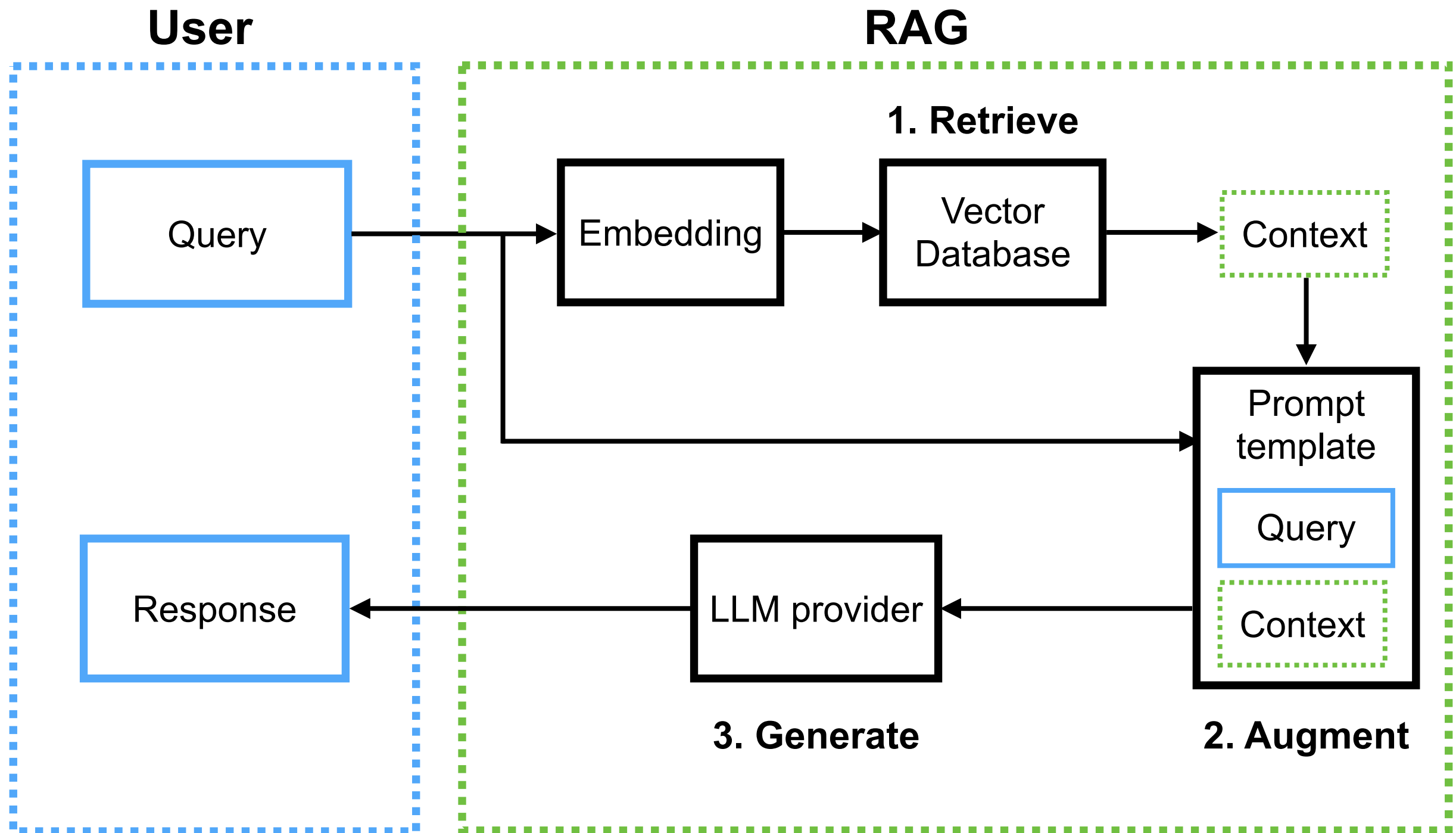
Data preparation

Search

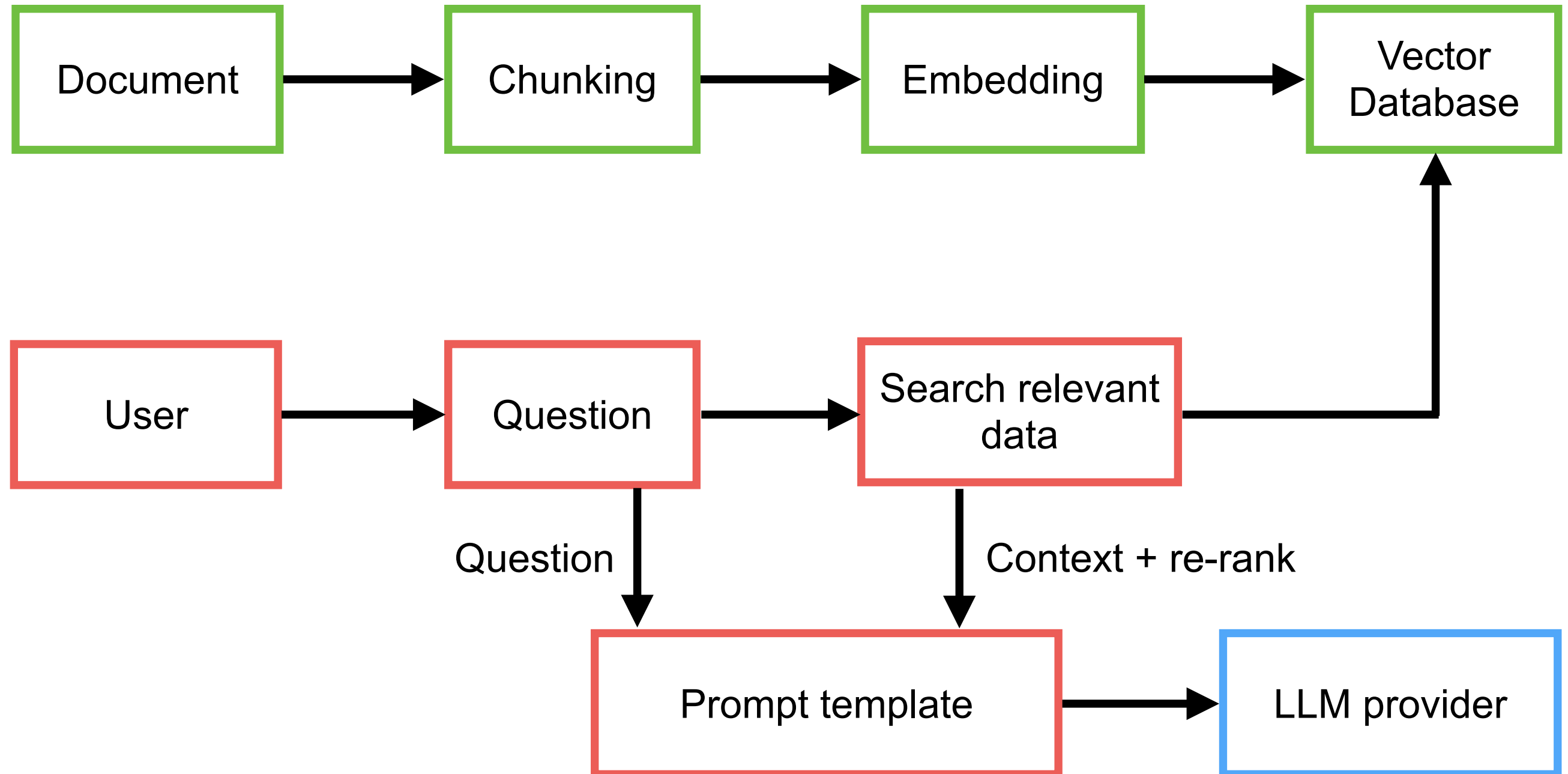
Search result



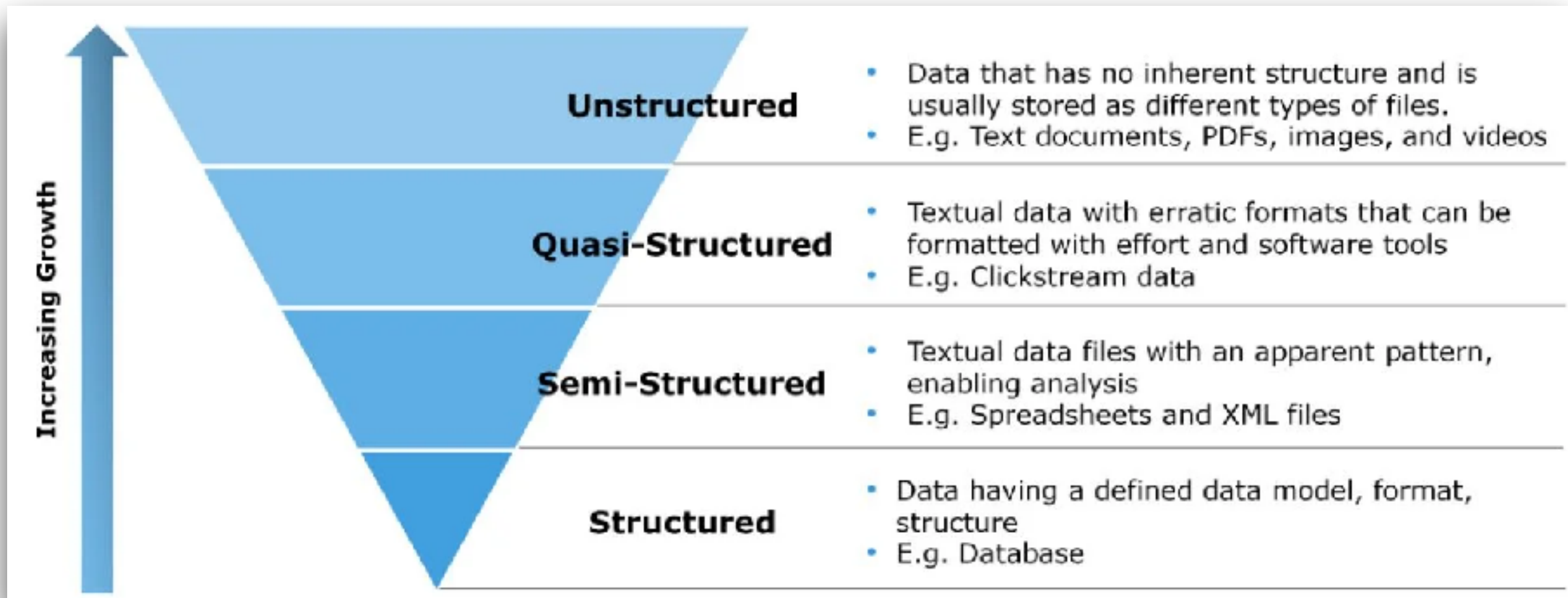
Basic RAG Architecture



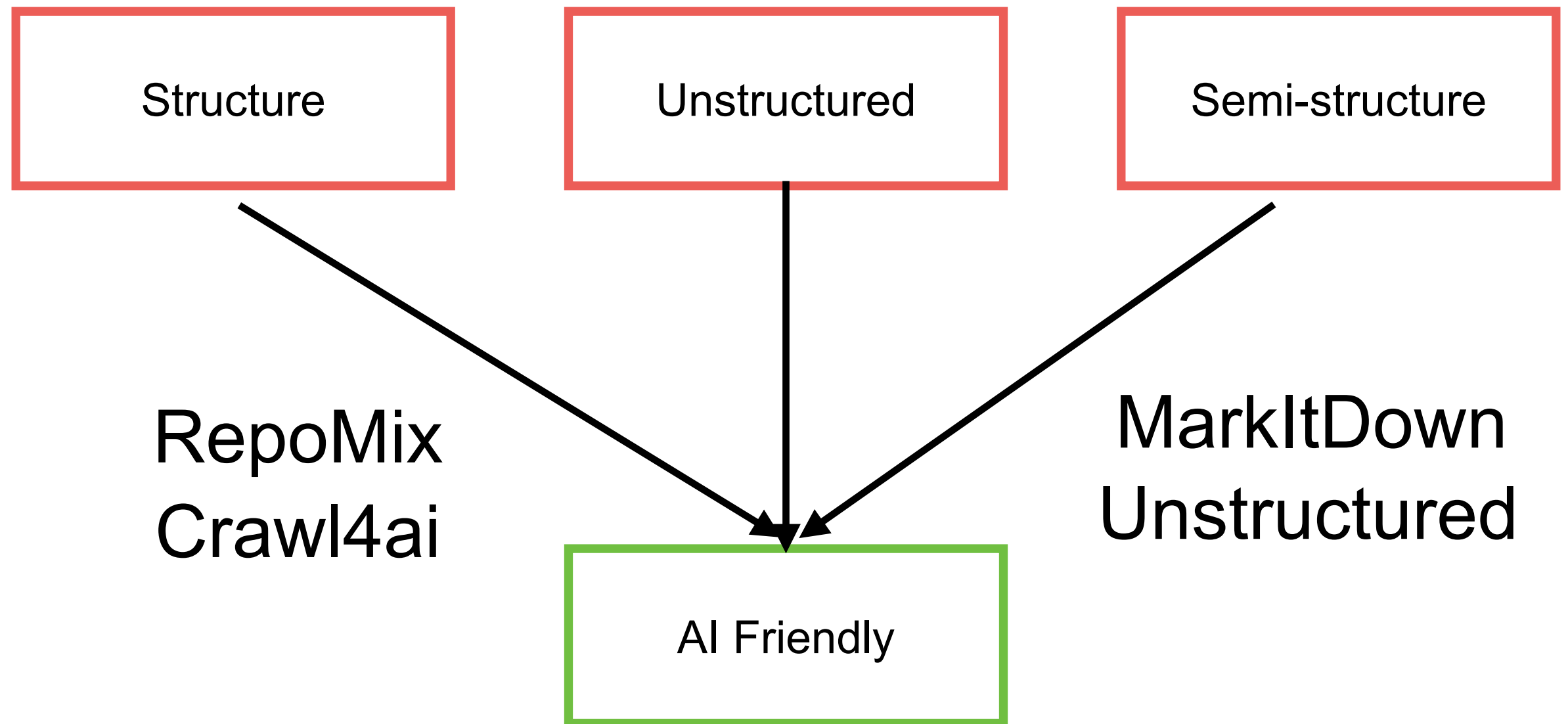
RAG process



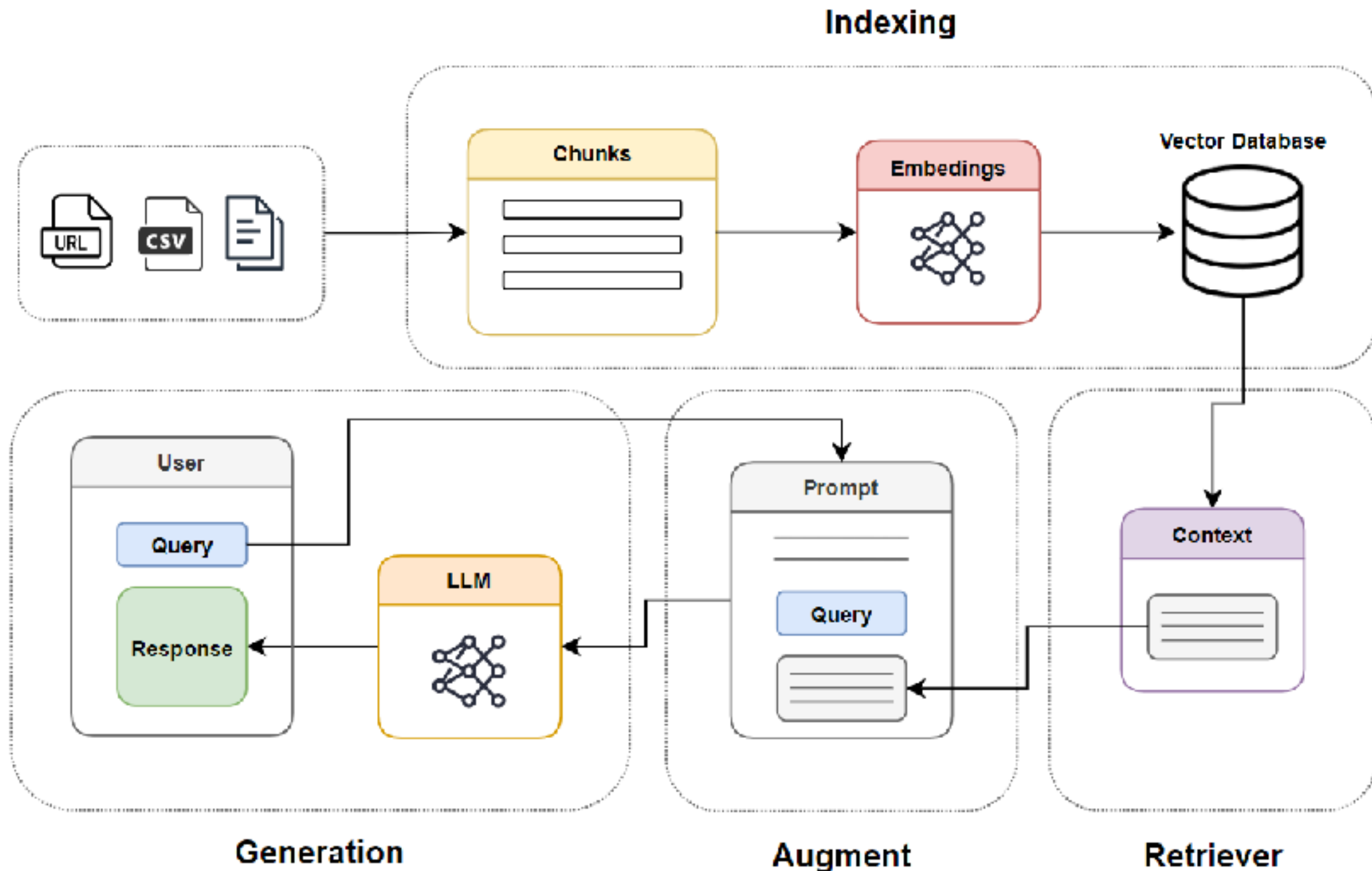
Structures of Data ?



Friendly Data for LLM/AI



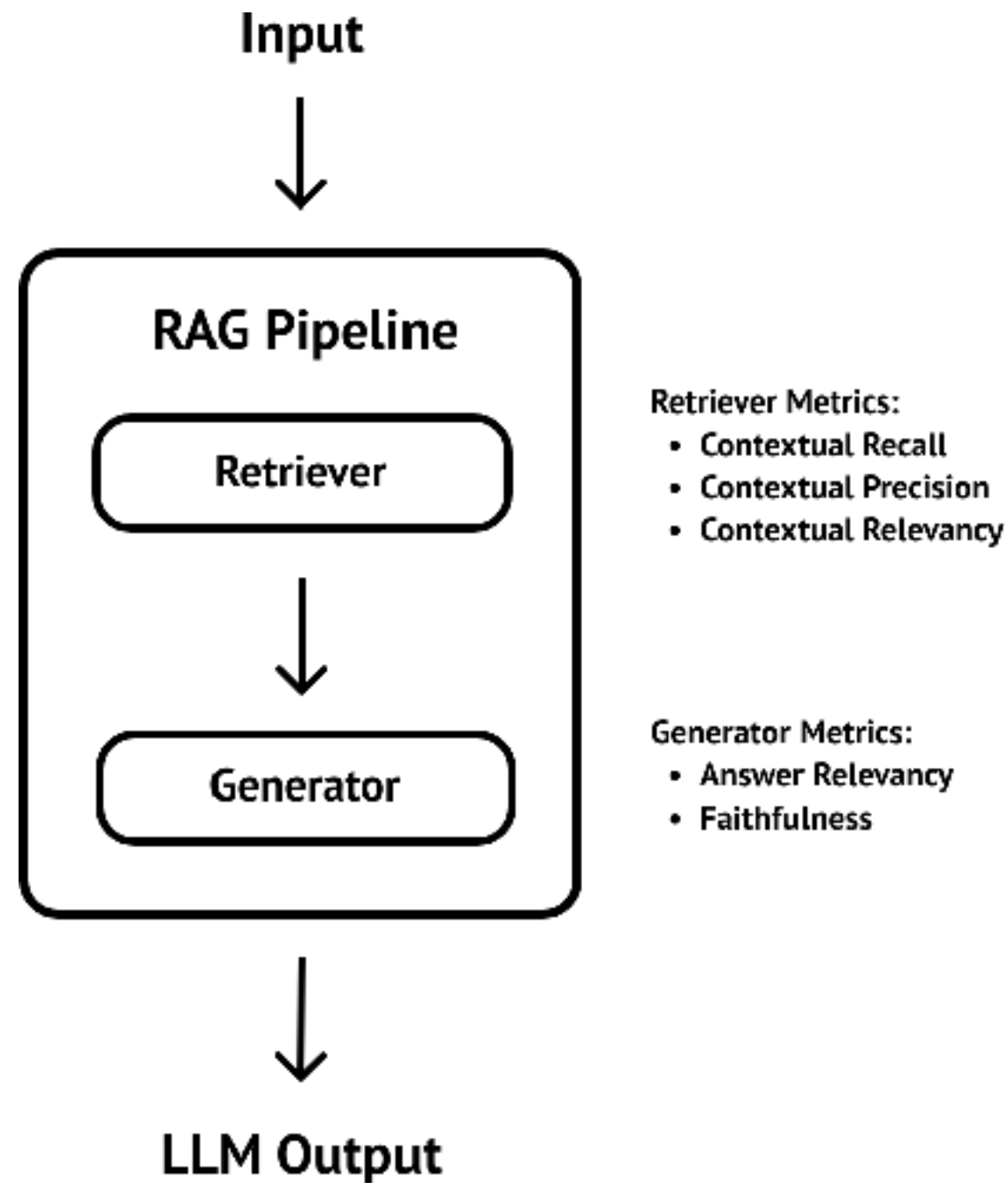
RAG Cookbook (techniques)



<https://github.com/athina-ai/rag-cookbooks>

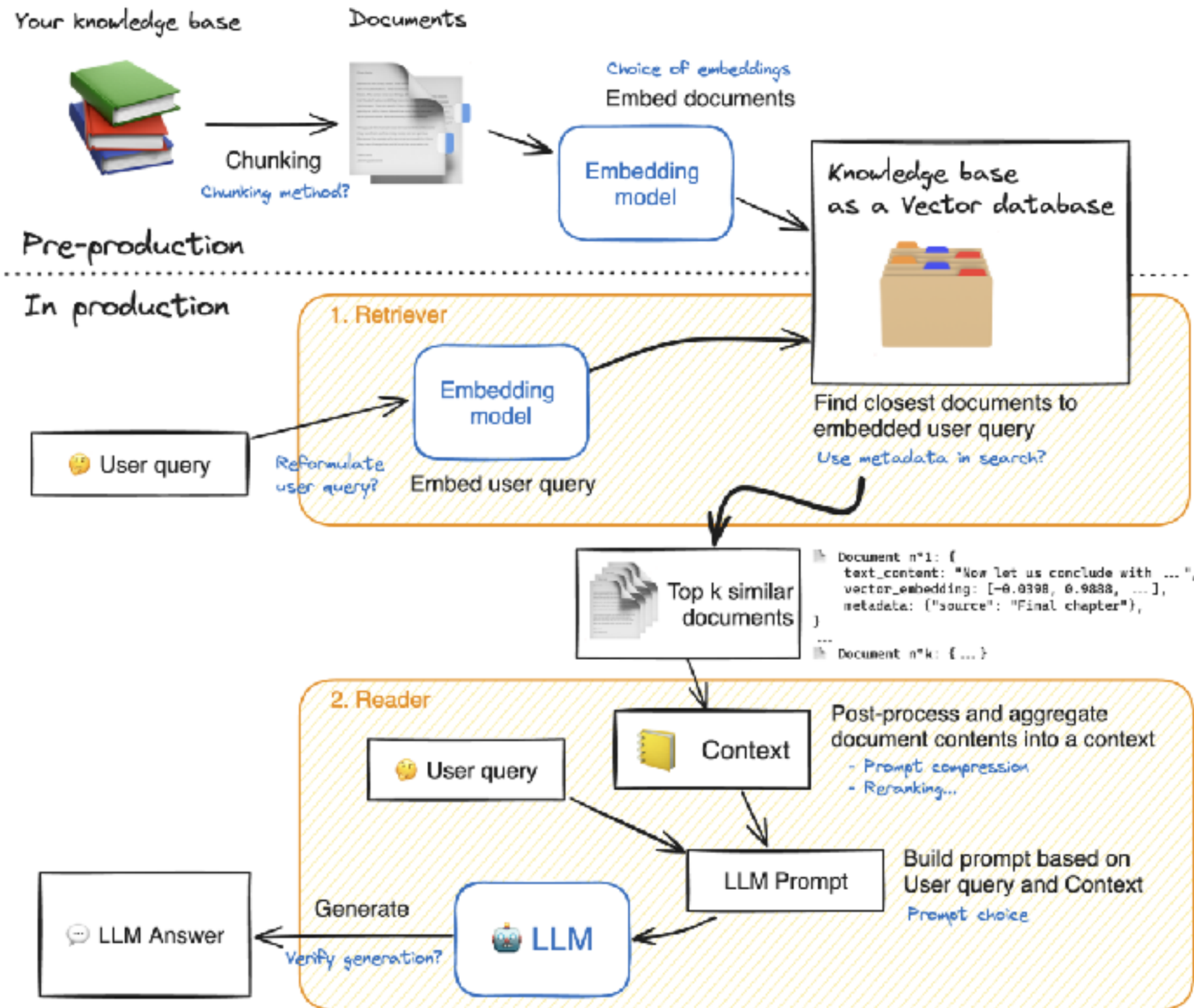


RAG Evaluation !!



<https://www.deepeval.com/guides/guides-rag-evaluation>





https://huggingface.co/learn/cookbook/en/rag_evaluation



RAG Implementation

Document
loading

Splitting
or
Chunking

Embedding

Store in
Vector DB

Retrieval and Re-ranking

Send Prompt to LLM

Output



Failure Points of RAG

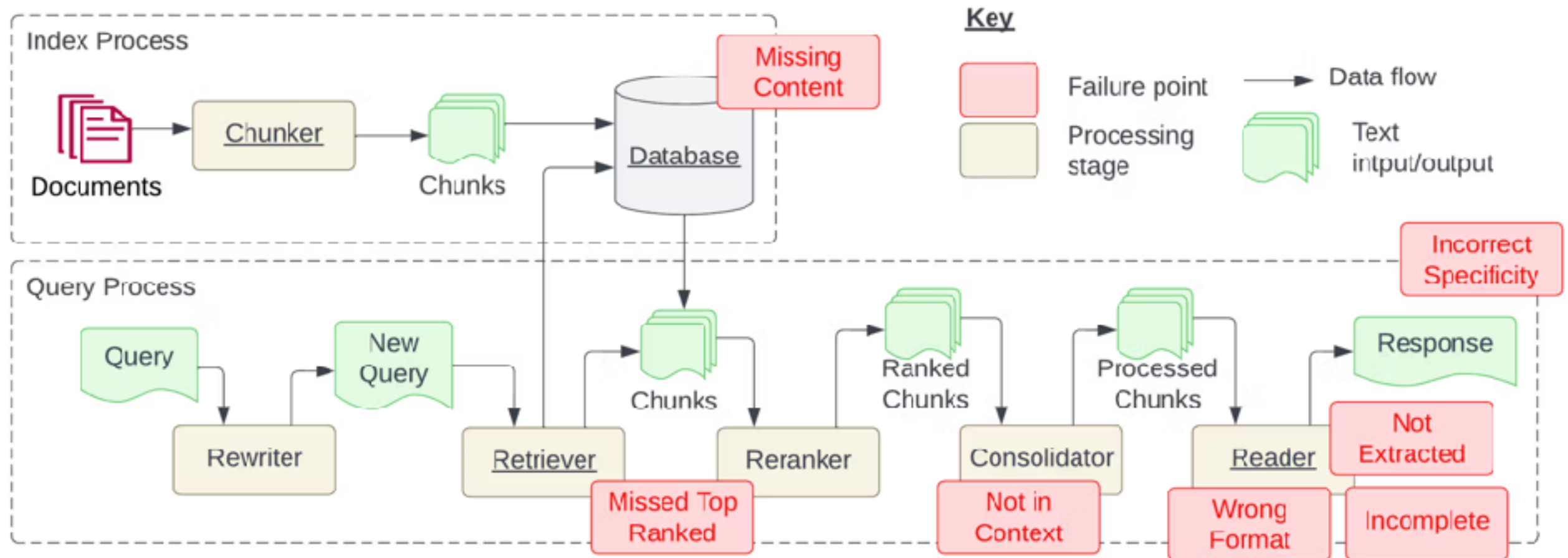


Figure 1: Indexing and Query processes required for creating a Retrieval Augmented Generation (RAG) system. The indexing process is typically done at development time and queries at runtime. Failure points identified in this study are shown in red boxes. All required stages are underlined. Figure expanded from [19].

<https://www.galileo.ai/blog/mastering-rag-how-to-architect-an-enterprise-rag-system>



RAG is better

But come with Cost !!

More latency

Retrieval errors

Accuracy !!

More
Complexity

Maintenance
overhead



RAG Techniques ?

Semantic
chunking

Chunk size
selector

Context chunk
header

Adaptive RAG

Re-ranking

Graph TAG

<https://github.com/FareedKhan-dev/all-rag-techniques>



Pre-Retrieval optimization ?

Preprocessing and cleansing data

Chunking strategies

Add metadata

Embedding model selection



Chunking strategies

Size of Document
> **context window size**



Chunking

Chunk
1

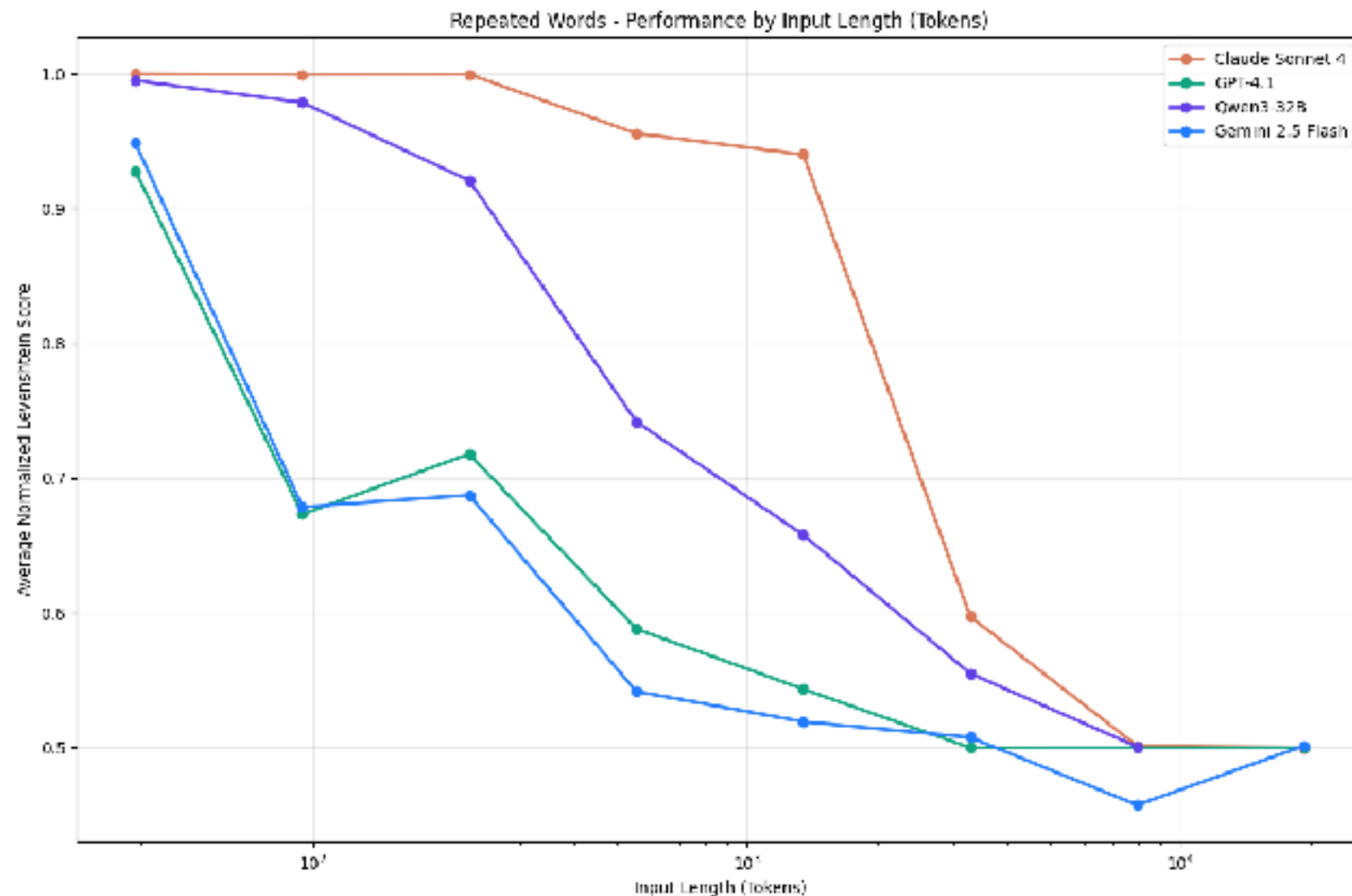
Chunk
2

Chunk
3



Context Window ?

Number of tokens in the context window increases,
the model's ability to accurately recall information from that context decreases



<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>



Chunking Strategies !!

Fixed size
Recursive characters
Document structure-based
Semantic chunking
Agentic chunking
...

<https://blog.dailydoseofds.com/p/5-chunking-strategies-for-rag>



Good Chunking

Efficiency

Relevance

Context
preservation

Improve content
generation

Reduce noise



Bad Chunking

Loss context

Redundancy

Inconsistency



Chunking Visualization

ChunkViz v0.1

Want to learn more about AI Engineering Patterns? Join me on [Twitter](#) or [Newsletter](#).

Language Models do better when they're focused.

One strategy is to pass a relevant subset (chunk) of your full data. There are many ways to chunk text.

This is an tool to understand different chunking/splitting strategies.

[Explain like I'm 5...](#)

One of the most important things I didn't understand about the world when I was a child is the degree to which the returns for performance are superlinear.

Teachers and coaches implicitly told us the returns were linear. "You get out," I heard a thousand times, "what you put in." They meant well, but this is rarely true. If your product is only half as good as your competitor's, you don't get half as many customers. You get no customers, and you go out of business.

It's obviously true that the returns for performance are superlinear in business. Some think this is a flaw of capitalism, and that if we changed the rules it would stop being true. But superlinear returns for performance are a feature of the world, not an artifact of rules we've invented. We see the same pattern in fame, power, military victories, knowledge, and

Upload text

Splitter: Character Splitter
Chunk Size: 25
Chunk Overlap: 0
Total Characters: 2658
Number of chunks: 107
Average chunk size: 24.8

One of the most important things I didn't understand about the world when I was a child is the degree to which the returns for performance are superlinear.

Teachers and coaches implicitly told us the returns were linear. "You get out," I heard a thousand times, "what you put in." They meant well, but this is rarely true. If your product is only half as good as your competitor's, you don't get half as many customers. You get no customers, and you go out of business.

<https://chunkviz.up.railway.app/>



Retrieval optimization ?

Re-ranking

Hybrid search

Query
transformation

Multi-vector
embedding

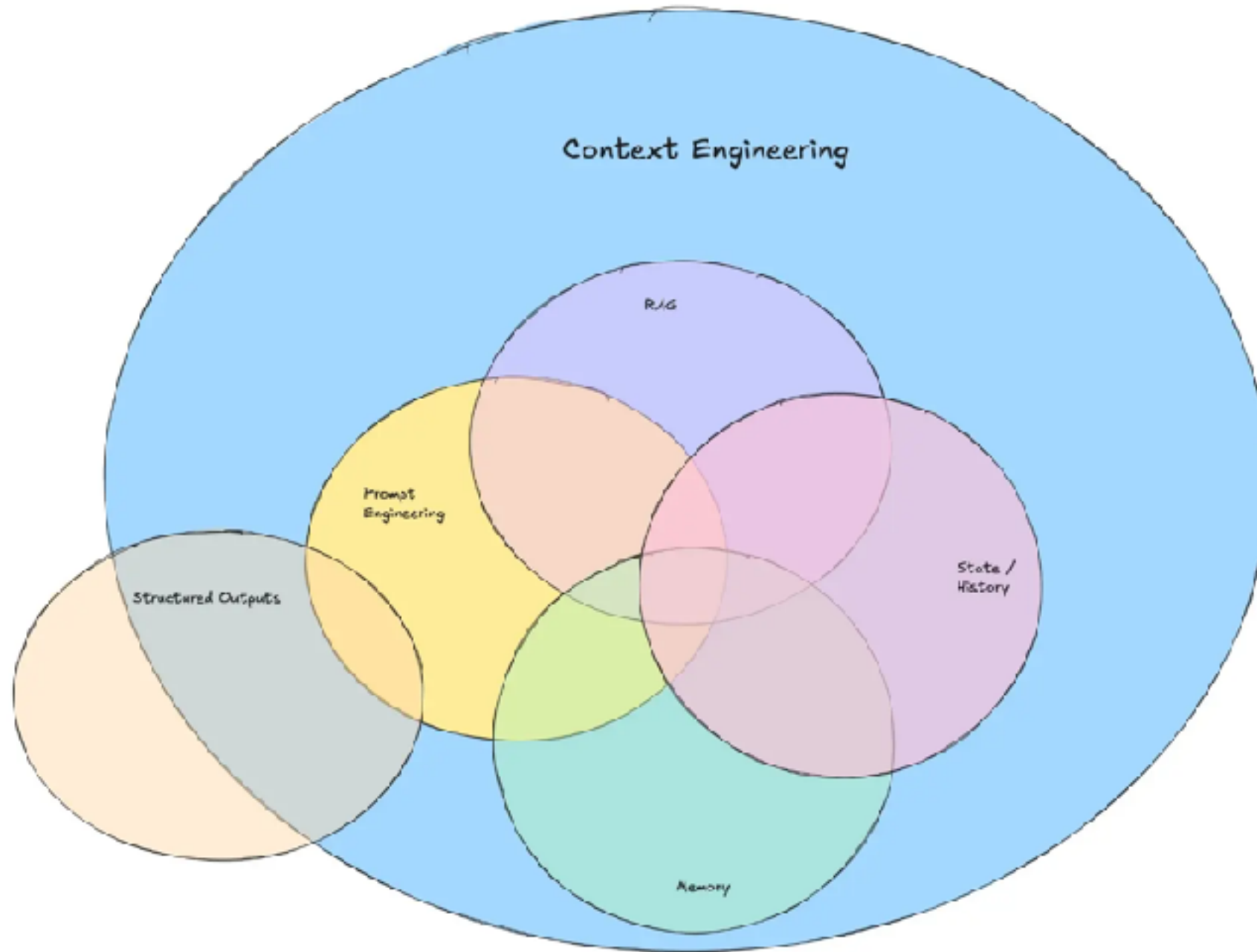
Contextual
retrieval

Vector database
Selection





Context Engineering



<https://www.promptingguide.ai/guides/context-engineering-guide>

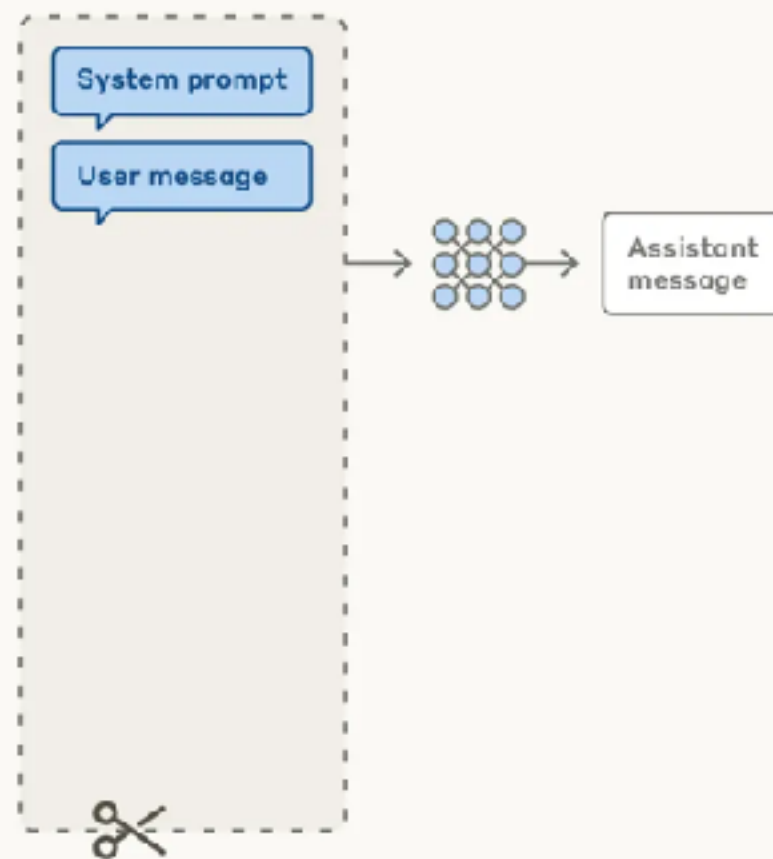


Effective Context Engineering

Prompt engineering vs. context engineering

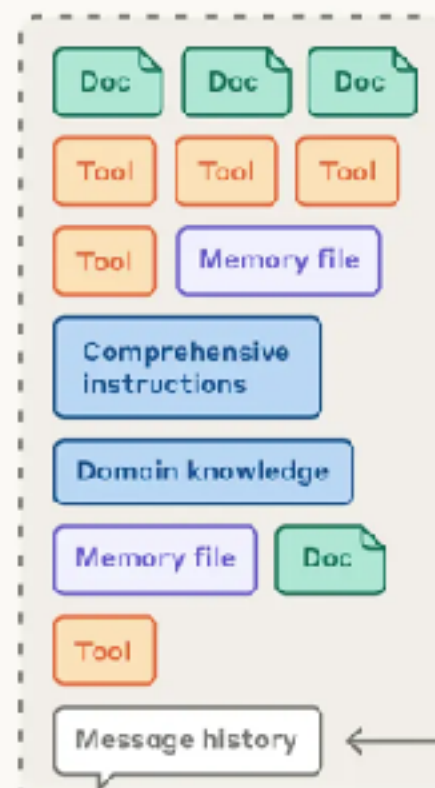
Prompt engineering for single turn queries

Context window



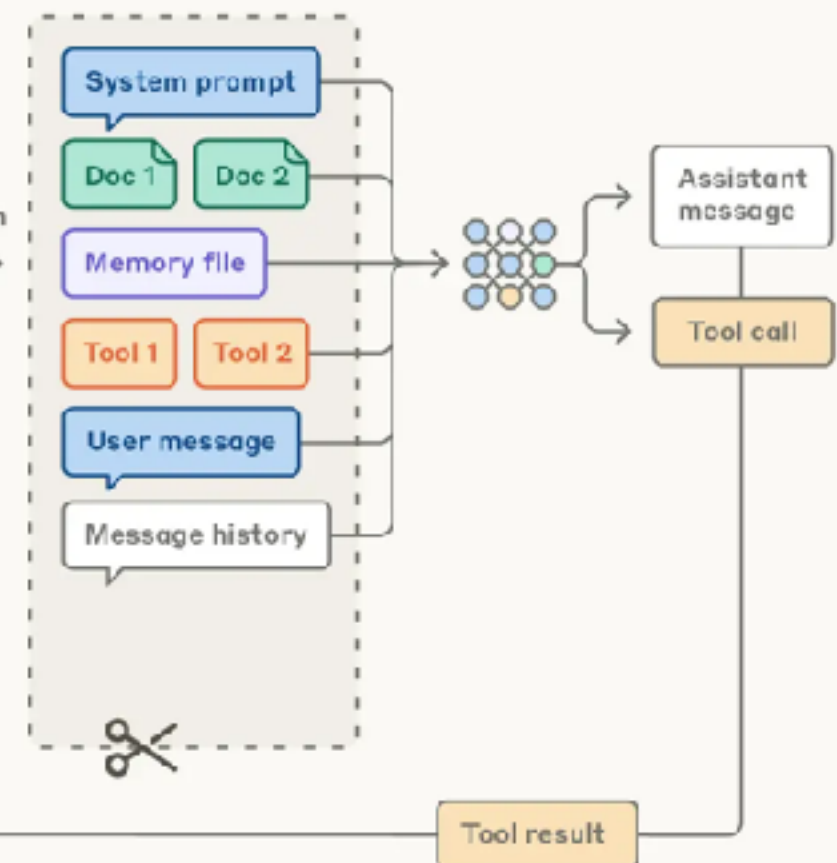
Context engineering for agents

Possible context to give model



Curation

Context window



<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

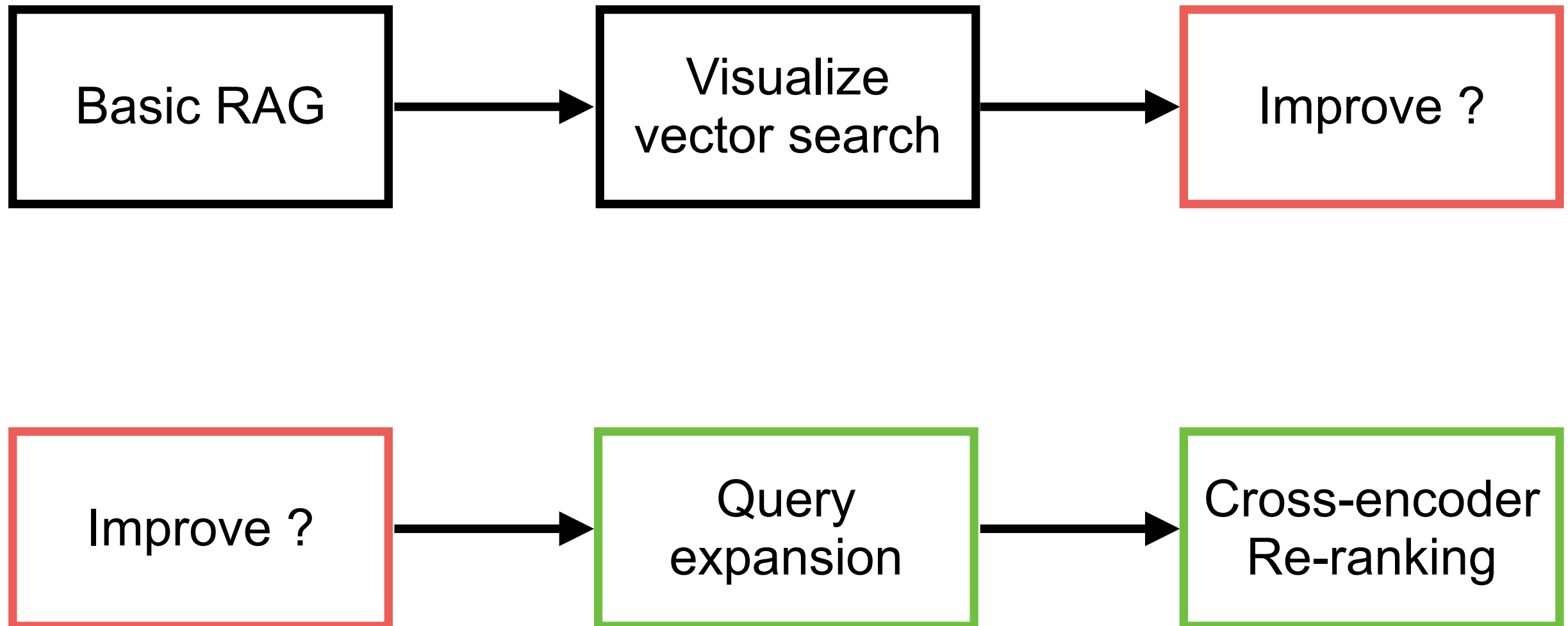


RAG Workshop

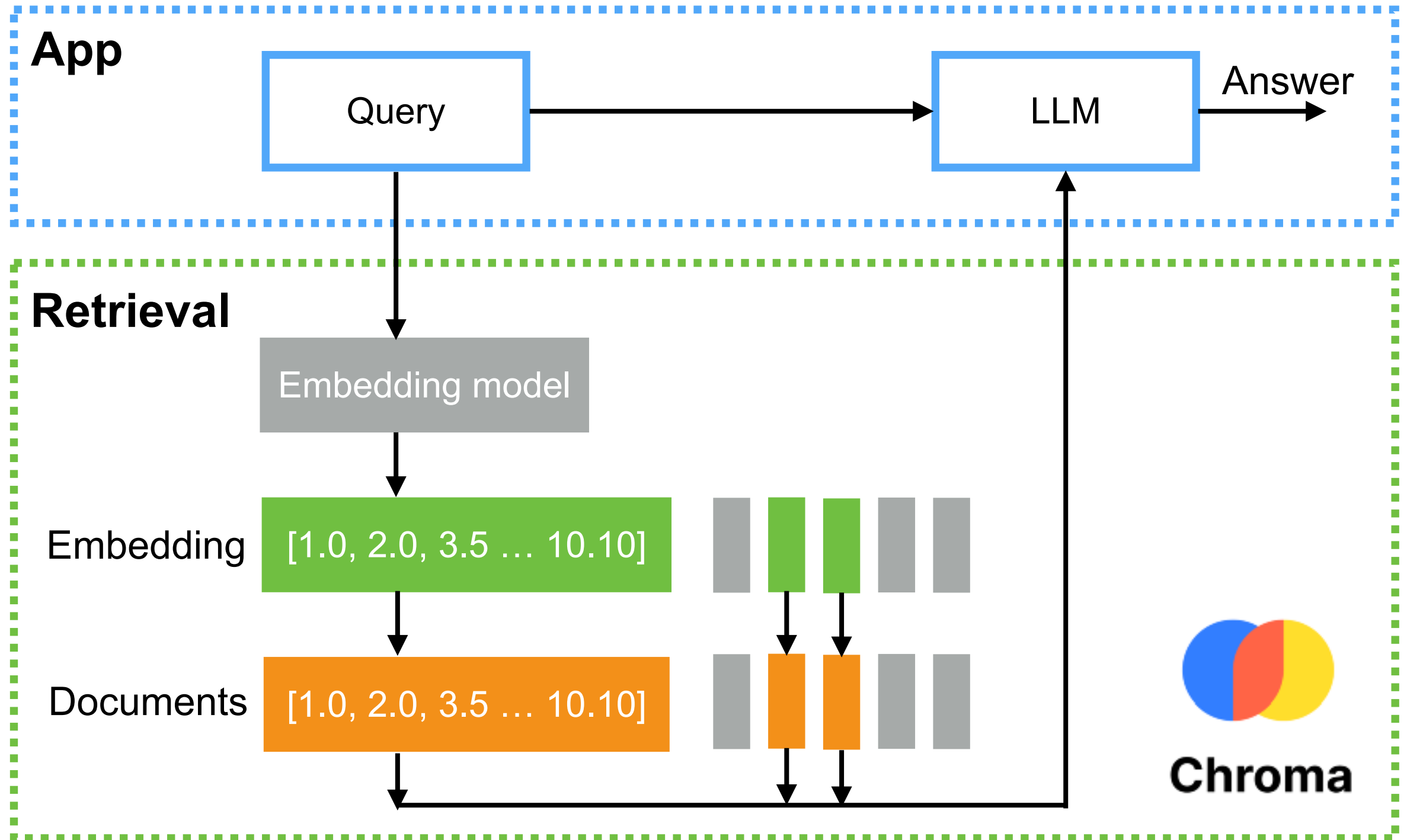
<https://github.com/up1/workshop-basic-llm/tree/main/workshop/basic-rag>



RAG workshop



Basic RAG



Query Expansion

Query expansion is a widely used technique to improve the recall of search systems

Ambiguity

Vocabulary
mismatch

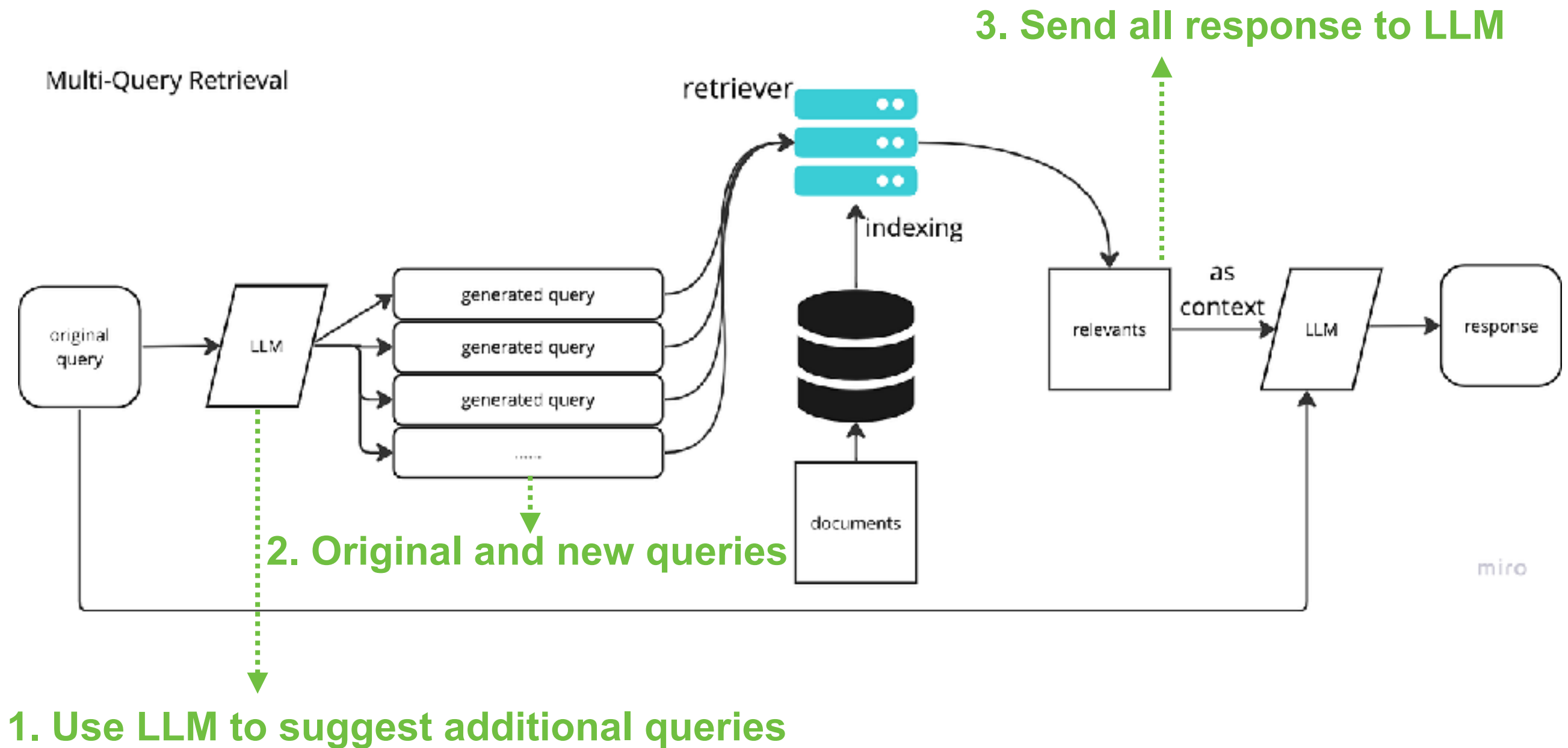
Lack of context

“Add synonyms, related context and contextual terms”

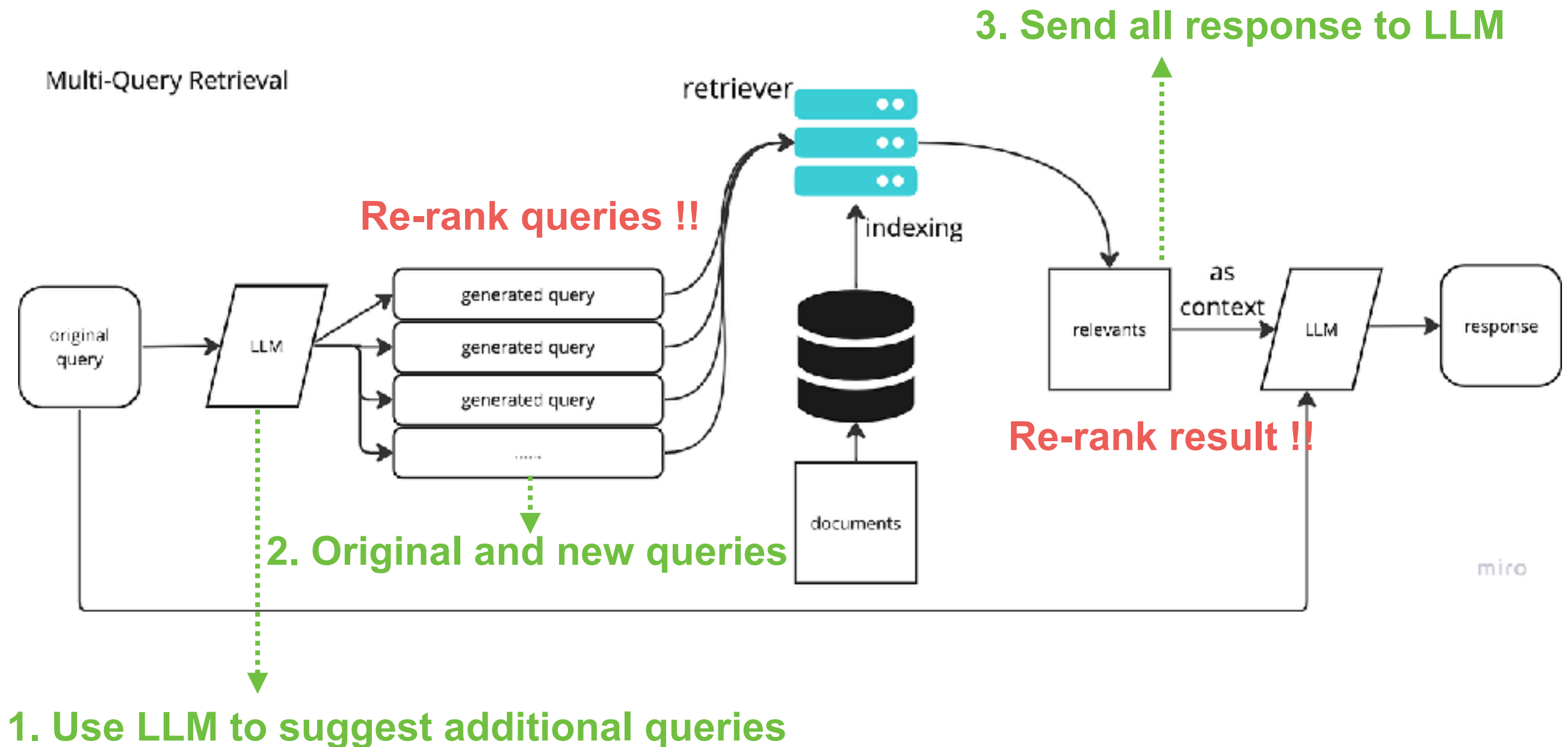
<https://arxiv.org/abs/2305.03653>



Query Expansion

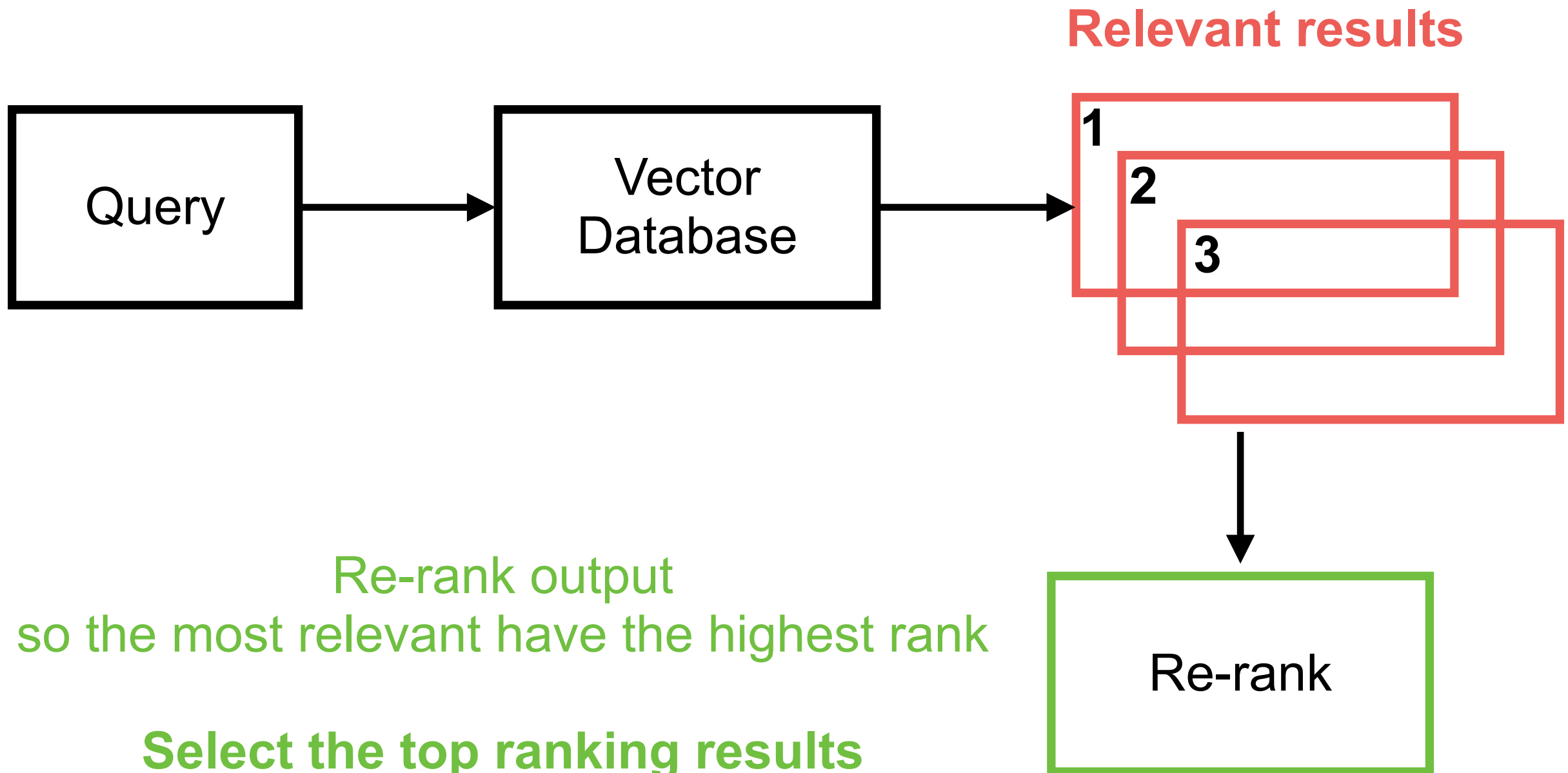


Re-rank !!

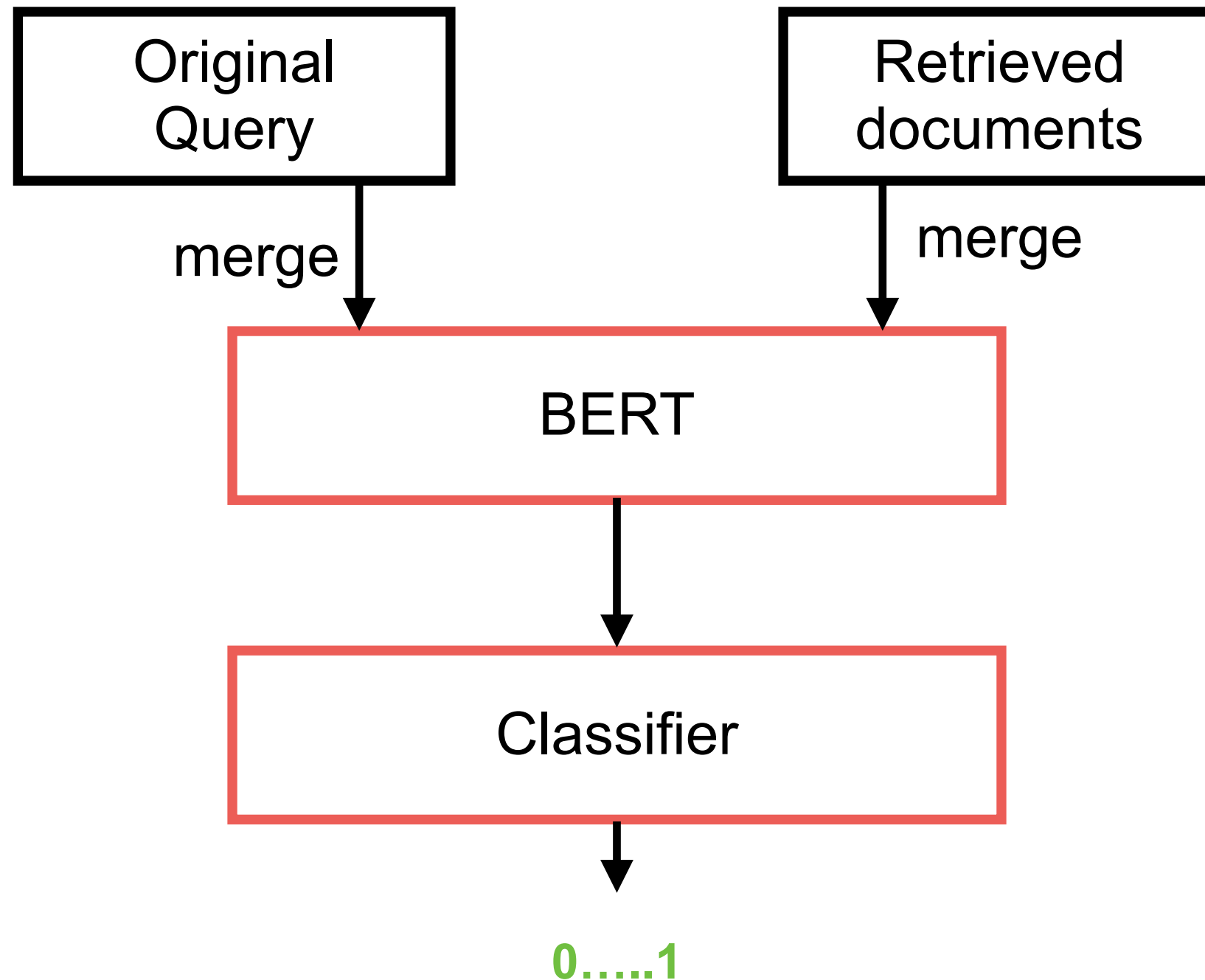


Cross-encoder Re-ranking

How to ordering relevant results !!

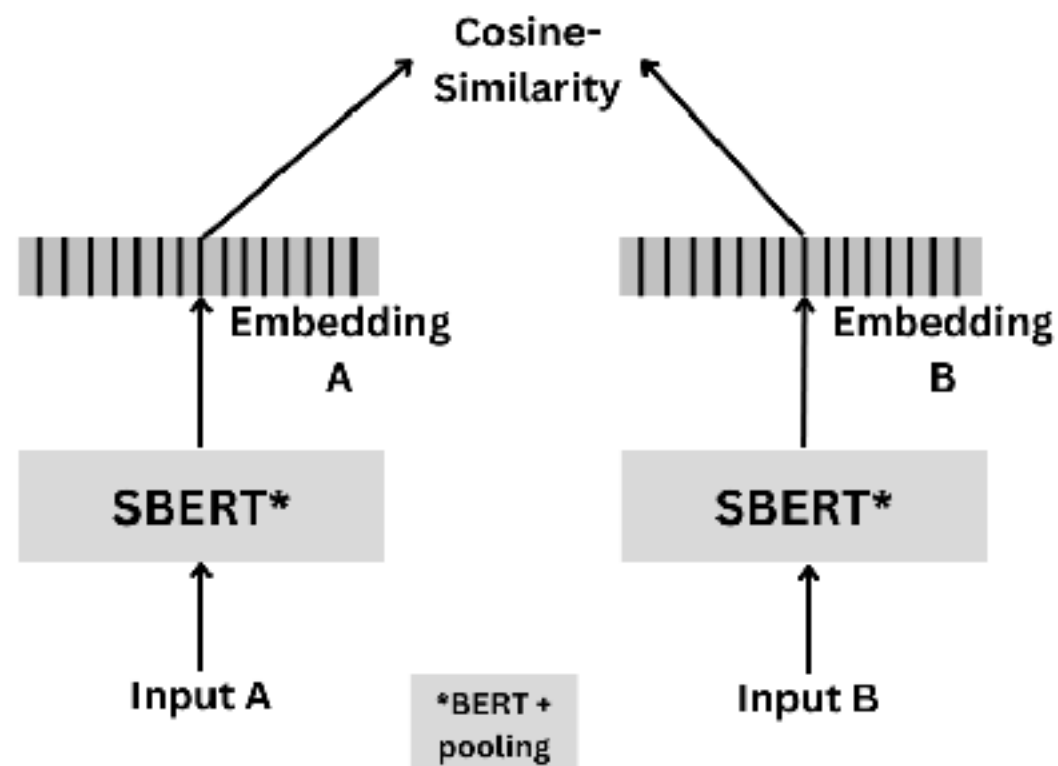


Cross-Encoder

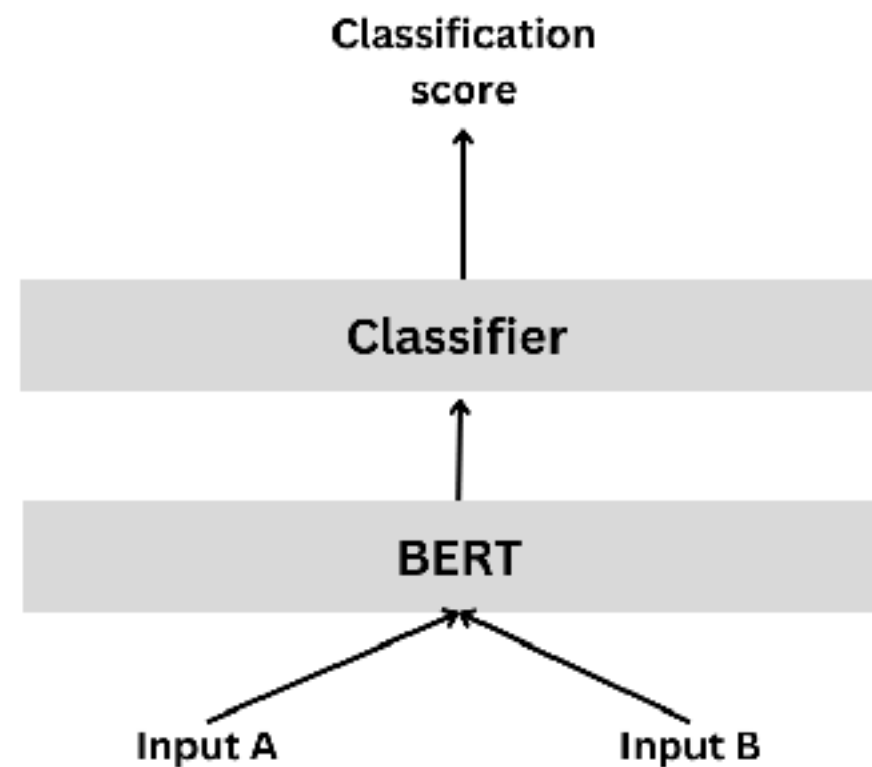


Cross-Encoder !!

Bi-encoder



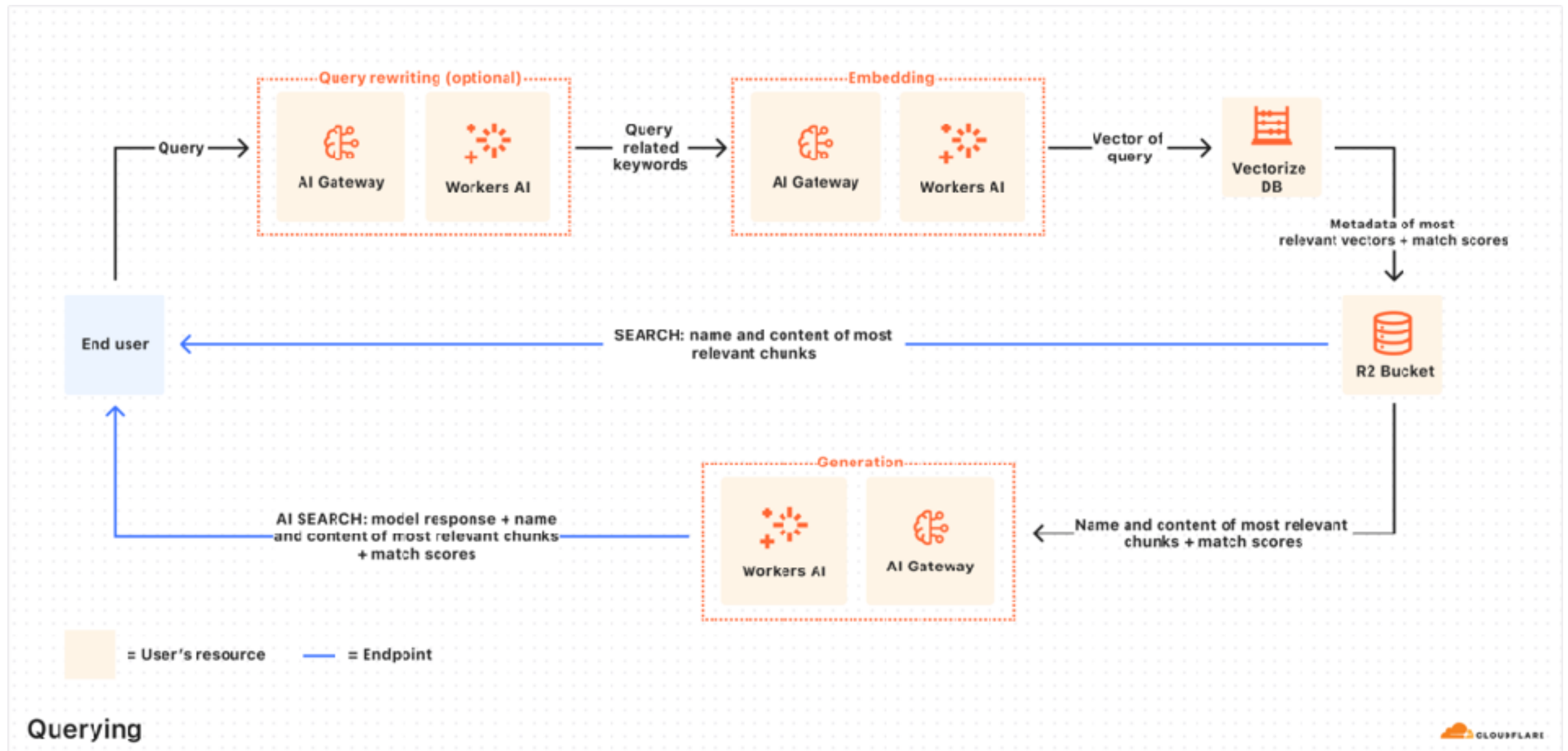
Cross Encoder



https://sbert.net/examples/cross_encoder/applications/README.html



Cloudflare AutoRAG



<https://developers.cloudflare.com/autorag/>



Guardrails



<https://github.com/guardrails-ai/guardrails>



Guardrails

Help to build reliable AI applications

Guard for
input and output

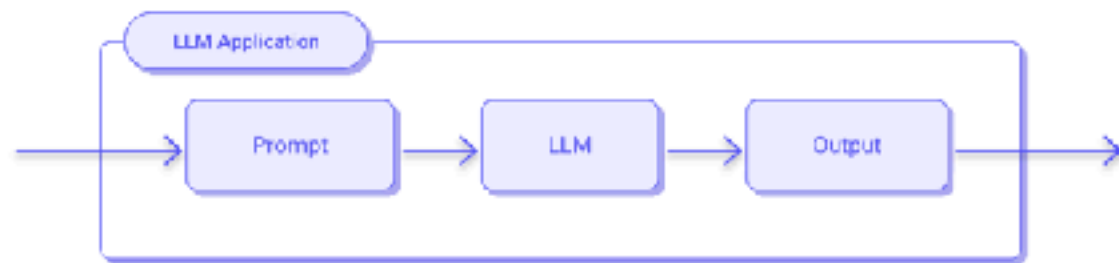
Help to generate
Structured output

<https://github.com/guardrails-ai/guardrails>

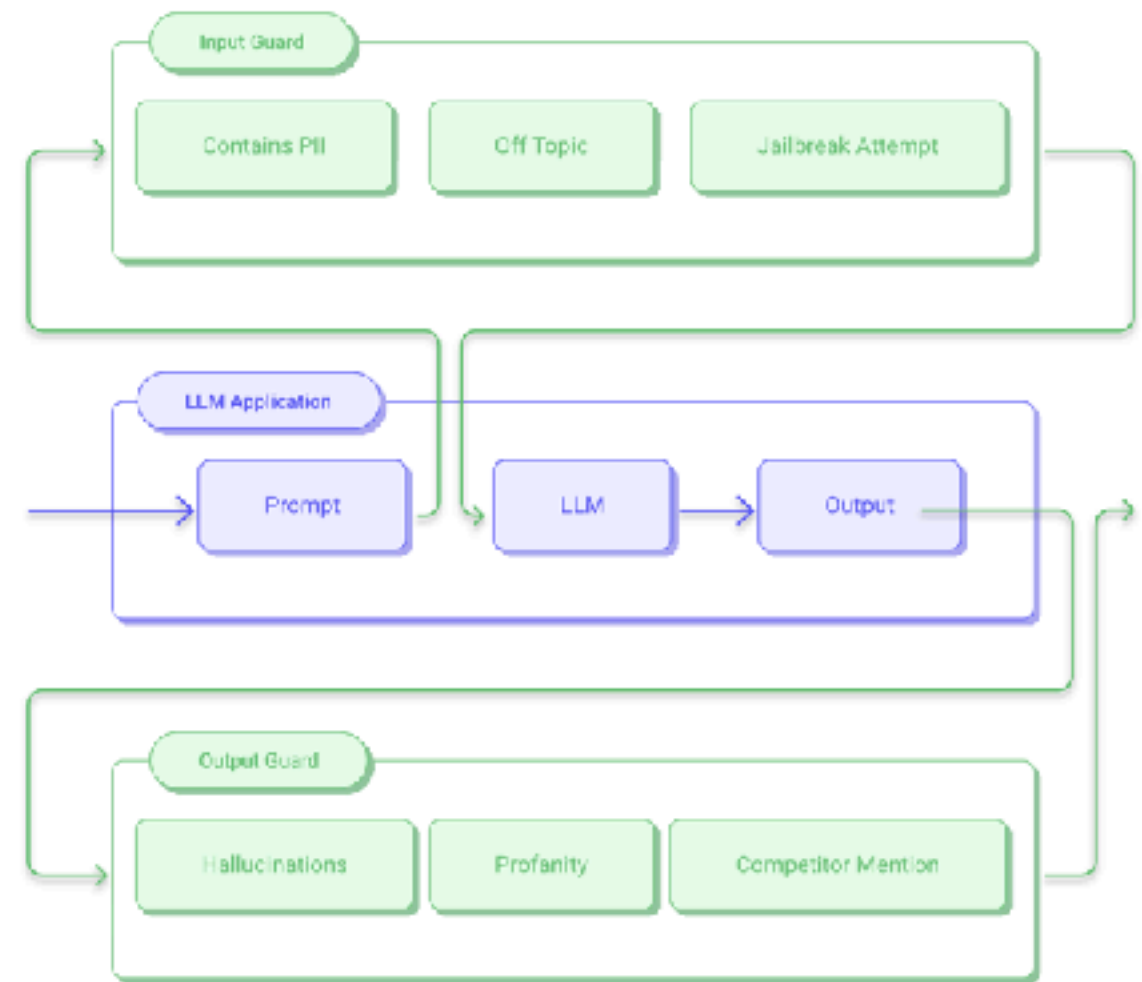


Guardrails

Without Guardrails



With Guardrails



Guardrails Hub

The screenshot displays the Guardrails AI Hub interface. On the left, there are several filter categories: **USE CASES** (Chatbots, Customer Support, Structured Data, RAG, Summarization, Codegen, Text2SQL), **RISK CATEGORY** (Etiquette, Brand Risk, Factuality, Formatting, Invalid Code, Jailbreaking, Code Exploits, Data Leakage), **INFRASTRUCTURE REQUIREMENTS** (ML, LLM, NA, Rule), **CONTENT TYPE** (String, Object, List, Integer, Float, SQL, Code, CSV, Python), **CERTIFICATION** (Guardrails Certified), and **LANGUAGE** (EN). The main section is titled **Validators** and includes a search bar, a 'Generate Code' button, and a list of validators. Each validator card shows its name, description, last update time, and associated risk categories. The validators listed are: Arize Dataset Embeddings, Ben List, Bespoke MiniCheck, Bias Check, Competitor Check, Correct Language, Cucumber Expression Match, and Detect PII.

Guardrails AI Hub Blog Docs [Sign in to get started](#)

Validators

Validators are basic Guardrails components that are used to validate an aspect of an LLM workflow. Validators can be used a to prevent end-users from seeing the results of faulty or unsafe LLM responses.

Search

Showing 66 of 66 validators

[Generate Code](#)

Arize Dataset Embeddings [Select](#)

Validates that user-generated input does not match the dataset of jailbreak...
Last updated 8 months ago

[STRING](#) [BRAND RISK](#) [ML](#)

Bespoke MiniCheck [Select](#)

Validates that the LLM-generated text is supported by the provided context using...
Last updated 7 months ago

[STRING](#) [BRAND RISK](#) [FACTUALITY](#) [ML](#)

Competitor Check [Select](#)

Flags mentions of competitors. Fixes responses by filtering out competitor names.
Last updated 5 months ago

[STRING](#) [BRAND RISK](#) [ML](#)

Cucumber Expression Match [Select](#)

Validates that the input string matches a specified cucumber expression.
Last updated 6 months ago

[STRING](#) [BRAND RISK](#) [ML](#)

Ben List [Select](#)

Validates that the output does not contain banned words, using fuzzy search.
Last updated 8 months ago

[STRING](#) [BRAND RISK](#) [ML](#)

Bias Check [Select](#)

Validates that the text is free from biases related to age, gender, sex, ethnicity,...
Last updated 1 week ago

[STRING](#) [BRAND RISK](#) [ML](#)

Correct Language [Select](#)

Validate that an LLM-generated text is in the expected language. If the text is not ...
Last updated 11 months ago

[STRING](#) [ETIQUETTE](#) [ML](#)

Detect PII [Select](#)

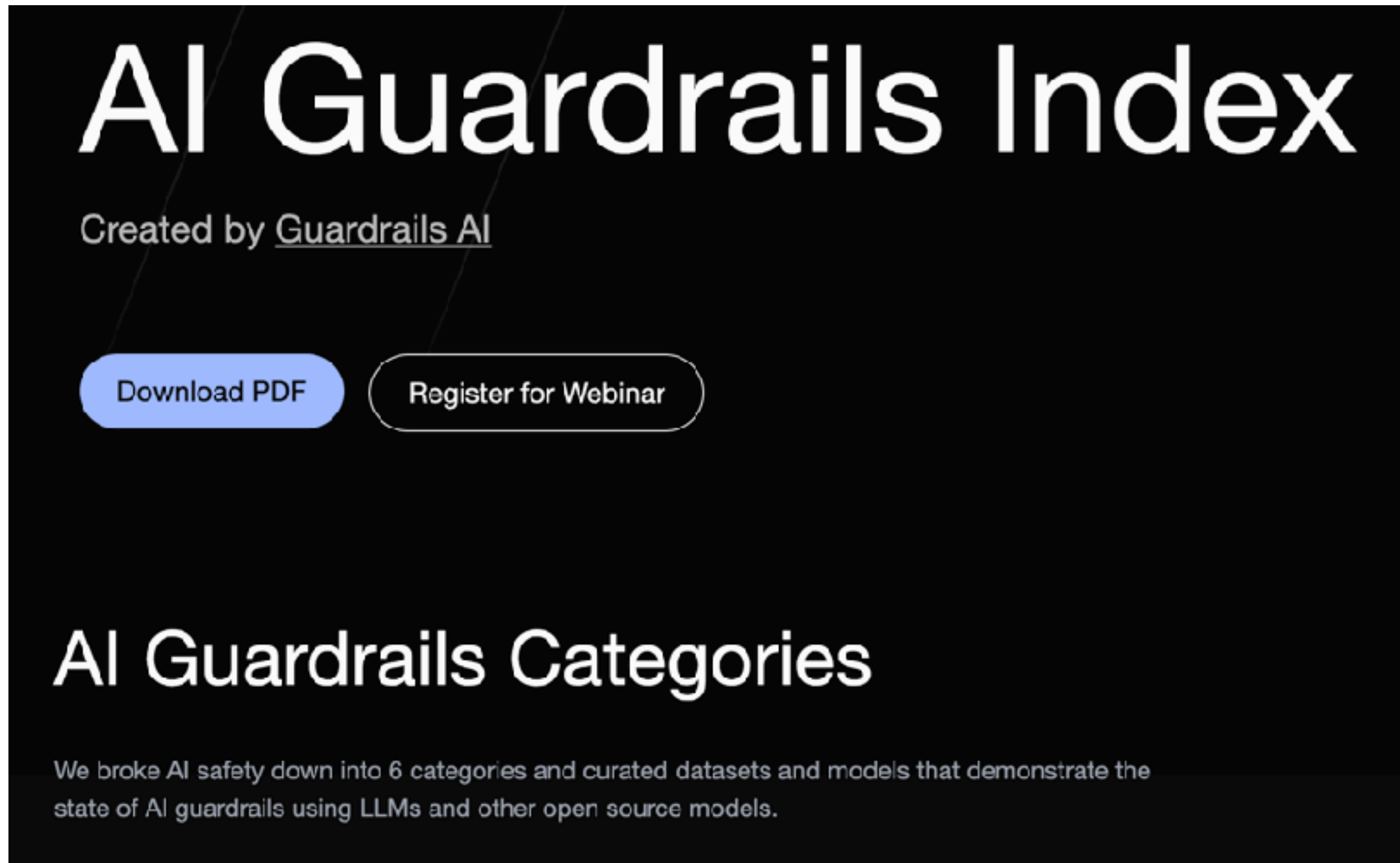
Detects personally identifiable information (PII) in text, using Microsoft Presidio.
Last updated 6 months ago

[STRING](#) [DATA LEAKAGE](#) [ML](#)

<https://hub.guardrailsai.com/>



Guardrails Index

A screenshot of the AI Guardrails Index website. The background is dark with white text. At the top, the title 'AI Guardrails Index' is in a large, bold font. Below it, in a smaller font, is 'Created by Guardrails AI'. There are two buttons: a blue one labeled 'Download PDF' and a white one with a black border labeled 'Register for Webinar'. Further down, the section 'AI Guardrails Categories' is shown in a large font. At the bottom of this section, a paragraph reads: 'We broke AI safety down into 6 categories and curated datasets and models that demonstrate the state of AI guardrails using LLMs and other open source models.'

<https://index.guardrailsai.com/>

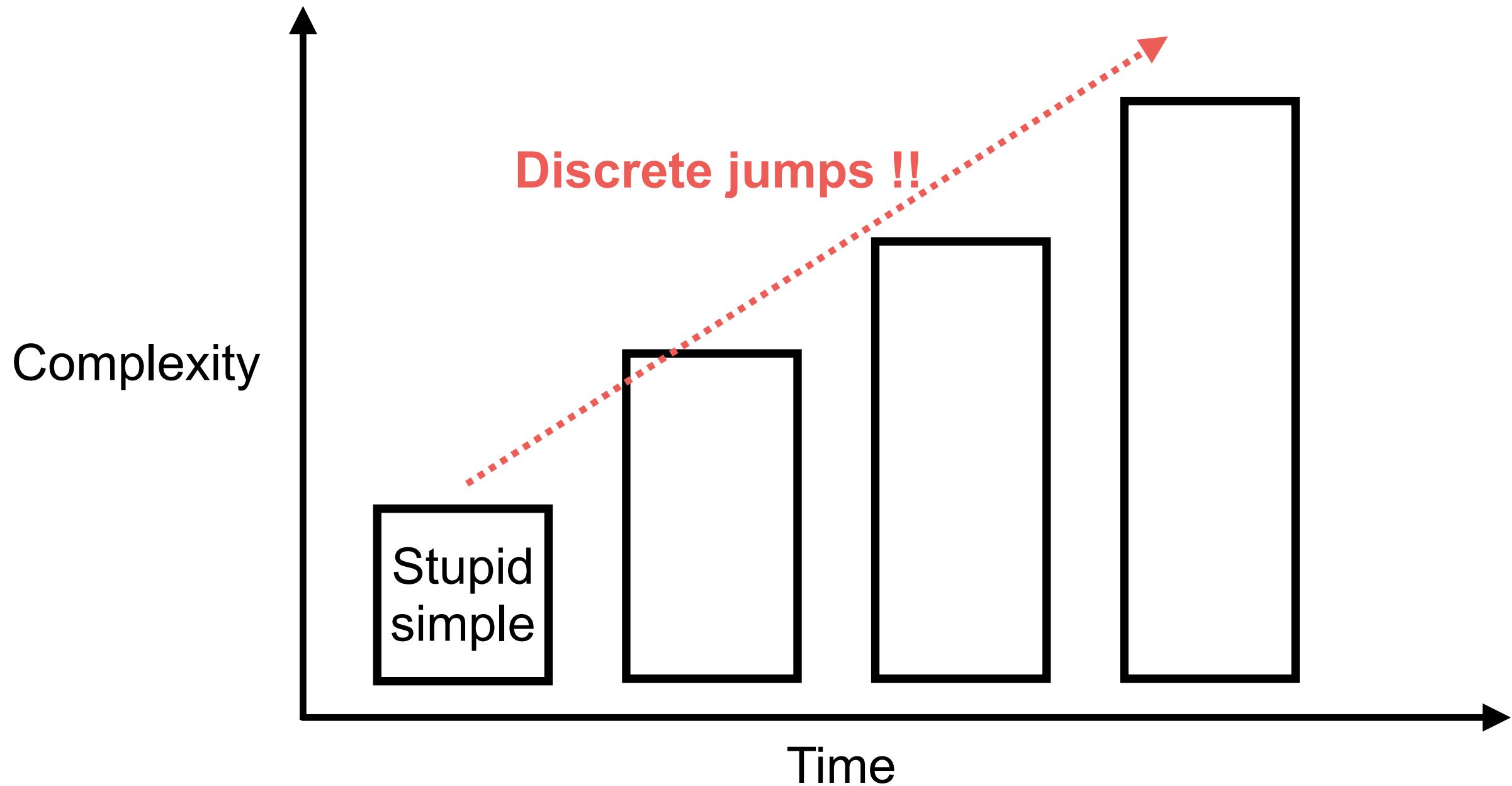


PDCA process

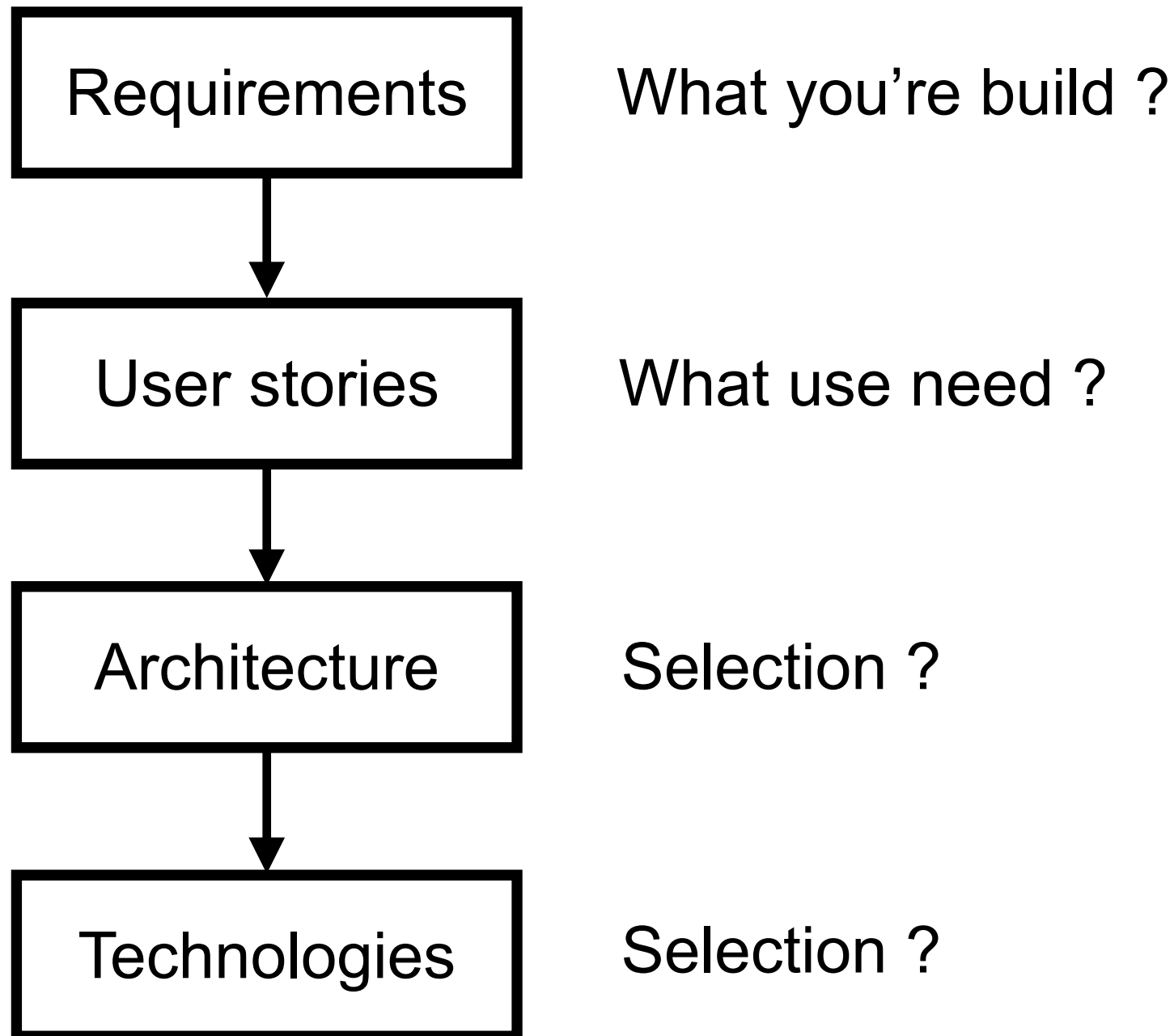


Staged Complexity

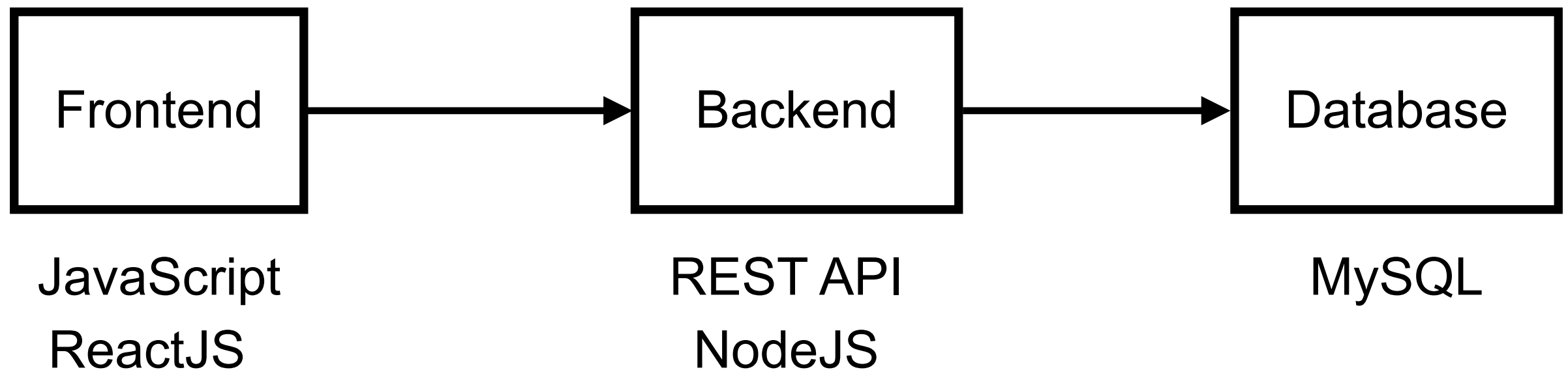
Start small and get a feature working first



Planning workflow



Architecture of project

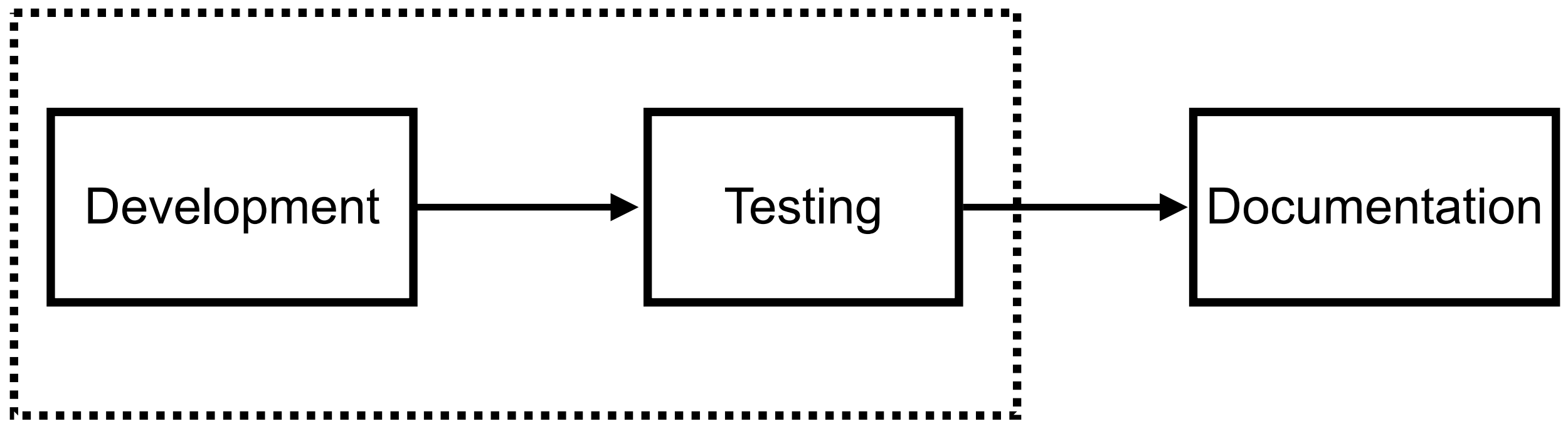


Choose your tech stack !!



Processes

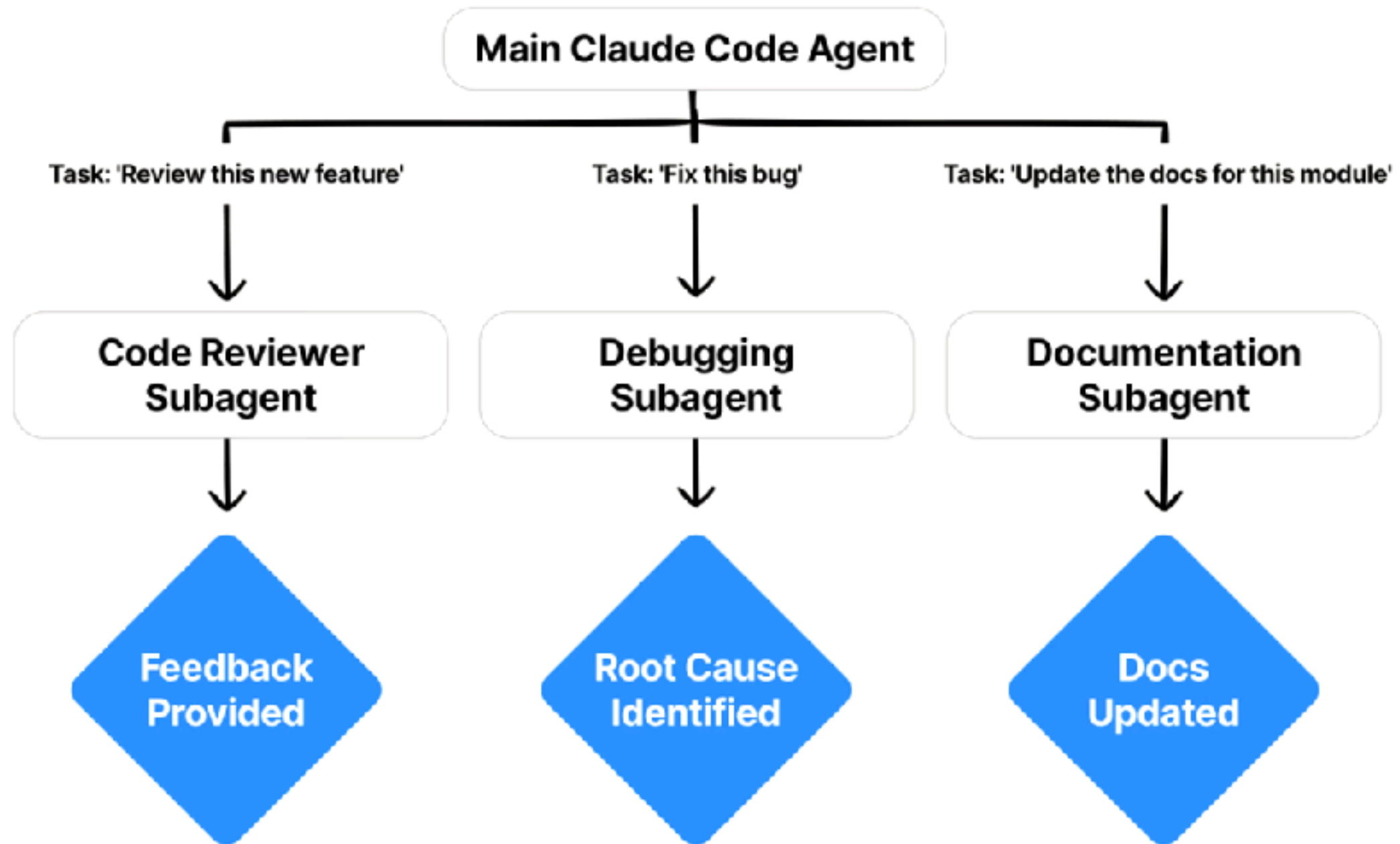
Iterative and incremental process



AI Subagent



AI Subagent



<https://code.claude.com/docs/en/sub-agents>



AI Subagent



<https://github.com/VoltAgent/awesome-claude-code-subagents>



Start your journey

